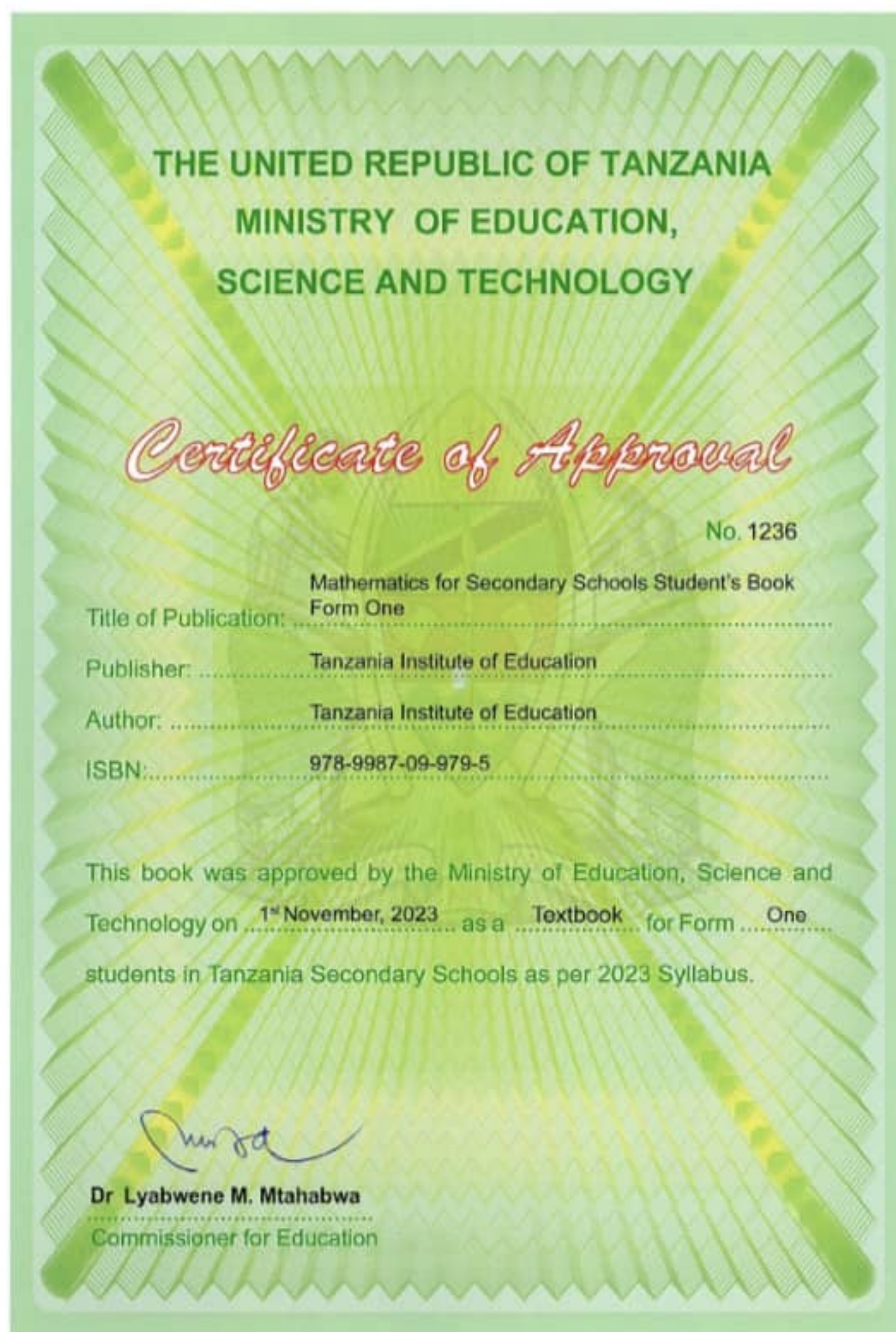


Mathematics

for Secondary Schools

Student's Book

Form One



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Tanzania Institute of Education

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Dr Aneth A. Komba

Director General

Tanzania Institute of Education

Preface

This textbook, *Mathematics for Secondary Schools*, is written specifically for Form One students in the United Republic of Tanzania. The book is prepared in accordance with the 2023 Mathematics Syllabus for Ordinary Secondary Education, Form I-IV, issued by the Ministry of Education, Science and Technology. It is a revised edition of Basic Mathematics for Secondary Schools Student's Book for Form One that was published in 2019 in accordance with the 2005 syllabus issued by the then, Ministry of Education and Vocational Training (MoEVT).

The book consists of six (6) chapters, namely: Concept of Mathematics, Numbers, Approximations, Ratios and proportions, Algebra, and Coordinate geometry. In addition to the contents, each chapter contains activities, illustrations, projects, and exercises. You are encouraged to do all the activities, projects and attempt all questions in the exercises. This will enhance your understanding and development of the intended competencies for this level.

Additional learning resources are available in the TIE library at <https://ol.tie.go.tz> or ol.tie.go.tz



Tanzania Institute of Education

Chapter One

Concept of Mathematics

Introduction

Mathematics is all around us. Almost everything we do involves Mathematics. It is applied in various fields such as science, engineering, technology, social studies, economics and many others. Numerous devices such as mobile phones, computers, television sets, and satellites are designed and manufactured on the basis of mathematical knowledge. In this chapter, you will learn about the meaning of Mathematics, branches of Mathematics, relationship between Mathematics and other subjects, and the importance of studying Mathematics. The competencies developed will enable you to apply Mathematics knowledge and skills to solve daily life problems in different fields such as Agriculture, Entrepreneurship, Botany, Zoology, Music, History, Business, and many other fields.

Meaning of Mathematics

The word 'Mathematics' comes from the Greek word '*mathema*', meaning 'which is learnt' or 'science, knowledge or learning'. Numbers, measurements, shapes of physical objects, and equations form a small part of it. Mathematics can be thought of as the science of the structures, orders, patterns, and relations that have evolved from elementary practices of counting, measuring, and describing the shapes of objects. It can also be thought of as a language of science because by using mathematical reasoning, one can develop an understanding and be able to predict nature. Mathematics has the ability to provide precise expression for every concept that can be formulated using mathematical symbols and structures. The knowledge and skills of Mathematics play a crucial role in understanding the concepts of other subjects, both in sciences and arts.

Branches of Mathematics

Mathematics can be categorized into different branches such as Arithmetic, Algebra, and Geometry.

Arithmetic

Arithmetic is a branch of Mathematics that deals with properties and manipulations of numbers. Manipulation of numbers is achieved through the use of basic mathematical operations namely: addition, subtraction, multiplication, and division.

Algebra

Algebra is a branch of Mathematics in which arithmetic operations are applied to symbols rather than specific numbers. The symbols or letters in algebra are called variables which represent quantities with no fixed values.

Geometry

The word geometry was derived from the Greek word 'Geo', which means 'earth' and 'metry', which means 'measurement'. Therefore, Geometry is a branch of Mathematics which deals with the study of the sizes, shapes, positions, angles, and dimensions of different physical objects. Moreover, properties of points, lines, planes, similarities, congruence, and shapes of different regular objects are also studied in Geometry.

Relationship between Mathematics and other subjects

Every aspect of our life makes use of Mathematics in one way or another. Mathematics plays a major role as a tool for effective understanding of other subjects. Numerous concepts in other subjects and fields are described precisely using Mathematics. Learning Mathematics can also benefit students through developing their problem solving and critical thinking skills. Note that, Mathematics formulas which are used to represent and describe concepts and scenarios in other fields are commonly referred to as mathematical models.

Activity 1.1: Discovering the use of Mathematics in daily lives

1. Recall your daily activities, identify and record applications of Mathematics in such activities.
2. Use a table or any method of your choice to represent the identified activities and its mathematical skills obtained in step 1.
3. Explain the mathematical aspects you have observed in step 2.

Mathematics and Agriculture

Agriculture is closely related to Mathematics. For instance, when farmers want to buy seeds, they need to understand the ratio of seeds that is sufficient per piece of land.

Similarly, the determination of the number of bags of fertilizers needed per acre requires some calculations. In these two examples, Mathematics enables farmers to avoid the wastage of financial resources by purchasing only the required amount of inputs. Mathematics is also used in Agriculture to determine suitable amount of water to be used in irrigation and the spacing between seedlings. Similarly, Mathematics is used to determine the investment, expenditure, and savings in cultivating a specific crop, dividing a piece of land, calculating the cost of labour, and so forth.

Mathematics and Biology

There is a direct relationship between Mathematics and Biology. For example, normal animal weights, rate of respiration, nutritive values of food, and transpiration are a few quantities in Biology that can be calculated using mathematical concepts. Mathematics can also be used to estimate the number of blood cells present in the body, measurement of blood pressure, and counting sex chromosomes, among many others.

Mathematics and Chemistry

Mathematics is widely applied in Chemistry to represent and solve various problems. Some Chemistry activities involve measurement of masses, volumes, lengths, temperatures and densities of matter. Mathematics is also used to measure the constituents of mixtures, balancing chemical equations, among many other applications.

Mathematics and Physics

Physics involves the study of laws, principles, and theorems which governs how matter works. Most of the quantities in Physics are expressed mathematically through formulas. In order to understand how to apply the formulas to solve some physics challenges, knowledge of Mathematics is needed. All the quantities in physics are expressed in numbers and units which have to be manipulated using the concepts and skills of Mathematics. For example, to determine the speed of a moving object, one has to find a ratio between the distance covered and the time taken.

Mathematics and Information and Communications Technology

Mathematics plays a big role in the field of Information and Communications Technology (ICT). Computer programs, applications, software, and computer languages make use of mathematical concepts. A common example of the application of numbers is in defining colours in web development languages. For instance, the HTML (hypertext markup language) and CSS (cascading style sheets) uses colour notations such as #000000 which represent Black, #008000 for Green and #800000 for Maroon. Furthermore, the digits 0 and 1 are commonly used in computing systems in computers.

Mathematics and Business Studies

Mathematics is widely applied in daily life activities related to Business Studies. For example, if one wants to determine profit or loss in business, a difference between the selling and buying prices is to be calculated. If the buying price is less than the selling price, the business makes profit. If the buying price is higher than the selling price, the business makes loss. Mathematics is also applied in other areas such as in loan borrowing, determining prices of items, among many other applications.

Mathematics and Geography

Geographers require mathematical calculations to find the distance from one place to another, finding gradients, altitudes of hills and mountains. Through mathematical calculations, geographical locations of different places are determined using latitudes and longitudes, and real-life objects such as buildings are represented on a map through the use of scales.

Mathematics and History

Mathematics helps in describing various historical activities such as duration of events and expressing historical events which happened at different points in time. For example, a simple mathematical operation is used to determine duration of historical events such as World Wars, colonial periods and time spent by leaders in positions, determining dates and ages of fossils by using some mathematical principles such as carbon-14 among many other applications.

Mathematics and Literature

Mastering basic concepts of arithmetic can enable a person to understand better and manage literary works. A kind of writing which can draw attention to readers depend on several factors such as number of words per sentence, the number of sentences per paragraph, and the number of paragraphs per page. For example, 20 words in a sentence are considered as average. Thus, in analysing writing style, one need to find an average number of words in each sentence, which requires Mathematics concepts.

Mathematics and Music

The knowledge of Mathematics plays a vital role in music. For instance, playing one note in piano has a different sound as when three harmonic notes are played together. Music notes are distinguished mathematically by their different values.

A semibreve has four beats, a minim has two beats, crotchet has one beat, quaver has $\frac{1}{2}$ a beat, semi quaver has $\frac{1}{4}$ a beat, and so on. Reading and playing music represented by notes in a music sheet depends on the mathematical ability to understand these values.

Mathematics in Sports and Games

Mathematics is important in all forms of sports and games. For example, in many sports and games, winners are those who have accumulated highest number of points. Furthermore, in most of the sports and games, there are specific number of players in the playing ground. For instance, a football team has 11 players while a netball team has 7 players. Moreover, players in some games such as football are given names in terms of numbers. In football, a goalkeeper is normally assigned number 1. In addition, all playing grounds are prepared with specific standards of measurements.

Activity 1.2: Exploring the use of Mathematics in other fields

Explore various resources such as books and internet to discover examples of mathematical concepts and skills that are used in 3 subjects other than those discussed in this chapter.

Importance of Mathematics

Learning Mathematics and using mathematical skills in daily life enhances problem solving, critical thinking, effective time management among many other importances.

Financial Management

Financial management requires mathematical skills during the preparation of budgets. Calculations are done to ensure the budget prepared correlates with the funds collected.

Problem-solving skills

Problem-solving is one of the most important skills in life. Mathematics is one of the most effective ways to increase analytical and logical thinking, which helps us to become better problem solvers.

Time management

Time management is a key to success for everyone. Therefore, time must be carefully managed. Mathematics helps to determine the time spent on every activity so as to maximize its efficiency.

Activity 1.3: Exploring the importance of Mathematics

Use different sources such as books and internet to explore the importance of Mathematics in daily life.

Exercise 1.1

1. What is Mathematics?
2. Explain five mathematical skills that are useful in your daily life activities.
3. Choose any five subjects of your choice other than Mathematics and explain with vivid examples how each of them is connected to Mathematics.
4. Write an essay arguing whether or not learning Mathematics is important in our lives.

Introduction

Numbers play a vital role in everyday life. Different activities are performed with the help of numbers. Numbers are used to quantify and measure quantities. For instance, length, mass, time, volume, and population are some of the quantities represented by numbers. In this chapter, you will learn about rational, irrational, and real numbers. You will also learn about repeating decimals, inequalities and absolute values of real numbers. The competencies developed will enable you to perform daily life activities such as counting things, managing money, distributing items, and interpreting numbers based on contexts, and many other applications.

Concept of numbers

Numbers are classified into different categories. Some categories of numbers that you have already learned include whole numbers, natural numbers, fractions, integers, and decimals. Other major categories of numbers include rational, irrational, and real numbers. Activity 2.1 enables you to identify categories of numbers from daily life activities.

Activity 2.1: Categorising numbers

1. Use different measuring tools such as tape measure, ruler, weighing balance, and measuring cylinders to measure lengths, masses, and volumes of different objects, respectively.
2. Record the measurements in task 1 and categorise them based on your understanding of categories of numbers.

Rational numbers

A rational number is any number that can be written in the form of $\frac{a}{b}$, where a and b are integers, and $b \neq 0$. The condition $b \neq 0$ is essential because division by zero is not defined. The set of rational numbers is denoted by the symbol $\mathbb{Q}$.

Fractions, integers, whole numbers, terminating and non-terminating decimals form the set of rational numbers. For instance,

$-\frac{34}{3}$, $-\frac{20}{3}$, -3 , $-\frac{4}{7}$, $-\frac{5}{6}$, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{4}{4}$, $\frac{9}{5}$, $\frac{100}{5}$, 0 , $\frac{30}{1}$, 0.45 , $0.\dot{3}$, $0.\dot{2}\dot{3}$, and $1.56\dot{7}$ are rational numbers.

Representation of rational numbers on a number line

Rational numbers can be represented on a number line. Positive rational numbers are represented on the right of zero (the origin) and the negative rational numbers on the left of the origin. The number line helps us to determine other rational numbers between any two rational numbers by increasing the number of divisions. Activity 2.2 enables you to locate numbers on a number line.

Activity 2.2: Locating numbers on a number line

1. Prepare a number line (2 to 3 metres long) using sticks and masking tapes or any other materials of your choice.
2. Cut manila cards in small sizes and write on them different types of numbers such as whole, integers, and fractions.
3. Give the cards to others to post each of the numbers on appropriate position on the number line.
4. While sticking the cards, explain to others why you think the positions are appropriate.

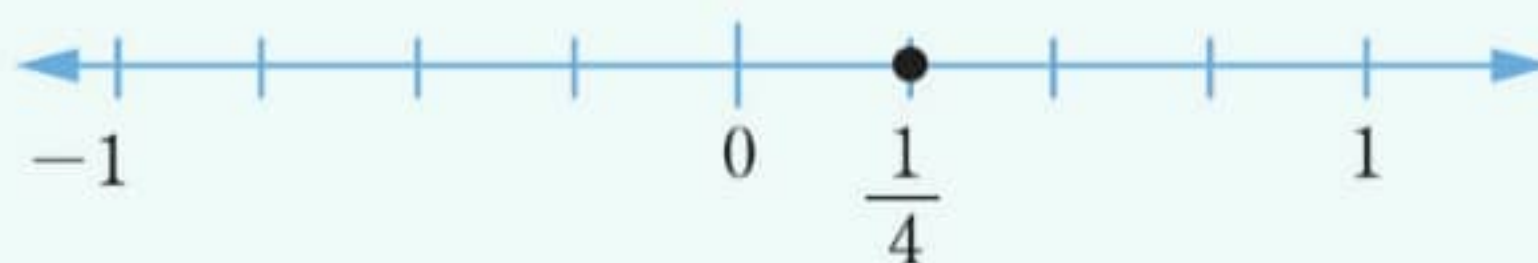
The representation of any positive rational number $\frac{a}{b}$ is done by dividing the unit interval into 'b' equal parts. The 'a' of these parts are taken along the number line to reach the point corresponding to $\frac{a}{b}$ on the right of zero if the number is positive and to the left of zero if the number is negative.

Example 2.1

Represent $\frac{1}{4}$ on a number line.

Solution

In order to represent $\frac{1}{4}$ on a number line, take one unit from 0 towards the right side and then divide that unit into 4 equal parts. Take one part out of the 4 parts to complete a part representing $\frac{1}{4}$. Therefore, $\frac{1}{4}$ on a number line is represented as shown in the following number line.

**Example 2.2**

Represent -2.4 on a number line.

Solution

$$\begin{aligned} -2.4 &= -2\frac{4}{10} \\ &= -2\frac{2}{5} \end{aligned}$$

To represent -2.4 on a number line, take 2 units from 0 towards the left side and then divide the third unit into 5 equal parts. Take 2 parts out of the 5 parts to complete a part representing -2.4 . Therefore, -2.4 on a number line is represented as shown in the following number line.

**Example 2.3**

Represent $\frac{13}{5}$ and $-\frac{13}{5}$ on the same number line.

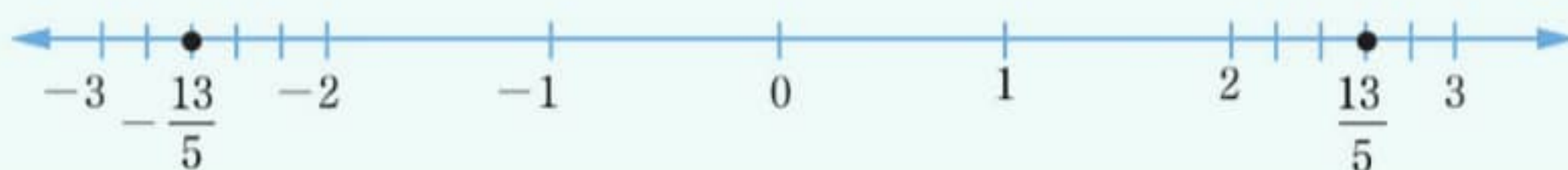
Solution

Convert $\frac{13}{5}$ into a mixed number. That is, $\frac{13}{5} = 2\frac{3}{5}$

$$= 2 + \frac{3}{5}.$$

To draw a number line, take 2 units from 0 towards the right side and then divide the third unit into 5 equal parts. Add the 2 units and the 3 parts out of the 5 parts to complete $\frac{13}{5}$. Also, convert $-\frac{13}{5}$ into a mixed number, that is, $-2\frac{3}{5}$.

Repeat the same procedure from 0 to the negative side of the number line to represent $-\frac{13}{5}$. Therefore, $\frac{13}{5}$ and $-\frac{13}{5}$ on a number line are represented as shown in the following number line.



Exercise 2.1

1. Represent each of the following rational numbers on a number line.

- | | | | |
|-------------------|-------------------|-------------------|--------------------|
| (a) 5 | (c) $\frac{1}{5}$ | (e) $\frac{3}{4}$ | (g) 0 |
| (b) $\frac{7}{3}$ | (d) 4.5 | (f) $\frac{3}{2}$ | (h) $3\frac{3}{4}$ |

2. Show the position of each of the following rational numbers on a number line.

- | | | | |
|--------------------|--------------------|---------------------|---------------------|
| (a) $2\frac{2}{3}$ | (c) $\frac{16}{8}$ | (e) $\frac{25}{6}$ | (g) $\frac{5}{5}$ |
| (b) $\frac{6}{1}$ | (d) $\frac{0}{3}$ | (f) $\frac{25}{50}$ | (h) $\frac{50}{25}$ |

3. Represent each of the following numbers on a number line.

- | | | |
|-------|--------------------|--------|
| (a) 3 | (b) $-\frac{5}{2}$ | (c) -3 |
|-------|--------------------|--------|

4. (a) Write all negative and positive rational numbers from the following list of numbers.

$-5, 0, 12, -3.416520..., 2.85, 7.14, 12.64646464, \frac{11}{5}, -\frac{3}{5}, -2\frac{1}{3}.$

- (b) Which numbers from (a) are not rational numbers? Justify your answer.
5. Write each of the following rational numbers in the form of $\frac{a}{b}$ in its simplest form.
- (a) 0.004 (c) -27 (e) -0.03 (g) $-6\frac{1}{8}$
- (b) 4 (d) 5.6 (f) $4\frac{1}{4}$ (h) $-2\frac{2}{3}$
6. Represent each of the following rational numbers on a number line.
- (a) -4 (e) $-\frac{5}{3}$
- (b) -2.5 (f) $-0.9, -0.8, 0.3, 0.2, 0.7$
- (c) 4 (g) $-\frac{3}{10}$
- (d) 1.8 (h) $-\frac{3}{2}, -\frac{5}{4}, -\frac{1}{2}, \frac{1}{4}, \frac{1}{2}, \frac{3}{2}, \frac{5}{4}, -\frac{3}{4}$
7. Write 5 rational numbers which are equivalent to $\frac{7}{10}$.
8. Write true or false in each of the following questions.
- (a) All whole numbers are rational numbers.
- (b) All integers are rational numbers.
- (c) The only rational numbers which are equal to their reciprocals are 1 and -1 .
9. Write down the rational number whose numerator is (-2×3) and whose denominator is $(23 - 16) \times (9 - 4)$.
10. A monthly salary of a certain worker is Tshs 250,000. One-fifth of his salary is spent in buying car fuels, half of remaining salary is spent in buying food and half of the rest is spent in miscellaneous expenses.
- (a) How much is his monthly savings?
- (b) State whether the answer in part (a) is a rational number or not.

Repeating decimals

Decimal numbers are part of rational numbers and are common in our daily life activities. The quantities such as length, height, age, volume and mass can be presented in decimals. Activity 2.3 guides you in expressing various quantities in fractions into decimals.

Activity 2.3: Expressing measurements of quantities in decimals

1. Take some fruits or similar objects and divide them into two, three, four, five, six, and seven equal parts.
2. Convert each fraction in task 1 into decimals in many decimal places as possible. You can work manually or use a calculator.
3. Study carefully the decimal part of the fractions and write down their unique characteristics.

A repeating decimal, also known as recurring decimal is a decimal number with at least one digit in the decimal part that repeats consecutively in a regular order without an end. For example, $1.6666666\ldots$ and $0.639639639639\ldots$ are repeating decimals because 6 and 639 digits from the two decimal numbers, respectively, in the decimal part repeat themselves without an end. The three dots indicate that the repeating digits continue infinitely.

Repeating decimals can also be represented by using a dot or a bar that is placed on top of a repeating digit.

Example 2.4

Write each of the following repeating decimals by using a dot and a bar.

- (a) $0.3333\ldots$ (b) $0.639639639\ldots$ (c) $0.474747474\ldots$

Solution

- (a) $0.3333\ldots$ is a repeating decimal which can be written as $0.\dot{3}$ or $0.\overline{3}$
 (b) $0.639639639\ldots$ can be written as $0.\dot{6}39$ or $0.\overline{639}$
 (c) $0.474747474\ldots$ can be written as $0.\dot{4}7$ or $0.\overline{47}$

In Example 2.4, the digits with a dot or a bar are repeating infinitely. In (b) and (c), it can be observed that if a group of digits is repeating, a dot should be put over the first and the last repeating digits.

Decimals are either terminating or non-terminating. Terminating decimals have a definite number of digits after the decimal point while non-terminating decimals have an endless number of digits after the decimal point. Thus, repeating decimals are non-terminating with one or more repeating digits in the decimal part.

Examples of terminating decimals are 0.5, 1.4, and 7.9 while non-repeating decimals are 3.1415926..., 1.4142135..., and 2.2360679...

Converting repeating decimals into fractions

When working with problems involving repeating decimals, it is important to convert them into simple fractions to maintain accuracy and avoid errors. A repeating decimal can be converted into fraction using the following steps:

1. Choose any variable to represent the required fraction.
2. Multiply both sides of the equation by a multiple of 10 depending on the number of repeating decimals. For example, $0.\dot{8}$ and $0.21\dot{3}$ have only one repeating digit which means they are multiplied by 10. Decimal numbers $0.\dot{1}\dot{1}$ and $1.2\dot{1}\dot{4}$ have two repeating digits, thus they are multiplied by 100, and $0.\dot{8}3\dot{5}$ will be multiplied by 1,000 since it has 3 repeating digits.
3. Subtract the equation in step 1 from the equation in step 2.
4. From the equation obtained in step 3, solve for the chosen variable and simplify where necessary.

Example 2.5

Convert each of the following decimals into fractions.

- (a) $0.\dot{3}$ (b) $0.\dot{8}\dot{3}$ (c) $0.\dot{8}3\dot{5}$ (d) $0.8\dot{3}$

Solution

- (a) Let $x = 0.\dot{3}$ (i)
 Multiply by 10 both sides of equation (i) to obtain,
 $10x = 3.\dot{3}$ (ii)
 Subtract equation (i) from equation (ii) as follows:
 $10x - x = 3.\dot{3} - 0.\dot{3}$
 Thus, $9x = 3$
 $x = \frac{1}{3}$
 Therefore, $0.\dot{3}$ into fraction is $\frac{1}{3}$.

- (b) $0.\dot{8}\dot{3}$
 Let $x = 0.\dot{8}\dot{3}$ (i)
 Multiply by 100 both sides of equation (i) to obtain
 $100x = 83.\dot{8}\dot{3}$ (ii)

Subtract equation (i) from equation (ii) as follows:

$$100x - x = 83.\dot{8}\dot{3} - 0.\dot{8}\dot{3}$$

$$99x = 83$$

$$x = \frac{83}{99}$$

Therefore, $0.\dot{8}\dot{3}$ into fraction is $\frac{83}{99}$.

(c) $0.\dot{8}3\dot{5}$

Let $x = 0.\dot{8}3\dot{5}$ (i)

Multiply by 1 000 both sides of equation (i) to get,

$$1000x = 835.\dot{8}3\dot{5} \quad (\text{ii})$$

Subtract equation (i) from equation (ii) and solve for x as follows:

$$1000x - x = 835.\dot{8}3\dot{5} - 0.\dot{8}3\dot{5}$$

$$999x = 835$$

$$x = \frac{835}{999}$$

Therefore, $0.\dot{8}3\dot{5}$ into a fraction is $\frac{835}{999}$.

(d) $0.8\dot{3}$

Let $x = 0.8\dot{3}$ (i)

Multiply by 10 both sides of equation (i) to get,

$$10x = 8.3\dot{3} \quad (\text{ii})$$

Subtract equation (i) from equation (ii) and solve for x as follows:

$$10x - x = 8.3\dot{3} - 0.8\dot{3}$$

$$9x = 7.5$$

$$x = \frac{7.5}{9}$$

$$= \frac{5}{6}$$

Therefore, $0.8\dot{3}$ into fraction is $\frac{5}{6}$.

Example 2.6

Convert each of the following decimals into fractions.

- (a) $9.\dot{2}$ (b) $5.\dot{2}\dot{3}$ (c) $2.\dot{1}0\dot{5}$

Solution

- (a) $9.\dot{2}$

$$\text{Let } x = 9.\dot{2} \quad \text{(i)}$$

Multiply by 10 both sides of equation (i) to get,

$$10x = 92.\dot{2} \quad \text{(ii)}$$

Subtract equation (i) from equation (ii) and solve for x as follows:

$$10x - x = 92.\dot{2} - 9.\dot{2}$$

$$9x = 83$$

$$x = \frac{83}{9}$$

Therefore, $9.\dot{2}$ into fraction is $\frac{83}{9}$.

- (b) $5.\dot{2}\dot{3}$

$$\text{Let } x = 5.\dot{2}\dot{3} \quad \text{(i)}$$

Multiply by 100 both sides of equation (i) to get

$$100x = 523.\dot{2}\dot{3} \quad \text{(ii)}$$

Subtract equation (i) from equation (ii) and solve for x as follows:

$$100x - x = 523.\dot{2}\dot{3} - 5.\dot{2}\dot{3}$$

$$99x = 518$$

$$x = \frac{518}{99}$$

Therefore, $5.\dot{2}\dot{3}$ into fraction is $\frac{518}{99}$.

- (c) $2.\dot{1}0\dot{5}$

$$\text{Let } x = 2.\dot{1}0\dot{5} \quad \text{(i)}$$

Multiply by 1,000 both sides of equation (i) to get

$$1000x = 2105.\dot{1}0\dot{5} \quad \text{(ii)}$$

Subtract equation (i) from equation (ii) and solve for x as follows:

$$1000x - x = 2105.\dot{1}0\dot{5} - 2.\dot{1}0\dot{5}$$

$$999x = 2103$$

$$x = \frac{2103}{999}$$

$$= \frac{701}{333}$$

Therefore, $2.\dot{1}0\dot{5}$ into fraction is $\frac{701}{333}$.

Exercise 2.2

- Convert each of the following decimals into fractions.

(a) $0.\dot{1}$	(c) $80.\dot{2}1\dot{7}$	(e) $13.0\dot{1}\dot{5}$	(g) $0.\dot{7}2\dot{3}$
(b) $0.\dot{8}$	(d) $3.1\dot{1}\dot{2}$	(f) $0.3\dot{4}$	(h) $2.\dot{4}$
- Which of the following rational numbers are repeating decimals?

(a) $\frac{1}{9}$	(b) $\frac{22}{7}$	(c) $3\frac{2}{3}$	(d) $\frac{20}{11}$
-------------------	--------------------	--------------------	---------------------
- If $x = 2.\dot{6}$ and $y = 2.8\dot{3}$, find the value of $\frac{x}{y}$.
- If $x = 0.\dot{3}$ and $y = 0.\dot{6}$, verify that $y^2 = x^2 + x$.
- Evaluate the following giving your answer as a fraction in its simplest form.

(a) $0.\dot{2} + 0.\dot{5}\dot{1}$	(b) $1.6\dot{7} + 2.\dot{1}$	(c) $0.6\dot{6} - 0.4\dot{1}$
------------------------------------	------------------------------	-------------------------------
- Show that $0.4\dot{6} = \frac{46}{99}$.
- Evaluate the following and give your answer in fractions.

$$1.2\dot{5}\dot{6} + 0.\dot{7} - 0.15$$
- The fraction, $a\frac{b}{c}$ is equivalent to $5.\dot{1}6\dot{5}$. Find the values of a , b , and c .
- What is the sum of the numerator and denominator of $0.\dot{2}\dot{7}$ when converted into fraction?
- Why learning the concept of recurring decimals is important in your daily life?

Converting fractions into repeating decimals

A fraction can be converted into a decimal by performing a long division. In this process, some fractions will be equivalent to either terminating or non-terminating decimals. If the resulting decimal is a non-terminating with recurring decimals, the repeating digits are indicated by the repeating decimal's notation.

Example 2.7

Convert the following fractions into decimals.

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{5}{33}$ (d) $\frac{17387}{909}$

Solution

- (a) Divide 1 by 3 using long division as follows and check your answer by a calculator.

$$\begin{array}{r}
 0.3333... \\
 3 \overline{) 1} \\
 \underline{- 0} \\
 10 \\
 \underline{- 9} \\
 10 \\
 \underline{- 9} \\
 10 \\
 \underline{- 9} \\
 10 \\
 \underline{- 9} \\
 1 \\
 \vdots
 \end{array}$$

Therefore, $\frac{1}{3} = 0.\dot{3}$.

- (b) Divide 2 by 3 using long division as follows and check your answer by a calculator.

$$\begin{array}{r}
 0.6666... \\
 3 \overline{) 2} \\
 \underline{- 0} \\
 20 \\
 \underline{- 18} \\
 20 \\
 \underline{- 18} \\
 20 \\
 \underline{- 18} \\
 20 \\
 \underline{- 18} \\
 2 \\
 \vdots
 \end{array}$$

Therefore, $\frac{2}{3} = 0.\dot{6}$.

- (c) Divide 5 by 33 by long division or by a calculator to get $0.1515151515\dots$. Note that in the answer, 1 and 5 repeats as you proceed to divide. Therefore, $\frac{5}{33} = 0.\dot{1}\dot{5}$ or $0.\overline{15}$.

- (d) Divide $\frac{17\,387}{909}$ by long division or by a calculator to obtain

$19.127612761276\dots$

Note that in the answer, the digits 1, 2, 7, and 6 repeats as you proceed to divide.

Therefore, $\frac{17\,387}{909} = 19.\dot{1}27\dot{6}$ or $19.\overline{1276}$.

Example 2.8

1. Convert each of the following mixed fractions into decimals.

(a) $1\frac{5}{9}$

(b) $1\frac{37}{45}$

(c) $9\frac{1}{6}$

Solution

- (a) Given $1\frac{5}{9}$.

Converting the mixed fraction into improper fraction gives, $1\frac{5}{9} = \frac{14}{9}$

Dividing 14 by 9 using the long division method gives $1.\dot{5}$.

Therefore, $1\frac{5}{9} = 1.\dot{5}$.

- (b) Given $1\frac{37}{45}$.

Converting the mixed fraction into improper fraction gives, $1\frac{37}{45} = \frac{82}{45}$

Dividing 82 by 45 using the long division method gives $1.8\dot{2}$.

Therefore, $1\frac{37}{45} = 1.8\dot{2}$.

- (c) Given $9\frac{1}{6}$.

Converting the mixed fraction into improper fraction gives, $9\frac{1}{6} = \frac{55}{6}$.

Divide 55 by 6 using the long division method to get $9.1\dot{6}$.

Therefore, $9\frac{1}{6} = 9.1\dot{6}$.

Example 2.9

A wire is divided into two pieces of lengths $2\frac{1}{3}$ cm and $2\frac{1}{2}$ cm. Find the total length of the wire in decimals.

Solution

Given the lengths of the two pieces of wire as $2\frac{1}{3}$ cm and $2\frac{1}{2}$ cm. Add the mixed fractions as follows:

$$\begin{aligned} 2\frac{1}{3} + 2\frac{1}{2} &= \frac{7}{3} + \frac{5}{2} \\ &= \frac{29}{6} \end{aligned}$$

Divide 29 by 6 using the long division method to obtain $4.8\dot{3}$.

Therefore, the total length of the wire is $4.8\dot{3}$ cm.

Exercise 2.3

1. Convert each of the following fractions into decimals.

(a) $\frac{6}{7}$

(c) $\frac{8}{11}$

(e) $\frac{25}{13}$

(b) $\frac{11}{6}$

(d) $\frac{102}{90}$

(f) $\frac{21}{13}$

2. Which of the following rational numbers are repeating decimals?

(a) $\frac{1}{9}$

(b) $\frac{22}{7}$

(c) $3\frac{2}{3}$

(d) $\frac{20}{11}$

3. Which is greater between $0.\dot{1}\dot{2}$ and $\frac{1}{3}$?

4. Write the following numbers in ascending order: $\frac{11}{23}$, $0.4\dot{7}\dot{2}$, and $\frac{5}{11}$.

5. Show that $\frac{23}{9} = 2.55555\dots$

6. Mwajuma has $\frac{1}{3}$ of a chocolate bar. She decides to share it equally among her 5 friends. How much chocolate will each friend get? Write your answer in fraction and decimal.

7. John wants to divide a 1 metre-long ribbon so that he can share it equally with his two friends. How much in metres will each get? Express your answer in decimals.
8. If $m = 0.\dot{2}$ and $n = 0.\dot{0}\dot{4}$, show that $m^2 = n(m+1)$.
9. A car travels $\frac{2}{3}$ of the total distance to its destination at a speed of 30 km/h, and then travels the remaining distance at a speed of 40 km/h. If the car took 4 hours for the whole journey, what is the total distance of the journey? Express your answer as a repeating decimal.
10. The base of a lid of bucket has an area of $\frac{25}{484}\pi \text{ m}^2$. Find its diameter in decimal.

Irrational numbers

Irrational numbers are special numbers in our life. A widely known and used irrational number is Pi (π) which appears in formulas for determining circumferences, areas, and volumes of circular shapes. Activity 2.4 allows you to use your experience in decimals to learn about irrational numbers.

Activity 2.4: Differentiating types of decimal numbers

1. Identify 10 different fractions which can be converted into terminating or repeating decimals.
2. Use a calculator to find answers to the square roots of at most 10 numbers that have no perfect squares and write your answers with at least 10 decimal places.
3. Compare the answers in tasks 1 and 2 and write down the differences observed.

An irrational number is a number which can be written as a non-terminating and non-repeating decimal. Also, these numbers cannot be expressed in the form of $\frac{a}{b}$, where a and b are integers and $b \neq 0$. The set of irrational numbers is denoted by $\mathbb{Q}'$. Irrational numbers cannot be represented exactly on a number line. However, they can always be approximated to rational numbers.

Example 2.10

Write any 6 irrational numbers.

Solution

The following is a list of some irrational numbers.

- (i) $\sqrt{2} = 1.414213562373\dots$ (iv) $-\sqrt{5} = -2.236067977499\dots$
 (ii) $e = 2.718281828459\dots$ (v) $\sqrt{11} = 3.316624790355\dots$
 (iii) $\pi = 3.141592653589\dots$ (vi) $-\sqrt{21} = -4.582575694956\dots$

From these examples, it can be observed that the digits after the decimal point continue infinitely without repeating.

Note:

The numbers such as π , e , $e\pi$, 2π , $e + \pi$, and other similar numbers are also called transidentals.

Example 2.11

Given the list of numbers

-6 , $-5\frac{3}{5}$, $-\sqrt{4}$, $-\frac{3}{5}$, $-\frac{3}{8}$, 0 , $\frac{4}{5}$, 1 , $1\frac{2}{3}$, $\sqrt{8}$, 3.01 , π , 8.47 . Identify all irrational numbers in the list.

Solution

The irrational numbers are $\sqrt{8}$ and π .

Exercise 2.4

1. Determine whether the following are rational or irrational numbers. Justify your answers.

$$\sqrt{11}, 0.65, \sqrt{12}, \frac{\pi}{2}, 1.101001000100001\dots, \sqrt{9}, 3+\sqrt{7}, 3\sqrt{3}-\sqrt{5}, -\sqrt{7}$$

2. Find an irrational number such that $x^2 = 7$.

3. Given the numbers -5 , -3.5 , 0 , $\frac{3}{4}$, $\sqrt{3}$, $\sqrt{5}$, 9 , $-4.3\dot{2}\dot{5}$, π , $\frac{\sqrt{2}}{2}$. List the numbers which are:

- | | |
|-------------------|------------------------|
| (a) Whole numbers | (c) Rational numbers |
| (b) Integers | (d) Irrational numbers |

4. State whether the following statements are true or false. Justify your answers.
 - (a) Every irrational number is a real number.
 - (b) All non-terminating decimals are irrational numbers.
 - (c) The square roots of all positive numbers are irrational numbers.
 - (d) The sum of two irrational numbers gives an irrational number.
5. Ramadhani is planning to build a circular pond whose base area is 100 square metres. Express the radius of the base of the pond as an irrational number.
6. Identify irrational numbers from the following list of numbers:
 $\sqrt{9}$, 0.54 , $\frac{91}{13}$, $1.2\dot{5}\dot{6}$, $\frac{\sqrt{2}}{2}$, $\frac{1}{11}$.
7.
 - (a) What is an irrational number?
 - (b) Explain the fact that the sum of two or more irrational numbers gives an irrational number.
8. Provide an example for each of the following statements.
 - (a) The product of rational and irrational number is an irrational number.
 - (b) The product of two irrational numbers is an irrational number.
9. Write five irrational numbers with denominators as irrational numbers.
10. In the following list of numbers, identify a list which does not contain irrational numbers.
 - (a) 0.1 , $\frac{2}{7}$, $5.33333\dots$
 - (b) $\sqrt{16}$, $\sqrt{1}$, $\sqrt{9}$, $3.844231028\dots$
 - (c) $\sqrt{4}$, $4\frac{2}{9}$, 0.6 , 7
 - (d) $-\frac{28}{9}$, 0 , $\sqrt{25}$, 3.45

Real numbers

Real numbers are the numbers which include both rational and irrational numbers. A set of real numbers is denoted by $\mathbb{R}$. Thus, all sets of numbers such as natural numbers, whole numbers, integers, rational numbers and irrational numbers are all real numbers. Natural numbers are the smallest set of numbers followed by whole numbers and integers which are all rational numbers. Rational and irrational numbers are opposite sets of numbers and are what makes the largest set of real numbers as shown in Figure 2.1.

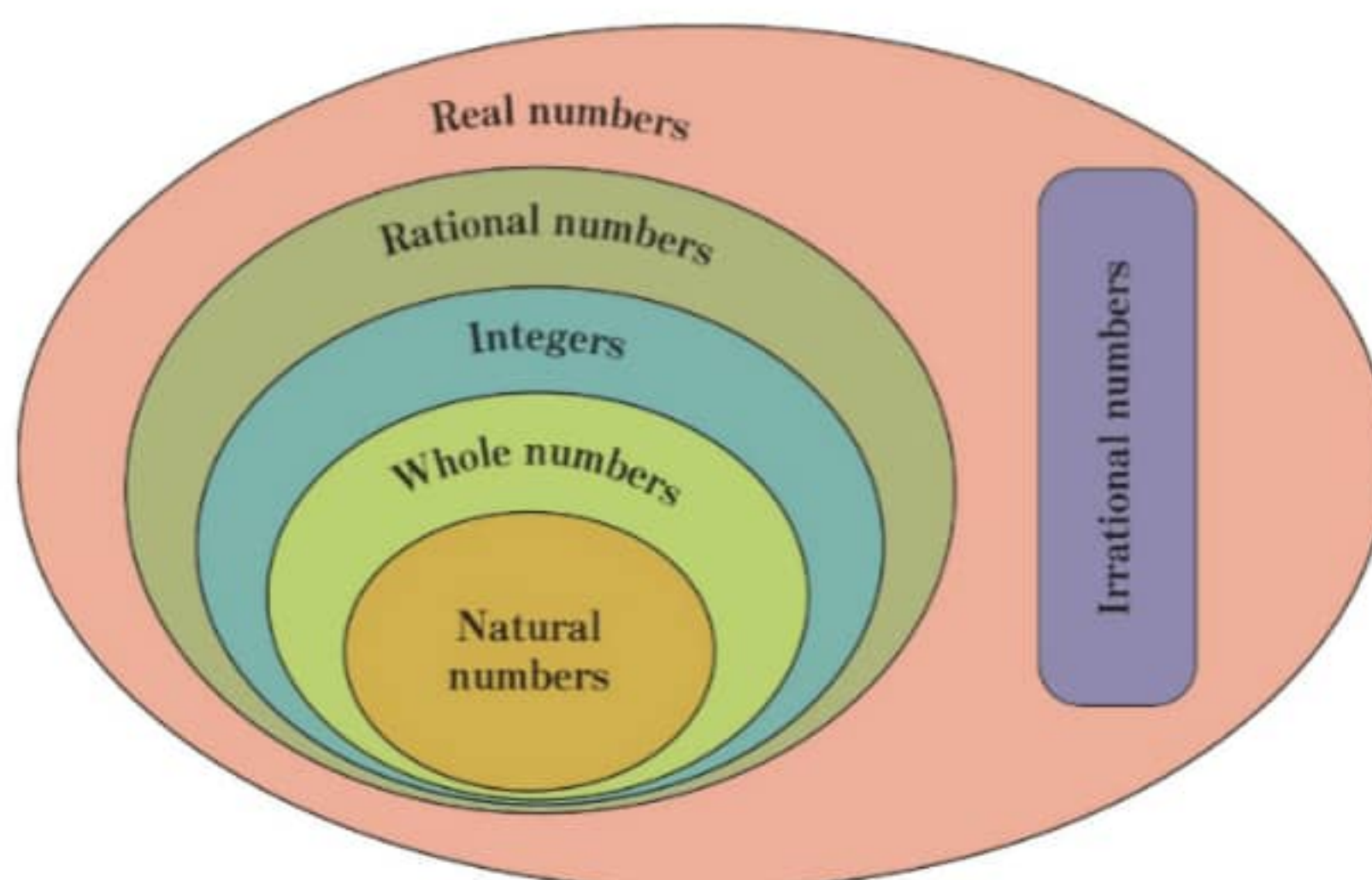


Figure 2.1: Categories of real numbers

If a and b are two real numbers, then either $a = b$, a is less than b or a is greater than b . For example, if:

- (i) $a = 3$ and $b = 7$, then a is less than b or b is greater than a , that is, 3 is less than 7 or 7 is greater than 3.
- (ii) $a = \frac{12}{3}$ and $b = 4$, then, $a = b$ that is, $\frac{12}{3} = 4$.
- (iii) $a = -3$ and $b = -5$, then a is greater than b , that is -3 is greater than -5 or b is less than a , that is, -5 is less than -3 .
- (iv) $a = -4$ and $b = 2$ then a is less than b , that is -4 is less than 2 or b is greater than a , that is, 2 is greater than -4 .

Examples of positive real numbers are 1, 2, 3, $\frac{2}{3}$, $\sqrt{7}$, 1.67, and 1.020020002 ... and examples of negative real numbers are:

$$-1, -2, -3, -\frac{3}{2}, -\sqrt{7}, -1.067, -1.020020002...$$

Note:

1. 0 is also a real number which is neither positive nor negative.
2. Every real number corresponds to a single point on the number line.
3. Every point on the number line corresponds to a certain real number.

Inequalities in real numbers

In daily life, people are faced with problems related to comparing quantities of the same item for the purpose of making decisions. Consider the following examples:

- (i) In some places, the speed of the car is limited to a certain maximum value due to the large number of pedestrians.
- (ii) Event organisers can set a maximum number of attendees for the event.
- (iii) Most banks limit the withdrawal amount to a certain minimum and maximum amount per day in automated teller machines.
- (iv) In schools, a minimum pass mark is set for students to be awarded certificates of completion of education.

Activity 2.5 guides you in comparing quantities in real life.

Activity 2.5: Comparing quantities in real life

1. Measure the lengths, widths, and heights of different objects of your choice.
2. Compare the measured values of a pair of objects in task 1.

In Mathematics, an inequality is a statement that compares or relates two values or expressions. The common terms involved in comparing the values or expressions are “less than”, “greater than”, “greater than or equal to”, “less than or equal to”, or “not equal to”. For instance, the following statements are examples of inequalities:

- (i) -5 is less than 0 .
- (ii) John’s age is greater than Jane’s age.
- (iii) x is less than or equal to 9 .
- (iv) y is greater than or equal to 0 .
- (v) 2.5 is not equal to 1.2 .

The mathematical statements can be written using symbols which represent inequalities as shown in the following table.

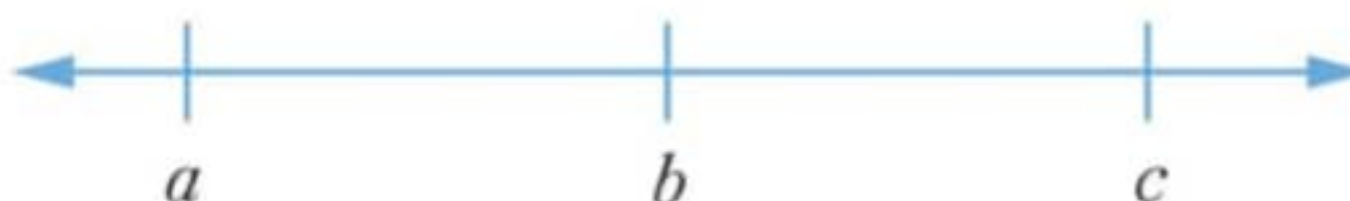
Word	Inequality symbols	Example
Less than	$<$	$-5 < 0$
Greater than	$>$	John’s age $>$ Jane’s age
Less than or equal	$\leq$	$x \leq 9$
Greater than or equal	$\geq$	$y \geq 0$
Not equal to	$\neq$	$2.5 \neq 1.2$

In general, mathematical statements which use $\neq$, $>$, $<$, $\geq$, or $\leq$ are called inequalities.

Note:

1. The sharp end of $>$ and $<$ always points to a smaller number.
2. The symbols $\leq$ and $\geq$ are sometimes referred to as at most and at least, respectively.
3. If the inequality is divided or multiplied both sides by a negative number, the direction of the inequality sign changes.

Suppose a , b , and c are real numbers which are represented on the number line shown in the following number line:



Generally, on a number line, a number to the right side of another number is always greater than a number to its left side. Thus, from the number line,

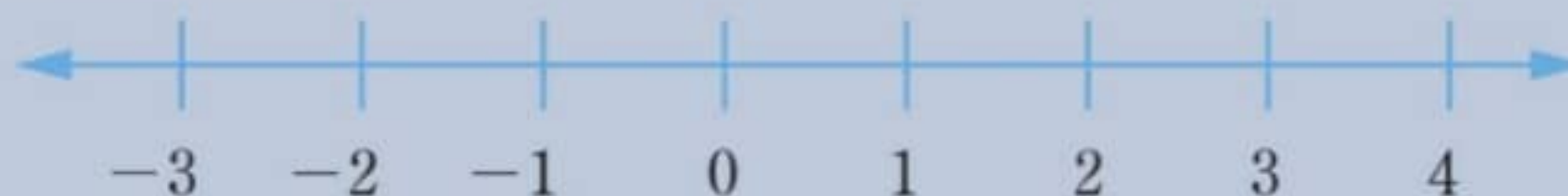
b is greater than a , which can also be written as $b > a$.

c is greater than b , which can also be written as $c > b$.

The vice versa is also true, that is, a number which is on the left side of another number on a number line is always less than the number on its right side. That is, $a < b$ and $b < c$.

Example 2.12

Study the following number line and use inequality symbols to answer the questions that follow.



- (a) Write all real numbers which are greater than zero.
- (b) Write all real numbers which do not exceed 2.
- (c) Write all real numbers between -3 and 4 .

Solution

- (a) Let x be any real number that is greater than 0. The symbol $>$ is used to represent the solution. There are infinity real numbers which are greater than zero. Therefore, $x > 0$.

- (b) Let x be any real number which does not exceed 2. Numbers on the left side of 2 on a number line are always less than 2. Also, 2 is included as it satisfies the given condition. Thus, the less than or equal inequality symbol should be used to present the solution. Therefore, $x \leq 2$.
- (c) Let x be a real number between -3 and 4 . This means that $-3 < x$ and $x < 4$. Combining the two inequalities gives, $-3 < x < 4$.

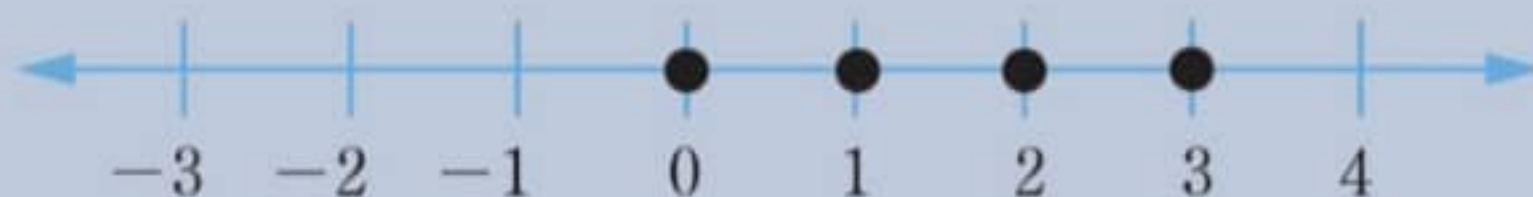
Example 2.13

If x is an integer, locate each of the following on a number line.

- (a) -1 is less than x and x is less than 4 .
 (b) -2 is less than or equal to x and x is less than or equal to 6 .

Solution

- (a) The statement is equivalent to $-1 < x < 4$, where x is an integer. On a number line, it is represented as follows:



- (b) The statement is equivalent to $-2 \leq x \leq 6$, where x is an integer. On a number line, it is represented as follows:



Example 2.14

Use a number line to compare each of the following pair of numbers.

- (a) 0.54 and 0.33
 (b) -0.54 and -0.33

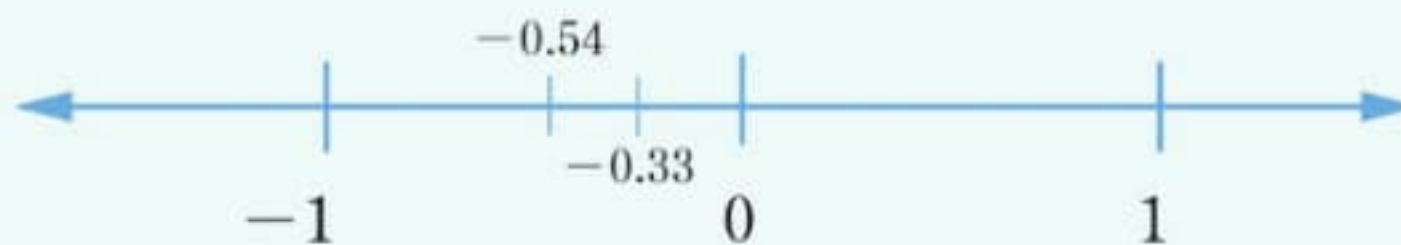
Solution

- (a) 0.54 and 0.33 are represented in the following number line.



From the number line, 0.33 is on the left side of 0.54, which implies that it is less than 0.54. Therefore, $0.33 < 0.54$ or $0.54 > 0.33$.

(b) -0.54 and -0.33 are located on a number line as follows:



From the number line, -0.33 is on the right side of -0.54 , hence -0.33 is greater than -0.54 . Therefore, $-0.33 > -0.54$ or $-0.54 < -0.33$.

Example 2.15

Compare the following numbers by using the inequality symbols $<$ and $>$.

(a) 0.432 and 0.437 (c) 3.7244 and 3.724

(b) -0.127 and 0.001 (d) $-\sqrt{7}$ and $\sqrt{5}$

Solution

(a) $0.432 < 0.437$ or $0.437 > 0.432$

(b) $-0.127 < 0.001$ or $0.001 > -0.127$

(c) $3.7244 > 3.724$ or $3.724 < 3.7244$

(d) $\sqrt{5} > -\sqrt{7}$ or $-\sqrt{7} < \sqrt{5}$

Exercise 2.5

1. Write true or false in each of the following inequalities.

- | | | | |
|---------------|----------------|------------------|--------------|
| (a) $3 < 30$ | (d) $6 \geq 2$ | (g) $-8 \geq -4$ | (j) $8 = -8$ |
| (b) $5 > -2$ | (e) $-3 < -2$ | (h) $-4 < -5$ | |
| (c) $3 < -30$ | (f) $2 > -4$ | (i) $5 < 10$ | |

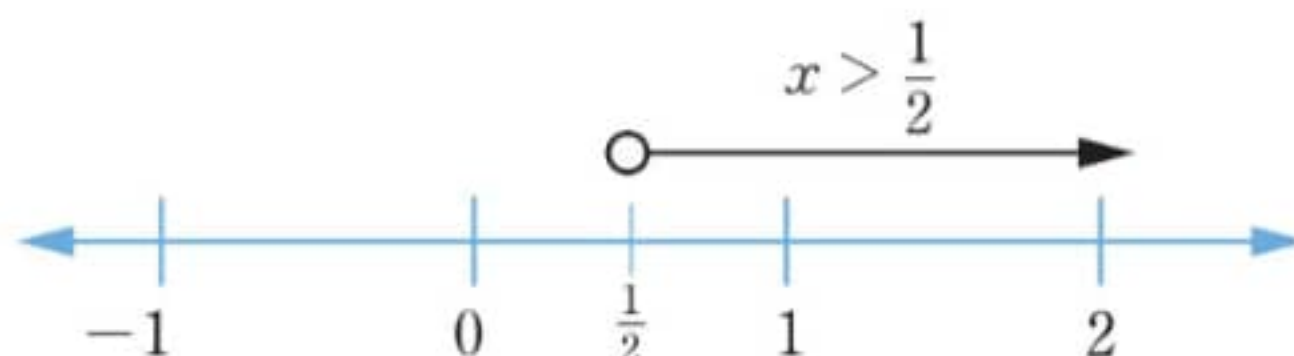
2. Which of the following inequalities are true?
 - (a) $-250 < -150$
 - (b) $2 \leq -2$
 - (c) $5 > -2$
 - (d) $6 \neq 6$
 - (e) $-3 = 3$
 - (f) $5 \geq -2$
3. List the numbers which satisfy each of the following conditions.
 - (a) $x < 6$ if x is a counting number.
 - (b) $x \leq 4$ if x is a whole number.
 - (c) $x > 3$ if x is an odd number.
 - (d) $x > -3$ and $x \leq 4$ if x is an integer.
4. In each of the following, insert a correct inequality symbols in the box to make a true statement.
 - (a) $-5 \times (-3) \square 8 \times (-3)$
 - (b) $13 \times (-2) \square 7 \times (-2)$
 - (c) $8 \times (5) \square 2 \times (-5)$
 - (d) $6 \times (-4) \square -3 \times (-4)$
 - (e) If $a > b$, then $a \times (-9) \square b \times (-9)$
5. Write all whole numbers in the interval $-4 \leq x < 6$.
6. Give a verbal statement equivalent to each of the following inequalities.
 - (a) $1 < x \leq 10$
 - (b) $x < -3$ or $x > 4$.
7. Ester is shorter than Nancy and Nancy is taller than Hyasinta. Express mathematically the relationship between their heights.
8. Anna has Tshs 500,000 in her bank account. Every week she withdraws Tshs 40,000 for various expenses. Without making any deposit, how many weeks can she withdraw this money if she wants to maintain a balance of at least Tshs 200,000?
9. The sum of two consecutive integers is at least negative fifteen. What are the smallest values of consecutive integers that will make this true?
10. Represent mathematically three real-life inequality scenarios of your choice.

Representing real numbers on a number line

Sometimes it is not easy to list all elements of a set of real numbers. For example, it is not possible to list all real numbers which are greater than $\frac{1}{2}$. However, the use of inequality symbols helps to define the elements in the given set. In this case, if x is a real number which is greater than $\frac{1}{2}$, then $x > \frac{1}{2}$ defines all the required elements in the set.

The inequality $x > \frac{1}{2}$ means that there are infinitely many real numbers satisfying this condition. Thus, they can be represented on a number line by using an arrow with a small circle at $\frac{1}{2}$ (initial point) since it is not included in the set.

The following number line shows all the real numbers which are greater than $\frac{1}{2}$.



Note:

1. A small circle (O) indicates that the number in a given set of real numbers is excluded in the given set. This is always true for a set of real numbers which involve $<$ or $>$ inequality symbols.
2. A thick dot (●) indicates that the number in a given set of numbers is included in the given set. This is always true for set of real numbers which involve $\leq$ or $\geq$ inequality symbols.

Example 2.16

Use a number line to represent all real numbers x in each of the following:

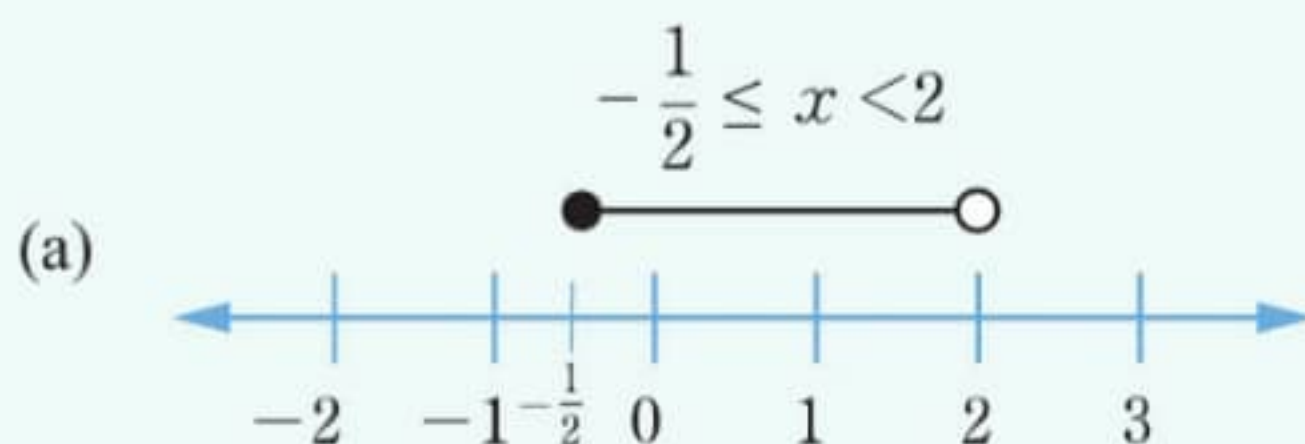
(a) $-\frac{1}{2} \leq x < 2$

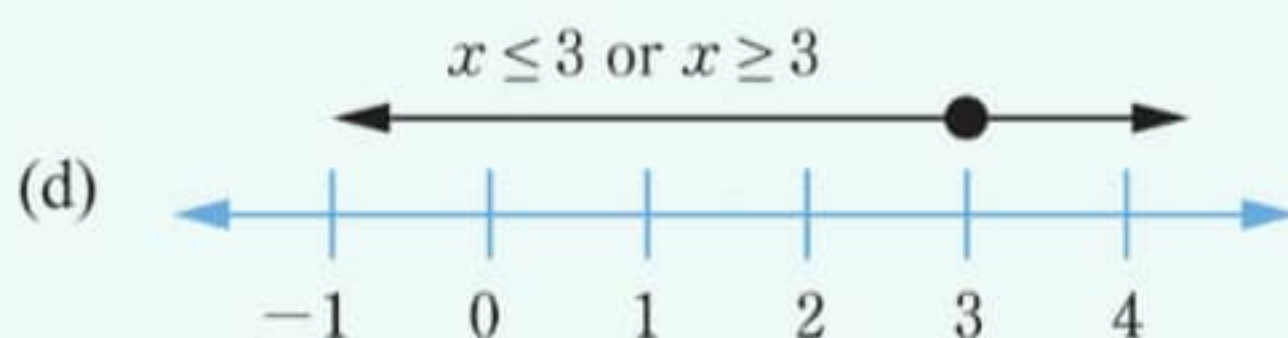
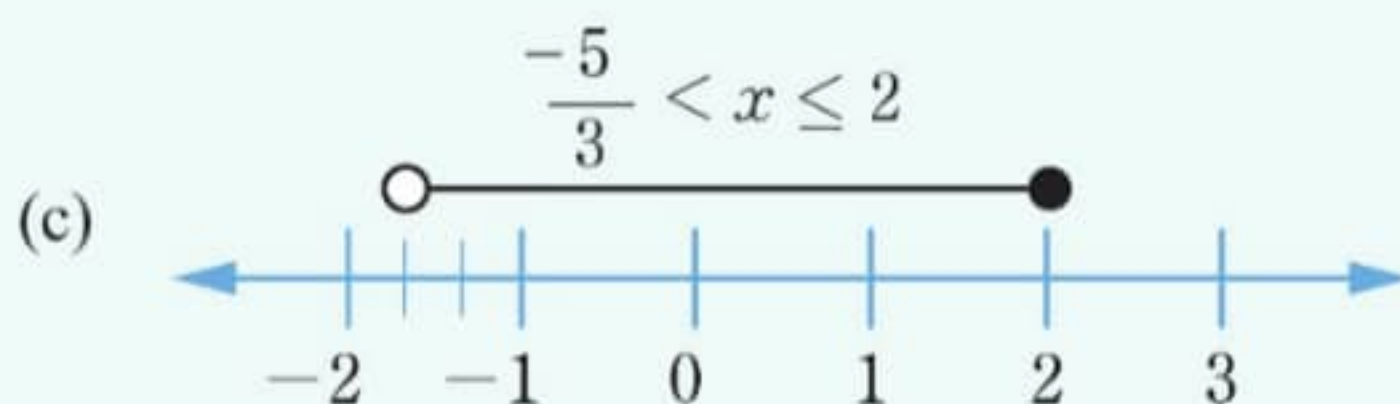
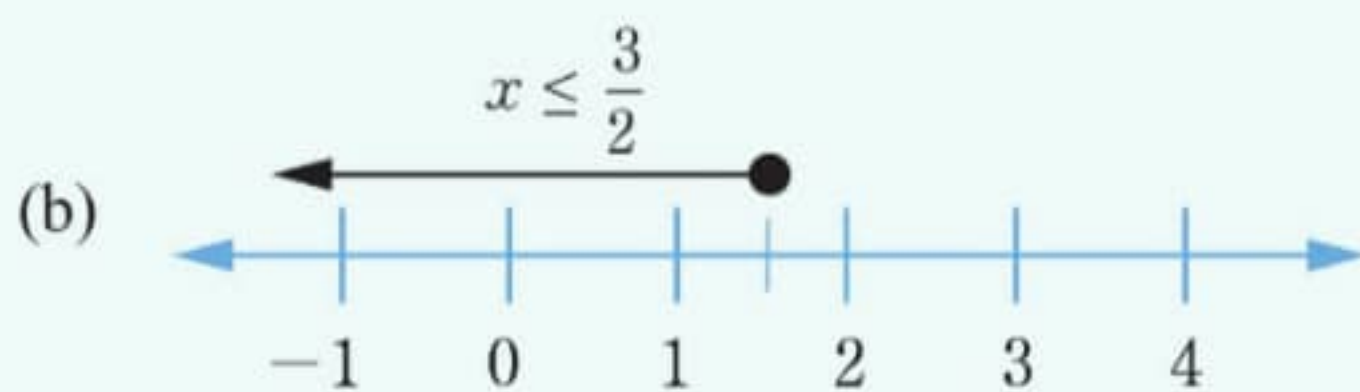
(c) $-\frac{5}{3} < x \leq 2$

(b) $x \leq \frac{3}{2}$

(d) $x \leq 3$ or $x \geq 3$

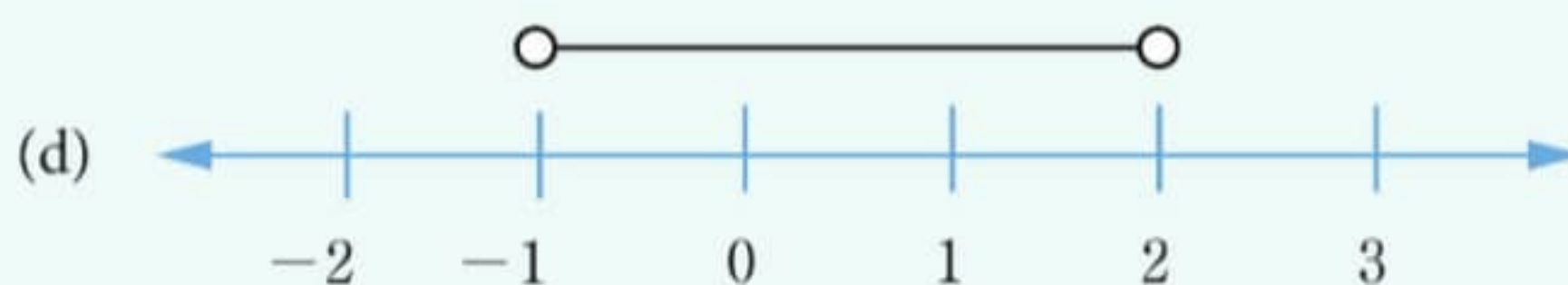
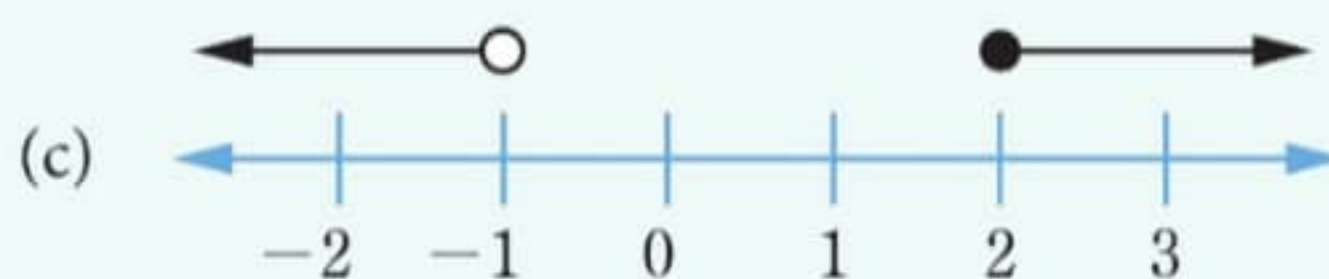
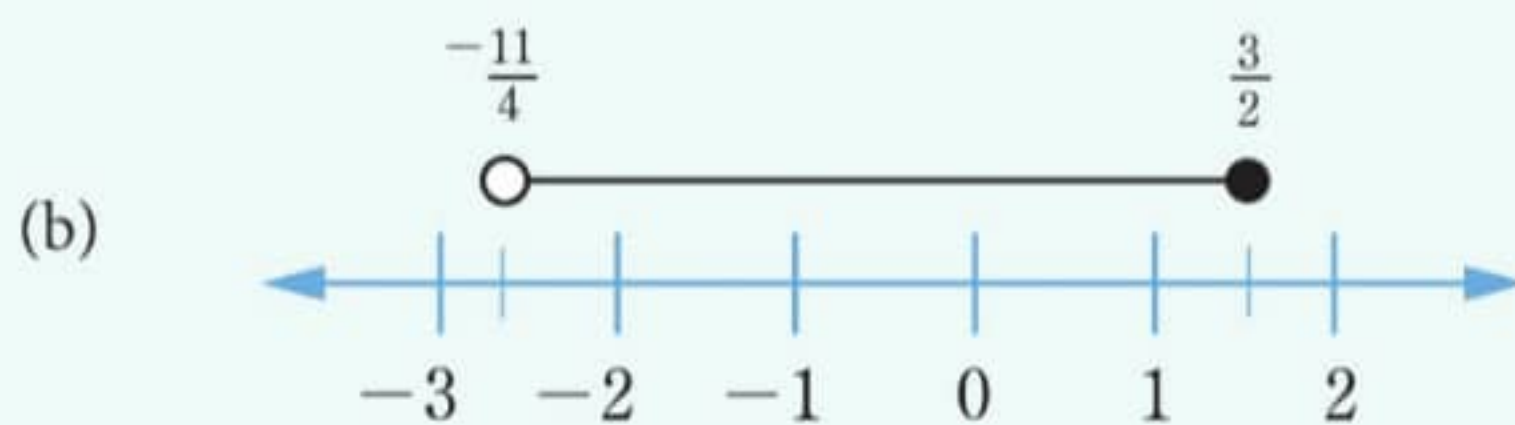
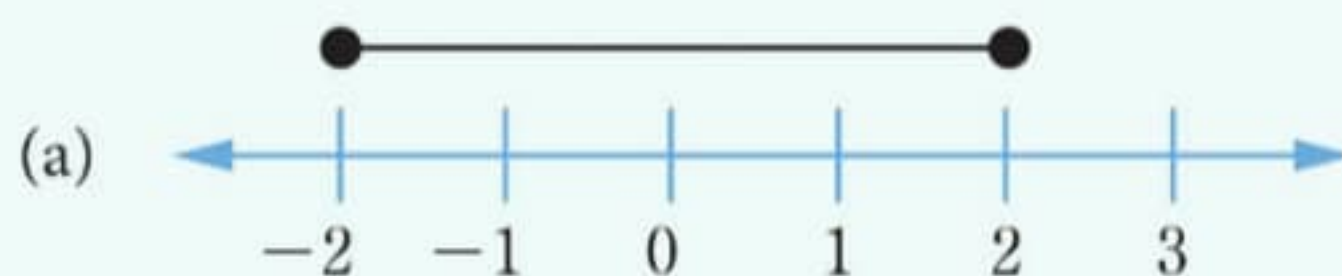
Solution





Example 2.17

Write the inequalities represented by each of the following number lines.



Solution

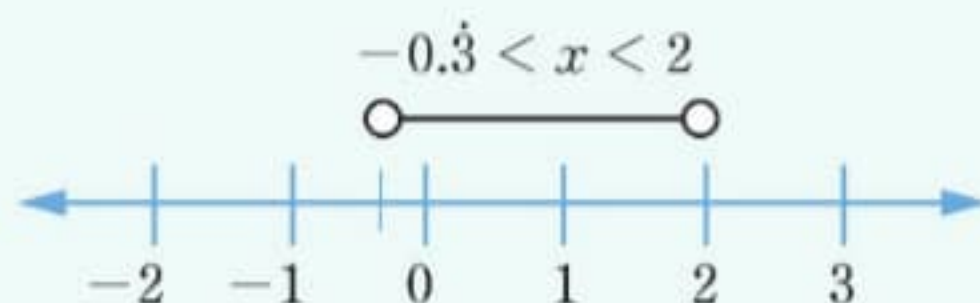
$$(a) -2 \leq x \leq 2 \quad (b) -\frac{11}{4} < x \leq \frac{3}{2} \quad (c) x < -1 \text{ or } x \geq 2 \quad (d) -1 < x < 2$$

Example 2.18

Represent on a number line all the real numbers between $-0.\dot{3}$ and 2.

Solution

Let x be a set of real numbers between $-0.\dot{3}$ and 2. This set of real numbers can be represented as $-0.\dot{3} < x < 2$ since, both $-0.\dot{3}$ and 2 are not included in the set. The number line representing this set of real numbers is shown in the following number line.

**Exercise 2.6**

- Represent each of the following numbers on a number line.
 - $-4, -2, 0, 2, 4$
 - $-2 \leq k \leq 1$ where k is a real number
 - $-5 < y \leq -2$, where y is a real number
 - $-2 < x < 4$, where x is a real number
- Represent each of the following integers on a number line under the given conditions.
 - $-6 < a < -2$
 - $-4 < b < -1$
 - $c < 5$ and $c > -3$
 - $-3 \leq e \leq 5$
 - $f > -9$ and $f \leq -4$.
- Draw a number line to represent each of the following real numbers.
 - $x < -1$ or $x > \frac{1}{2}$
 - $-6 \leq x$ and $x \leq -2$
- Use a number line to determine the distance between -3 and 2 .

5. Represent the given set of real numbers on a number line: 0.2, 0.4, 0.6, and 0.8.
6. Locate three consecutive integers on a number line which are immediate greater than -5 .
7. Represent the following numbers on a number line.
 $0.2, 0.4, -0.6, \frac{23}{5}, 6, \frac{32}{7}$
8. Represent each of the following inequalities on a number line.

(a) $-\frac{5}{2} < x < -\frac{1}{2}$	(d) $x < -\frac{5}{2}$ or $x > -2\frac{1}{3}$
(b) $\frac{1}{2} \leq x < \frac{7}{2}$	(e) $2 < x < 5\frac{1}{4}$
(c) $-\frac{2}{3} < x \leq 4$	(f) $-3 \leq x \leq 4$
9. Represent each of the following information on a number line.
 - (a) The expected mass of a human at birth is at least 3 kg and less than 5 kg.
 - (b) A manufacturing company produces buckets of at least 20 cm³.
 - (c) The mass of students in a class is at most 50 kg and not less than 35 kg.
10. On a certain number line, numbers are placed at an interval of 1 unit.
 - (a) What number is 6 units to the right of -4 ?
 - (b) What number is 3 units to the left of 2?

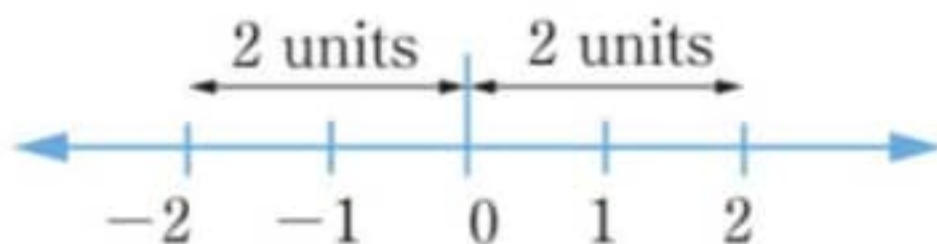
Activity 2.6 guides you in representing real numbers on a number line by using graphing tools.

Activity 2.6: Representing real numbers on a number line by using graphing tools

1. Identify different mathematical software which can be used in drawing and representing real numbers on a number line.
2. Study the basic procedures for drawing and representing numbers by using the software.
3. Use the software to draw and locate real numbers on a number line.

Absolute value of a real number

The absolute value of a real number x , denoted by $|x|$ is the non-negative value of x regardless of its sign. For example $|2| = 2$ and $|-2| = 2$. Consider the following number line.



The distance from 0 to -2 and that from 0 to 2 is the same, that is 2 units. Thus, 2 is an absolute value of both 2 and -2 . The absolute value of a number is the magnitude of that number regardless of its sign.

The absolute value of a number x is written as $|x|$ such that $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$, and its distance is represented on a number line from 0. This distance is always positive or zero.

For example; $|0.75| = 0.75$, $|-0.143| = 0.143$, $|\frac{1}{2}| = \frac{1}{2}$, $|0| = 0$, and $|-2| = 2$.

In general, for any two positive and negative numbers x and $-x$, their absolute values are equal, that is, $|x| = |-x|$.

Example 2.19

Evaluate each of the following:

- | | | | |
|------------|---------------|---------------------|---------------|
| (a) $ -2 $ | (c) $ 0 $ | (e) $ -3 \times 6 $ | (g) $ 5 + 2 $ |
| (b) $ 10 $ | (d) $ 8 - 2 $ | (f) $ 2 - 6 $ | |

Solution

- | | | |
|-----------------|----------------------------------|-------------------------|
| (a) $ -2 = 2$ | (d) $ 8 - 2 = 6 = 6$ | (g) $ 5 + 2 = 7 = 7$ |
| (b) $ 10 = 10$ | (e) $ -3 \times 6 = -18 = 18$ | |
| (c) $ 0 = 0$ | (f) $ 2 - 6 = -4 = 4$ | |

Example 2.20

Find the absolute value of $-\frac{13}{10} + \frac{4}{5}$

Solution

$$\begin{aligned}\left|-\frac{13}{10}+\frac{4}{5}\right| &= \left|\frac{-13+8}{10}\right| \\ &= \left|-\frac{5}{10}\right| \\ &= \left|-\frac{1}{2}\right| \\ &= \frac{1}{2}\end{aligned}$$

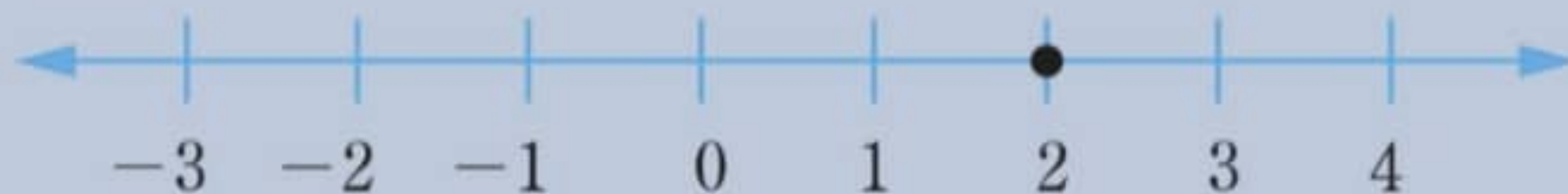
Therefore, the absolute value of $-\frac{13}{10}+\frac{4}{5}$ is $\frac{1}{2}$.

Example 2.21

Represent the absolute value of -2 on a number line.

Solution

$|-2| = 2$. Thus, the representation of 2 on a number line is shown in the following number line.



Exercise 2.7

- Find the absolute value in each of the following:
(a) 62 (b) $\sqrt{2}$ (c) -8 (e) -12.5 (d) $-\frac{4}{7}$ (f) $-pq$
- Determine the absolute value in each of the following:
(a) $\frac{4}{3}-\frac{3}{2}$ (d) $-\frac{1}{3}\times 11+\frac{10}{3}$
(b) $-3.68+3.89$ (e) $0.111-0.889$
(c) $\frac{3}{2}-\frac{4}{3}$ (f) $-0.111-(-0.01001)$
- Evaluate each of the following and arrange the answers in ascending order.
 $-|-4|$, $|-12|$, $|-5|$, $|-9|$, $|-37|$, $|10|$, $-|12|$, $-|2|$.

4. Evaluate each of the following and arrange the answers in descending order.
 $-|-34|$, $|23|$, $|29|$, $|-80|$, $|-17|$, $|25|$.
5. Evaluate each of the following expressions.

(a) $-3 - -3 $	(d) $ 4 - 10 $
(b) $ -5 - -3 - -4 $	(e) $- -4(6 - 3(7 - 10)) $
(c) $ -3 + 4(-2) - 6$	
6. In a certain country, the temperature on the certain day was 30°C and on the following day the temperature was 12°C use the concept of absolute value to determine the temperature change in these two days.
7. If x is an integer, represent the absolute values of the solutions to each of the following on a number line.

(a) $-\frac{5}{2} < x < \frac{1}{2}$	(c) $-4 < x < \frac{2}{3}$	(e) $2 \leq x < 5\frac{1}{4}$
(b) $-\frac{1}{2} < x < \frac{7}{2}$	(d) $-\frac{10}{3} < x < \frac{5}{2}$	(f) $-3 \leq x < 3$
8. The mean income of a certain company is 77 million Tanzanian shillings per year. If the income varies by 1.2 million Tanzanian shillings, what are the minimum and maximum income of the company?
9. An ice cream machine fills 160 g of ice cream in a cone. If the mass of ice cream varies by 5 g, find the heaviest and lightest cone the machine can fill.
10. In a part time job, one person target to collect Tshs 105,000 per month. Last month she was within Tshs 25,000 of her target. What is the minimum amount that she might have made last month?

Chapter summary

1. A rational number is any number written in the form of $\frac{a}{b}$, where both a and b are integers and $b \neq 0$.
2. All terminating and repeating decimals are rational numbers.
3. A decimal number which neither terminates nor repeats is an irrational number.
4. An irrational number cannot be written in the form of $\frac{a}{b}$, where a and b are integers and $b \neq 0$.

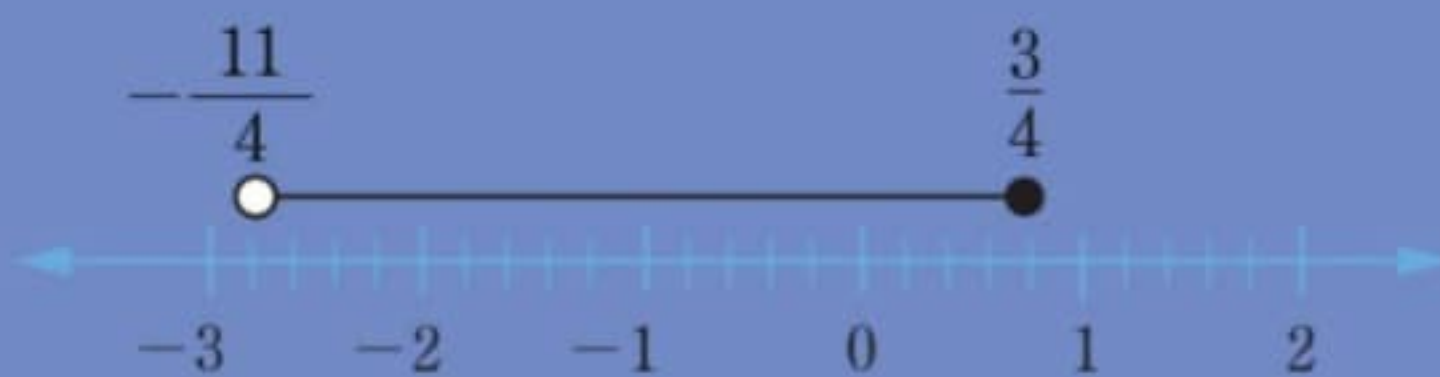
5. Rational and irrational numbers together form the set of real numbers.
6. For any two real numbers a and b , only one of the following is true; Either $a = b$ or $a < b$ or $a > b$.
7. Multiplying or dividing both sides of an inequality by a negative number reverses the inequality symbols.
8. The absolute value of a number is its positive value, regardless of its sign.

Revision exercise 2

1. Write each of the following rational numbers in the form of $\frac{a}{b}$, where a and b are integers and $b \neq 0$.

(a) $6\frac{1}{2}$	(c) $0.\dot{8}\dot{1}$	(e) 23	(g) -4.3	(i) 0.012
(b) $-9\frac{3}{11}$	(d) 193.5	(f) $245\frac{2}{9}$	(h) $114.9\dot{7}$	(j) $0.\dot{1}2\dot{3}$
2. List twelve rational numbers which lie between:

(a) -1 and 0	(b) 4 and 5	(c) -3 and -4
------------------	-----------------	-------------------
3. Represent on a number line all real numbers x such that $x < \frac{5}{3}$.
4. Represent $-4 \leq x < 4$ on a number line, where x is a real number.
5. Write an inequality for real numbers x represented on the following number line.



6. Find the absolute value in each of the following expressions.

(a) $\frac{3}{8} - \frac{1}{16} + \frac{5}{4}$

(d) $62.15 - 88.765 + 45.678$

(b) $1\frac{2}{3} + 2\frac{3}{4} + 3\frac{4}{5}$

(e) $2\frac{1}{3} + 4\frac{1}{5} + \frac{1}{2}$

(c) $-\frac{17}{2} + \frac{1}{10} - 3\frac{2}{7}$

(f) $\left(7\frac{5}{8} + 3\frac{1}{4}\right) - \left(2\frac{2}{5} + 4\frac{1}{2}\right)$

7. Find the absolute value in each of the following expressions.

$$\begin{array}{lll}
 \text{(a)} \quad \frac{2}{3} \times 3\frac{9}{16} & \text{(d)} \quad 6\frac{1}{4} + \left(4\frac{1}{2} \div 2\right) & \text{(g)} \quad \frac{\frac{1}{2} + \frac{1}{3}}{\frac{2}{3} + \frac{4}{5} + 1\frac{1}{4}} \\
 \text{(b)} \quad 2\frac{1}{2} \times \frac{7}{8} \div \left(-3\frac{3}{4}\right) & \text{(e)} \quad -\frac{4}{5} \times \left(\frac{3}{7} + 4\frac{7}{10}\right) & \text{(h)} \quad \frac{6\frac{2}{3} \times 2\frac{4}{5} - 3}{\left(2\frac{2}{3} \times 2\frac{4}{5}\right) - 3} \\
 \text{(c)} \quad -\frac{4}{5} \div \frac{33}{25} & \text{(f)} \quad 5\frac{1}{2} - \left(-7 + \frac{3}{11}\right) & \text{(i)} \quad -\frac{16}{5} \times \left(-\frac{3}{8} \times -\frac{2}{9}\right)
 \end{array}$$

8. Write the following using the recurring decimal notation, hence convert each of the decimals into fractions.

$$\begin{array}{ll}
 \text{(a)} \quad 0.22442244\ldots & \text{(d)} \quad 0.1010101010\ldots \\
 \text{(b)} \quad 0.123123123123\ldots & \text{(e)} \quad 2.153846153846\ldots \\
 \text{(c)} \quad 1.578578578\ldots
 \end{array}$$

9. Write each of the following fractions into decimals.

$$\begin{array}{lllll}
 \text{(a)} \quad \frac{5}{9} & \text{(c)} \quad \frac{29}{6} & \text{(e)} \quad \frac{5}{12} & \text{(g)} \quad \frac{40}{33} & \text{(i)} \quad \frac{7}{15} \\
 \text{(b)} \quad \frac{14}{11} & \text{(d)} \quad \frac{2}{27} & \text{(f)} \quad \frac{5}{11} & \text{(h)} \quad \frac{4}{27} & \text{(j)} \quad \frac{1}{13}
 \end{array}$$

10. If $x = 0.\dot{1}\dot{5}$ and $y = 0.\dot{3}$, find the value in each of the following in fraction.

$$\text{(a)} \quad x + y \quad \text{(b)} \quad \frac{x}{y} \quad \text{(c)} \quad \frac{y}{x}$$

11. Identify each of the following as rational or irrational numbers.

$$0.75, \frac{90}{12007}, \sqrt{7}, 12, \frac{1}{\sqrt{2}}$$

12. Identify whether or not, a mixed number $-3\frac{1}{2}$ is rational.

13. Write all real numbers from 5 to 6.

14. A rational number N is equal to $\frac{2}{5}$ times the sum of $\frac{24}{7}$ and $\frac{1}{14}$. Find the number N.

15. Identify fractions which are equivalent to each of the following decimals.

$$\text{(a)} \quad 2.344444444\ldots \quad \text{(b)} \quad 21.305305305305\ldots \quad \text{(c)} \quad 0.17171717\ldots$$

16. If $x = 1.\dot{3}$ and $y = 0.\dot{2}\dot{1}$, find the value of xy and $x - y$.
17. The average number of tomatoes in a heap is 28. The number of tomatoes vary by three. What is the maximum number of tomatoes that could be in a heap.
18. The mean distance from Morogoro to Chalinze is 90 km. However, due to errors in measurements, it varies by 1.6 km from different travel data companies. What is the possible maximum and minimum distance from Morogoro to Chalinze?
19. Runa bought a land for Tshs 2,562,000 and made a profit of Tshs 1,225,600 after selling it. How much did he sell the land?
20. How many years are in 5,840 days if a year has 365 days?
21. If a pond has a capacity of 35,000 litres of water, how long does it take to be filled by a hose with a capacity of filling 13 litres per minute?
22. Write down the inequalities which correspond to each of the following number lines.



23. Represent each of the following information on a number line.
 - (a) The number of football players allowed on the pitch for match is not less than 8 and does not exceed 11.
 - (b) Maximum volume of a container is at least 1.5 litres
 - (c) In a class, the tallest student is 160 cm and the shortest is 100 cm.
 - (d) The lowest temperature in April was 4°C .

24. If $x = 0.\dot{2}$, $y = 0.\dot{3}$, and $z = 0.\dot{9}$, evaluate each of the following expressions.

(a) xy (b) $x + yz$ (c) $\frac{xz}{y} + y$ (d) $xy + xz + yz$

25. Evaluate each of the following expressions.

(a) $1.0\dot{2}\dot{5} - 0.80\dot{5}$ (c) $0.\dot{6}\dot{8} - 0.0\dot{3}\dot{1}$
(b) $0.0\dot{6} - 0.0\dot{5}$ (d) $0.\dot{4} \div 0.\dot{2}$

Project: Appreciating numbers in daily live

Write an essay that describe with real-life examples on how numbers are used in various real-life activities.

Chapter Three

Approximations

Introduction

Understanding approximations is important in our daily life. Approximations are used in various real life situations like estimating cost of items, distances, and ingredients for a recipe. In this chapter, you will learn about rounding off numbers by place values, number of decimal places and by significant figures. Also, you will learn how to perform approximation in calculations. The competencies developed will help you to perform real-life activities such as estimating number of things, analysing large data, making quick calculations, preparing budgets, and many other applications.

Meaning of approximations

Approximation is a process of finding a number that is closest to the exact value. It involves expressing a number into a higher or lower value which is close to the exact value. Approximation can be termed as estimation and is represented by the symbol ' $\approx$ '.

Activity 3.1: Estimating challenge

1. In your surrounding environment (class or home), estimate the number of objects or students available or measurements of objects.
2. Record your answers and perform actual counting or measurements and find the difference.
3. Compare the differences you have noted between the estimated and actual measurement and comment on the differences.

Rounding off numbers

Rounding off numbers is a process of making a number simpler, but keeping its value closer to the original value. The outcome is less precise but more user-friendly. Rounding off the number can be done by considering place values, number of decimal places, and number of significant figures.

Rounding off numbers by place values

Rounding off numbers by place values is done by considering the given place value for both whole and decimal parts of a number. Thus, for whole numbers, it can be done to the nearest ones, tens, hundreds, thousands or any other place value. For decimal numbers, it can be done to the nearest tenths, hundredths, thousandths and so on. The following are basic procedures for rounding off numbers by place values.

- (a) If the digit to the right of the rounding digit is either 0, 1, 2, 3, or 4, then the digit at the required place value remains unchanged, and each digit to the right of it become zeros.

Note:

A rounding digit is the digit at a place where a number is supposed to be rounded off.

Example 3.1

Round off each of the following numbers according to the given instruction.

- (i) 591,622 to the nearest hundreds.
- (ii) 35.4 to the nearest ones.
- (iii) 234 to the nearest hundreds.

Solution

- (i) The digit of 591,622 in the hundreds place is 6 and the digit on the right of 6 is 2, which is less than 5. In this case, maintain 6 and replace each digit to the right of 6 with zero. Thus, 591,622 to the nearest hundreds is 591,600.
- (ii) 35.4 to the nearest ones is 35.
- (iii) 234 to the nearest hundreds is 200.

Example 3.2

Round off;

- (i) 35.49643 to the nearest thousandths.
- (ii) 274 to the nearest tens.
- (iii) 3.2743 to the nearest hundredths.
- (iv) 856,145 to the nearest thousands.
- (v) 42.245468 to the nearest thousandths.

Solution

- (i) 35.49643 rounded off to the nearest thousandths is 35. 496.
- (ii) 274 rounded off to the nearest tens is 270.
- (iii) 3.2743 rounded off to the nearest hundredths is 3.27.
- (iv) 856,145 rounded off to the nearest thousands is 856,000.
- (v) 42.245468 rounded off to the nearest thousandths is 42.245.

- (b) If the digit to the right of a rounding digit is greater than 5, then 1 is added to the rounding digit and each digit to the right is replaced by zero.

Example 3.3

Round off:

- (i) 0.267 to nearest tenths.
- (ii) 17.82 to the nearest ones.
- (iii) 63 504 to the nearest thousands.

Solution

- (i) 0.267 to the nearest tenths is 0.3.
- (ii) 17.82 rounded off to the nearest ones is 18.
- (iii) 63 504 rounded off to the nearest thousands is 64 000.

Example 3.4

The population of a certain country in 1967 census showed that there were 5,834,875 men and 6,111,188 women. Round off the figures to the nearest millions.

Solution

5,834,875 men is rounded off to the nearest millions as 6,000,000 men (1 is added to 5 because the digit on the right of 5 is 8).

6,111,188 women is rounded off to the nearest millions as 6,000,000 women (6 remain unchanged since the digit on the right of 6 is 1 which is less than 5 and each digit on the right of 6 is replaced by zero).

- (b) If the digit to the right of a rounding digit is 5, then the rounding digit is treated in three main ways.
- (i) Add 1 to the rounding digit if it is odd.
 - (ii) Add 1 to the rounding digit (whether even or odd) if there are other digits that follow 5 which are all not zero.
 - (iii) The digit remains unchanged if all the digits following 5 are zeros and the rounding digit is even.

Example 3.5

Round off each of the following numbers according to the given instructions.

- (a) 2.635 to the nearest hundredths.
- (b) 42,243,568 to the nearest thousands.
- (c) 8,115,684 to the nearest ten thousands.

Solution

- (a) 2.635 rounded off to the nearest hundredths is 2.64 since the rounding digit is odd and the digit to its immediate right is 5.
- (b) 42,243,568 rounded off to the nearest thousands is 42,244,000 since digits after 5 (which is on the right of 3 as a rounding digit) are all not zero.
- (c) 8,115,684 rounded to the nearest ten thousands is 8,120,000 since digits after 5 (which is on the right of 1 as a rounding digit) are all not zero.

Example 3.6

Round off:.

- (a) 6.645 to the nearest hundredths.
- (b) 8,125,000 to the nearest ten thousands.

Solution

- (a) 6.645 rounded off to the nearest hundredths is 6.64 since digits after 5 are all zeros.
- (b) 8,125,000 rounded off to the nearest ten thousands is 8,120,000 since all the digits after 5 are all zero.

Example 3.7

Round off the following numbers according to the instruction.

- (a) 3.6265 to the nearest thousandths.
 (b) 2,500,000 to the nearest millions.

Solution

- (a) 3.6265 to the nearest thousandths is 3.626 since the digit to the right of the rounding digit (6) is 5 and all digits to the right of 5 are zeros.
 (b) 2,500,000 to the nearest millions is 2,000,000 since the digit to the right of the rounding digit (2) is 5 and all digits to the right of 5 are zeros.

Exercise 3.1

- Round off each of the following numbers to the nearest thousands.
 (a) 8,259 (d) 100,998 (g) 60,500
 (b) 12,222 (e) 17,501 (h) 9,999
 (c) 13,709 (f) 2,349,673 (i) 1,234,567
- Round off each of the following numbers to the nearest ones.
 (a) 41.4 (d) 0.8
 (b) 0.49 (e) 0.379
 (c) 2.613 (f) 2.55
- Round off each of the following numbers to the nearest tens.
 (a) 25.12 (d) 10.0089 (g) 20.17
 (b) 16.15 (e) 50.408 (h) 311.114
 (c) 28.929 (f) 33.456
- The total mass of cotton harvested in a certain district was 17,816,273 kg. Round off the mass to the nearest.
 (a) Millions of kilograms (b) Thousands of kilograms
- In 1983, the number of primary school pupils in a certain region was 237,268. Round off the number of pupils to the nearest thousands.
- Find the value of each of the following, and round off the answers to the nearest hundredths:
 (a) 38.5×4.1 (c) 2.5×43.642 (e) 72×0.98
 (b) 9.9×9.9 (d) 10.3×4.4 (f) 0.048×20.08

7. Find the value of each of the following, and round off the answers to the nearest tenths.
 - (a) $2,684 \div 17$ (c) $0.33 \div 1.1$
 - (b) $4,314 \div 430$ (d) $5,301 \div 18$
8. Convert each of the following fractions into decimals and round off the answers to the nearest tenths.
 - (a) $\frac{2}{9}$ (b) $\frac{3}{7}$ (c) $\frac{17}{3}$ (d) $\frac{1}{4}$
9. Consider the following numbers: 9, 2.1, 42.045, and 635.7891.
 - (a) Add all the given numbers.
 - (b) Round off the number obtained in (a) to the nearest:
 - (i) Hundredths (ii) Tenths (iii) Hundreds
10. A factory got a profit of Tshs 67,459,853 after selling its products. How much profit did the factory get to the nearest thousands?
11. An electrician used 12,789 m, 13,952 m, and 9,374 m lengths of wire in 3 different months, respectively.
 - (a) Find the average amount of wire used to the nearest hundreds for the three months
 - (b) What is the difference between the actual amount and the estimated amount?

Rounding off by number of decimal places

Rounding off decimals involves a similar process of rounding numbers by place values. When rounding off decimals, the digits to the right of the given decimal place are dropped rather than being replaced by zeros.

In the process of rounding off, the following steps are used:

- (a) If the digit after the rounding digit is either 0, 1, 2, 3, or 4, then the rounding digit remain unchanged, and all the other digits to the right are dropped.

Example 3.8

Round off 0.2464 to 3 decimal places.

Solution

The digit in the third decimal place is 6 and the nearest digit to the right is 4 which is less than 5. In this case, 6 will remain unchanged and 4 is dropped. Therefore, $0.2464 \approx 0.246$ to 3 decimal places.

- (b) If the digit to the right of a rounding digit is greater than 5, then 1 is added to the rounding digit and other digits to the right are dropped.

Example 3.9

Round off 0.97381 to 3 decimal places.

Solution

The digit in the third decimal place is 3 and the nearest digit to the right is 8 which is greater than 5. In this case, 1 is added to the rounding digit and each digit to the right is dropped. Therefore, $0.97381 \approx 0.974$ to 3 decimal places.

- (c) If the digit to the right of a rounding digit is 5, then the rounding digit is considered as follows:
- Add 1 to the rounding digit if it is odd.
 - Add 1 to the rounding digit (whether even or odd) if there are other digits that follow 5 which are all not zero.
 - The digit remains unchanged if all the digits following 5 are zeros and the rounding digit is even.

Example 3.10

Round off the following numbers.

- 6.2358 to two decimal places.
- 3.4655 to two decimal places.
- 0.2475 to three decimal places.

Solution

- 6.2358 rounded off to two decimal places is 6.24 since the rounding digit is odd and the digit to the right of the rounding digit is 5.
- 3.4655 is rounded off as 3.47 to two decimal places since the rounding digit is followed by 5 and the digit after 5 are not all zero.
- 0.2475 is rounded off as 2.248 to three decimal places since the rounding digit is odd.

Example 3.11

Round off the following numbers.

- (i) 1.88500 to two decimal places.
- (ii) 2.682450 to four decimal places.

Solution

- (i) 1.88500 rounded off to two decimal places is 1.88 since the digits after 5 are all zero.
- (ii) 2.682450 rounded off to four decimal places is 2.6824 as the digit after 5 is 0.

Example 3.12

- (i) Round off 8.6521 to one decimal place.
- (ii) Round off 81.26584 to two decimal places.

Solution

- (i) 8.6521 is rounded off as 8.7 to one decimal place.
- (ii) 81.26584 is rounded off as 81.27 to two decimal places.

Example 3.13

- (i) Round 3.485 to two decimal places.
- (ii) Round 0.65 to one decimal place.
- (iii) Round 0.2465 to three decimal places.

Solution

- (i) 3.485 is rounded off as 3.48 to two decimal places.
- (ii) 0.65 is rounded off as 0.6 to one decimal place.
- (iii) 0.2465 is rounded off as 0.246 to three decimal places.

Exercise 3.2

- Round off the following numbers correct to the given number of decimal places.
(a) 4000.001 (2 decimal places) (d) 25.35447 (4 decimal places)
(b) 19.86 (1 decimal place) (e) 0.01001 (4 decimal places)
(c) 0.0075 (3 decimal places) (f) 10.0100165 (3 decimal places)
- Find the value of each of the following correct to 2 decimal places.
(a) $3.142 \times 15.25 \times 15$
(b) 7.9909×0.011
(c) 0.0073×0.73
- Write each of the following numbers correct to 2 decimal places.
(a) 0.0817 (d) 0.7153 (g) 3.6149
(b) 5.0744 (e) 12.047 (h) 0.00825
(c) 1.70007 (f) 3.3456 (i) 72.7946
- Convert each of the following fractions into decimals correct to 3 decimal places.
(a) $\frac{10}{7}$ (b) $\frac{4}{9}$ (c) $\frac{2}{3}$ (d) $\frac{4}{11}$ (e) $\frac{11}{7}$ (f) $2\frac{11}{29}$
- Write each of the following numbers correct to 1 decimal place.
(a) 3.142 (d) 0.09178
(b) 0.6667 (e) 0.7159
(c) 250.707 (f) 10.445
- Write each of the following numbers correct to 3 decimal places.
(a) 0.7526 (d) 8.4999
(b) 34.7007 (e) 5.5555
(c) 3.14159 (f) 2.66732
- Divide 8.51 by 0.472, then give the answer correct to 2 decimal places.
- Multiply 9.0017 by 0.0987, then give the answer correct to 2 decimal places.
- Round off the following numbers correct to the given number of decimal places.
(a) 0.002752 (4 decimal places) (d) 6.0097 (3 decimal places)
(b) 20.04416 (2 decimal places) (e) 2.14678 (1 decimal place)
(c) 0.17244 (3 decimal places)

10. Using a calculator or otherwise, evaluate the following and give the answer in specified number of decimal places.
- (a) $\frac{27 \times 3.648}{4.8954}$ (2 decimal places)
- (b) $\frac{10.64 \times 13.28}{14.32 \times 4.893}$ (3 decimal places)
- (c) $\frac{2.84 + 1.2}{6.28 - 3.49}$ (1 decimal place)
- (d) $153.4 \div 6.5$ (1 decimal place)
- (e) $44.0561 \div 11$ (2 decimal places)
- (f) $0.001144 \div 0.056$ (3 decimal places)
11. If the length of a square plate is 3.14 m, calculate its area correct to 3 decimal places.
12. Evaluate each of the following expressions and write your answer according to the given instructions.
- (a) $531.7 \div 4.3$ (1 decimal place)
- (b) $21.0561 \div 14$ (1 decimal place)
- (c) $0.001432 \div 0.076$ (3 decimal places)

Significant figures

A significant figure is a digit in a number that gives information about the precision and accuracy of the measured value.

Rules for significant figures

1. Any digit from 1 to 9 appearing in a number is a significant figure.
2. Each zero appearing in a number between digits 1 to 9 is a significant figure. For example, 0 in 602 is a significant figure. That is, 602 has 3 significant figures.
3. When zeros are written to the right of the last non-zero digit, the zeros are not significant figures. For example, 72,000 has only 2 significant figures.
4. In decimals, any zero to the left of the first non-zero digit is not a significant figure. For example, the zeros in 0.025 are not significant figures. Thus, 0.025 has 2 significant figures.
5. All zeros that are on the right of a decimal point are significant only if a non-zero digit does not follow them. For example, 0.040 has two significant figures while 0.6700 has four significant figures, and 10.000 has five significant figures.

6. When zero is written at the end of an approximated number, including decimals, it is a significant figure. For example, $2.73 \approx 3.0$, and in this case, 0 is a significant figure. Also, $45,961 \approx 46,000$, the zero at the third-place value is a significant figure.
7. All the zeros that are on the right of the non-zero digit are significant if they come from measurements. For instance, 20 m has two significant figures and 300 g has 3 significant figures.

Note:

- (i) Significant figures are also known as significant digits.
- (ii) Significant figures are always counted from left towards the right of a number.

Rounding off numbers by significant figures

Rounding off numbers to a specified number of significant figures is one of the methods of approximations. Similar rules for rounding off numbers are used in writing numbers to a given number of significant figures.

Example 3.14

Determine the number of significant figures in each of the following numbers.

- | | | |
|------------|-------------------|----------------|
| (a) 2.3004 | (c) 0.00002 | (e) 0.00165000 |
| (b) 0.0804 | (d) 4,002,000,000 | (f) 3,000 km |

Solution

- (a) 2.3004 has 5 significant figures.
- (b) 0.0804 has 3 significant figures.
- (c) 0.00002 has 1 significant figure.
- (d) 4,002,000,000 has 4 significant figures.
- (e) 0.00165000 has 6 significant figures.
- (f) 3,000 km has 4 significant figures.

Example 3.15

Write 38,176 correct to 1 significant figure.

Solution

Counting from left to right, the given number has 5 significant figures. The first digit, 3 is increased by one to become 4 (since 8 is greater than 5), and each of the right-side digits become zero. Thus, $38,176 \approx 40,000$ correct to 1 significant figure.

Example 3.16

Find the value of 0.3143×6.06 giving the answer correct to 3 significant figures.

Solution

$$\begin{aligned} 0.3143 \times 6.06 &= 1.904658 \\ &\approx 1.90 \end{aligned}$$

Therefore, $0.3143 \times 6.06 \approx 1.90$ correct to 3 significant figures.

Example 3.17

In an experiment for determining the value of pi (π) the following values were obtained:

3.142, 3.1578, 3.099867, 3.14103, and 3.15009. Calculate an average value of pi correct to 3 significant figures.

Solution

Find the average of the given values as follows.

$$\begin{aligned} \text{The average value of pi } (\pi) &= \frac{3.142 + 3.1578 + 3.099867 + 3.14103 + 3.15009}{5} \\ &= \frac{15.690787}{5} \\ &= 3.1381574 \\ &\approx 3.14 \text{ correct to 3 significant figures} \end{aligned}$$

Therefore, the average value of pi (π) is 3.14.

Exercise 3.3

- Determine the number of significant figures from each of the following numbers.

(a) 5.0372	(d) 292.00044	(g) 0.001110000
(b) 97168.90000	(e) 1.0000678	(h) 12030 km
(c) 0.0000678	(f) 0.101	

2. Approximate each of the following numbers correct to the required number of significant figures.
 - (a) 0.285173 (3 significant figures)
 - (b) 88092.7 (4 significant figures)
 - (c) 2.007138 (3 significant figures)
 - (d) 74.471 (1 significant figure)
 - (e) 10.6987 (3 significant figures)
 - (f) 126.306 (2 significant figures)
3. Write each of the following fractions in decimals according to the given instruction:
 - (a) $\frac{25}{13}$ correct to 4 significant figures.
 - (b) $\frac{5}{7}$ correct to 3 significant figures.
4. Find the sum of the following numbers, and give your answers correct to 3 significant figures.
 - (a) 8917, 201392, 500099, 1983
 - (b) 0.5021, 21.1108, 0.00472, 3.079
 - (c) 98.528, 8.888, 0.337, 24.82
 - (d) 0.07637, 72.0976, 4.7135, 25.4556
5. Find the value of each of the following expressions. Write your answers correct to the given number of significant figures.
 - (a) 0.83×0.737 (3 significant figures)
 - (b) 5.178×20 (2 significant figures)
 - (c) 7.351×4.83 (4 significant figures)
 - (d) $7.31 \div 0.983$ (3 significant figures)
 - (e) $149.001 \div 8.99$ (3 significant figures)
 - (f) 6.814×72.35 (4 significant figures)
6.
 - (a) Write $\frac{22}{3}$ in decimals correct to 4 significant figures.
 - (b) Find the value of the following expression in decimal, correct to 2 significant figures, $\frac{1}{2} + 3\frac{1}{3} + 4$.
 - (c) Write the number 0.48965 correct to 4 significant figures.

7. Round off each of the following according to the given number of significant figures.
 - (a) 12.7831 correct to 4 significant figures.
 - (b) 321.045 correct to 2 significant figures.
 - (c) 0.007138 correct to 3 significant figures.
 - (d) 37.205 correct to 4 significant figures.
 - (e) 37.207 correct to 1 significant figure.
 - (f) 399.603 correct to 4 significant figures.
8. A car is equipped with two front bulbs, two rear parking bulbs and a license bulb. Each front bulb consumes 4.328 watts, the rear bulb consumes 3.432 watts and the license bulb consumes 5.634 watts. Calculate the total power consumed by the bulbs, giving your answer correct to 3 significant figures.
9. A total of 24,582 people attended a political rally in a certain country. Round this number to 1 and 2 significant figures and state the most correct figure you could report to a newspaper company.
10. The following data shows the number of people who travelled by bus in seven days in a certain country:
20192, 18800, 23468, 25477, 18991, 26060, and 22645.
Find the total number of passengers correct to 2 significant figures.

Approximations in calculations

When working out some calculations involving numbers, sometimes it is important to find an estimate of the answer. To find an estimated answer, take a suitable approximation by rounding off the numbers involved. A suitable estimation can be to round off a number to the nearest whole number or in simple decimals which enables you to estimate the given problem easily and quickly.

Example 3.18

Estimate the value of 38×71 .

Solution

For easy estimation, round off 38 and 71 to the nearest tens.

That is, $38 \times 71 \approx 40 \times 70 = 2\,800$.

Therefore, $38 \times 71 \approx 2\,800$.

Example 3.19

Estimate the value of $256.5 \div 63.5$

Solution

Round off both numbers to the nearest ones.

$256.5 \approx 256$ to the nearest ones and
 $63.5 \approx 64$ to the nearest ones. It follows that,

$$\begin{aligned} 256.5 \div 63.5 &\approx 256 \div 64 \\ &\approx 4 \end{aligned}$$

Therefore, $256.5 \div 63.5 \approx 4$.

Example 3.20

Estimate the value of $\frac{3.485 \times \sqrt{35.82}}{351.25}$ correct to 1 decimal place.

Solution

$$\frac{3.485 \times \sqrt{35.82}}{351.25} \approx \frac{3.5 \times \sqrt{36}}{350}$$

$$\approx \frac{3.5 \times 6}{350}$$

$$\approx 0.06$$

$$\approx 0.1 \text{ correct to 1 decimal place.}$$

Example 3.21

Estimate the value of $\frac{58.34 \times 109.8}{2.23 \times 47.56}$.

Solution

$$\begin{aligned} \frac{58.34 \times 109.8}{2.23 \times 47.56} &\approx \frac{60 \times 110}{2 \times 50} \\ &\approx \frac{6600}{100} \\ &\approx 66 \end{aligned}$$

Example 3.22

A box can contain 7 pencils. There are 251 pencils to be packed in these boxes. Approximate the required number of boxes and determine the number of pencils which will be left unpacked.

Solution

The number of boxes can be obtained by dividing the total number of pencils to be packed by the number of pencils which a box can hold.

$$\begin{aligned} \text{Thus, number of boxes} &= \frac{251}{7} \\ &= 35.857 \end{aligned}$$

If the number of boxes is approximated to 36, then $36 \times 7 = 252$ which exceeds the available number of pencils. If the number of boxes is rounded down to 35, it gives 245 packed pencils, which means 6 pencils will remain unpacked.

Therefore, 35 boxes are required and 6 pencils will be left unpacked.

Exercise 3.4

In question 1 to 10, estimate the value of each of the given expressions.

1. 43×28
2. $2\,912 \times 32$
3. 82×61
10. $9\,110\,218\,800 \div 4\,081$
4. 868×31
5. 2.94×248
6. $171\,220 \div 79$
7. $35\,164 \times 23.04$
8. $1.029 \div 0.021$
9. $4\,981 \div 6\,438$
11. A shopkeeper sold 192 t-shirts at a price of 5,950 Tanzanian shillings each. By estimation, how much money did the shopkeeper get?
12. A village received 40,376 bags of fertilizer to be distributed to 392 farmers. Estimate the number of bags each farmer got.
13. Estimate the answers of each of the following expressions.
 - (a) $25.45 \div 268$
 - (b) $1.0031 + 56.241$
 - (c) $528 \times 3\,902$
 - (d) $61 \div 24$
 - (e) $824 - 325$
14. Find the approximate answers to each of the following:
 - (a) $\frac{21.3}{0.11}$
 - (b) 4.93×2.14
 - (c) $67.3 \times \frac{1}{4.96}$
 - (d) $\frac{42.3 \times 2.81}{20.5}$
15. Amina and her 17 friends bought 212 bottles of soft drink at a price of Tshs 635 per bottle. If the total cost was shared equally among themselves, estimate how much each person paid.
16. Estimate the number of 46 cm by 46 cm tiles which will be required to cover a room measuring:
 - (a) 3.84 m by 2.98 m
 - (b) 6.41 m by 3.28 m
 - (c) 4.29 m by 4.6 m

Chapter summary

1. Approximating or rounding off a number is a process of writing a number close enough to the exact number.
2. During approximation, if the digit to the right is 5, the rounding digit is:
 - (i) increased by 1 if it is odd.
 - (ii) left unchanged if it is even and has zero digits after 5 or has no digit after 5.

- (iii) Any digit from 1 to 9 appearing in a number is a significant figure.
- (iv) When zeros are written to the right of the last non-zero digit of an exact number, the zeros are not significant figures.

3. In decimals, any zero written to the left of the first non-zero digit is not a significant figure.
4. All zeros that are on the right of the decimal point are significant, only if a non-zero digit does not follow them.
5. When a zero is written at the end of an approximated decimal or number, it is a significant figure.
6. All the zeros that are on the right of the last non-zero digit are significant if they come from measurements.

Revision exercise 3

1. Round off each of the following numbers to the nearest hundredths.

(a) 8.648	(d) 31.7842
(b) 1.0544	(e) 19.6723
(c) 0.341	(f) 0.453
2. Round off each of the following numbers to the nearest millions and thousands.

(a) 78,911,393	(d) 1,350,095,450
(b) 1,114,562	(e) 20,781,233
(c) 22,878,130	
3. The population of children under the age of fifteen years in a certain country in 2004 was 5,267,910. Round off the population to the nearest.

(a) Millions	(b) Thousands	(c) Hundreds
--------------	---------------	--------------
4. Express each of the following fractions in a decimal, correct to 4 significant figures.

(a) $\frac{5}{11}$	(b) $\frac{3}{70}$	(c) $\frac{1}{6}$
--------------------	--------------------	-------------------
5. Write 86.463 correct to:

(a) 1 decimal place	(c) 1 significant figure
(b) 2 decimal places	(d) 2 significant figures

6. Write 0.00607049 correct to:
- (a) 3 decimal places (c) 5 decimal places
(b) 4 decimal places (d) 6 decimal places
7. (a) Write each of the following numbers correct to 1 decimal place.
(i) 17.84 (ii) 17.084 (iii) 2.045 (iv) 0.048
(b) Write each of the following numbers correct to 2 decimal places.
(i) 23.748 (ii) 23.0845 (iii) 0.0485 (iv) 0.0803
8. Write the following numbers correct to 2 significant figures.
- (a) 8 034 000 (d) 0.82500 (g) 0.008
(b) 0.47 (e) 1.0271 (h) 0.307
(c) 0.0785 (f) 6.0450 (i) 0.379
9. Convert each of the following fractions into decimals and write your answer correct to 2 decimal places.
- (a) $\frac{9}{13}$ (b) $\frac{18}{19}$ (c) $\frac{17}{180}$
10. Determine the number of significant figures in each of the following numbers.
- (a) 2.73 (c) 0.006 (e) 0.12000
(b) 400 780 (d) 0.1089 (f) 4 800 kg
11. Write:
- (a) 34.996 correct to 2 decimal places.
(b) 35.0482 correct to 3 significant figures.
12. Write each of the following numbers correct to three decimal places.
- (a) 0.00606 (d) 2.6047
(b) 3.199281 (e) 72.87247
(c) 8.27491
13. Approximate the answers of each of the following:
- (a) $\frac{11.3}{0.11}$ (c) $67.3 \times \frac{1}{4.96}$
(b) 4.93×2.14 (d) $\frac{42.3 \times 2.81}{20.5}$

14. Find the value of each of the following expressions.
- (a) $69\,875 \times 12 + 789\,852 \div 12$ (to the nearest thousands)
 - (b) $134\,048 \div 568 - 96\,045 + 279\,455 \times 18$
(to the nearest hundred thousands)
 - (c) $155 \times (98654168 - 92652149)$ (to the nearest millions)
 - (d) $2618954 + (45260 \div 365) \times 68 - (1614312 - 1594069)$
(to the nearest thousands)
 - (e) $895 + (5\,894\,325 \div 85\,425) - 794$ (to the nearest tens)
15. (a) A certain boys' school with 2,485 students has decided to make uniforms for all students. If a shirt and a pair of trousers use 2 m and 1.5 m, respectively, how much materials of cloth in total will the school need? Give your answer to the nearest hundreds.
- (b) Two hundred and forty-six university students were given Tshs 21,069,900 to share equally among themselves. How much money did each student get to the nearest thousands?
- (c) A train can carry 150 passengers in one trip. If it makes 16 trips a day
- (i) Estimate to the nearest ten thousands the number of passengers the train can carry in 5 days.
 - (ii) How much money to the nearest millions will be collected for 16 days when each passenger pays a fare of Tshs 1,050?
16. A man gave $\frac{7}{25}$ of his money to his son, $\frac{2}{9}$ of the remaining amount to his daughter, and the rest amount to his wife. If his wife got Tshs 11,990,300;
- (a) How much money to the nearest ten thousands did the man have in total?
 - (b) Estimate the amount of money to the nearest hundred thousands collected by his daughter.
17. A tablet contains 0.1763688 milligrams of a certain chemical. A dose requires a child to take $\frac{1}{2}$ a tablet, three times a day for 7 days. Find the total amount of chemical in the dose correct to hundredths of a milligram.

18. Write 47628.02415 correct to:
- (a) 3 significant figures
 - (b) 2 decimal places
 - (c) 3 decimal places
 - (d) 2 significant figures
 - (e) 4 decimal places
 - (f) 4 significant figures
19. There are 64 pipes each with a length 4.22 m arranged horizontally end to end. Estimate the length of all the pipes.
20. A car uses 1.13 litres of petrol to travel a distance of 11.65 km.
- (a) Estimate the rate of fuel consumption of the car in km per litre.
 - (b) Estimate the amount of fuel the car will consume for travelling:
 - (i) 40.28 km
 - (ii) 348 km
21. The XYZ cooperative society produces 94 jars of liquid soap every day. The cost of production of each jar consists of Tshs 4,015 for labour and Tshs 2,856 for materials. Estimate:
- (a) The total cost of producing a jar of liquid soap.
 - (b) The cost of materials for producing 94 jars.
 - (c) The cost of labour for producing 94 jars.
22. Estimate the values obtained from each of the given expressions.
- | | |
|------------------------|------------------------|
| (a) 8.7×410 | (f) $24.4 \div 0.673$ |
| (b) 4.17×730 | (g) $9.05 \div 18.2$ |
| (c) 28×0.83 | (h) 0.633×425 |
| (d) 54.5×1.96 | (i) $0.136 \div 8.45$ |
| (e) $430 \div 31.2$ | (j) 7.40×36.4 |

Project: Estimating the value of pi

1. Collect different cylindrical objects such as bottle tops, buckets, rods, pipes, and related objects.
2. Use a piece of thread to measure and carefully record the circumference and the diameter of at least 5 different cylindrical shapes.
3. Divide the circumference by the diameter obtained in task 2 and record your answers to 5 decimal places.
4. Calculate the average value of the results obtained in task 3.
5. Estimate the value obtained in task 4 correct to 3 decimal places.
6. The resulting number you obtained in task 5 is the approximated value of π which in fraction is $\frac{22}{7}$.
7. Explore various sources such as books and the internet to confirm the value of π .
8. Comment on your accuracy in estimating the value of π and discuss any possible ways you could reduce errors as much as possible.

Chapter Four

Ratios and proportions

Introduction

Ratios and proportions play an important role in our everyday life. They are used in various daily life activities such as in cooking, financial activities, medicine prescription, construction, arts and design, and many more applications. Understanding proportions helps us manage various daily life activities such as balancing diets, distributing things, and planning of events. In this chapter, you will learn about ratios and proportions. The competencies developed will enable you to ensure equal and proportional sharing and distribution of things among other applications.

Ratios

Sharing things is a common practice in our life. For example, people can share portions of land, wealth, and many other things. A required portion of water, fruit paste and sugar can result into a nice juice. Activity 4.1 introduces you with some everyday uses of ratios.

Activity 4.1: Ratios in daily life

1. Use various resources to explore carefully the relationship that exists between:
 - (i) Human body weight and height
 - (ii) Circumference of the neck and waist
2. Represent these facts mathematically and provide your interpretation.
3. Identify 3 more similar facts in everyday life and describe them mathematically.

A ratio is a quantitative comparison between two or more quantities of the same units, normally expressed as a fraction. It shows the number of times one value contains the other. A ratio between the numbers p and q is written as $p : q$, which is equivalent to $p \div q$ or $\frac{p}{q}$ and is read as p to q . It should be noted that multiplying

or dividing both terms of a ratio by the same number does not change the ratio. In other words, $2:4 = 4:8$ (multiplying each term of the ratio by 2). The ratio can be reduced to its lowest terms by expressing the ratio as a fraction and simplify it to its lowest term.

Example 4.1

Simplify $48:36$.

Solution

Given $48:36$

The Given ratio can be written in fraction as $\frac{48}{36}$

Dividing the numerator and denominator by 12 gives,

$$\frac{48}{36} = \frac{4}{3}$$

Therefore, $48:36$ in its simplified form is $4:3$.

Example 4.2

Given $a:b = 1:2$ and $b:c = 3:5$, find $a:b:c$ and $a:c$.

Solution

Given $a:b = 1:2$ and $b:c = 3:5$. b is in both ratios, so make the two ratios equivalent by multiplying $a:b = 1:2$ by 3 and $b:c = 3:5$ by 2 as follows:

$$a:b = (1:2) \times 3$$

$$b:c = (3:5) \times 2$$

$$a:b = 3:6$$

$$b:c = 6:10$$

Therefore, $a:b:c = 3:6:10$ and $a:c = 3:10$.

Example 4.3

Jaha and Siwema shared 40,000 Tanzanian shillings. Jaha received 15,000 Tanzanian shillings and Siwema received 25,000 Tanzanian shillings. Find the ratio of the amount of money each of them received.

Solution

The ratio is written as 15,000 to 25,000 or

$$15,000 \div 25,000 = \frac{15,000}{25,000} = \frac{3}{5}$$

Therefore, the ratio of the amount of money received by Jaha and Siwema is $3:5$.

Example 4.4

A man's monthly income is 1,274,000 Tanzanian shillings. He spends 1,078,000 Tanzanian shillings every month. Find the ratio between his income to expenditure and savings to income.

Solution

Income to expenditure ratio:

Given, income = Tshs 1,274,000

and expenditure = Tshs 1,078,000
Hence, income : expenditure

$$\begin{aligned} &= \frac{1274000}{1078000} \\ &= \frac{13}{11} \\ &= 13:11 \end{aligned}$$

Therefore, the ratio of income to expenditure is 13:11.

To find savings to income ratio, start to calculate savings as follows:

$$\begin{aligned} &= \text{Tsh } (1274000 - 1078000) \\ &= \text{Tsh } 196,000 \end{aligned}$$

$$\begin{aligned} \text{Thus, savings: income} &= \frac{196000}{1274000} \\ &= 2:13 \end{aligned}$$

Therefore, the ratio of savings to income is 2:13.

Example 4.5

A Mathematics club has 21 members of which 13 are females and the rest are males. Find the ratio of males to all club members.

Solution

Given, total number of club members = 21,

number of females = 13

Thus, number of males

= total number of club members
– number of females

$$\begin{aligned} &= 21 - 13 \\ &= 8. \end{aligned}$$

Therefore, the ratio of males to all club members is 8:21.

Example 4.6

Express the ratio 200 cm:1 m in its lowest form.

Solution

Using comparison of metric units, it is known that 1 m : 100 cm. Since the units must be the same, then it follows that,

$$\begin{aligned} 200 \text{ cm}:1 \text{ m} &= 200 \text{ cm}:100 \text{ cm} \\ &= 2:1. \end{aligned}$$

Therefore, the ratio 200 cm:1 m in its lowest form is 2:1.

Example 4.7

A special cereal mixture contains rice, wheat, and corn in the ratio 2:3:5, respectively. If a bag of the mixture contains 3 kilograms of rice, how much corn does it contain?

Solution

Let x be the amount of corn in kilograms that a special cereal mixture contains.

The ratio of rice to corn is given by $2:5 = 3 \text{ kg}:x \text{ kg}$

The ratio can be written in fraction as $\frac{\text{rice}}{\text{corn}} = \frac{2}{5} = \frac{3}{x}$

$$\text{Thus, } \frac{2}{5} = \frac{3}{x}$$

Cross multiplication gives,

$$2x = 3 \times 5$$

$$x = \frac{3 \times 5}{2}$$

$$= 7.5$$

Therefore, the special cereal mixture contains 7.5 kilograms of corn.

Example 4.8

A clothing shop sells t-shirts which are in three colours: red, blue, and green. The colours are in the ratio of 3 to 4 to 5, respectively. If the shop has 20 blue t-shirts, how many t-shirts are there in the shop?

Solution

Let x be the number of red t-shirts and y be the number of green t-shirts.

Writing the items in ratios gives,

$$\text{green: blue} = 5 : 4 = y : 20 \text{ and}$$

$$\text{red: blue} = 3 : 4 = x : 20$$

Writing in fraction form gives,

$$\frac{\text{green}}{\text{blue}} = \frac{5}{4} = \frac{y}{20} \quad (\text{i})$$

$$\frac{\text{red}}{\text{blue}} = \frac{3}{4} = \frac{x}{20} \quad (\text{ii})$$

From equation (i), it follows that,

$$\frac{5}{4} = \frac{y}{20}$$

Cross multiplication gives,

$$4y = 5 \times 20$$

$$y = \frac{5 \times 20}{4}$$

$$= 25$$

From equation (ii), it follows that,

$$\frac{3}{4} = \frac{x}{20}$$

Cross multiplication gives,

$$4x = 3 \times 20$$

$$x = \frac{3 \times 20}{4}$$

$$= 15$$

Thus, the total number of t-shirts is $15 + 20 + 25 = 60$.

Therefore, the clothing shop has 60 t-shirts.

Exercise 4.1

1. Express each of the following ratios in its lowest form.

- (a) $8:12$
- (b) $40:50$
- (c) $100:50$
- (d) $77:35$
- (e) $48:36$
- (f) $160:96$
- (g) $144:81$
- (h) $3\frac{1}{2}:1\frac{1}{4}$
- (i) $14:\frac{1}{2}$
- (j) $1\frac{1}{2}:2\frac{1}{2}$
- (k) $\frac{1}{4}:\frac{1}{7}$
- (l) $\frac{2}{5}:1\frac{3}{5}$
- (m) $0.17:1$
- (n) $3.5:0.07$
- (o) $7:0.007$
- (p) $120:105$
- (q) Tshs 136 to Tshs 4
- (r) 50 s to 90 s
- (s) 0.75 km to 0.5 km
- (t) 120 km to 105 km

2. Complete each of the following ratios.
- (a) $25: \underline{\hspace{1cm}} = 5:6$
 - (b) $15:20 = 18: \underline{\hspace{1cm}}$
 - (c) $24:8 = \underline{\hspace{1cm}}:7$
 - (d) $30:6 = \underline{\hspace{1cm}}:9$
 - (e) $4:7 = 20: \underline{\hspace{1cm}}$
 - (f) $4:9 = \underline{\hspace{1cm}}:63$
 - (g) $12: \underline{\hspace{1cm}} = 3:7$
 - (h) $\underline{\hspace{1cm}}:8 = 21:24$
3. A herd of 52 horses consists of white and black horses. If it contains 12 white horses, find the ratio of white horses to black horses.
4. The ratio of girls to boys in a swimming club was $2:4$. If 14 members were girls, how many members were there in the club?
5. If $4M = 5N = 6P$, find $M:N:P$.
6. Find $A:B:C$, when:
- (a) $A:B = 3:5$ and $A:C = 6:7$
 - (b) $B:C = \frac{1}{2}:\frac{1}{6}$ and $A:B = \frac{1}{3}:\frac{1}{5}$
7. David, Amina, and Nuru shared some oranges. The ratio of the number of oranges received by David and Amina was $7:3$. The number of oranges David received to that of Nuru was in the ratio $4:5$. If Nuru received 21 more oranges than David, find the number of oranges that Amina received.

Proportions

Proportion is essential in solving real-life problems involving scaling, ratios, and percentages. Such problems involve comparing and relating different quantities. Thus, proportion is a part, a share, or a number considered in comparison to a whole. Engage in Activity 4.2 to recall some real-life activities on proportions.

Activity 4.2: Relating quantities

1. In your daily life, recall things or events for which the occurrence of one affects the occurrence of another. Some examples include number of litres of fuel used by a car per kilometre and number of people and time used to complete a task.
2. Write your own three examples and show mathematically how the two things or events relate to each other.

The concept of equivalent fractions describes correctly what it means by a proportion. For example, $\frac{1}{2}$ and $\frac{2}{4}$ are equivalent, and hence the two fractions are proportional. Thus, a proportion is a statement which shows that two ratios are equal. That means, if two ratios are proportional, the corresponding parts of each ratio have the same relationship.

Proportions can be written in two ways:

- (a) As two equal fractions; $\frac{a}{b} = \frac{c}{d}$
- (b) Using colons;

When two ratios are equal, that is

$$a : b = c : d, \text{ it follows that } \frac{a}{b} = \frac{c}{d}$$

Thus, $a \times d = b \times c$.

When three numbers a , b , and c are in continued proportion, then c is called the third proportional. The third proportional of two numbers a and b is

defined as $a : b = b : c$, that is, $\frac{a}{b} = \frac{b}{c}$.

Example 4.9

Find the proportional parts of 156 in the ratio 3 : 4 : 5.

Solution

Given the total number of parts = 156, ratio = 3 : 4 : 5.

The sum of the terms of the ratio is $3 + 4 + 5 = 12$.

The required proportional parts are:

$$\frac{3}{12} \times 156 = 39$$

$$\frac{4}{12} \times 156 = 52$$

$$\frac{5}{12} \times 156 = 65$$

Therefore, the required proportional parts are 39, 52, and 65.

Example 4.10

The first, second, third, and fourth terms are proportional. If the first, second, and third terms are 42, 36, and 35, respectively, find the fourth term.

Solution

Let x be the fourth term.

Thus, 42, 36, 35, x are in proportion.

Writing the numbers in ratios gives,

$$\frac{42}{36} = \frac{35}{x}$$

Cross multiplication gives,

$$\begin{aligned} 42 \times x &= 35 \times 36 \\ x &= \frac{35 \times 36}{42} \\ &= 30. \end{aligned}$$

Therefore, the fourth term is 30.

Example 4.11

The ratio of the number of girls to that of boys in a school is 7:12. If the number of boys is 1,380, find:

- the number of girls.
- the total number of students in the school.

Solution

(a) Given

Number of girls: number of boys = 7:12

Number of boys = 1,380

Let x be the number of girls. It follows that $7:12 = x:1380$.

Writing in fraction gives, $\frac{7}{12} = \frac{x}{1380}$

Cross multiplication gives,

$$\begin{aligned} 12x &= 7 \times 1380 \\ x &= \frac{7 \times 1380}{12} \\ &= 805. \end{aligned}$$

Therefore, the number of girls is 805.

- The total number of students
= number of girls + number of boys
= $805 + 1380$
= 2,185.

Therefore, the total number of students in the school is 2,185.

Example 4.12

In a road safety campaign, it was revealed that 1 out of every 10 drivers in a certain city attended safety driving seminars. If there were 400 drivers in the city, how many drivers have attended the seminars?

Solution

Let y be the number of drivers who are likely to have attended driving seminars.

Given

The ratio of drivers who have attended safety driving seminars = 1:10

The ratio of the drivers who are likely to have attended the seminar = $y:400$

According to proportions, the two ratios are equal. Thus,

$$\frac{1}{10} = \frac{y}{400}$$

Cross-multiplication gives,

$$10y = 1 \times 400$$

$$y = \frac{1 \times 400}{10}$$

$$= 40$$

Therefore, 40 drivers out of 400 have attended the driving safety seminar.

Exercise 4.2

- Divide each of the following in the given ratios.
 - 100 in ratio 7:3
 - 75 in the ratio 3:2
 - 16 in the ratio 3:3:2
 - $\frac{1}{2}$ in the ratio 4:1
- If 800 kilograms of rice are shared between two families in the ratio 3:2, how much does each family get?
- Divide 60,000 Tanzanian shillings among Juma, Ali, and Aisha in the ratio 5:3:2.
- Musa, Josephine, Mary, and Kato have 300, 100, 500, and 600 shares in a cooperative shop, respectively. Divide 150,000 Tanzanian shillings among them in the ratio of their shares.
- Magnesium combines with oxygen in the ratio 3:2 by mass to form a new substance. What mass of magnesium will be needed to combine with 1.4 kg of oxygen?
- Divide 28.6 kg of meat among four families in the ratio 4:5:6:7.
- A powdery mixture is made up of powders A and B in the ratio 5:4. If 72 kg of this mixture are required, how much of each type should be used?
- An alloy is made up of metals x and y in the ratio 2.4:1 by mass. How much mass of y is required if 8 kg of x is used to make the alloy?
- If the ratio of lead to tin in a solder is 1.8:1, how much mass of tin is needed when 90 kg of lead is used?
- A line segment AB, which is 24 cm long, is divided at a point P (between A and B) in the ratio 7:5. Find the lengths of $\overline{AP}$ and $\overline{PB}$.
- The volumes of two tanks are in the ratio 2:5. If the volume of the first tank is 2,600 litres, find the total volume of the two tanks.
- A real horse is 1.8 m high. The statue of the horse is 3 m high. What is the ratio of the height of the horse to the height of the statue?
- Jason has 500, 200, and 100 Tanzanian shilling coins. The ratio of the total amount of these coins is 2:3:5, and the total amount is 780,000 Tanzanian shillings. Find the number of the coins of each kind.
- Find the fourth proportional of 6, 9, and 12.
- Find the third proportional of $2n^2$ and $3n^2$.

Chapter summary

1. A ratio is a comparison between two or more quantities which are in the same unit and can be simplified as a fraction.
2. A ratio can be expressed as a fraction, using colon or in words.
3. A proportion is a statement that two ratios are equal.
4. In general, a proportion describes a part, a share or a number considered in comparison to a whole.

Revision exercise 4

1. Express each of the following ratios as percentages.

(a) 1:20	(c) 49:500	(e) $1\frac{1}{2}:2\frac{1}{4}$
(b) 1:200	(d) $\frac{3}{8}:\frac{1}{2}$	
2. Express each of the following percentages as ratios in their lowest form.

(a) 72%	(c) 0.25%	(e) $8\frac{1}{3}\%$
(b) $27\frac{1}{2}\%$	(d) 28%	
3. An entrepreneur has a capital of 1,520,000 Tanzanian shillings. If she increases it by 20%, how much will she have?
4. Three relatives share 180,950 Tanzanian shillings so that the first receives twice as much as the second, and the second receives twice as much as the third. How much does the first receive?
5. Divide 270 kg in the ratio 1:4:5.
6. Divide 360° in the ratio 1:2:5:7.
7. Calcium and chlorine combine in the ratio 9:16. Calculate the mass of chlorine that will combine with 5 grams of calcium.
8. Ashura is making a juice containing water and lemon juice. She needs 3 cups of water for every cup of lemon juice. If she wants to make 10 cups of the juice, how many cups of water and lemon juice does she need?
9. Two numbers are in the ratio of 3:5. Find the larger number if the smaller number is 36.
10. Three numbers are in the ratio of 3:4:5. Find the largest number if their sum is 48.
11. Ambele weighs 65.8 kg. He wants to reduce his mass so that the ratio of his present mass to his new mass is 8:7. Find his new mass.

12. The ages of a father and his son are in the ratio of 4:2, respectively. How old is the father if the son is 20 years old?
13. The ratio of the heights in centimetres of two goats is 8:18. What is the height of the smaller goat if the larger goat is 45 cm tall?
14. The scale on a map is 1:30,000. What actual distance does a length of 12 cm on a map represent if a scale is given as a ratio of the distance on the map to the corresponding actual distance on the ground?
15. A Tanzanian map was drawn on a scale of 1:10,000,000. If the distance between Mbeya and Dodoma is approximately 589 km, how far apart in centimetres are the two regions on the map?
16. A piece of carpet is 28 m long. A shopkeeper divided it into three pieces in the ratio 1:2:5. Find the length of the smallest and largest piece of carpet.
17. At a football match, the ratio of children to adults watching the match is 1:5. There are 80 children in the stadium and each adult ticket cost Tshs 10,000 and that of children is quarter of the adult's ticket. Find the total amount of money obtained from ticket sales.
18. An athlete runs 200 metres in 50 seconds. If he runs constantly, how much time will he use to run 1,000 metres?
19. Juma read 81 pages of a book containing 144 pages. Find:
 - (a) the ratio of a number of pages remaining to the pages read in its lowest form.
 - (b) the ratio of the percentage of the pages read to the total number of pages.
20. Express the ratio of 18 hours to 2 days in its lowest terms.
21. Two consecutive numbers are in the ratio 2:7. If 6 is added to their sum, the result is 32. Find the numbers.
22. Find $(5x - 2y) : (3x - y)$ in its lowest form if $x : y = 4 : 9$.
23. The denominator of a fraction exceeds the numerator by 5. If 4 is subtracted from the numerator and the denominator remains unchanged, the resulting fraction is $\frac{2}{5}$. Find the original fraction.
24. In an isosceles triangle, the ratio of each of the equal sides to the third side is 3:1. The perimeter of the triangle is 49 cm, find the length of each side.
25. Six copies of Mathematics text books cost Tshs 66,000. Find the cost of 8 copies of the same textbooks.

Project: Planning for a catering of an event

You have been appointed in a catering committee for preparation of a graduation ceremony. A total number of 250 participants are expected to attend the graduation.

- (a) Plan for the amount of different varieties of food needed for the ceremony.
- (b) Show all the necessary mathematical calculations in determining the total amount of each type of variety of food for the meal.

Chapter Five

Algebra

Introduction

Algebra is a branch of Mathematics that helps to solve real-life problems by using letters, numbers, and symbols. It unifies all branches of Mathematics, and thus, it is a tool used in daily life activities. Algebra is used to understand and work with unknown quantities in various situations. For instance, it helps to calculate expenditure when shopping, plan travel routes, cooking or preparing recipes for different servings. In this chapter, you will learn about algebraic expressions, equations with one unknown, equations with two unknowns, and inequalities with one unknown. The competencies developed will enable you to solve daily life problems such as planning for recipes, scheduling routes, estimating profit and loss, currency conversions, among many other applications.

Algebraic expressions

In algebra, letters or symbols are used to represent numbers. For instance, it is common to represent the length of an object by the letter l , radius by letter r , and diameter by letter d . In addition, letters such as x , y , z , α and β can be used to represent numbers.

An algebraic expression is a mathematical statement which contains numbers, letters and mathematical operations such as $+$, $-$, $\times$, and $\div$. An algebraic expression is made up of terms.

A term can be a number, a variable, a number multiplied by a variable, or a variable multiplied by a variable. For example, in the algebraic expression $5x - 3y + 8$, terms are $5x$, $-3y$ and 8 . Terms with the same variables and exponents are called like terms. For example, $8y$ and $9y$ or $8xy$ and $3xy$ or $6x^2y$ and $2x^2y$ are like terms. Terms with different variables or exponents are called unlike terms. For example, $6x$ and $6z$ or $3mn$ and $4vy$ or $6u^2y$ and $2uy^2$ are unlike terms.

Simplification of algebraic expressions

Simplification of algebraic expressions involves combining like terms by adding or subtracting their coefficients in order to obtain a simple term. It may also involve multiplying and dividing terms as well as working with fractions and brackets.

Example 5.1

Multiply $5a + 2b - 3c$ by 4.

Solution

$$\begin{aligned}(5a + 2b - 3c) \times 4 &= 4(5a + 2b - 3c) \\ &= 4 \times 5a + 4 \times 2b + 4 \times (-3c) \\ &= 20a + 8b - 12c\end{aligned}$$

Therefore, $5a + 2b - 3c$ multiplied by 4 gives $20a + 8b - 12c$.

Example 5.2

Multiply $2x - 3y$ by $-2a$

Solution

$$\begin{aligned}(2x - 3y) \times (-2a) &= (-2a)(2x) + (-2a)(-3y) \\ &= -4ax + 6ay\end{aligned}$$

Therefore, when $2x - 3y$ is multiplied by $-2a$, the result is $-4ax + 6ay$.

Example 5.3

Re-write $-5a(m + 2n - 2)$ without brackets.

Solution

$$\begin{aligned}-5a(m + 2n - 2) &= -5am + (-5a \times 2n) + (-5a \times -2) \\ &= -5am - 10an + 10a\end{aligned}$$

Therefore, $-5a(m + 2n - 2) = -5am - 10an + 10a$.

Example 5.4

Divide $4ax + 6ay - 10az$ by $2a$.

Solution

$$\begin{aligned}(4ax + 6ay - 10az) \div 2a &= \frac{4ax + 6ay - 10az}{2a} \\ &= \frac{4ax}{2a} + \frac{6ay}{2a} - \frac{10az}{2a} \\ &= 2x + 3y - 5z\end{aligned}$$

Therefore, $4ax + 6ay - 10az$ divided by $2a$ is $2x + 3y - 5z$.

Example 5.5

Simplify each of the following algebraic expressions.

$$(a) \frac{1}{4}(8b + 12b - b) \qquad (b) \frac{1}{5}(15g + 5g)$$

Solution

$$\begin{aligned}(a) \frac{1}{4}(8b + 12b - b) &= \frac{1}{4}(19b) \\ &= \frac{19b}{4}\end{aligned}$$

$$\text{Therefore, } \frac{1}{4}(8b + 12b - b) = \frac{19b}{4}.$$

$$\begin{aligned}(b) \frac{1}{5}(15g + 5g) &= \frac{1}{5}(20g) \\ &= \frac{20g}{5} \\ &= 4g\end{aligned}$$

$$\text{Therefore, } \frac{1}{5}(15g + 5g) = 4g.$$

Example 5.6

Simplify each of the following algebraic expressions.

(a) $\frac{5t + 7t}{9}$

(b) $\frac{17c - 3c}{4}$

Solution

$$\begin{aligned} \text{(a)} \quad \frac{5t + 7t}{9} &= \frac{12t}{9} \\ &= \frac{4}{3}t \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \frac{17c - 3c}{4} &= \frac{14c}{4} \\ &= \frac{7}{2}c \end{aligned}$$

Therefore, $\frac{5t + 7t}{9} = \frac{4}{3}t$.

Therefore, $\frac{17c - 3c}{4} = \frac{7}{2}c$.

Example 5.7

Express each of the following algebraic expressions as a single fraction.

(a) $\frac{h-1}{3} + \frac{h+1}{2}$

(b) $\frac{b-2a}{4b} - \frac{b+3a}{20b}$

Solution

$$\begin{aligned} \text{(a)} \quad \frac{h-1}{3} + \frac{h+1}{2} &= \frac{2(h-1) + 3(h+1)}{6} \quad (\text{Multiply by LCM of 2 and 3 and simplify}) \\ &= \frac{2h - 2 + 3h + 3}{6} \\ &= \frac{5h + 1}{6} \end{aligned}$$

Therefore, $\frac{h-1}{3} + \frac{h+1}{2} = \frac{5h+1}{6}$.

$$\begin{aligned} \text{(b)} \quad \frac{b-2a}{4b} - \frac{b+3a}{20b} &= \frac{5(b-2a) - (b+3a)}{20b} \quad (\text{multiply by LCM of } 4b \text{ and } 20b \text{ and simplify}) \\ &= \frac{5b - 10a - b - 3a}{20b} \end{aligned}$$

$$= \frac{4b - 13a}{20b}$$

Therefore,

$$\frac{b - 2a}{4b} - \frac{b + 3a}{20b} = \frac{4b - 13a}{20b}.$$

Example 5.8

Simplify the expression
 $3n - 7n + 12n$.

Solution

$$\begin{aligned} 3n - 7n + 12n &= 3n + 12n - 7n \\ &= 15n - 7n \\ &= 8n \end{aligned}$$

Therefore, $3n - 7n + 12n = 8n$.

Example 5.9

Simplify the expression
 $6m - 4 - 2m + 15$ and state the coefficient of the variable m in the simplified expression.

Solution

$$\begin{aligned} 6m - 4 - 2m + 15 &= 6m - 2m + 15 - 4 \\ &= 4m + 11 \end{aligned}$$

The coefficient of m in the final expression is 4.

Example 5.10

If $m = 2$, $n = 4$, and $p = -5$, find the value of each of the following expressions.

(a) $3mn - 6mp + 2n^2p^3$

(b) $m^2n^2 - p^2n^3 + 7nmp$

Solution

(a) $3mn - 6mp + 2n^2p^3$

$$\begin{aligned} 3mn - 6mp + 2n^2p^3 &= (3 \times 2 \times 4) + (-6 \times 2 \times (-5)) + (2 \times 4^2 \times (-5)^3) \\ &= 24 + 60 - 4000 \\ &= -3916 \end{aligned}$$

Therefore, the value of $3mn - 6mp + 2n^2p^3 = -3\,916$.

(b) $m^2n^2 - p^2n^3 + 7nmp$

$$\begin{aligned} m^2n^2 - p^2n^3 + 7nmp &= (2)^2(4)^2 - (-5)^2(4)^3 + 7 \times 4 \times 2 \times (-5) \\ &= 64 - 1\,600 - 280 \\ &= -1816 \end{aligned}$$

Therefore, the value of $m^2n^2 - p^2n^3 + 7nmp$ is $-1\,816$.

Example 5.11

Simplify $8m^2n - 6a^2c + 14 - 4m^2n + 3a^2c - 8$ and then, state:

- (a) The coefficient of m^2n .
- (b) The number of terms.
- (c) The constant term.

Solution

$$\begin{aligned} 8m^2n - 6a^2c + 14 - 4m^2n + 3a^2c - 8 &= 8m^2n - 4m^2n + 3a^2c - 6a^2c + 14 - 8 \\ &= 4m^2n - 3a^2c + 6 \end{aligned}$$

- (a) The coefficient of m^2n is 4.
- (b) There are 3 terms.
- (c) The constant term is 6.

Exercise 5.1

1. Simplify each of the following algebraic expressions and then state the number of terms and its coefficients.

(a) $n + n + n + n + n + k + k + k + x + x$

(b) $3x + 4y - 7z + 3x - 7y + 2z$

(c) $3\frac{1}{2}x + 7x - \frac{1}{4}x$

2. Simplify each of the following algebraic expressions.

(a) $12m + 13m$

(b) $-5hk(2r - 3d - 1)$

(c) $9z + 6z - 8z + 5z$

(d) $0.5m(3a - 2b)$

(e) $9a - 6b + 3a$

(f) $15n - 9n$

(g) $x + 4 - 5x$

(h) $4k - k + 3k$

(i) $r + 2r + 3r + 4$

(j) $8y - 3 - 7y + 48y - 3$

(k) $5(5x + 6m + 4y)$

(l) $-2m(-13x + y - 6z)$

3. Simplify each of the following algebraic expressions where possible.

(a) $\frac{1}{q}(qz + 4qz)$

(e) $\frac{a+b}{4} - \frac{a-b}{8}$

(b) $\frac{1}{5}(10x - 15y)$

(f) $\frac{m-2}{4n} - \frac{m-1}{6n}$

(c) $\frac{1}{3}(2b + 3) - \frac{1}{2}(4b - 1)$

(g) $\frac{5(1+y)}{3} + \frac{3y+1}{5}$

(d) $6 + \frac{3(d-7)}{4}$

4. Simplify each of the following algebraic expressions and use the given values to evaluate the simplified expression.

(a) $6n + 8m + 4n - 5m$, if $m = -3$ and $n = 4$

(b) $14x + 8 - 3x + 2$, if $x = -2$

(c) $26 + 2b - 6c - 3b + 6c$, if $c = 5$ and $b = -4$

(d) $3a - 5b - 7a + 6c + 7a$, if $a = 3$, $b = 2$, and $c = 5$

(e) $5a - b - a + 9b + 3a$, if $a = 4$ and $b = -2$

(f) $4x - 6y + 7x + 2y$, if $x = 10$ and $y = -2$

(g) $9y + 5 - 4y + 3x + 2$, if $x = 4$ and $y = -4$

(h) $13x + 14 + 8x - 4 - 11x$, if $x = -6$

(i) $7x + 3y - 4 - 3y$, if $x = 5$ and $y = -3$

(j) $8m + 0.4m - 2 - 6m + 8$, if $m = -4$

(k) $4x^2y^3 - 6k^3l + 10xy^3 - 4kl^3$, given that $x = 1$, $y = 2$, $k = -2$, and $l = -3$

5. Simplify each of the following expressions.

(a) $5(2x - 3) - (3 - 2x) + 2x^2$

(b) $2p^2q - 3q + 3qp^2$

(c) $xy^2 + 4xy + 3y^2x - 2xy$

(d) $4(x - y) - 2(-3x + 6y)$

6. Simplify each of the following rational expressions.

(a) $\frac{xy^2 + 8xy + 3y^2x - 2xy}{2xy}$

(c) $\frac{7(x-2y) - 3(-x-2y)}{10x-8y}$

(b) $\frac{-2(-3x+7y) - 14x}{2}$

(d) $\frac{3}{8}x^2 - \frac{5}{12}x^2$

7. Evaluate each of the following:

(a) $b^2 - 3ab$ when $a = 3$ and $b = -2$

(b) $\frac{-2xy}{2y-x}$ when $x = -4$ and $y = 6$

(c) $u^2 - 3v$ when $u = 2$ and $v = -4$

(d) $2bc \div (a+7)$ when $a = 3$, $b = -5$, and $c = 4$.

8. Write each of the following statements into a variable expression:

(a) The sum of one-half of x and x .

(b) Ten divided by the difference between y and three.

(c) The difference between six and the quotient of a number and twelve.

9. Simplify each of the following expressions.

(a) $(-8a)\left(-\frac{3}{4}\right)$

(c) $-3(2y^2 - 7) + 8(3y^2 - 12)$

(b) $\frac{1}{7}(14x) - \frac{1}{5}(10x) - \frac{1}{8}(16x)$

(d) $(-10n)\left(-\frac{1}{10}\right) + \frac{1}{3}(9n) - \frac{1}{2}(-2n)$

10. Evaluate each of the following variable expressions when $a = -2$, $b = 4$, $c = -1$ and $d = 3$.

(a) $\frac{b+2d}{3b}$

(d) $-\frac{3}{4}b + \frac{1}{2}(ac + bd)$

(b) $\frac{bd}{a} \div c$

(e) $-\frac{2}{3}d - \frac{1}{5}(bd - ac)$

(c) $\frac{abc}{b-d}$

(f) $\frac{b^2 - a}{ad + 3c}$

Engage in Activity 5.1 to explore and represent real-life activities by using algebraic expressions.

Activity 5.1: Using algebraic expressions in daily life experiences

1. Explore different resources to discover various daily life activities whose processes can be represented by algebraic expressions.
2. Express algebraically the scenarios explored in task 1 and provide justifications for your answers.

Algebraic equations

An algebraic equation is a mathematical statement connecting two algebraic expressions with an equal sign ($=$). It expresses a statement or a problem in a clear and short way. For example, $2x = 16$ and $3x + 4y = 8$ are algebraic equations. Therefore, an equation has two equal sides, the left-hand side and the right-hand side.

Linear equations

A linear equation is an algebraic equation in which each variable has an exponent of one. A linear equation can have one or more unknown variables. For example, $3x - 2 = 7$ and $2z + 1 = 0$ are linear equations with one variable x and z , respectively, whereas $5x + 2y = 9$ and $2x - 3y = 1$ are linear equations with two variables x and y .

Formulation of linear equations

A linear equation can be formulated from a real-life activity or from a given word problem. Formulation of a linear equation from a real-life scenario or word problem involves identifying the unknown, assigning the unknown with a variable, and finally writing the equation based on the given conditions.

The following table shows some common words used in word problems which involve mathematical operations.

Words	Mathematical symbol
Addition, sum, increased by, plus, total, more than, exceed	+
Difference, subtract, decreased, reduced by, minus, less than	−
Multiplication, times, product, multiplied by	×
Division, divided by, quotient, ratio	÷
Equals, result is, is equal to, is, gives, the same as	=

Example 5.12

The sum of two numbers is 20. If one of the numbers is 8, form a linear equation connecting the two numbers.

Solution

Let x be the unknown number. Since the known number is 8, then the algebraic equation connecting the two numbers is $8 + x = 20$.

Example 5.13

The difference between 24 and another number is 8. Form a linear equation connecting the numbers.

Solution

Let y be the unknown number. Since the known number is 24, then the algebraic equation connecting the two numbers is either $24 - y = 8$ or $y - 24 = 8$.

Example 5.14

The difference between two numbers is 48. Form a linear equation which describes the relationship between the two numbers.

Solution

Let z be one of the numbers and p be the other number.

Since the difference between the two numbers is 48, and it is not specified which number is larger or smaller than the other, then the equations which relates the numbers is either $z - p = 48$ or $p - z = 48$.

Example 5.15

A product from a company requires 4 and 3 units of materials of type A and B, respectively. If the company has a total of 20 units of materials, formulate the equation relating the types of materials.

Solution

Let: x be the number of materials of type A.

y be the number of materials of type B.

Total units of the materials available are 20.

Total units of materials of type A and B are $4x$ and $3y$ respectively. Therefore, the equation relating the two types of materials is $4x + 3y = 20$.

Exercise 5.2

Formulate a linear equation in each of the following statements.

1. Five times a number gives twenty.

2. The difference between 123 and another number is 150.
3. The sum of 21 and another number is 125.
4. When a certain number is increased by 15, the result is 88.
5. When 99 is decreased by a certain number, the result is 63.
6. The product of 12 and another number is the same as two times the sum of 12 and the number.
7. A number is such that, when it is doubled and 8 is added to it, the result is the same as multiplying the number by 3, and subtracting 7.
8. When 36 is added to a certain number, the result is the same as multiplying the number by 5.
9. Janeth's mother is five times as old as Janeth. After 8 years, her mother will be 4 times as old as Janeth.
10. When the sum of n and $(n + 3)$ numbers is multiplied by 5, the result is two times the difference between the larger number and the smaller number.
11. The sum of one-third of a number and two-fifth of another number is 1.
12. The difference between one-half of a number and one-third of another number is 2.
13. Four times the value of a quantity equals the sum of 2 and a quarter of the value of another quantity.

Solving linear equations with one unknown

Solving an equation with one unknown means finding the value of the unknown variable which satisfy the equation. It involves several steps of simplification to obtain the value of the variable which satisfies the given equation.

Note: Whatever operation is done on the left-hand side of the equation must be done on the right-hand side.

Example 5.16

Find the solution of $x + 5 = 8$.

Solution

Given $x + 5 = 8$.

Collect the like terms by subtracting 5 from both sides. That is,

$$\begin{aligned} x + 5 - 5 &= 8 - 5 \\ &= 3 \end{aligned}$$

Therefore, $x = 3$

Example 5.17

Find the value of x that satisfies the equation $x - 8 = 16 - 2x$.

Solution

Given $x - 8 = 16 - 2x$.

Collect the like terms by performing the following:

$$x - 8 + 8 = 16 - 2x + 8$$

(Add 8 on both sides)

$$x = 16 - 2x + 8$$

$$x + 2x = 24 - 2x + 2x$$

(Add $2x$ on both sides)

$$3x = 24$$

Divide by 3 on both sides of the equation to get,

$$\begin{aligned} x &= \frac{24}{3} \\ &= 8 \end{aligned}$$

Therefore, $x = 8$.

Example 5.18

Find the value of y if $3y - 5 = 7$

Solution

Given $3y - 5 = 7$.

Add 5 on both sides of the equation to obtain,

$$3y - 5 + 5 = 7 + 5$$

$$3y = 12$$

Dividing by 3 on both sides of the equation gives,

$$\frac{3}{3}y = \frac{12}{3}$$

$$y = 4$$

Therefore, $y = 4$.

Example 5.19

Solve for the value of x in the equation

$$\frac{3x}{2} + 3 = 12.$$

Solution

Given $\frac{3x}{2} + 3 = 12$.

Subtract 3 from both sides of the equation get,

$$\frac{3x}{2} + 3 - 3 = 12 - 3$$

$$\frac{3x}{2} = 9$$

Multiply both sides of the equation by 2 to get,

$$\frac{3x}{2} \times 2 = 9 \times 2$$

$$3x = 18$$

Divide by 3 on both sides of the equation to get,

$$x = \frac{18}{3}$$

$$= 6$$

Therefore, $x = 6$.

Example 5.20

Solve for the value of z if

$$\frac{5}{3}(3z - 2) = 10.$$

Solution

Given $\frac{5}{3}(3z - 2) = 10$.

Multiply by 3 on both sides of the equation. That is,

$$3 \times \frac{5}{3}(3z - 2) = 3 \times 10$$

$$5(3z - 2) = 30$$

Opening the brackets gives,

$$15z - 10 = 30$$

$$15z = 40$$

Dividing by 15 on both sides of the equation gives,

$$z = \frac{40}{15}$$

$$= \frac{8}{3}$$

Therefore, $z = \frac{8}{3}$.

Example 5.21

Solve for the value of y , if $\frac{8}{3y-2} = 2$.

Solution

Given $\frac{8}{3y-2} = 2$.

Multiplying by $(3y-2)$ on both sides of the equation gives,

$$\frac{8}{(3y-2)} \times (3y-2) = 2(3y-2)$$

$$8 = 2(3y-2)$$

$$8 = 6y - 4$$

$$12 = 6y$$

$$y = \frac{12}{6}$$

$$= 2$$

Therefore, $y = 2$.

Example 5.22

Find the values of x if $|x| = 4$.

Solution

Given $|x| = 4$.

Apply the definition of absolute value. That is,

$$\pm x = 4$$

Either, $+x = 4$ or $-x = 4$

Therefore, $x = 4$ or $x = -4$.

Example 5.23

Solve for the value of x if $|6-x| = 1$.

Solution

Given $|6-x| = 1$.

Apply the definition of absolute value. That is, $\pm(6-x) = 1$

Either,

$$+(6-x) = 1 \text{ or } -(6-x) = 1.$$

It follows that,

$$6-x = 1 \text{ or } -6+x = 1$$

$$x = 5 \text{ or } x = 7$$

Therefore, $x = 5$ or $x = 7$.

Example 5.24

Solve for the value of x if $|x+2| = 2$ and represent the solution on a number line.

Solution

Given $|x+2| = 2$. Apply the definition of absolute value to get,

$$\pm(x+2) = 2$$

Either $+(x+2) = 2$ or $-(x+2) = 2$.

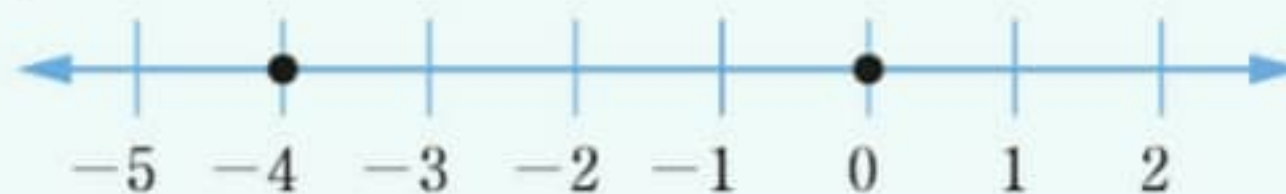
It follows that,

$$x + 2 = 2 \text{ or } -x - 2 = 2$$

$$x = 0 \text{ or } -x = 4$$

Therefore, $x = 0$ or $x = -4$.

The solution is presented on a number line as follows.



Example 5.25

Solve for the value of x if $|x + 4| = |2x - 1|$ and represent the solution on a number line.

Solution

Given $|x + 4| = |2x - 1|$.

Applying the definition of absolute value of a number gives,

$$\pm(x + 4) = \pm(2x - 1)$$

$+(x + 4) = (2x - 1)$ or $-(x + 4) = -(2x - 1)$ or $+(x + 4) = -(2x - 1)$ or $-(x + 4) = +(2x - 1)$ will give the solution. Thus, there are four possible alternatives that give the solution.

However, the equations $+(x + 4) = +(2x - 1)$ and $-(x + 4) = -(2x - 1)$ are equivalent. Similarly, $+(x + 4) = -(2x - 1)$ and $-(x + 4) = +(2x - 1)$ are also equivalent. Thus, there are only two possible alternatives, which are $+(x + 4) = +(2x - 1)$ and $-(x + 4) = +(2x - 1)$.

Proceed to solve the two equations as follows:

$$x + 4 = 2x - 1 \text{ or } -x - 4 = 2x - 1$$

Collecting like terms and solving for the value of x gives,

$$x = 5 \text{ or } x = -1$$

Therefore, $x = 5$ or $x = -1$

The solution is presented on the following number line.



Engage in Activity 5.2 to learn how to use mathematical software in solving different linear equations.

Activity 5.2: Solving linear equations by using mathematical software

1. Identify different mathematical software which can be used in solving systems of linear equations algebraically or graphically.
2. Study the basic procedures for solving linear equations by using the identified software.
3. Use the software to solve different equations and check your answers by substitution.

Exercise 5.3

1. Find the value of x in each of the following equations.

(a) $x + 12 = 25$	(h) $\frac{x}{5} - 3 = 5$	(o) $\frac{3}{5}x - 1 = 5$
(b) $x - 8 = 8$	(i) $0.2x + 7 = 9$	
(c) $3x = 15$	(j) $0.6x - 5 = 7$	
(d) $\frac{x}{5} = 5$	(k) $\frac{x-6}{7} - \frac{2x+3}{4} = \frac{x-5}{2}$	
(e) $2x + 12 = 25$	(l) $3x - 3 = 15$	
(f) $2x - 8 = 8$	(m) $\frac{7x+1}{2} = 14$	
(g) $\frac{x}{5} + 3 = 5$	(n) $\frac{3x+2}{5} = 4$	

2. Find the solution in each of the following equations.

(a) $ y = 3$	(d) $ 10 - y = 1$
(b) $ 2 - y = 5$	(e) $ 6 - y = 3$
(c) $ 2y = 6$	(f) $\left y - \frac{1}{3}\right = \frac{5}{3}$

3. Show the solution in each of the following equations on a number line.

(a) $ x - 5 = 3$	(c) $ 3 - x = \frac{10}{3}$
(b) $ -2 - x = 4$	(d) $\left \frac{1}{2}x + 2\right = 6$

4. Find the solution in each of the following equations.

(a) $3|z-3|=4$ (c) $|6-z|=4$ (e) $|3x-2|=2|8+x|$
 (b) $2|z|=6$ (d) $|2-x|=2|x+1|$

5. Determine the solution of each of the following equations.

(a) $0.09p + 0.11p = 3$ (c) $\frac{8x-3}{6x+9} = -\frac{2}{3}$
 (b) $3(5.4+r) = 5.4r$ (d) $\frac{1}{3}y + \frac{1}{4} = \frac{1}{2}y + 0.25 + \frac{1}{6}y$

6. Find the value of x in each of the following equations.

(a) $ax = b$ (d) $y = \frac{7}{3}x + z$
 (b) $x + n = m$ (e) $a(x+b) = d$
 (c) $px - 5 = n$

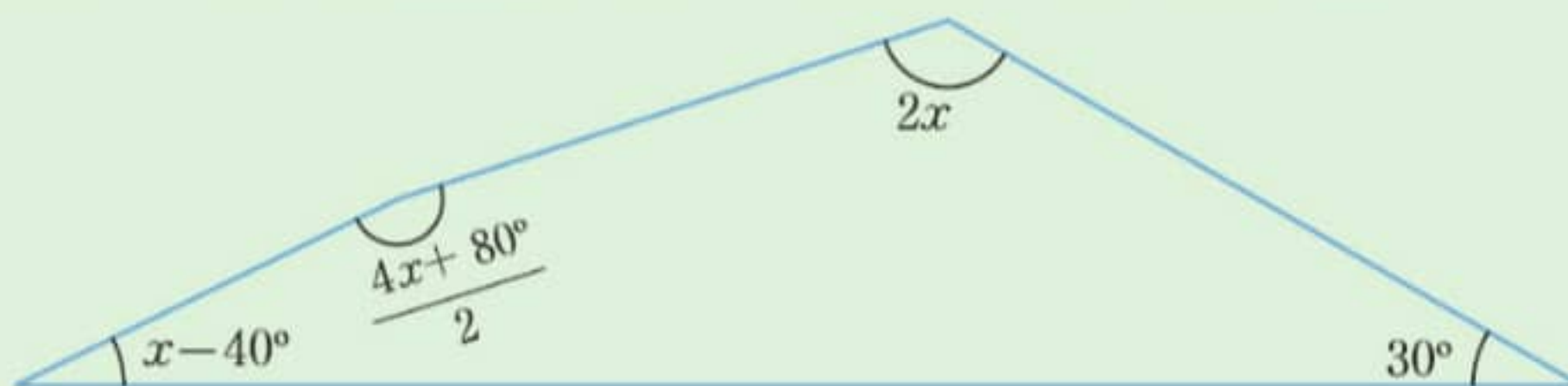
7. Solve for the value of x in each of the following equations.

(a) $\frac{2x+8}{3x-2} = \frac{5}{4}$ (b) $7x+11 = 8-5x$ (c) $\frac{2}{x} + 3 = 4$

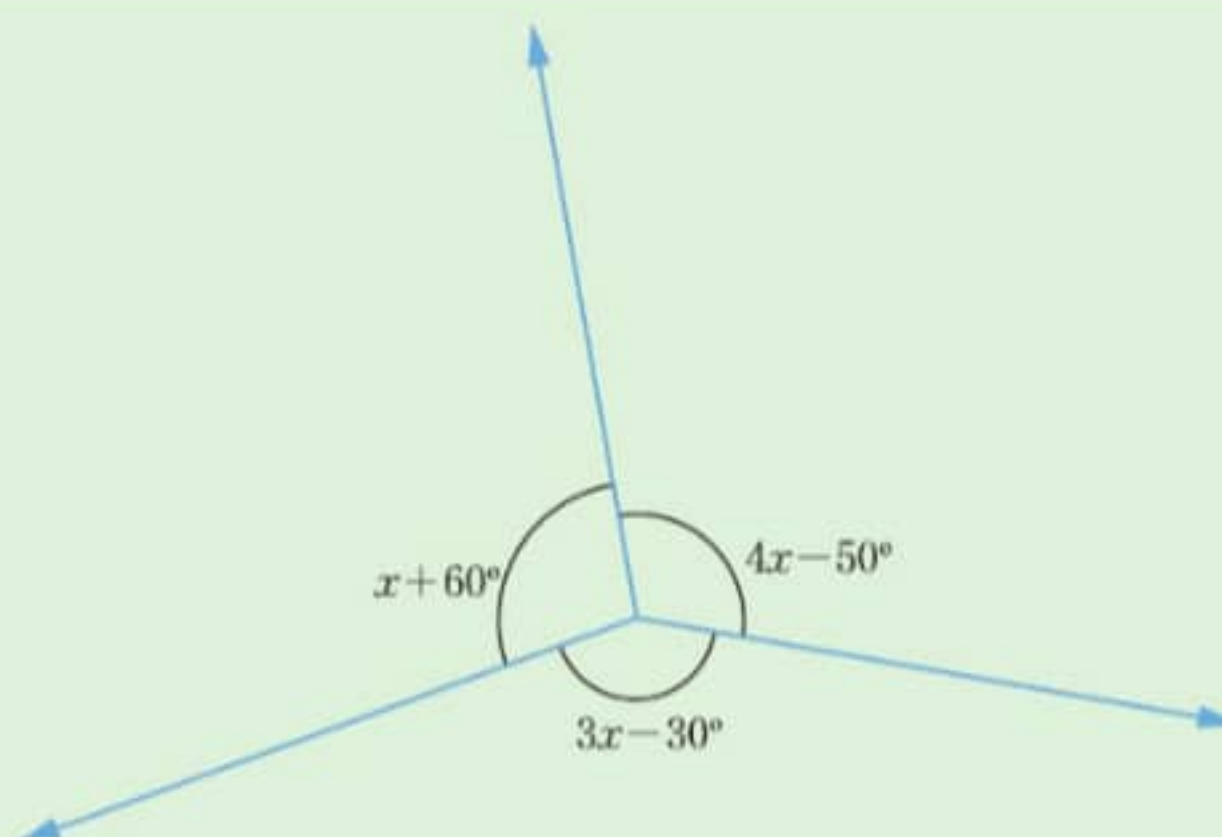
8. If the angles of a triangle are $2x$, $(x+20^\circ)$, and $(x+30^\circ)$, find the value of x .

9. In each of the following figures, calculate the value of x in degrees.

(a)



(b)



Solving word problems involving linear equations

Solving word problems on linear equations involve formulating the corresponding equations. The formulation of an equation requires careful interpretation of the given word problem. Once the equation has been formulated, the usual procedures for solving the equation are used.

Example 5.26

If John has 5,600 Tanzanian shillings, how many oranges can he buy if each orange costs 200 Tanzanian shillings?

Solution

Let x be the number of oranges that John can buy. The cost of x oranges is Tshs $200x$.

Thus, $200x = 5600$

$$\begin{aligned} x &= \frac{5600}{200} \\ &= 28. \end{aligned}$$

Therefore, John can buy 28 oranges.

Example 5.27

The age of a father is four times the age of his son. If the sum of their ages is 50 years, find the age of the son.

Solution

Let x be the age of the son. Thus, the age of the father is $4x$ years. It follows that,

$$x + 4x = 50$$

$$5x = 50$$

$$\begin{aligned} x &= \frac{50}{5} \\ &= 10 \end{aligned}$$

Therefore, the age of the son is 10 years.

Example 5.28

The sum of two consecutive integers is 31. Find the smaller number.

Solution

Since the two numbers are consecutive, then if the first integer is n , the second integer is $n + 1$

Thus, $n + (n + 1) = 31$

$$2n + 1 = 31$$

$$2n = 30$$

$$\frac{2}{2}n = \frac{30}{2}$$

$$n = 15$$

Therefore, the smaller number is 15.

Example 5.29

A father is 32 years older than his son. After 4 years, the father's age will be twice the age of his son. Find their present ages.

Solution

Let x be the son's present age.

Thus, the father's present age is $(x + 32)$ years.

The son's age after 4 years will be $(x + 4)$ years.

The father's age after 4 years will be $(x + 32 + 4)$ years $= (x + 36)$ years.

After 4 years, the father's age will be twice the age of his son. That is,

$$x + 36 = 2(x + 4)$$

$$x + 36 = 2x + 8$$

$$2x - x = 36 - 8$$

$$x = 28$$

$$\begin{aligned}\text{The father's present age} &= x + 32 \\ &= 28 + 32 \\ &= 60\end{aligned}$$

Therefore, the son's present age is 28 years and the father's present age is 60 years.

Engage in Activity 5.3 to use algebra in cooking as among the daily-life activities.

Activity 5.3: Algebra in cooking

Cooking rice dish requires an optimum ratio between the amount of rice to be cooked and amount of water to be used. On average, 2 cups of water are enough to cook 1 same cup of rice.

1. Determine an algebraic equation which relates the amount of rice and water to be used.
2. If you are to decide on the amount of rice to be cooked in a ceremony with 300 participants, determine the average amount of rice needed assuming that on average, 1 kg of rice can be served to 4 people.
3. Use the algebraic equation in task 1 to determine the average amount of water needed to cook the amount of rice you have determined in task 2.

Exercise 5.4

1. John is 6 years older than Amina. If the sum of their ages is 30 years, find Amina's age.
2. Asha's age is five times the age of Martin. If the sum of their ages is 36 years, find Martin's age.
3. If the sum of two consecutive numbers is 37, find the numbers.
4. When a certain number is added to 7, the result is $12\frac{3}{4}$. Find the number.
5. If the product of $2\frac{1}{2}$ and y is 25, find the value of y .
6. The difference between 48 and another number is 17. Find the number.

7. When the difference between 24 and m is multiplied by 5, the result is -20 . Find the value of m .
8. One-third of a certain number is added to three-fifths of the same number to give the sum of 14. Find the number.
9. The product of half of a certain number and 6 is 48. Find the number.
10. The sum of a number and its 40 percent is 28. Find the number.
11. A mother is 28 years older than her daughter. After 10 years, the mother will be three times older than her daughter. Find their present ages.
12. A boy bought y balls at 500 Tanzanian shillings each. This number of balls could decrease by 4 if the price for each ball were 700 Tanzanian shillings each. Find the value of y .
13. When a number is multiplied by 8 and 9 is subtracted from the product, the result is 45 more than twice the number. Find the number.
14. Julius is four times older than Amina. Five years ago, the sum of their ages was 50. Find their present age.
15. If 18 is decreased by the sum of a number and 4, the result is 2 less than 3 times the number. Find the number.

Linear simultaneous equations

Linear simultaneous equations, also referred to as a system of linear equations, are two or more algebraic equations which share the variables and the solutions. They are called simultaneous equations because they are solved at once (simultaneously) and have common solution. Consider the following linear equation.

$$x + y = 20 \quad (1)$$

Equation (1) involves two unknowns x and y . Different values of x and y can be used to make the equation true. The values of x and y that satisfy equation (1) in ordered pairs (x, y) are $(1, 19)$, $(2, 18)$, $(3, 17)$, $(10, 10)$, $(13.4, 6.6)$ and so on. Therefore, in order to get unique values of x and y in equation (1), another linear equation is needed. For instance, if it is also known that

$$x - y = 8 \quad (2)$$

It is possible to use equation (1) and (2) to find the unique values of x and y which satisfy both equations. These equations are such that the value of x in equation (1) is the same as the value of x in equation (2) and the same applies to y , that is, the equations have common solution.

Solving linear simultaneous equations

There are several methods of solving simultaneous equations. This section discusses how solutions of linear simultaneous equations are obtained by elimination and substitution methods.

Solving linear simultaneous equations by elimination method

Elimination method involves performing suitable mathematical operations to reduce the two equations into a single equation with one unknown. The following are the steps for solving linear simultaneous equations by elimination method:

Step 1: Choose the easiest variable to be eliminated. Multiply the given equations by suitable numbers so as to make the coefficients of the chosen variable in both equations equal.

Step 2: Add the new equations if the coefficients of the chosen variable to be eliminated are opposite in signs, otherwise subtract them.

Step 3: Solve the obtained equations. This gives the value of one of the unknowns.

Step 4: Repeat steps 1 to 3 for the value of the second unknown or substitute the solution obtained in step 3 into one of the equations to get the value of the other unknown.

Note: Some simultaneous equations can be converted into linear equations by changing the variables.

Example 5.30

Solve the following linear simultaneous equations by elimination method.

$$\begin{cases} 6x + y = 15 \\ 3x + y = 9 \end{cases}$$

Solution

Label the equations, say (i) and (ii), respectively. That is,

$$\begin{cases} 6x + y = 15 & \text{(i)} \\ 3x + y = 9 & \text{(ii)} \end{cases}$$

Choose the easiest variable to be eliminated. In this case, y can be eliminated easily.

Eliminate y to form an equation with variable x only.

The coefficient of y in equation (i) is 1 and in equation (ii) is 1.

Since the signs of the variable y in (i) and (ii) are the same, subtract equation (ii) from (i) as follows.

$$\begin{array}{r} 6x + y = 15 \\ - (3x + y = 9) \\ \hline 3x = 6 \end{array}$$

It follows that,

$$3x = 6$$

$$x = 2$$

Substitute the value of x into one of the equations to obtain the value of y .

Substitute $x = 2$ into the equation (ii) to obtain,

$$3(2) + y = 9$$

$$6 + y = 9$$

$$y = 3$$

Therefore, $x = 2$ and $y = 3$ is the solution of the given simultaneous equations.

Example 5.31

Solve the following simultaneous equations by elimination method.

$$\begin{cases} 2x + y = 10 \\ 3x - 2y = 1 \end{cases}$$

Solution

Label the given simultaneous equations.

$$\begin{cases} 2x + y = 10 & \text{(i)} \\ 3x - 2y = 1 & \text{(ii)} \end{cases}$$

Choose the variable to be eliminated. Eliminate y to form an equation with variable x only.

The coefficient of y in equation (i) is 1 and in equation (ii) is -2 .

Multiply equation (i) by -2 to get a new equation.

$$\begin{cases} -2(2x + y) = 10 \times -2 \\ 3x - 2y = 1 \end{cases}$$

$$-4x - 2y = -20 \quad \text{(iii)}$$

Subtract equation (ii) from equation (iii) as follows:

$$\begin{array}{r} -4x - 2y = -20 \\ - \quad 3x - 2y = 1 \\ \hline -7x \quad \quad = -21 \end{array}$$

Solve for the value of x in the equation of one unknown. That is,

$$\begin{aligned} -7x &= -21 \\ x &= \frac{21}{7} \\ &= 3 \end{aligned}$$

Substitute the value of x into (i) or (ii) or (iii) to find the value of y .

From equation (i), substituting the value of x gives

$$2(3) + y = 10$$

$$6 + y = 10$$

$$y = 10 - 6$$

$$= 4$$

Therefore, $x = 3$ and $y = 4$.

Example 5.32

Solve the following linear simultaneous equations by elimination method.

$$\begin{cases} 3x - 2y = 8 \\ 2x + 5y = -1 \end{cases}$$

Solution

Label the two equations as follows.

$$\begin{cases} 3x - 2y = 8 & \text{(i)} \\ 2x + 5y = -1 & \text{(ii)} \end{cases}$$

Eliminate y by multiplying equation (i) by 5 and equation (ii) by 2 as follows:

$$\begin{cases} 5(3x - 2y) = 8 \times 5 \\ 2(2x + 5y) = -1 \times 2 \end{cases}$$

$$\begin{cases} 15x - 10y = 40 & \text{(iii)} \\ 4x + 10y = -2 & \text{(iv)} \end{cases}$$

Since the coefficients of y in equations (iii) and (iv) are equal with opposite signs, add the two equations. That is,

$$\begin{array}{r} + \begin{cases} 15x - 10y = 40 \\ 4x + 10y = -2 \end{cases} \\ \hline 19x \quad \quad = 38 \end{array}$$

Solve for the value of x in the equation of one unknown. That is,

$$\begin{aligned} 19x &= 38 \\ x &= \frac{38}{19} \\ &= 2 \end{aligned}$$

Substitute $x = 2$ into equation (ii) to obtain the value of y .

From $2x + 5y = -1$, it follows that,

$$2(2) + 5y = -1$$

$$4 + 5y = -1$$

$$5y = -5$$

$$y = -1$$

Therefore, $x = 2$ and $y = -1$ is the solution of the simultaneous equation.

Example 5.33

Solve the following simultaneous equations by elimination method.

$$\begin{cases} \frac{x}{2} + \frac{y}{5} = 4 \\ \frac{x}{4} + \frac{y}{2} = 6 \end{cases}$$

Solution

Label the two equations.

$$\begin{cases} \frac{x}{2} + \frac{y}{5} = 4 & \text{(i)} \\ \frac{x}{4} + \frac{y}{2} = 6 & \text{(ii)} \end{cases}$$

For simplicity, transform the two equations, (i) and (ii) into equations with integer coefficients of the variables. In order to remove the fractions, multiply the equations by the LCM of the denominators.

That is, the LCM in equation (i) is 10 and the LCM in equation (ii) is 4. It follows that,

$$\begin{cases} 10\left(\frac{x}{2} + \frac{y}{5}\right) = 4 \times 10 \\ 4\left(\frac{x}{4} + \frac{y}{2}\right) = 6 \times 4 \end{cases}$$

$$\begin{cases} 5x + 2y = 40 & \text{(iii)} \end{cases}$$

$$\begin{cases} x + 2y = 24 & \text{(iv)} \end{cases}$$

Eliminate y to obtain an equation with variable x only.

Since the coefficients of y in equations (iii) and (iv) are the same, subtract equation (iv) from equation (iii).

$$\begin{array}{r} \begin{cases} 5x + 2y = 40 \\ - \quad x + 2y = 24 \end{cases} \\ \hline 4x \quad \quad = 16 \end{array}$$

$$x = \frac{16}{4}$$

$$x = 4$$

Substitute $x = 4$ into equation (iv) to get the value of y .

$$\text{From } x + 2y = 24,$$

$$4 + 2y = 24$$

$$2y = 20$$

$$y = 10$$

Therefore, $x = 4$ and $y = 10$ is the solution of the simultaneous equation.

Example 5.34

Find the solution of the following simultaneous equations by elimination method.

$$\begin{cases} \frac{2}{x} - \frac{3}{y} = -3 \\ \frac{1}{x} + \frac{6}{y} = 1 \end{cases}$$

Solution

Convert the simultaneous equations into linear equations.

$$\text{Let } a = \frac{1}{x} \text{ and } b = \frac{1}{y}.$$

The simultaneous equation will be equivalent to the following:

$$\begin{cases} 2a - 3b = -3 & \text{(i)} \end{cases}$$

$$\begin{cases} a + 6b = 1 & \text{(ii)} \end{cases}$$

Eliminate a to obtain an equation with variable b only.

Multiply equation (ii) by 2. That is,

$$2(a + 6b) = 1 \times 2$$

$$2a + 12b = 2 \quad \text{(iii)}$$

Subtract equation (iii) from equation (i) as follows:

$$\begin{array}{r} \begin{cases} 2a - 3b = -3 \\ - \quad 2a + 12b = 2 \end{cases} \\ \hline -15b = -5 \end{array}$$

Solve for the value of b as follows:

$$-15b = -5$$

$$b = \frac{-5}{-15}$$

$$= \frac{1}{3}$$

Substitute $b = \frac{1}{3}$ into equation (ii) to obtain the value of a .

From $a + 6b = 1$. It implies that,

$$a + 6\left(\frac{1}{3}\right) = 1.$$

$$a + 2 = 1.$$

$$a = -1$$

But, $a = \frac{1}{x}$ and $b = \frac{1}{y}$. It follows that, $-1 = \frac{1}{x}$ and $\frac{1}{3} = \frac{1}{y}$.

Solving for the values of x and y gives,

$$x = -1 \text{ and } y = 3.$$

Therefore, $x = -1$ and $y = 3$ is the solution of the simultaneous equation.

Exercise 5.5

Solve the following simultaneous equations by using elimination method.

$$1. \begin{cases} 2x + y = 5 \\ 4x - y = 7 \end{cases}$$

$$2. \begin{cases} 3p + q = 6 \\ 5p + q = 8 \end{cases}$$

$$3. \begin{cases} 5x - 2y = 16 \\ x + 2y = 8 \end{cases}$$

$$4. \begin{cases} 8x + 5y = 40 \\ 9x - 5y = 5 \end{cases}$$

$$5. \begin{cases} 7x - 4y = 17 \\ 5x - 4y = 11 \end{cases}$$

$$6. \begin{cases} 0.7x - 0.5y = 2.5 \\ 0.7x + 0.3y = 2.9 \end{cases}$$

$$7. \begin{cases} 2x + y = 0 \\ x - 2y = -5 \end{cases}$$

$$8. \begin{cases} \frac{4}{x} + \frac{1}{y} = 3 \\ \frac{1}{x} + \frac{2}{y} = -1 \end{cases}$$

$$9. \begin{cases} \frac{8}{x} - \frac{12}{y} = -2 \\ \frac{5}{x} + \frac{3}{2y} = 1 \end{cases}$$

$$10. \begin{cases} 0.2x + 0.3y = 0.1 \\ 0.3x - 0.1y = 0.7 \end{cases}$$

$$11. \begin{cases} 0.05x + 0.02y = 0.01 \\ 0.02x - 0.03y = -0.11 \end{cases}$$

$$12. \begin{cases} \frac{1}{2}x + 3y = 3 \\ \frac{3}{4}x - 2y = -2 \end{cases}$$

$$13. \begin{cases} u + 2v = 1 \\ 3u - 4v = 8 \end{cases}$$

$$14. \begin{cases} 2p + q = 4 \\ -3p + 5q = -19 \end{cases}$$

$$15. \begin{cases} 2m + \frac{1}{3}n = 3 \\ -3m + n = 6 \end{cases}$$

Solving linear simultaneous equations by substitution method

Substitution method involves choosing one equation and transposing one of its variables by making it the subject of the other. The resulting equation is substituted into the other to obtain a single equation with one unknown. The following are the steps for solving simultaneous equations of two unknowns by substitution method.

Step 1: Express one of the variables in terms of the other from one of the equations.

Step 2: Substitute the equation obtained in Step 1 into the second equation to obtain an equation in one unknown.

Step 3: Solve the equation in one unknown obtained in Step 2.

Step 4: Substitute the value of the variable obtained in Step 3 to one of the equations to get the value of the other unknown.

Example 5.35

Solve the following simultaneous equations by substitution method.

$$\begin{cases} 3y - 2x = -7 \\ y + 2x = 11 \end{cases}$$

Solution

Label the two equations as follows.

$$\begin{cases} 3y - 2x = -7 & \text{(i)} \\ y + 2x = 11 & \text{(ii)} \end{cases}$$

Using equation (ii), express y in terms of x as follows.

From $y + 2x = 11$, it implies that,

$$y = 11 - 2x \quad \text{(iii)}$$

Substituting equation (iii) into equation (i) gives,

$$3(11 - 2x) - 2x = -7$$

Expanding and collecting like terms gives,

$$-8x = -40$$

$$x = \frac{40}{8} = 5$$

Substitute $x = 5$ into equation (iii), that is,

$$\begin{aligned} y &= 11 - (2 \times 5) \\ &= 11 - 10 \\ &= 1 \end{aligned}$$

Therefore, $x = 5$ and $y = 1$.

Example 5.36

Solve the following simultaneous equations by substitution method.

$$\begin{cases} 6x + 5y = 3 \\ 7x + 8y = 10 \end{cases}$$

Solution

Label the two equations:

$$\begin{cases} 6x + 5y = 3 & \text{(i)} \\ 7x + 8y = 10 & \text{(ii)} \end{cases}$$

Use equation (i) to express x in terms of y . That is,

$$x = \frac{3 - 5y}{6} \quad \text{(iii)}$$

Substitute equation (iii) into equation (ii) to obtain,

$$7\left(\frac{3-5y}{6}\right) + 8y = 10$$

Simplification gives,

$$\frac{21-35y}{6} + 8y = 10$$

Further simplification gives,

$$\left(\frac{21-35y}{6} + 8y\right) \times 6 = 10 \times 6$$

$$21 - 35y + 48y = 60$$

$$13y = 60 - 21$$

$$13y = 39$$

$$y = \frac{39}{13}$$

$$= 3$$

Hence, $y = 3$.

Substitute $y = 3$ into equation (iii) to obtain the value of x . That is,

$$x = \frac{3-5(3)}{6}$$

$$x = \frac{3-15}{6}$$

$$= \frac{-12}{6}$$

$$= -2$$

Therefore, $x = -2$ and $y = 3$.

Example 5.37

Find the solution of the following simultaneous equations by substitution method.

$$\begin{cases} 0.2x + 0.5y = 2 \\ -0.4x + 0.3y = -1.4 \end{cases}$$

Solution

Transform the equations into equations with integer coefficients. Multiply by 10 both sides of the equations.

$$\begin{cases} 10(0.2x + 0.5y) = 2 \times 10 \\ 10(-0.4x + 0.3y) = -1.4 \times 10 \end{cases}$$

$$\begin{cases} 2x + 5y = 20 & \text{(i)} \end{cases}$$

$$\begin{cases} -4x + 3y = -14 & \text{(ii)} \end{cases}$$

Use equation (i) to make y the subject. That is,

$$5y = 20 - 2x$$

$$y = 4 - \frac{2}{5}x \quad \text{(iii)}$$

Substitute equation (iii) into (ii). Thus,

$$-4x + 3\left(4 - \frac{2}{5}x\right) = -14$$

$$-4x + 12 - \frac{6}{5}x = -14$$

Further simplification gives,

$$-\frac{26}{5}x = -26$$

$$\frac{1}{5}x = 1$$

$$x = 5$$

Substitute $x = 5$ in equation (iii) to obtain the value of y . That is,

$$y = 4 - \frac{2}{5}x$$

$$\begin{aligned}
 &= 4 - \frac{2}{5}(5) \\
 &= 4 - 2 \\
 &= 2
 \end{aligned}$$

Therefore, $x = 5$ and $y = 2$

Exercise 5.6

Solve the following simultaneous equations by using substitution method.

$$1. \begin{cases} 3x - 2y = 5 \\ 2x + y = 8 \end{cases}$$

$$2. \begin{cases} 5a + b = 23 \\ 3a - 2b = 6 \end{cases}$$

$$3. \begin{cases} x - 3y = 2 \\ 4x + 2y = 36 \end{cases}$$

$$4. \begin{cases} 7x - y = 14 \\ 8x - 2y = 16 \end{cases}$$

$$5. \begin{cases} 7x + y = 17 \\ 8x - 2y = 10 \end{cases}$$

$$6. \begin{cases} 2x - y = 9 \\ x + y = 9 \end{cases}$$

$$7. \begin{cases} 3y - x = 4 \\ y + 2x = 6 \end{cases}$$

$$8. \begin{cases} 8m - n = 38 \\ m - 3n = -1 \end{cases}$$

$$9. \begin{cases} 2a + b = 21 \\ a + 2b = 45 \end{cases}$$

$$10. \begin{cases} 0.3x - 0.2y = 2.8 \\ 1.5x - 0.4y = 7 \end{cases}$$

$$11. \begin{cases} \frac{x}{2} - \frac{y}{3} = 5 \\ \frac{x}{3} - \frac{y}{6} = 3\frac{1}{2} \end{cases}$$

$$12. \begin{cases} p - q = 5 \\ 3p - q = p + 13 \end{cases}$$

$$13. \begin{cases} 5x - 2y = 10 \\ -x + 3y = 24 \end{cases}$$

$$14. \begin{cases} 2x + y = 10 \\ x - 2y = 1 \end{cases}$$

$$15. \begin{cases} \frac{x}{2} + y = 9 \\ 3x - \frac{y}{5} = 23 \end{cases}$$

Solving word problems on linear simultaneous equations

Simultaneous equations are useful in solving real-life problems involving more than one unknown which can be solved simultaneously. Such problems can be formulated from different real-life scenarios such as the routes of the journey, managing multiple loans, recipe proportions and many others. Formulating and solving word problems on simultaneous equations involve the following steps:

1. Let the unknowns be represented by variables.

- Write the two simultaneous equations according to the requirements of the problem.
- Solve the linear simultaneous equations.
- Give the conclusion according to the demand of the problem.

Example 5.38

The age of a father is four times the age of his son. If the sum of their ages is 60 years, find the age of the son and that of the father.

Solution

Let x be the age of the son and y be the age of the father.

It follows that,

$$y = 4x \quad (i)$$

$$x + y = 60 \quad (ii)$$

Substituting equation (i) into equation (ii) gives,

$$x + 4x = 60$$

$$5x = 60$$

$$x = \frac{60}{5}$$

$$= 12.$$

Hence, $x = 12$.

Substituting $x = 12$ into equation (i) gives,

$$\begin{aligned} y &= 4 \times 12 \\ &= 48 \end{aligned}$$

Hence, $y = 48$

Therefore, the son's age is 12 years and the father's age is 48 years.

Example 5.39

The sum of two numbers is 30. The difference between the larger number and three times the smaller number is 2. Find the two numbers.

Solution

Let x be the larger number and y be the smaller number.

Thus,

$$\begin{cases} x + y = 30 & (i) \end{cases}$$

$$\begin{cases} x - 3y = 2 & (ii) \end{cases}$$

Solve equations (i) and (ii) simultaneously using elimination method.

Eliminate x to obtain an equation in one variable. That is,

$$\begin{array}{r} \begin{cases} x + y = 30 \\ x - 3y = 2 \end{cases} \\ \hline \end{array}$$

$$4y = 28$$

$$y = \frac{28}{4}$$

$$= 7$$

Hence, $y = 7$.

Substitute the value of y in equation (i) or (ii). From $x + y = 30$,

$$x + 7 = 30$$

$$x = 23$$

Hence, $x = 23$.

Therefore, the larger number is 23 and the smaller number is 7.

Example 5.40

Dashi has 83,000 Tanzanian shillings for shopping. If he buys 2 ties and 2 shirts, he remains with 9,000 Tanzanian shillings. If he buys 1 tie and 3 shirts he spends all the money. Find the price of a tie and a shirt.

Solution

Let: p be the price of a tie in Tanzanian shillings.

r be the price of a shirt in Tanzanian shillings.

It implies that for the first option, the cost for 2 ties is Tshs $2p$ and the cost for 2 shirts is Tshs $2r$.

Thus, the total cost for 2 ties and 2 shirts is $2p + 2r$.

For the second option, the cost for 1 tie is Tshs p and the cost for 3 shirts is Tshs $3r$.

Therefore, the total cost for 1 tie and 3 shirts is $p + 3r$.

From this information, form a system of linear simultaneous equations as follows:

$$\begin{cases} 83\,000 - (2p + 2r) = 9\,000 \\ 83\,000 - (p + 3r) = 0 \end{cases}$$

Simplifying simultaneous equations gives,

$$\begin{cases} p + r = 37,000 \\ p - 3r = 83,000 \end{cases}$$

Use substitution method to solve equations (i) and (ii) as follows. From equation (ii), expressing p in terms of r gives

$$p = 83000 - 3r \quad \text{(iii)}$$

Substitute equation (iii) into equation (i), that is,

$$\begin{aligned} 83000 - 3r + r &= 34000 \\ 83000 - 2r &= 34000 \\ -2r &= -46000 \\ r &= \frac{-46000}{-2} \\ &= 23,000 \end{aligned}$$

Hence, $r = 23,000$

Substitute $r = 23,000$ into equation (iii) to obtain the value of p . That is

$$\begin{aligned} p &= 83,000 - (3 \times 23,000) \\ &= 83,000 - 69,000 \\ &= 14,000 \end{aligned}$$

Therefore, the price of each tie is 14,000 Tanzanian shillings and the price of each shirt is 23,000 Tanzanian shillings.

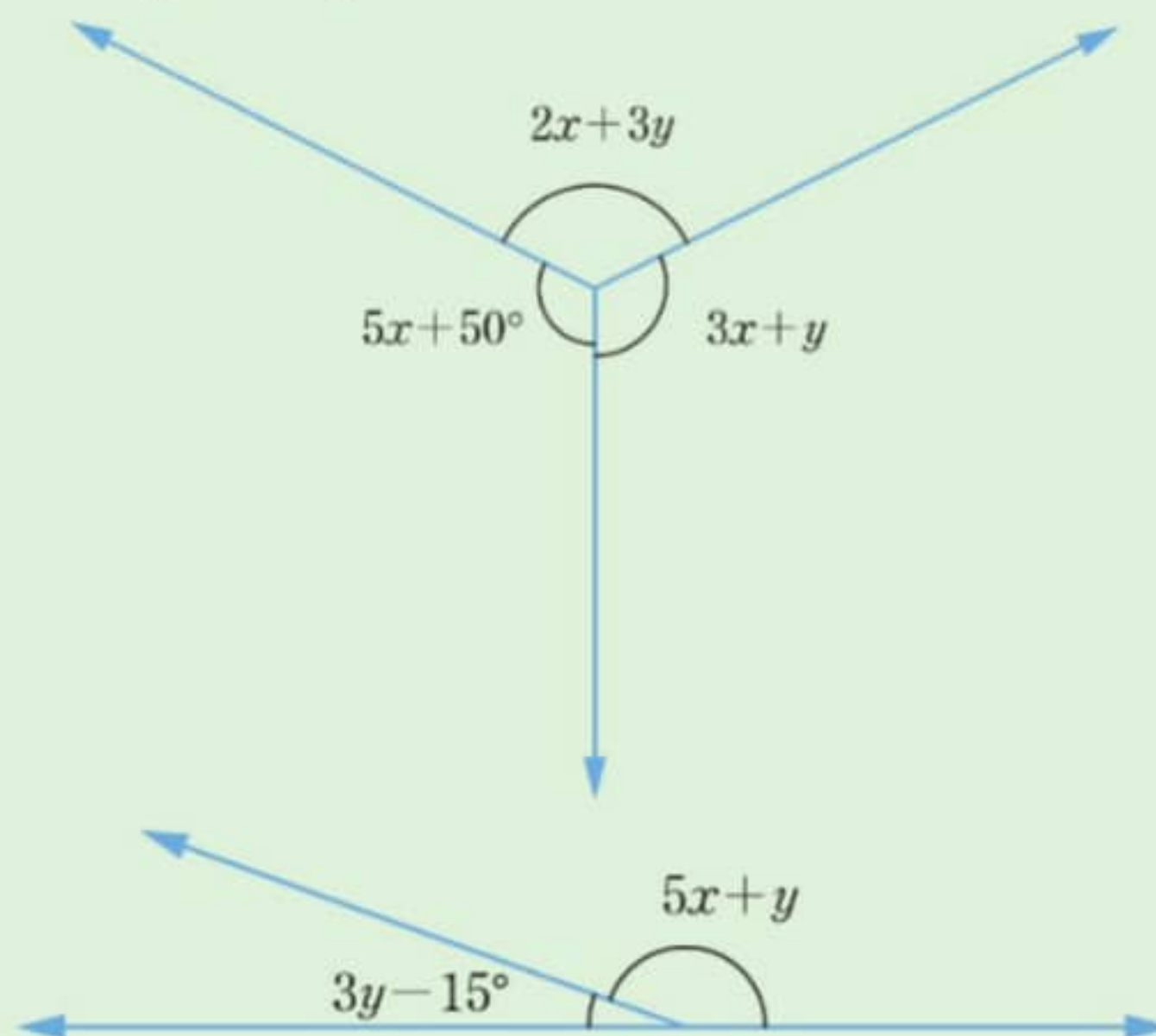
Activity 5.4: Demonstrating occurrence of simultaneous events

1. Use various resources such as offline and online libraries, applications and games, and real objects to discuss real life scenario which apply the concept of simultaneous equations.
2. Represent the scenarios using systems of linear simultaneous equations, describe the variables, solve them and interpret the scenarios.

Exercise 5.7

1. The sum of two numbers is 109 and their difference is 29. Find the numbers.
2. Two numbers are such that the first number plus three times the second number is 1, and the first number minus three times the second number is $\frac{1}{7}$. Find the two numbers.
3. The sum of the number of boys and girls in a class is 36. If twice the number of girls exceeds the number of boys by 12, find the number of girls and boys in the class.
4. Twice the length of a rectangle exceeds three times its width by one centimetre. If one third of the difference of the length and the width is one centimetre, find the dimensions of the rectangle.
5. The cost of 4 pencils and 5 pens together is 6,000 Tanzanian shillings, while the cost of 6 pencils and 8 pens together is 9,400 Tanzanian shillings. Calculate the cost of one pencil and one pen.
6. Half of Paul's money plus one fifth of Hamisa's money is 14,000 Tanzanian shillings. Three quarters of Paul's money plus two-thirds of Hamisa's money is 26,500 Tanzanian shillings. How much does each one has?
7. One-third of the sum of two numbers is 50 and one fifth of their difference is 2. Find the numbers.
8. One pair of the opposite sides of a parallelogram is $4x - 3y$ units and $(3x - 5)$ units while the other pair is $(2x + y)$ units and $2\frac{1}{2}$ units. Find the values of x and y .
9. The sides of an equilateral triangle are given as $\left(3\frac{1}{2}x + y\right)$ centimetres, $(2y - 6x)$ centimetres, and $x + y + 2$ centimetres. Find the values of x and y .

10. Use the following two figures to calculate the values of x and y .



Solution of inequalities with one unknown

Solutions of inequalities with one variable are determined by using similar approach as that used in solving linear equations. However, there are some specific aspects to consider when solving inequalities. In most cases, solutions in inequalities are presented in range or as a set. Thus, solving linear inequalities requires adhering to the following rules:

- Adding or subtracting an equal value to both sides does not change the inequality sign.
- Multiplying or dividing on both sides by a positive number does not change the inequality sign.

- Multiplying or dividing on both sides by a negative number changes the inequality sign.

Example 5.41

Solve for the values of x if $x - 3 < \frac{1}{2}$.

Solution

Given $x - 3 < \frac{1}{2}$.

Add 3 to both sides of the inequality to obtain

$$\begin{aligned} x - 3 + 3 &< \frac{1}{2} + 3 \\ x &< 3\frac{1}{2} \end{aligned}$$

Therefore, $x < 3\frac{1}{2}$.

Example 5.42

Solve for the value of x if $5x + 2 > 1$.

Solution

Given $5x + 2 > 1$.

Subtract 2 from both sides of the inequality. That is,

$$5x + 2 - 2 > 1 - 2$$

$$5x > -1$$

Divide by 5 on both sides

$$\frac{5x}{5} > -\frac{1}{5}$$

$$x > -\frac{1}{5}$$

Therefore, $x > -\frac{1}{5}$.

Example 5.43

Find the value of x which satisfy the inequality $-4x + 3 \geq \frac{1}{2}$.

Solution

Given $-4x + 3 \geq \frac{1}{2}$.

Subtracting 3 from both sides of the inequality to obtain,

$$-4x + 3 - 3 \geq \frac{1}{2} - 3$$

$$-4x \geq -\frac{5}{2}$$

Divide by -4 on both sides of the inequality to get,

$$x \leq \frac{5}{8}$$

Therefore, $x \leq \frac{5}{8}$.

Example 5.44

Solve the inequality

$$5(x + 2) - 3x \leq 4 + 2x + 3(x - 1).$$

Solution

Given

$$5(x + 2) - 3x \leq 4 + 2x + 3(x - 1).$$

Opening brackets and simplifying gives,

$$2x + 10 \leq 5x + 1$$

Collecting like terms gives,

$$2x - 5x \leq 1 - 10$$

$$-3x \leq -9$$

Divide by -3 on both sides of the inequality to get,

$$x \geq 3$$

Therefore, $x \geq 3$

Example 5.45

Indicate the solution of $|x| \leq 2$ on a number line.

Solution

Given $|x| \leq 2$, it implies that $\pm x \leq 2$

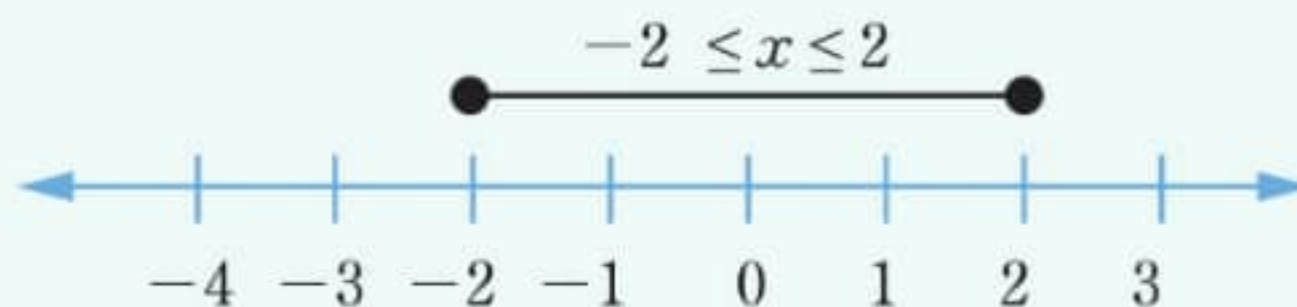
Thus, either $+x \leq 2$ or $-x \leq 2$.

$$x \leq 2 \text{ or } x \geq -2$$

Combining the two inequalities gives,

$$-2 \leq x \leq 2$$

Therefore, the solution of the inequality is represented on the following number line.



Example 5.46

Find the solution of $|2x+1| > 3$ and represent it on a number line.

Solution

Given $|2x+1| > 3$. It follows that, $\pm(2x+1) > 3$

Thus, either $+(2x+1) > 3$ or $-(2x+1) > 3$.

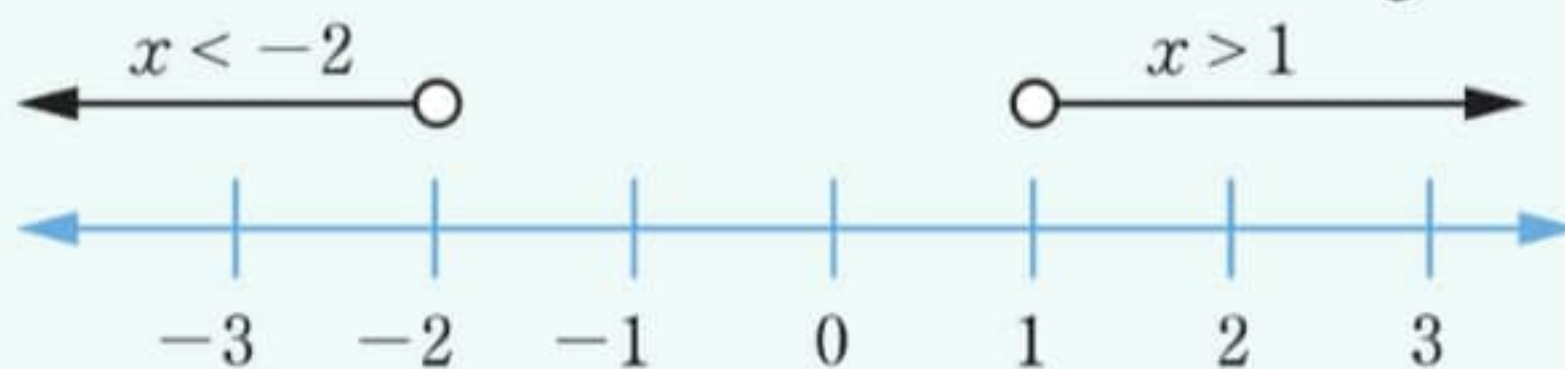
$$2x+1 > 3 \text{ or } -2x-1 > 3$$

$$2x > 2 \text{ or } -2x > 4$$

$$x > 1 \text{ or } x < -2$$

Therefore, the solution of the inequality is $x < -2$ or $x > 1$.

The representation of this solution is shown on the following number line.



Example 5.47

Find the values of real numbers x which satisfy the inequality $\sqrt{2x+3} < 3$.

Solution

Given $\sqrt{2x+3} < 3$, it implies that the inequality is valid if and only if $2x+3 \geq 0$ because the square root of a negative number is not a real number. It follows that,
 $2x \geq -3$

$$x \geq \frac{-3}{2} \quad (\text{i})$$

Squaring both sides of the given inequality gives,

$$(\sqrt{2x+3})^2 < 3^2$$

$$2x+3 < 9$$

$$2x < 6$$

$$x < 3 \quad (\text{ii})$$

Combining the two solutions (i) and

$$\text{(ii) gives } -\frac{3}{2} \leq x < 3.$$

Therefore, the solution of the inequality is $-\frac{3}{2} \leq x < 3$.

Exercise 5.8

1. List the numbers which satisfy each of the following conditions.

(a) $x < 6$ if x is a counting number.

(b) $x \leq 4$ if x is a whole number.

(c) $x > 3$ if x is an odd number.

(d) $x > -3$ and $x \leq 4$ if x is an integer.

2. Solve for x in each of the following inequalities.

(a) $5x > 12$

(b) $4 - x < 10$

(c) $2 - x > 8$

(d) $2x - 5 \leq \frac{1}{2}$

(e) $\frac{x}{2} - \frac{1}{2} < \frac{1}{2}$

(f) $6x + 4 \geq 2$

(g) $4x - \frac{3}{4} > 2x + \frac{1}{4}$

3. Find the solution for each of the following inequalities.

(a) $|y| > 2$

(b) $|-2 - y| \leq 1$

(c) $|y| \leq 3$

(d) $|y| < \frac{5}{2}$

(e) $|y + 2| > 2$

4. Represent the solution of each of the following on a number line.

(a) $|x - 4| \leq 5$

(b) $|x| > 2.5$

(c) $|7 - x| > 0$

(d) $\left| -\frac{2}{3}x - 4 \right| < 2$

(e) $|4x - 2| \geq 8$

(f) $\left| \frac{1}{2} - x \right| > 3$

5. Find the solution of the following inequalities.

(a) $2(2x + 3) - 10 < 6(x - 2)$

(b) $\frac{2x + 3}{4} + 6 \geq 2 + \frac{4x}{3}$

$$(c) \quad 3 - x \geq 5(3 - x)$$

$$(d) \quad \frac{y-3}{4} - 1 > \frac{y}{2}$$

6. Represent the solution of each of the following inequalities on a number line.

$$(a) \quad \frac{2x}{5} - \frac{1}{2}(x-3) \leq \frac{2x}{5} - \frac{3}{10}(x+2)$$

$$(b) \quad \frac{2}{3}(x+7) - \frac{x}{4} > \frac{1}{2}(3-x) + \frac{x}{6}$$

$$(c) \quad 3 - m < 4(m-3)$$

$$(d) \quad 2(1-u) \geq 5u.$$

7. Find the solution of each of the following inequalities.

$$(a) \quad 2|2x-1| \leq \frac{1}{2}$$

$$(b) \quad 3|1-x| \geq \frac{1}{3}$$

$$(c) \quad \frac{1}{2}|y+3| > \frac{2}{3}$$

$$(d) \quad -2 - \frac{b}{4} \leq \frac{1+b}{3}$$

8. Solve and represent the solution of each of the following on a number line.

$$(a) \quad |7-3x| \leq 2$$

$$(b) \quad |2x-1| < 3$$

$$(c) \quad \left| \frac{1}{2}x - \frac{3}{4} \right| < 2$$

$$(d) \quad |0.2m + 1.7| \geq 0.5$$

9. Solve each of the following inequalities.

$$(a) \quad |3x-7| \leq 4$$

$$(b) \quad |4-2y| > 6$$

$$(c) \quad \left| \frac{9}{5}c + 32 \right| < 31$$

$$(d) \quad |5x+2| \geq 8$$

10. Prove that:

$$(a) \quad |-x| = |x|$$

$$(b) \quad |x| = |y| \text{ if and only if } x = y \text{ or } x = -y.$$

Word problems involving inequalities

In real-life activities, there are some situations which require setting limits of quantities. Some of these activities include describing maximum number of people to attend a ceremony, minimum pass marks, maximum amount of salt required and so on. Such relationships can be represented and solved by using concepts of inequalities.

Example 5.48

If 5 is subtracted from twice a certain number, the result is less than 11. Find the number.

Solution

Let y be the required number.
The corresponding inequality is

$$2y - 5 < 11.$$

Adding 5 on both sides gives

$$2y - 5 + 5 < 11 + 5$$

$$2y < 16$$

Divide by 2 on both sides of the equation to get

$$\frac{2y}{2} < \frac{16}{2}$$

$$y < 8$$

Therefore, $y < 8$.

Example 5.49

A minimum average mark for a candidate to pass an examination is 61 in two tests. If Vivian obtained 54 marks in the first test, find the lowest possible marks she should obtain in the second test in order to pass the examination.

Solution

Let x be the minimum score in the second test for Vivian to pass the examination. This implies that, the average of the scores in the two tests should be greater or equal to 61.

Thus,

$$\frac{1}{2}(54 + x) \geq 61$$

Solve the inequality to obtain the values of x . That is,

$$27 + \frac{1}{2}x \geq 61$$

$$\frac{1}{2}x \geq 34$$

$$x \geq 68.$$

Solving for x gives $x \geq 68$.

Therefore, the lowest possible marks that Vivian should obtain in the second test is 68.

Example 5.50

In a village water project, a deep well is to be drilled. The temperature at x metres below the Earth's surface is given by $T = 30 + 25(x - 3)$, $5 \leq x \leq 75$ where T is the temperature in degrees of Celsius. What is the minimum depth for the temperature to be greater or equal to 155° ?

Solution

Given $T = 30 + 25(x - 3)$. It implies that $T \geq 155^\circ\text{C}$ is the inequality required to obtain the depth. That is,

$$30 + 25(x - 3) \geq 155$$

Solve the inequality to obtain the values of x . Thus,

$$25(x - 3) \geq 155 - 30$$

$$25(x - 3) \geq 125$$

$$x - 3 \geq 5$$

$$x \geq 8$$

Therefore, the minimum depth is 8 metres.

Exercise 5.9

1. A rectangle has a length of 9 centimetres and an area of not less than 45 square centimetres. Find the possible values of its width.
2. Mgeni sold 200 coconuts at x Tanzanian shillings each. If he got more than 30,000 Tanzanian shillings, find the value of x .
3. Three times a certain number is less than 2. Find the number.
4. If 9 is added to twice a certain whole number, the result is greater than 63. Find the number.
5. A number is such that, 2 more than thrice the number is not more than 4 less than 5 times the number.

Form an inequality to represent this information and solve it.

6. When 5 is added to an integer n , the result is greater than 7. Find the smallest possible value of n .
7. If two-quarters of a certain number is subtracted from 1, the result is always greater than 0. Formulate the inequality representing this statement.
8. The width of a rectangle is 10 cm and its length is $(2x + 4)$ cm. If its area is less than 200 cm^2 , find the maximum length of the rectangle if x is an integer.
9. The difference between 5 and thrice of a number is less than 8. Find the smallest and largest integers satisfying this inequality.
10. The difference between twice a number and 7 is less or equal to the sum of 9 and 6 times the number. Formulate the inequality which describes the relationship.

Chapter summary

1. Algebra uses letters or symbols to represent numbers in mathematical statements called algebraic expressions and equations.
2. An algebraic expression is formed by one or more terms.

3. A term of an algebraic expression can be a single letter, or letters multiplied by numbers, or a constant separated by addition or subtraction.
4. The number which multiplies a variable is called the coefficient of the variable.
5. Simplification of an algebraic expression is a process of combining like terms to obtain a single term as possible.
6. An equation is a statement which connects two expressions with an equal sign (=).
7. Simultaneous equations are a set of two or more related equations with multiple unknowns.
8. Inequalities are used to compare two values or expressions.
9. Common inequalities are less than (<), greater than (>), less than or equal to ($\leq$), greater than or equal to ($\geq$) and not equal ($\neq$).

Revision exercise 5

1. Write each of the following expressions in the simplest form.
 - (a) $14a + 7a - 5a$
 - (b) $9m - 4m + 2m$
 - (c) $0.5x + 1.7y - 1.2x + 2.4y$
 - (d) $\frac{5}{2}x - \frac{3}{2} + \frac{5}{2}x + \frac{9}{4}$
2. Simplify each of the following expressions and identify the number of terms and coefficients.
 - (a) $16m + 3n - 9m - 9n + 3m$
 - (b) $0.7x + 2.3y - 4.4x + 3.6y$
 - (c) $\frac{1}{2}x + \frac{3}{2}y - \frac{3}{4}z + \frac{3}{2}x - \frac{1}{4}y + z$
 - (d) $4ab + 5ac - 7ad - 7ad + 3ac$
3. Simplify each of the following algebraic expressions.
 - (a) $7m - 9n + 6m + 8n - 3m$
 - (b) $5k - 5n + 6k - 17k - 5$
 - (c) $5x + 9y + 7x - 8y - 3x$
 - (d) $(3x - y) - 2(4x - 3y)$
4. Multiply each of the following algebraic expressions.
 - (a) $5x - 7y$ by 4
 - (b) $4a - 5c + 2$ by -5
 - (c) $11p + 7q - 5$ by $2a$
 - (d) $\frac{3}{2}m - \frac{5}{2}$ by $-6n$
 - (e) $7x - 2y + 3z$ by 0.5
5. Find the answers to the following algebraic expressions.
 - (a) $8ab \times 5mn \times 3$
 - (b) $3n \times (-2m) \times 7$
 - (c) $(-2x) \left(\frac{a}{2} - 3b \right)$
6. Complete each of the following statements.
 - (a) $5ax - 2ay + 3az = a(\dots\dots\dots)$
 - (b) $9ax - 15ac - 6a = -3a(\dots\dots\dots)$

- (c) $axy - bxy + cxy = xy(\dots\dots\dots)$
 (d) $25mn + 5mp - 5mq = 5m(\dots\dots\dots)$
 (e) $ax + bx + ay - by = x(\dots\dots\dots) + y(\dots\dots\dots)$

7. Evaluate the value of each of the following expressions when $m = 5$ and $n = 2$

(a) $6mn - 2mn + 4mn$ (b) $\frac{m}{5} - \frac{mn}{2}$ (c) $\frac{2m}{4} + \frac{5mn}{10}$

8. Express each of the following statements in an algebraic form.

- (a) The sum of a number m and three times number n is 225.
 (b) The difference between 100 and number n is m .
 (c) When the sum of x and y is multiplied by 3, the result is 49.
 (d) The product of twice a certain number x and three times a certain number y is 249.

9. Solve each of the following equations.

(a) $5x - 7\frac{1}{2} = -\frac{3}{2}x$ (d) $\frac{3}{5}x - 3 = 3$ (g) $\frac{3}{2}x + \frac{x}{5} = \frac{5}{2}x - 5$
 (b) $3 - 2x = 15$ (e) $\frac{7}{x} = 8$ (h) $5x - \frac{x-2}{3} = 5 + \frac{3x+1}{2}$
 (c) $\frac{x}{5} - 3 = 3$ (f) $5x - 7 = 2x + 3$

10. Abdallah is 5 centimetres taller than Jane, and the sum of their heights is 307 centimetres. Find their heights.

11. The age of James is one-fifth the age of his father. If the sum of their ages is 66 years, find their ages.

12. If the four angles of a quadrilateral are given in degrees as $(4x - 20^\circ)$, $(3x + 40^\circ)$, $(3x + 50^\circ)$, and $(x + 30^\circ)$, find the value of x .

13. Solve each of the following simultaneous equations by using substitution method.

(a) $\begin{cases} y - 2x = 0 \\ 2y + 3x = 21 \end{cases}$ (c) $\begin{cases} x - y = -4 \\ 5x - y = 4 \end{cases}$
 (b) $\begin{cases} 2x - y = 0 \\ x + 2y = 11 \end{cases}$ (d) $\begin{cases} 2x - 3y = -1 \\ -2x + y = -13 \end{cases}$

14. Solve each of the following simultaneous equations by using elimination method.

(a)
$$\begin{cases} 3x - 2y = 17 \\ x + 2y = 11 \end{cases}$$

(b)
$$\begin{cases} 5x - 3y = 19 \\ 4x - 3y = 14 \end{cases}$$

(c)
$$\begin{cases} 0.1x + 0.2y = 1.1 \\ 1.5x - 0.2y = 10.1 \end{cases}$$

(d)
$$\begin{cases} x - 2y = -1 \\ 3x - 4y = 3 \end{cases}$$

15. Solve each of the following pairs of simultaneous equations by using a method of your choice.

(a)
$$\begin{cases} 2x + y = 31 \\ 3x + 2y = 52 \end{cases}$$

(b)
$$\begin{cases} 9(x - y) - (x + y) = 14 \\ (x + y) + 5(x - y) = 28 \end{cases}$$

(c)
$$\begin{cases} 4y - 2x = 2 \\ 6y - x = 6 \end{cases}$$

(d)
$$\begin{cases} \frac{x}{2} - \frac{y}{3} = -1 \\ \frac{x}{4} + \frac{y}{3} = \frac{5}{2} \end{cases}$$

16. If the sum of the ages of a father and his daughter is 60 years, and the age of the father is five times the age of the daughter, find their ages.
17. Two numbers x and y are such that, the sum of $2x$ and y is 62, while the sum of x and $2y$ is 79. Find the values of x and y .

18. Two numbers are such that, one number is greater than the other. If 25 is subtracted from the greater number, the result is half the smaller number. When the two numbers are added, the result is 91. Find the two numbers.

19. Solve for the values of x in each of the following:

(a) $x - 2 > \frac{1}{2}$

(b) $3x + 1 \leq 4$

(c) $-5x - 2 > 3$

(d) $\frac{1}{2} - 4x \leq -\frac{1}{2}$

(e) $\frac{x}{3} - \frac{1}{4} \geq \frac{1}{8}$

(f) $10x + 3 < 2\frac{1}{2}x - 12$

20. Twice the sum of two numbers is at most 87. If one of the numbers is 13, what is the largest possible value of the second number?
21. Solve the following system of linear simultaneous equations.

$$\begin{cases} \frac{3}{2}x - 0.2y = 4 \\ -x + 3y = \frac{9}{2} \end{cases}$$

22. Formulate an algebraic equation of the total number of students basing on the following statement: In a class of 42 students, some of them came late, and the rest came early.

23. If 1 is subtracted from both the numerator and the denominator of a fraction, the result is $-\frac{1}{2}$. If 2 is added to twice the numerator and 6 is added to thrice the denominator of the same fraction, the result is $\frac{4}{15}$. Find the fraction.
24. A certain school placed two orders with the stationery dealers. The first order was 50 boxes of marker pens and 20 boxes of correction fluid for the total amount of 40,000 Tanzanian shillings. The second order was 10 boxes of marker pens and 20 boxes of correction fluid for the total amount of 24,000 Tanzanian shillings.
- Formulate a system of simultaneous equations that describes the cost of a box of marker pen and a box of correction fluid.
 - Calculate the cost of a box of marker pens and a box of correction fluid.
25. When Juma went to a national park, he was interested with two kinds of wild animals only; zebras and elephants. If the total number of zebras and elephants he saw was 102, formulate the equation representing the scenario.

26. If $r = 5$, find all possible solutions to the following equations:
- $|x - r| = |r - 2x|$
 - $|4x - r| = |2x + r|$
27. Find all possible solutions to the following equations:
- $|2x - 6| = |3x - 10|$
 - $|6x + 2| = |13x - 1|$
 - $|5x + 5| = |2x + 12|$
 - $|9x + 18| = |2x - 1|$
28. If one side of a triangle is one-third of its perimeter, the second side is one-sixth of the perimeter, and the third side is 9 centimetres, find the perimeter of the triangle.
29. Find the solution of each of the following simultaneous equations.
- $$\begin{cases} 6x - y = 3 \\ 4x - 3y = -5 \end{cases}$$
 - $$\begin{cases} \frac{1}{3}x - \frac{1}{2}y = 0 \\ \frac{1}{2}x - \frac{1}{3}y = \frac{5}{6} \end{cases}$$
 - $$\begin{cases} x + \frac{y}{2} = 5 \\ \frac{x}{2} - \frac{y}{3} = \frac{1}{6} \end{cases}$$
 - $$\begin{cases} 3x + 2y = 8 \\ 2x + 3y = 12 \end{cases}$$

30. Find the solution of each of the following simultaneous equations.

$$(a) \begin{cases} \frac{5}{x} - \frac{1}{2y} = \frac{1}{2} \\ \frac{2}{y} + \frac{15}{x} = 5 \end{cases}$$

$$(b) \begin{cases} \frac{3}{x} + \frac{1}{y} = -4 \\ \frac{1}{x} - \frac{3}{y} = 2 \end{cases}$$

$$(c) \begin{cases} \frac{3}{y} - \frac{4}{x} = -5 \\ \frac{2}{x} - \frac{5}{y} = 6 \end{cases}$$

31. Find all possible solutions in each of the following inequalities.

$$(a) |7x - 3| < 25$$

$$(b) 5|x + 1| \geq 2$$

$$(c) 3|p - 4| \leq 2$$

32. Find the solution of each of the following equations.

$$(a) |3x - 1| = |x + 1|$$

$$(b) 4|x - 3| = |3x + 2|$$

$$(c) 2|1 - x| = |1 + x|$$

$$(d) |2 - 3x| = 5|1 - 2x|$$

33. If one side of a triangle is one-fourth of the perimeter, the second side is 7 cm and the third side is two-fifths of the perimeter. What is its perimeter?

34. Find four consecutive even integers such that the sum of the first three exceeds the fourth by 8.

35. Represent the solution of each of the following inequalities on a number line.

$$(a) |x - 3| > 5$$

$$(b) |2x + 1| \geq 3$$

$$(c) |2 - 5x| > 12$$

$$(d) |x + 3| < 4$$

36. Represent the solution of each of the following inequalities on a number line.

$$(a) 6x - 2 < x + 3$$

$$(b) 11x + 2 \geq 6x - 3$$

$$(c) 3(2x + 1) \leq 2(x - 1)$$

$$(d) 4(x - 2) > 6x$$

37. Determine the solution of each of the following inequalities.

$$(a) \sqrt{x - 1} > 1$$

$$(b) \sqrt{3x + 2} < 2$$

$$(c) \sqrt{5x - 1} \leq 3$$

$$(d) \sqrt{2(x + 1)} \geq 0$$

38. Represent each of the following inequalities on a number line.

$$(a) -1 \leq p < 2$$

$$(b) \frac{-1}{2} < m \leq 1$$

$$(c) -5 \leq x < 5$$

39. Find the solution of each of the following simultaneous equations.

$$(a) \quad \begin{cases} 2x - 3y = -11 \\ y = -2 - 5x \end{cases}$$

$$(c) \quad \begin{cases} 3x + \frac{2}{5}y = 3 \\ x = \frac{5}{7}y + 1 \end{cases}$$

$$(b) \quad \begin{cases} x = y + 3 \\ \frac{1}{2}x + 3y = -2 \end{cases}$$

$$(d) \quad \begin{cases} \frac{2}{3}x - \frac{1}{2}y = -1 \\ y = \frac{4}{5}x + 2 \end{cases}$$

40. Show that $|x - y| = |y - x|$ for all real numbers x and y .

Project: Exploring uses of Algebra in everyday life

Use various sources such as books and the internet to explore different ways algebra is used in solving various daily life problems.

Introduction

Coordinate geometry is the branch of Mathematics which helps to uniquely determine the positions and movements of objects. It also helps to understand the relationship between points on a plane using ordered pairs of coordinates. In this chapter, you will learn about gradient or slope of a straight line, equation of a straight line, graphing straight lines as well as solving linear simultaneous equations graphically. The competencies developed will enable you in solving problems such as plotting and analysing graphs, surveying and locating places, locating plot boundaries, interpreting traffic information, constructing buildings and roads and many other applications.

Basic concepts of coordinate geometry

Coordinate geometry uses ordered pairs of numbers to describe the positions of points on the xy -coordinate plane. The xy -coordinate plane or xy -plane is made up of two lines intersecting at right angles as shown in Figure 6.1.

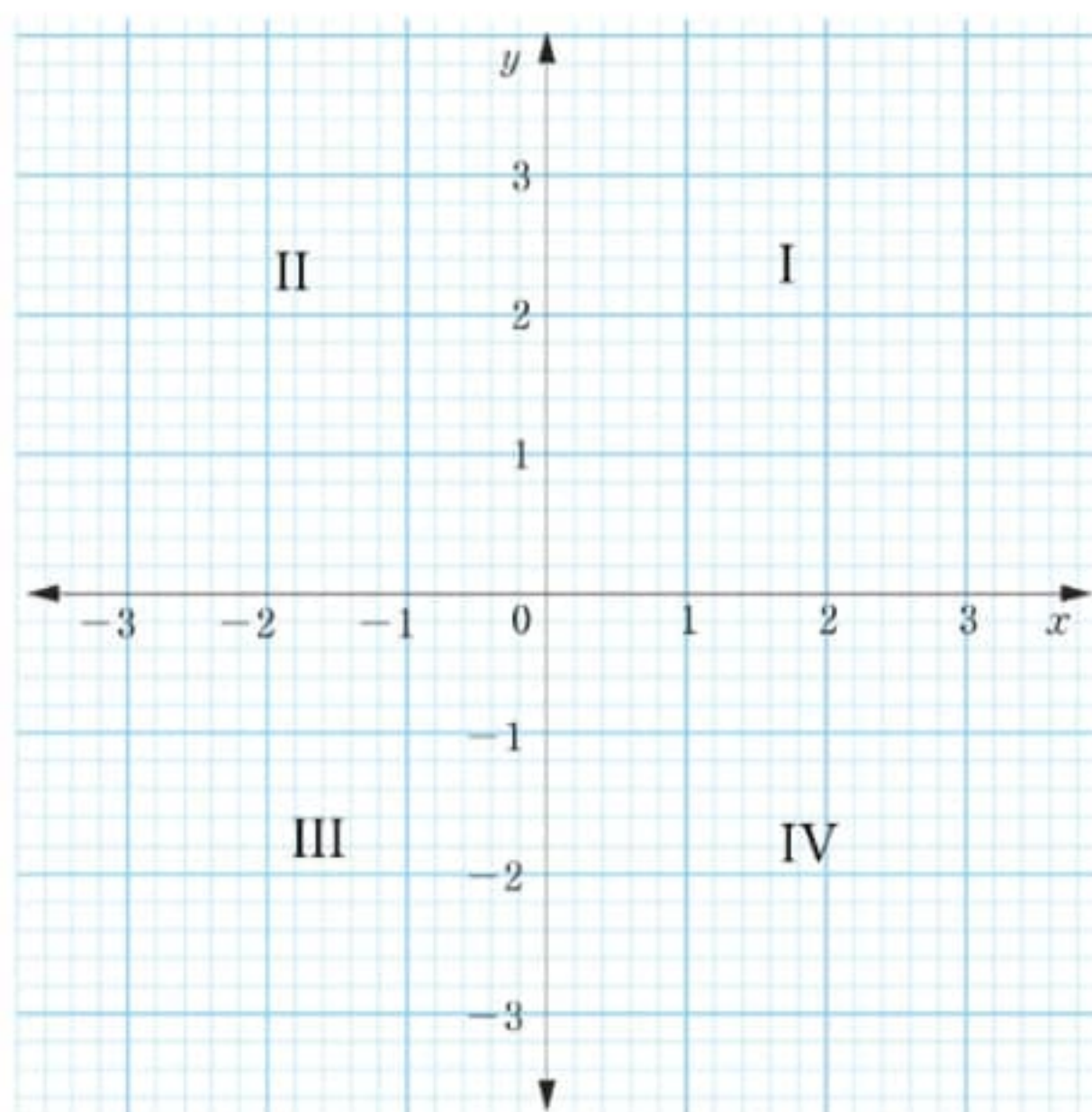


Figure 6.1: The xy -coordinate plane

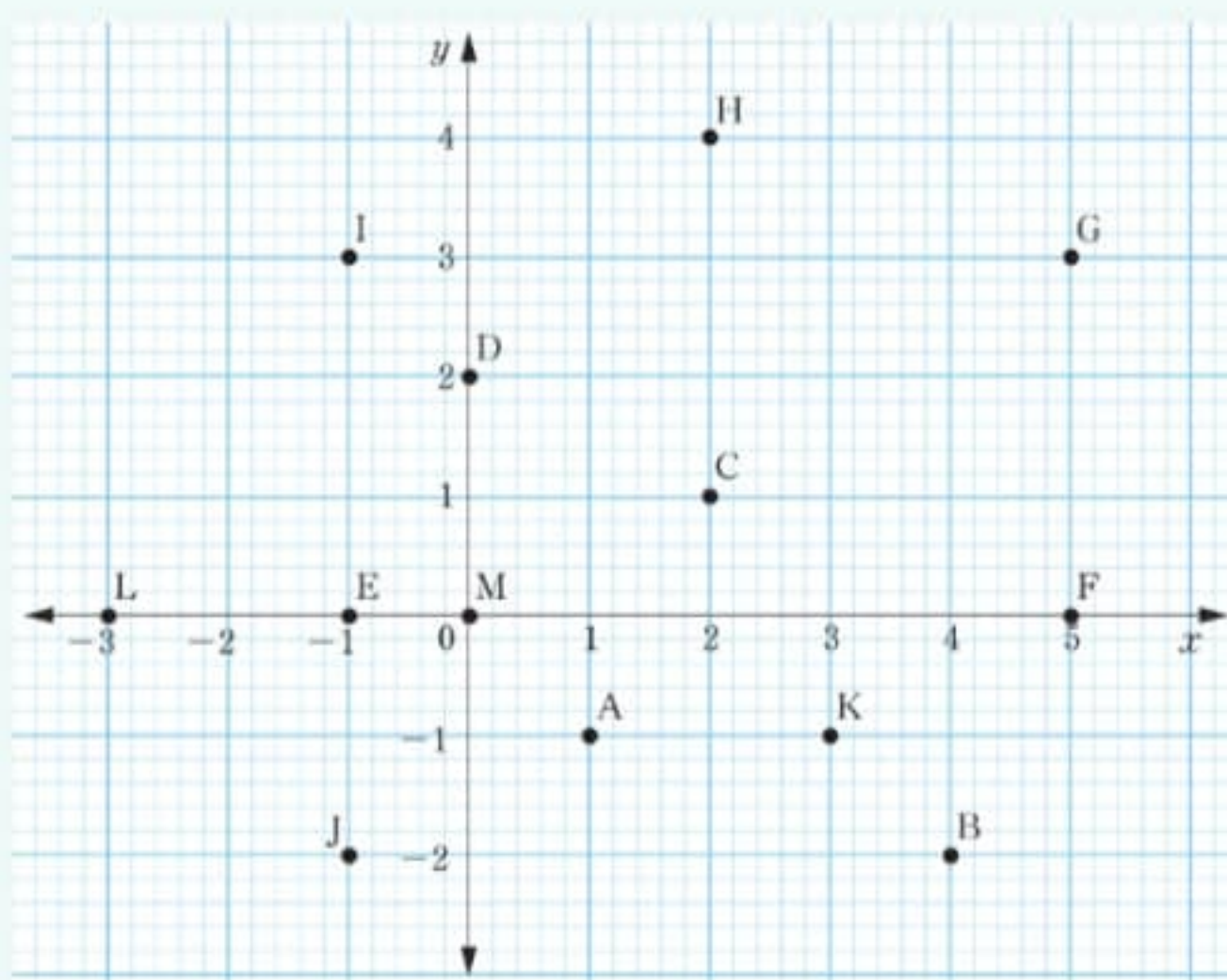
Each point on a plane is uniquely represented by means of an ordered pair of real numbers called the coordinates of a point. The xy -plane is divided into four parts called quadrants as shown in Figure 6.1. The first quadrant (I) has positive values of x and y , the second quadrant (II) has negative values of x and positive values of y . The third quadrant (III) has all negative values of x and y and the fourth quadrant (IV) has x positive values and y negative values. Recall that, the positions of points on the xy -plane are determined by using the x and y coordinates. For example, for the point $(4, 6)$, the first number 4 represents a distance from the origin along the x -axis and 6 represents a distance from the origin along the y -axis. Engage in Activity 6.1 to review the basic concepts in coordinate geometry.

Activity 6.1: Reading coordinates of points on xy -plane

1. Use your class sitting arrangement to locate at least 5 positions of different students.
2. Use the positions of students in task 1 to take the following measurements:
 - (a) Measure the perpendicular distances from their position to both the left and front walls.
 - (b) Record the measurements obtained in (a) in the form of (x, y) where x represents the distance from the left wall and y represents the distance from the front wall.
3. Repeat task 2 for all the positions of students identified in task 1.
4. Represent the coordinates of the points in tasks 2 and 3 on the xy -plane.

Example 6.1

Write the ordered pair for each point shown in the following figure.



Solution

From the graph given, the ordered pairs are as follows:

A(1, -1), B(4, -2), C(2, 1), D(0, 2), E(-1, 0), F(5, 0), G(5, 3),
H(2, 4), I(-1, 3), J(-1, -2), K(3, -1), L(-3, 0), and M(0, 0).

Example 6.2

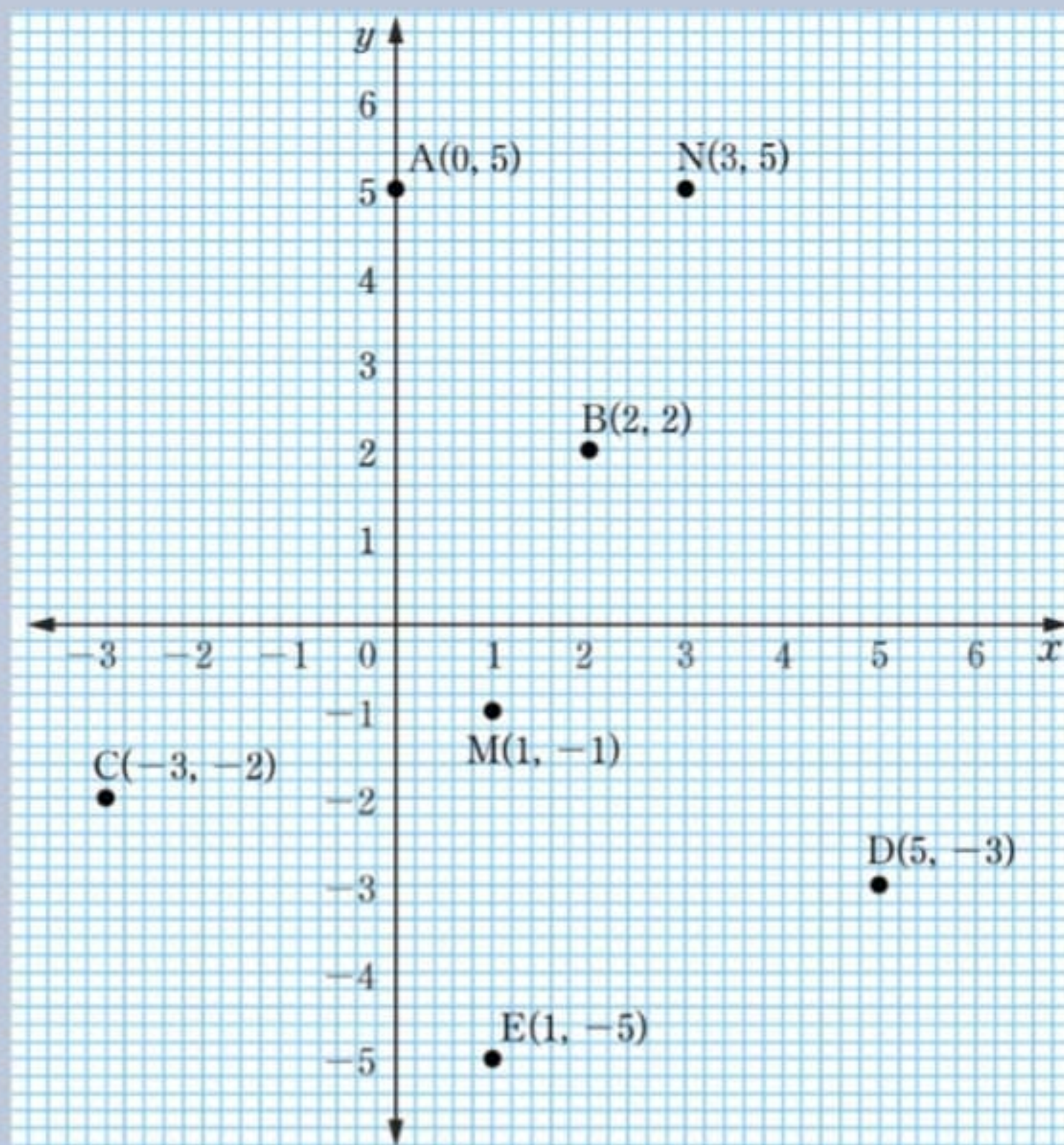
(a) Use a graph paper to locate each of the following points on the xy -plane.

- | | |
|-----------------|---------------|
| (i) M(1, -1) | (v) E(1, -5) |
| (ii) B(2, 2) | (vi) N(3, 5) |
| (iii) C(-3, -2) | (vii) A(0, 5) |
| (iv) D(5, -3) | |

(b) State the quadrant in which each point in (a) is located.

Solution

(a) The given points are located in the xy -plane as shown in the following figure.

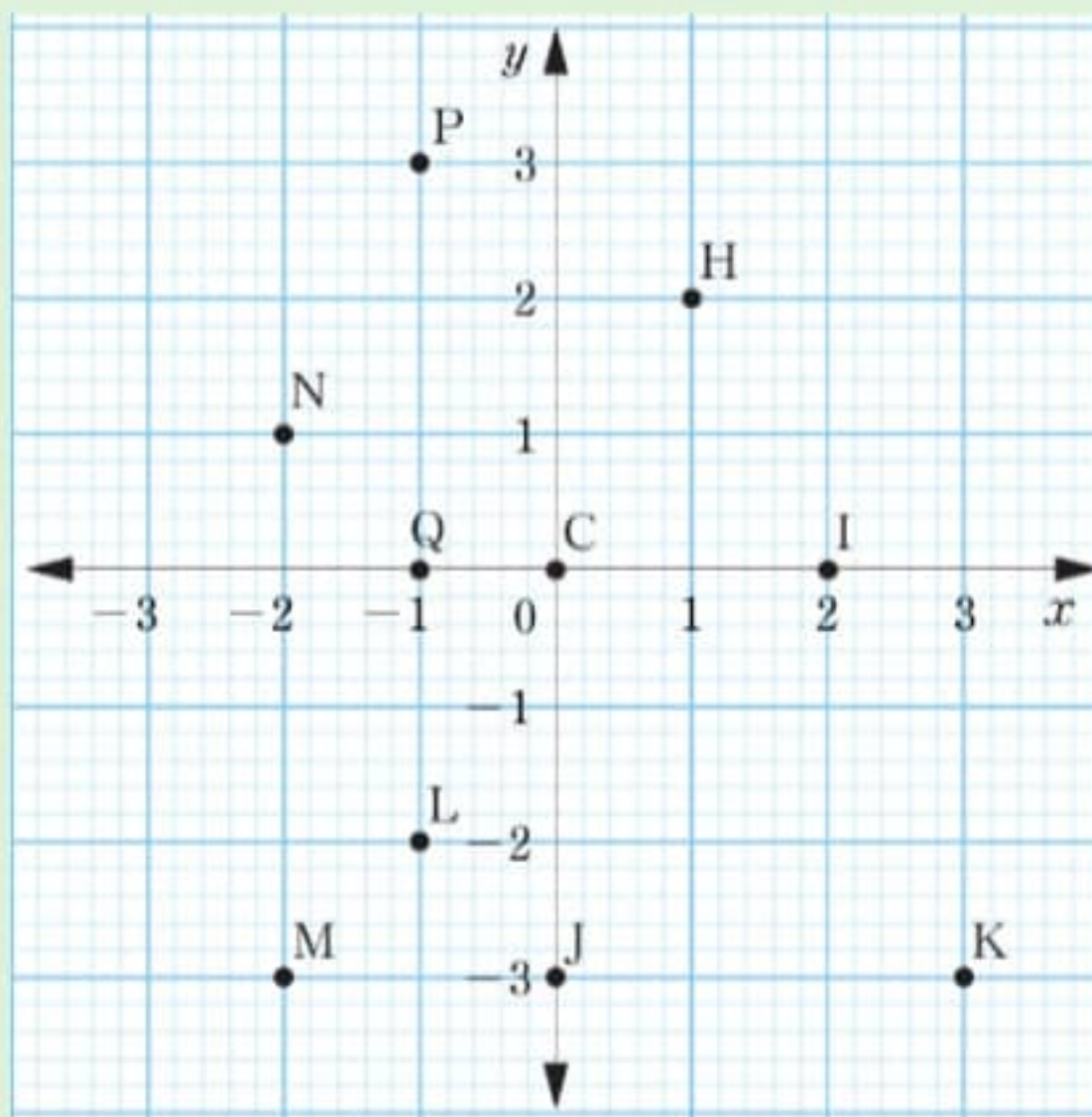


- (a) (i) $M(1, -1)$ is in the fourth quadrant
 (ii) $B(2, 2)$ is in the first quadrant
 (iii) $C(-3, -2)$ is in the third quadrant
 (iv) $D(5, -3)$ is in the fourth quadrant
 (v) $E(1, -5)$ is in the fourth quadrant
 (vi) $N(3, 5)$ is in the first quadrant
 (vii) $A(0, 5)$ is not in any quadrant, but on the y -axis.

Exercise 6.1

- Locate each of the following points on the same coordinate plane.

(a) $M(-2, 3)$	(c) $R(-3, -2)$	(e) $Q(5, -1)$
(b) $P(1, -4)$	(d) $N(3, 4)$	
- Study the following figure and answer the questions that follow.



- Write down the coordinates of each of the labelled points.
 - State the quadrant to which the points H, K, L, and N belong.
- Draw the coordinate axes on a graph paper and plot the following points. Join the points with line segments to form polygons. Name each of the polygons.
 - $(1, 1), (3, 1), (3, 3), (1, 3)$

- (b) $(-2, 1), (2, 5), (2, -2)$
 (c) $(1, 2), (3, 4), (5, 2), (4, 0), (2, 0)$
 (d) $(1, 2), (3, 4), (5, 2), (4, 0), (2, 0)$
 (e) $(1.5, -3), (6, 3), (-1.5, 3), (-6, -3)$
4. In which quadrant does each of the following points lie?
 (a) $P(-4, -10)$ (c) $R(16, -12)$
 (b) $T(-16, 8)$ (d) $S(4, 2)$
5. Locate the following points on the xy -plane:
 $A(1, 2), B(-2, 4), C(0, 0), D(3, -4), E(-3, -2), F(-5, 0),$
 $G(4, 0), H(0, 5), I(0, -4)$
 (a) Join the points A and B, B and E, A and E by line segments.
 (b) Use a protractor to measure angle BAE and check your answer with any suitable graphing mathematical software of your choice.
6. Plot each of the following points on the xy -plane and join them with straight line in the order given and state the name of the shape obtained. Use a graphing mathematical software to cross-check your answer.
 $(-1, 0), (-2, 2), (0, 1), (2, 2), (1, 0), (2, -2), (0, -1), (-2, -2).$
7. Plot each of the following points in the xy -plane, join the points and name the figures obtained.
 (a) $P(0, -2), Q(-3, -2), R(-3, 4)$
 (b) $A(1, 2), B(-3, 2), C(-3, -1), D(1, -1)$
 (c) $A(3, 4), B(0, 4), C(-2, 1), D(4, 1)$
8. Plot each of the following points on the xy -plane and join the points with line segments to form a polygon. Use a suitable mathematical software to cross-check your answer.
 (a) $A(-2, 3), B(1, 0), C(3, 2)$
 (b) $A(1, 1), B(1, 4), C(4, 5), D(5, 2)$
9. Plot the following points on the xy -plane and determine a missing coordinate of a point to complete a polygon stated in brackets.
 (a) $A(1, 1), B(2, 3), C(4, 2)$ (square)
 (b) $A(4, 4), B(9, 4), D(2, 1)$ (parallelogram)

10. The trapezium is formed using the points $A(-2, -2)$, $B(5, -2)$, $B(5, -2)$, and $D(4, 3)$. Use the given points to draw the trapezium and determine its height by counting the number of unit boxes.

Gradient of a straight line

A straight line is formed by joining two distinct points. Any two points lying on the straight line can be used to define it. The nature of the straight line is described by its gradient or slope. That is, it can be horizontal, vertical or slanted. The concept of gradient or slope is applicable in many real-life activities such as lifting heavy loads, climbing mountains, determining steepness during road construction and many other applications. Engage in Activity 6.2 to learn how to recognise the gradient of a straight line.

Activity 6.2: Recognising the gradient of a straight line

1. Use a graph paper to plot the points $A(2, 1)$, $B(5, 6)$, $C(2, 7)$, and $D(6, 1)$ on the same xy -plane.
2. Find the difference between the x -coordinates and between y -coordinates of $\overline{AB}$, $\overline{CD}$, $\overline{BA}$, and $\overline{DC}$.
3. Find the ratio of the difference of y -coordinates to x -coordinates obtained in task 2.
4. The relationship between the ratios obtained in step 3 is called the gradient of a line.
5. Explain how the value obtained in task 3 determines the nature of a line.

A gradient or slope is a measure of steepness of a straight line and it is usually denoted by m . In coordinate geometry, the standard way to define the gradient of a straight line is by finding the ratio between the change in y (vertical increase) to the change in x (horizontal increase), that is,

$$\text{Gradient, } m = \frac{\text{change in } y \text{ coordinates}}{\text{change in } x \text{ coordinates}}$$

Consider a line passing through the points $A(x_1, y_1)$ and $B(x_2, y_2)$. Also, consider a right-angled triangle ABC with the straight line $\overline{AB}$ shown in Figure 6.2. The gradient of a line passing through points A and B is determined by the ratio of the length of the vertical side of the triangle to the length of the horizontal side of

the triangle. The length of the vertical side $\overline{BC}$ of the triangle is the difference in y -coordinates of the points B and C, that is, $y_2 - y_1$ or $y_1 - y_2$. The length of the horizontal side of the triangle is the difference in x -coordinates of the points A and C, that is $x_2 - x_1$ or $x_1 - x_2$.

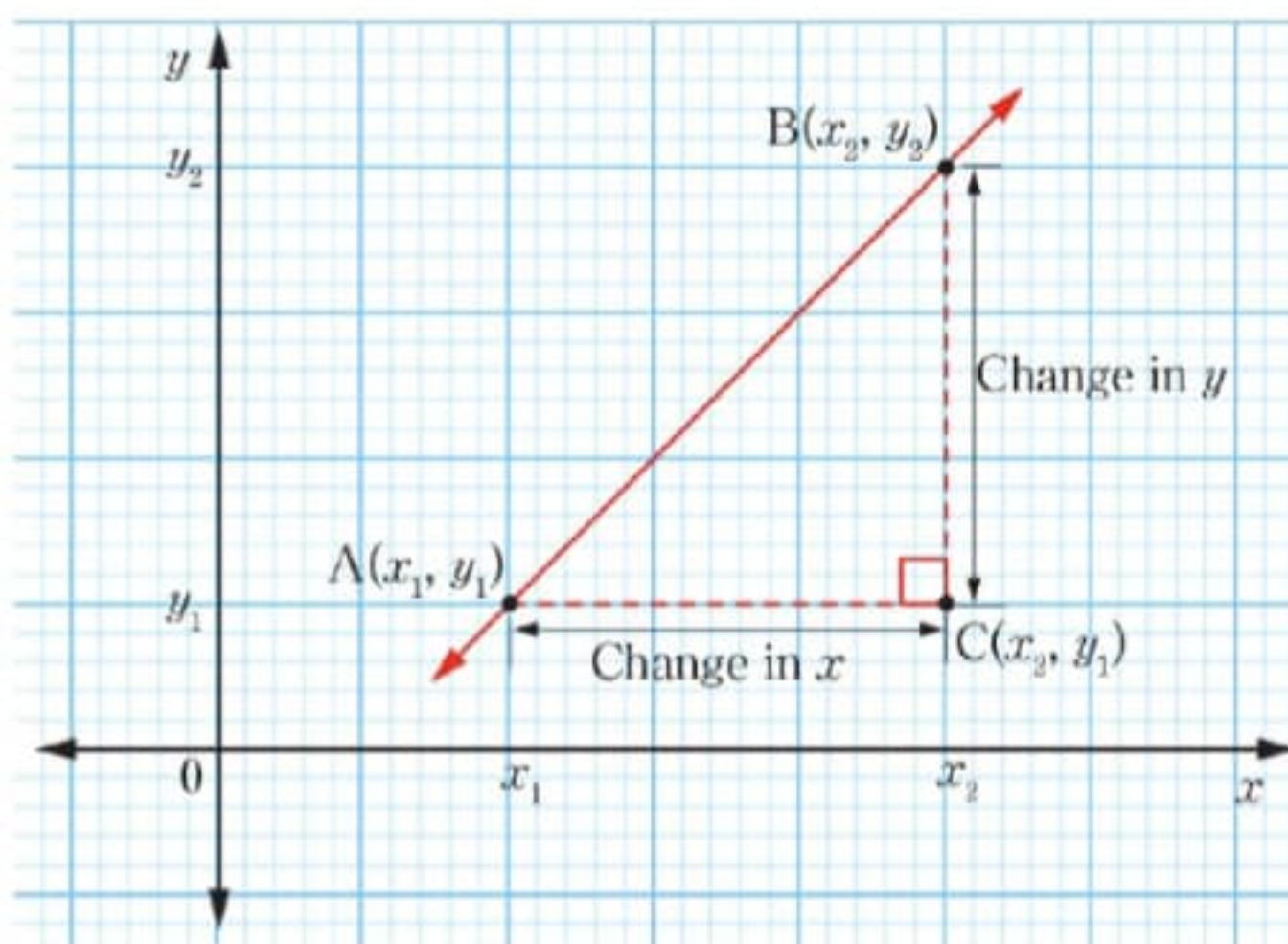


Figure 6.2: A straight line AB with horizontal and vertical changes

From Figure 6.2, the gradient m of the straight line AB is determined by using the following formula:

$$= \frac{\text{change in (vertical increase)}}{\text{change in (horizontal increase)}}$$

$$\text{Thus, } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}.$$

The following are the procedures for calculating the gradient of a straight line joining two points $A(x_1, y_1)$ and $B(x_2, y_2)$.

1. Write down the formula for calculating the gradient, $m = \frac{y_2 - y_1}{x_2 - x_1}$.
2. Assign values to (x_1, y_1) and (x_2, y_2) .
3. Substitute the values into the formula to obtain the gradient m .
4. Write the final answer which is the gradient m .

Note:

When finding the gradient, the order of finding the difference between the coordinate values in the formula does not matter. One may choose to find the difference from the second point or may choose to find the difference from the first point provided that the coordinates are subtracted correspondingly.

Example 6.3

Find the gradient of the line joining points C (2, 7) and D (6, 1).

Solution

Given the points C (2, 7) and D (6, 1). Let $(x_1, y_1) = (6, 1)$ and $(x_2, y_2) = (2, 7)$.

From $m = \frac{y_2 - y_1}{x_2 - x_1}$, it follows that

$$= \frac{7 - 1}{2 - 6}$$

$$= \frac{6}{-4}$$

$$= -\frac{3}{2}$$

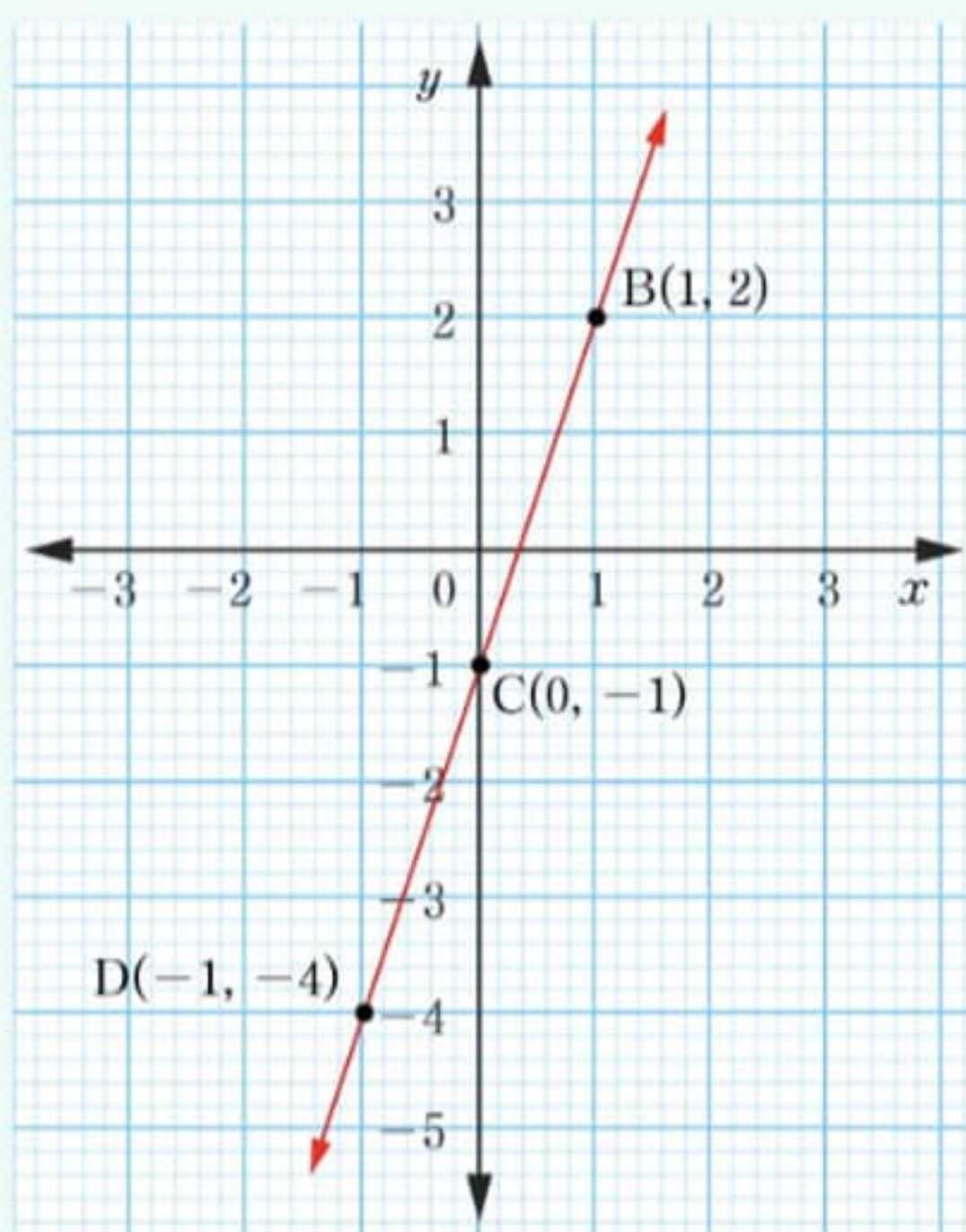
Therefore, the gradient of the line joining the points C and D is $-\frac{3}{2}$.

Example 6.4

Use the following figure to find the gradient or slope of the straight line joining the points:

(a) B and C

(b) B and D



Solution

- (a) The slope of the line joining B(1, 2) and C(0, -1) is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$.

Let $(x_1, y_1) = (0, -1)$ and $(x_2, y_2) = (1, 2)$.

$$\begin{aligned}\text{Thus, } m &= \frac{2 - (-1)}{1 - 0} \\ &= \frac{3}{1} \\ &= 3\end{aligned}$$

Therefore, the gradient of line BC is 3.

- (b) The slope of the line joining B (1, 2) and D (-1, -4) is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1}. \text{ Let } (x_1, y_1) = (1, 2) \text{ and } (x_2, y_2) = (-1, -4)$$

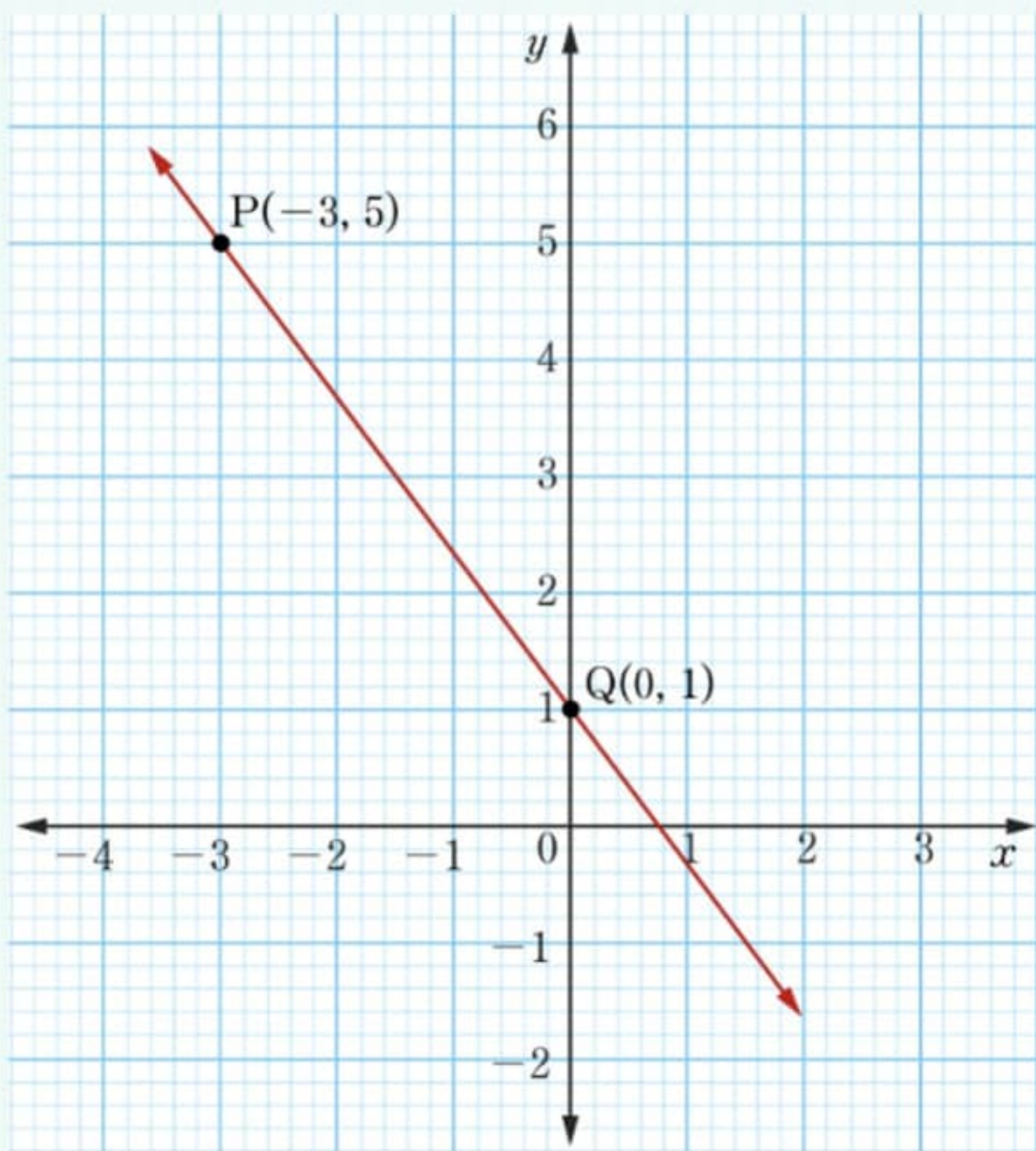
$$\begin{aligned}\text{Thus, } m &= \frac{-4 - 2}{-1 - 1} \\ &= \frac{-6}{-2} \\ &= 3\end{aligned}$$

Therefore, the slope of $\overline{BD}$ is 3.

In this example, it can be observed that the gradient of a straight line does not change when different pair of points are used provided that all the points lie on the same line.

Example 6.5

Find the gradient of a straight line given in the following figure.



Solution

Given points $P(-3, 5)$ and $Q(0, 1)$.

Let $(x_1, y_1) = (-3, 5)$ and $(x_2, y_2) = (0, 1)$.

From $m = \frac{y_2 - y_1}{x_2 - x_1}$, it follows that,

$$\begin{aligned} m &= \frac{1 - 5}{0 - (-3)} \\ &= -\frac{4}{3} \end{aligned}$$

Therefore, the gradient of a line PQ is $-\frac{4}{3}$.

Example 6.6

The gradient of the line passing through the points P(3, 5) and Q(x, 1) is $\frac{4}{3}$. Find the value of x.

Solution

Given the points P (3, 5) and Q (x, 1). The gradient of the line joining the two points is given by,

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

But, $x_1 = 3$, $x_2 = x$, $y_1 = 5$, $y_2 = 1$ and $m = \frac{4}{3}$. Substitute these values into the formula for slope to get,

$$\frac{4}{3} = \frac{1 - 5}{x - 3}$$

$$\frac{4}{3} = \frac{-4}{x - 3}$$

Cross multiplication gives,

$$4(x - 3) = -16$$

$$4x - 12 = -16$$

$$4x = -4$$

Therefore, the value of x is -1.

In Examples 6.3, 6.4, and 6.5, it can be observed that the values of the gradient are non-zero. Thus, only slanted lines have non-zero gradients. Moreover, a line slants towards right ($\nearrow$) if its gradient is positive as in Example 6.4 and slants towards left ($\nwarrow$) if the gradient is negative as in Example 6.5.

However, horizontal and vertical lines have unique gradients. If the line is horizontal, the change in y-values is zero and the gradient of the line is zero, as shown by the line DE in Figure 6.3.

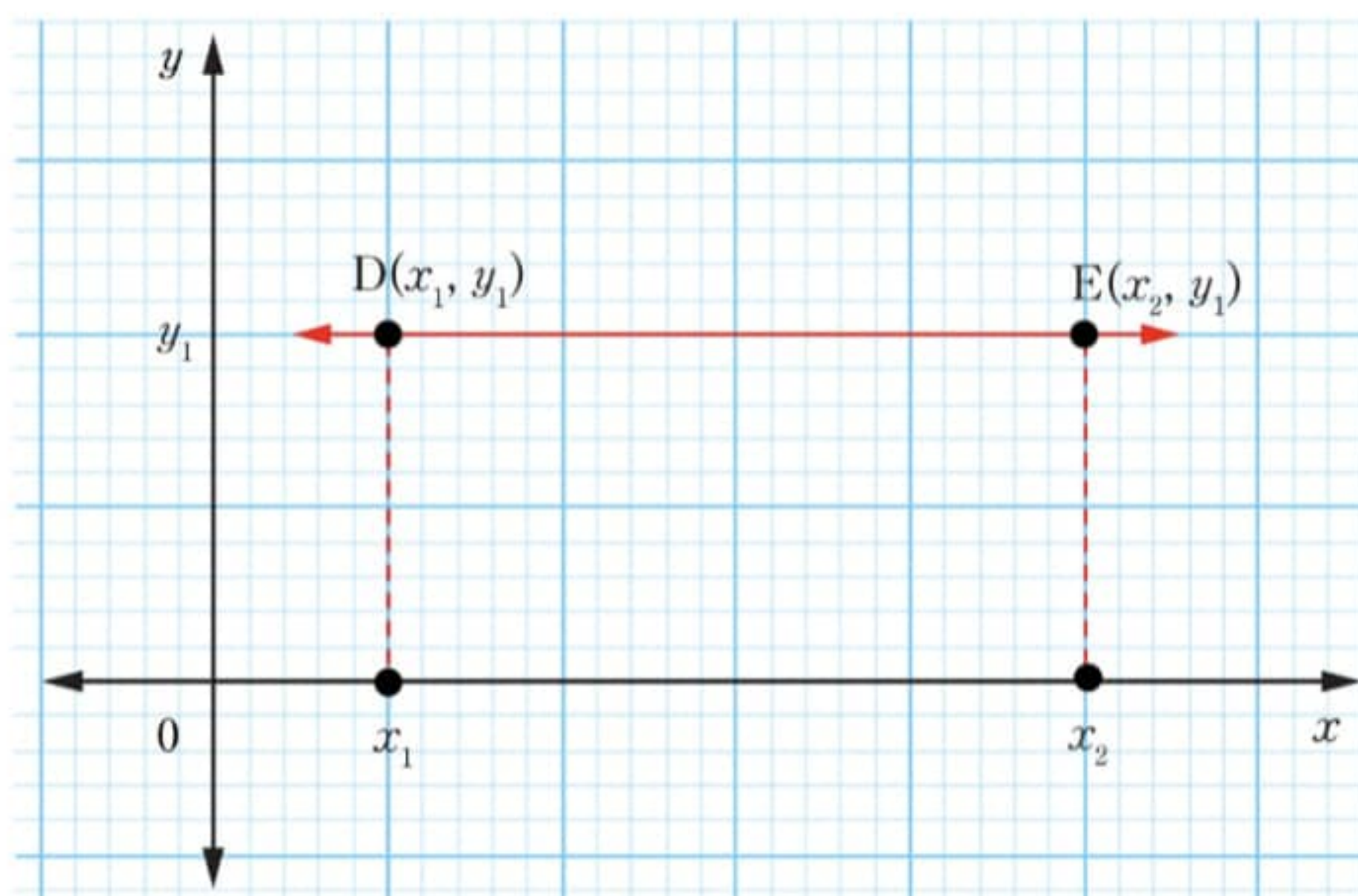


Figure 6.3: A horizontal line DE with zero gradient

The gradient of the straight line joining the two points $D(x_1, y_1)$ and $E(x_2, y_1)$ can be obtained as follows:

$$\text{From } m = \frac{y_2 - y_1}{x_2 - x_1}.$$

But $y_2 = y_1$ and $x_1 \neq x_2$. It implies that,

$$\begin{aligned} m &= \frac{y_1 - y_1}{x_2 - x_1} \\ &= \frac{0}{x_2 - x_1} \\ &= 0 \end{aligned}$$

Therefore, the gradient of a horizontal line DE is 0.

Since the gradient of the line segment DE is zero, then the line DE is horizontal. On the other hand, if the line is vertical, the change in x -values is zero which makes the denominator zero. Consider the line PN in Figure 6.4.

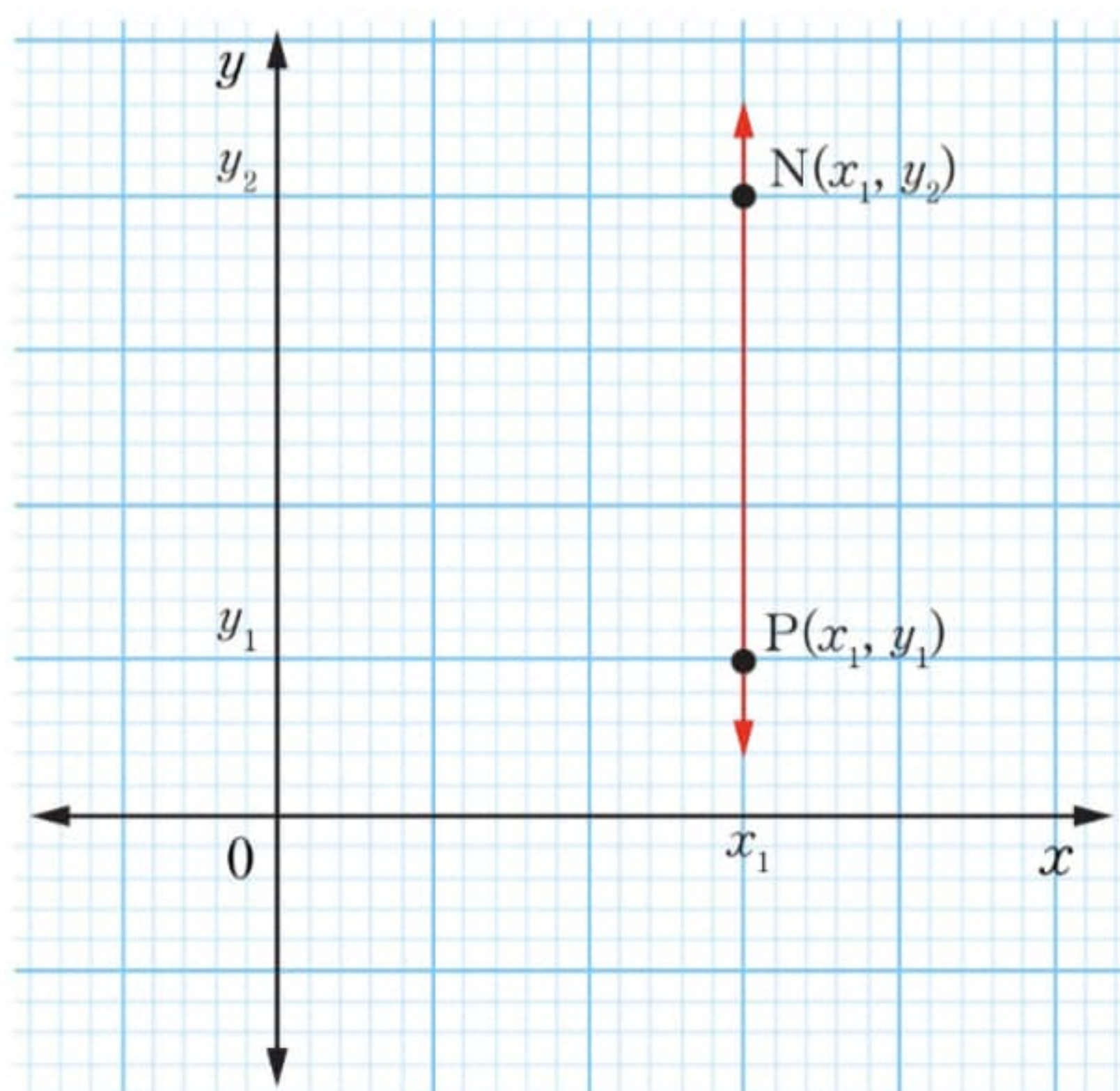


Figure 6.4: A vertical line PN with undefined gradient

The gradient of the line joining the two points $P(x_1, y_1)$ and $N(x_1, y_2)$ can be obtained as follows:

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

But $x_1 = x_2$ and $y_1 \neq y_2$. It implies that,

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_1 - x_1} \\ &= \frac{y_2 - y_1}{0} \end{aligned}$$

Note that, there is no real answer when a number is divided by 0. This is generally referred to as undefined. Since the gradient of PN is undefined, then the line PN is vertical.

Example 6.7

Show that a straight line passing through the points $M(3,1)$ and $N(3,4)$ is vertical.

Solution

Given the points $M(3,1)$ and $N(3,4)$. The gradient of the line joining the two points is given by;

$$m = \frac{y_2 - y_1}{x_2 - x_1}. \text{ Let } (x_1, y_1) = (3, 1) \text{ and } (x_2, y_2) = (3, 4).$$

It follows that,

$$\begin{aligned} m &= \frac{4-1}{3-3} \\ &= \frac{3}{0} \text{ which is undefined} \end{aligned}$$

Since the gradient of the line joining M and N is undefined, then the line MN is vertical.

Example 6.8

Determine the nature of the straight line joining the points $P(1, 2)$ and $Q(5, 2)$.

Solution

Given the points $P(1, 2)$ and $Q(5, 2)$. The gradient of the line joining the two points is given by;

$$\begin{aligned} m &= \frac{y_2 - y_1}{x_2 - x_1}. \text{ Let } (x_1, y_1) = (1, 2) \text{ and } (x_2, y_2) = (5, 2). \\ \text{Thus, } m &= \frac{2-2}{5-1} \\ &= \frac{0}{4} \\ &= 0 \end{aligned}$$

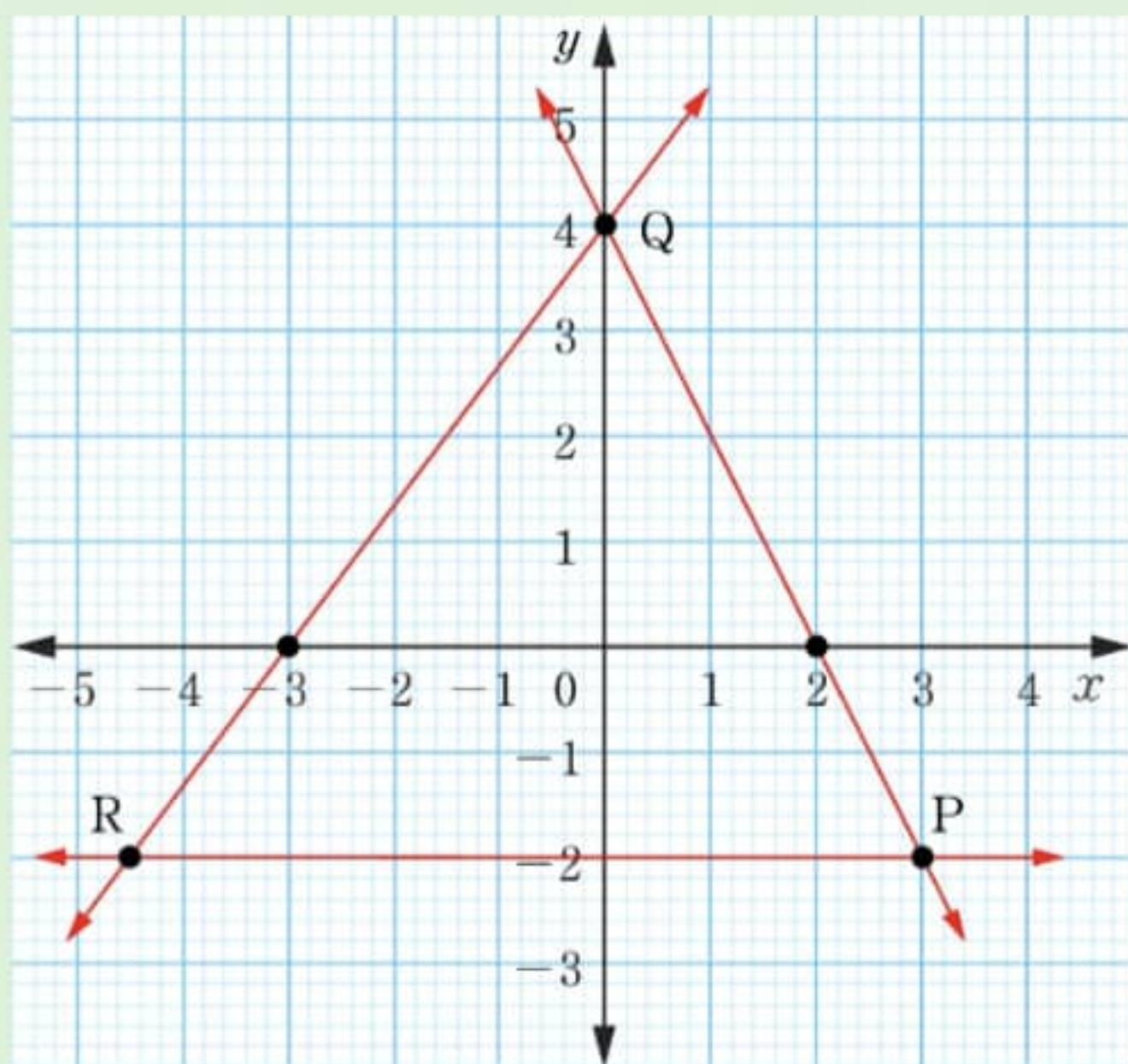
Since the gradient of the line PQ is 0, then the nature of the line is horizontal.

Exercise 6.2

- Plot each of the following pair of points on a graph paper and join them to obtain a straight line, hence state whether the gradient is positive, negative, zero or undefined.

(a) $(0, 3), (2, 5)$	(d) $(2, 6), (5, 3)$
(b) $(5, 8), (4, 1)$	(e) $(1, 6), (3, -1)$
(c) $(3, 6), (-2, -1)$	
- Calculate the gradient of the straight line passing through each of the following pairs of points:

(a) $(2, 1), (7, 6)$	(g) $(-2, 1), (4, 3)$
(b) $(5, 12), (3, 2)$	(h) $(-4, 4), (-3, -3)$
(c) $(3, 7), (1, -1)$	(i) $(2, 6), (0, 3)$
(d) $(-1, 5), (4, 1)$	(j) $\left(3\frac{1}{2}, \frac{1}{2}\right), \left(-7\frac{1}{2}, 2\right)$
(e) $(-1, 6), (4, 1)$	(k) $(99, 6), (119, 1)$
(f) $(-4, 4), (1, -4)$	(l) $(0.64, -0.62), (-1.36, -0.62)$
- Study the following graph and answer the questions that follow.



- (a) What are the coordinates of the points P, Q, and R?
 - (b) Find the gradients of $\overline{PQ}$, $\overline{PR}$, and $\overline{QR}$.
 - (c) Write the name of the polygon formed by joining the points P, Q, and R.
4. Catherine is a civil engineer who works at the road construction project. The road needs to have a slope of 0.04 for proper drainage. If the horizontal length is 600 metres, what should be the vertical length of elevation?
 5. The slope of the line joining the points $(1, p)$ and $(4, 2p)$ is $\frac{1}{3}$. Find the value of p .
 6. Determine the nature of the straight line which passes through the points $K(-3, -2)$ and $L(2, 1)$.
 7. A line passes through points $P(k, 4)$ and $Q(3, k + 2)$. What value of k will make the line to be (a) horizontal (b) vertical?
 8. Calculate the gradient of each of the following pair of points and state the nature of the line.

(a) $(3, 6), (-2, -1)$	(d) $(2, 10), (2, 0)$
(b) $(0, 2), (6, 2)$	(e) $(1, 0), (4, 0)$
(c) $(-2, 3), (-1, -3)$	
 9. The points $A(3, y)$, $B(0, 4)$, and $C(-2, 8)$ are on the same straight line. Determine the value of y .
 10. Compare the values of the gradients of $\overline{AB}$ and $\overline{PQ}$ if the points representing these lines are $A(6, -4)$, $B(8, 4)$, $P(6, 1)$, and $Q(7, 5)$.
 11. A line joining $(x, 0)$ and $(5, -2)$ has a slope of $\frac{3}{4}$. Find the value of x .
 12. Find the value of x so that the straight line joining the points $(x, -5)$ and $(2, 1)$ has the same gradient as the line passing through points $(1, 5)$ and $(-2, -1)$.
 13. Show whether line AB and PQ have the same gradients given that $A(-3, 1)$, $B(1, 2)$, $P(0, -1)$, and $Q(-4, 2)$.

Equation of a straight line

Suppose that two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ lie on a straight line, then the relationship between x and y coordinates of a point $N(x, y)$ lying on $\overline{PQ}$ can be determined. If $x_1 \neq x_2$, then point N will join $\overline{PQ}$ to form a straight line only if the gradient of line PN is the same as the gradient of line PQ as shown in Figure 6.5.

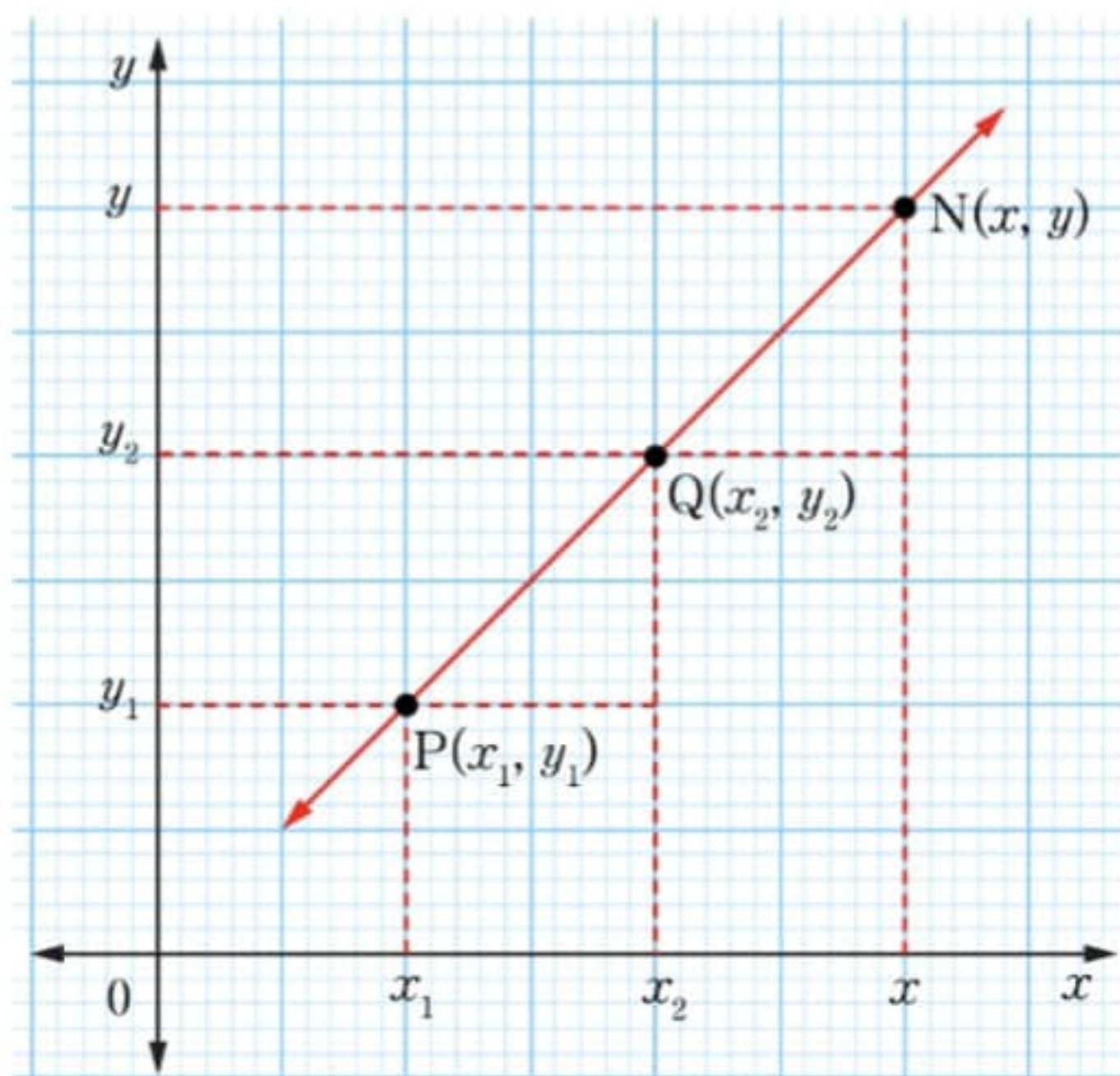


Figure 6.5: A straight line joining points P , Q , and N

$$\text{The gradient of } \overline{PQ} = \frac{y_2 - y_1}{x_2 - x_1} = m \quad (1)$$

$$\text{The gradient of } \overline{PN} = \frac{y - y_1}{x - x_1} = m \quad (2)$$

Equations (1) and (2) gives the same results.

From equation (2), it implies that $y - y_1 = m(x - x_1)$. Making y the subject gives

$$y = m(x - x_1) + y_1$$

Further simplification gives $y = mx - mx_1 + y_1$.

Rearrangement gives $y = mx + (y_1 - mx_1)$.

Let $c = y_1 - mx_1$, it follows that $y = mx + (y_1 - mx_1)$ is simplified to

$$y = mx + c \quad (3)$$

Equation (3) is an equation of a straight line passing through points $P(x_1, y_1)$ and $Q(x_2, y_2)$ with m as the gradient and c is the value of y at the point where the line crosses y -axis. In this case, y -intercept is point $(0, c)$ since the line will intersect the y -axis if the x -coordinate is 0.

The x -intercept (a point where the line intersects the x -axis) is found when the y -coordinate is 0. Setting $y = 0$ in $y = mx + c$ and solving for x gives,

$$y = mx + c$$

$$0 = mx + c$$

$$x = -\frac{c}{m}$$

Thus, the x -intercept is $\left(-\frac{c}{m}, 0\right)$.

In general, if two points (x_1, y_1) and (x_2, y_2) lie on a straight line, then any other point (x, y) on a line can be used to obtain the equation of the straight line using the formula,

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

Example 6.9

Find the equation of a straight line passing through the points $(-4, 3)$ and $(1, -7)$.

Solution

The equation of a straight line passing through points (x_1, y_1) and (x_2, y_2) is obtained by using the relation: $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$, where (x, y) is any other point on the line.

Let $(x_1, y_1) = (-4, 3)$ and $(x_2, y_2) = (1, -7)$. It follows that

$$x_1 = -4, x_2 = 1, y_1 = 3, \text{ and } y_2 = -7.$$

$$\text{Thus, } \frac{y - 3}{x - (-4)} = \frac{-7 - 3}{1 - (-4)}$$

$$\frac{y - 3}{x + 4} = \frac{-10}{5}$$

$$\frac{y-3}{x+4} = -2$$

Cross multiplication gives,

$$y-3 = -2(x+4)$$

$$y-3 = -2x-8$$

$$y = -2x-8+3$$

$$y = -2x-5$$

Therefore, the equation of the straight line is $y = -2x - 5$.

Example 6.10

Find the equation of a straight line which passes through the points A (3, 7) and B (-2, -3).

Solution

Using the formula for finding an equation of a straight line,

$$\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$$

Let $(x_1, y_1) = (3, 7)$ and $(x_2, y_2) = (-2, -3)$. Substitute these values into the equation as follows:

$$\frac{y-7}{x-3} = \frac{-3-7}{-2-3}$$

Simplification gives,

$$\frac{y-7}{x-3} = \frac{-10}{-5}$$

$$\frac{y-7}{x-3} = 2$$

Cross multiply and rearrange to get, $y = 2x + 1$.

Therefore, the equation of the line passing through the given points is $y = 2x + 1$.

Example 6.11

A straight line passes through the points $P(-4, 5)$ and $Q(-2, 1)$. Find its:

- (a) Gradient (b) Equation (c) x and y -intercepts

Solution

- (a) The gradient of the line is given by:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Let $(x_1, y_1) = (-4, 5)$ and $(x_2, y_2) = (-2, 1)$. It implies that,

$$\begin{aligned} m &= \frac{1 - 5}{-2 - (-4)} \\ &= \frac{-4}{2} \\ &= -2 \end{aligned}$$

Therefore, the gradient of a straight line is -2 .

- (b) Let $A(x, y)$ be any point on the line. Take the points $P(-4, 5)$ and $A(x, y)$ to determine the equation of the line, that is,

$$m = \frac{y - 5}{x - (-4)}. \text{ But } m = -2.$$

$$\text{Thus, } -2 = \frac{y - 5}{x + 4}$$

$$\frac{-2}{1} = \frac{y - 5}{x + 4}$$

Cross multiplication gives,

$$(y - 5) = -2(x + 4)$$

$$y - 5 = -2x - 8 + 5$$

$$y = -2x - 8 + 5$$

$$y = -2x - 3$$

Therefore, the equation of the straight line is $y = -2x - 3$.

- (c) From the equation $y = -2x - 3$, the y -intercept is obtained by setting $x = 0$ in the equation.

From $y = -2x - 3$, it implies that,

$$\begin{aligned} y &= -2(0) - 3 \\ &= -3 \end{aligned}$$

Therefore, the y -intercept is $(0, -3)$.

For x -intercept, set $y = 0$, that is

$$\begin{aligned} 0 &= -2x - 3 \\ 2x &= -3 \\ &= \frac{-3}{2} \end{aligned}$$

Therefore, the x -intercept is $\left(-\frac{3}{2}, 0\right)$.

Example 6.12

Verify that $y - 7 = 3(x - 3)$ represents an equation of the straight line passing through the point $(3, 7)$, hence determine its gradient and y -intercept.

Solution

A line passes through a point if its coordinates satisfies the equation representing the line.

Thus, substitute the coordinates of $(3, 7)$ in the equation of the line. That is,

$$\begin{aligned} 7 - 7 &= 3(3 - 3) \\ 0 &= 0 \end{aligned}$$

Therefore, the line with given equation passes through the point $(3, 7)$.

Write the equation in the form of $y = mx + c$, as follows:

$$\begin{aligned} y - 7 &= 3(x - 3) \\ y &= 3x - 2 \end{aligned}$$

Comparing with $y = mx + c$, $m = 3$ and $c = -2$.

Therefore, the gradient of the line is 3 and y -intercept is $(0, -2)$.

Exercise 6.3

- Find the equation in each of the lines passing through the following pair of points.

(a) (7, 9), (2, 5)	(f) (-2, -2), (3, -4)
(b) (5, 3), (0, 1)	(g) (-4, 3), (8, -6)
(c) (2, 5), (5, 7)	(h) (-7, 2), (-1, 6)
(d) (6, 1), (1, 6)	(i) (-4, 0), (3, -3)
(e) (4, 0), (3, 0)	(j) (0, 0), (-2, 2)
- Find the y -intercept in each of the lines joining the following pair of points.

(a) (0, 0), (-3, -4)	(c) (2, 3), (-3, 1)	(e) (-5, 2), (3, 2)
(b) (-4, -4), (4, 4)	(d) (0, 4), (4, 0)	(f) (-1, -2), (-5, -5)
- Find the equation of a straight line passing through each of the following points and state x and y -intercepts.
 - A (4, 8) and B (5, -2)
 - P (1, 3) and Q(-3, -2)
 - R (1, 6) and S (5, 9)
 - C (3, 2) and D (7, -2)
 - $T\left(\frac{1}{2}, 2\right)$ and $J\left(3, \frac{1}{3}\right)$
 - $T\left(-2\frac{1}{2}, 7\right)$ and $L\left(8, -2\frac{1}{2}\right)$
- Verify whether the points (-2, 2) and (6, 0) lie on the line joining the points A (-4, 1) and B (2, 4).
- Determine the value of k in order that the straight line whose equation is $kx - y = 5$ passes through the point (3, 5).
- What should be the value of t for the straight line represented by the equation $3x - ty = 16$ to pass through the point (5, -4)?
- Nyanda applies 2 bags of fertilizers to his 4 acres farm while Myoma applies 6 bags to her 12 acres farm. Taking x to be the number of bags of fertilizers and y to be the size of the farm in acres;

- (a) Represent the above information as points in the xy -plane and draw a line through the points.
 - (b) Determine;
 - (i) The gradient of the line and state what it signifies
 - (ii) The equation of the line and x and y -intercepts
 - (c) Using the equation of the line obtained in b(ii), determine;
 - (i) The number of bags Nyanda will need for his 24 acres farm
 - (ii) The size of the Myoma farm if she bought 75 bags of fertilizers.
8. The sewage and water authority total charges (y) include a fixed service charge and variable charge depending on the number of units of water consumed (x). In a certain month, Kelvin paid Tshs 70 for 10 units and in another month he paid Tshs 95 for 15 units.
- (a) Determine the equation representing the information and by using a suitable scale, draw the graph of the equation.
 - (b) Find:
 - (i) the y -intercept and state what it represent
 - (ii) the slope of the line and state what it represent

General equation of a straight line

Equation in the form $y = mx + c$ is sometimes referred to as slope-intercept equation. Equation of straight line can also be expressed in a general form. Referring equations (1) and (2) which have same gradient, that is:

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1} \quad (4)$$

Equation (4) can be rearranged to get,

$$(y_2 - y_1)x - (x_2 - x_1)y + y_1x_2 - x_1y_2 = 0$$

Since x_1 , y_1 , x_2 , and y_2 are given real numbers, then $(y_2 - y_1)$, $-(x_2 - x_1)$, and $y_1x_2 - x_1y_2$ are also real numbers. Thus, they can be replaced by constants a , b , and c , respectively. Therefore, the equation can be reduced to

$$ax + by + c = 0 \quad (5)$$

Equation (5) is called the general equation of a straight line.

Example 6.13

Find the equation of a straight line joining the points A (1, 6) and B (5, 9).
Write the answer in form of $ax + by + c = 0$.

Solution

Let $(x_1, y_1) = (1, 6)$ and $(x_2, y_2) = (5, 9)$.

Using $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$, it follows that,

$$\frac{y - 6}{x - 1} = \frac{9 - 6}{5 - 1}$$

$$\frac{y - 6}{x - 1} = \frac{3}{4}$$

Cross multiplication gives, $4y - 24 = 3x - 3$.

Rearranging gives, $3x - 4y + 21 = 0$.

Therefore, the equation of the line is $3x - 4y + 21 = 0$.

Example 6.14

Find the equation of a line passing through the points (4, 7) and (-2, -5),
giving your answer in the form $ax + bx = c$.

Solution

Let $(x_1, y_1) = (4, 7)$ and $(x_2, y_2) = (-2, -5)$.

From $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$, it follows that,

$$\frac{y - 7}{x - 4} = \frac{-5 - 7}{-2 - 4}$$

$$\frac{y - 7}{x - 4} = 2$$

Cross multiplication gives,

$$y - 7 = 2x - 8$$

$$2x - y = 1$$

Therefore, the equation of the line is $2x - y = 1$.

Example 6.15

A straight line is given by an equation $4x + y + 6 = 0$. Find:

- (a) Gradient
- (b) x and y -intercepts

Solution

Given $4x + y + 6 = 0$. Express the equation in the form $y = mx + c$. That is,
 $y = -4x - 6$. Thus,

(a) gradient $= -4$

(b) $c = -6$ and therefore y -intercept is $(0, -6)$.

From $y = -4x - 6$, substitute $y = 0$. It follows that,

$$0 = -4x - 6,$$

$$4x = -6$$

$$x = \frac{-6}{4} = \frac{-3}{2}$$

Therefore, x -intercept is $\left(-\frac{3}{2}, 0\right)$.

Example 6.16

Find the x and y -intercepts of the straight line $3x - 2y + 10 = 0$.

Solution

Given $3x - 2y + 10 = 0$. When $x = 0$, it implies that $2y = 10$. Thus, $y = 5$.

Therefore, the y -intercept is $(0, 5)$.

The x -intercept is obtained when $y = 0$. It implies that $3x = -10$, thus,

$$x = -\frac{10}{3}.$$

Therefore, x -intercept is $\left(-\frac{10}{3}, 0\right)$.

Exercise 6.4

- Find the equation of the straight line joining each of the following pair of points giving your answer in the form of $ax + by + c = 0$.
 - $(2, 4)$ and $(4, -2)$
 - $(3, 2)$ and $(7, -2)$
 - $(-3, 4)$ and $(8, 4)$
 - $(-1, -4)$ and $(4, -3)$
 - $(2, 4)$ and $(-3, 1)$
- Find the gradient and x and y -intercepts in each of the following lines.
 - $14x + 7y = 25$
 - $3y = 4x - 12$
 - $15 = 3x - 5y$
 - $3x - y = 12$
 - $7x - 2y + 3 = 0$
 - $ax - by + c = 0$
 - $\frac{x}{a} + \frac{y}{b} = 1$
- Find the equations of the following straight lines in the form $ax + by + c = 0$.
 - The line joining the points $(2, -4)$ and $(3, 1)$.
 - The line through the point $(3, 1)$ having the same gradient as the line $3x + 5y = 6$.
 - The line through the point $(3, -2)$ having the same y -intercept as the line $6x - 3y = 39$.
 - The line through the point $(7, -1)$ having the same x -intercept as the line $3x + 5y = 6$.
- The points $(4, -2)$ and $(-2, -5)$ lie on the straight line $\frac{x}{a} + \frac{y}{b} = 1$. Find the values of a and b .
- Rewrite the following equations in the form $y = mx + c$ and then determine the gradient and the y -intercept of each equation without drawing their graphs.
 - $7x + 4y = 11$
 - $14x + 3y = 12$
 - $2x = 5 + y$
 - $4x + 5y = 40$
 - $8x - y = 0$
 - $39x - 13y + 70 = 0$

6. A line has an equation $ax + (a-1)y + 2a = 0$.
- Rewrite the equation in the form of $y = mx + c$.
 - If the line has y -intercept at $(0, 2)$, find the value of a .
 - Using the value of a obtained in (b), find the gradient of the line.

Equation of a straight line when given a gradient and a point

The equation of a straight line can be determined when given a gradient m and a point $A(x_1, y_1)$ lying on the line. Choosing another point $P(x, y)$ which lie on the line, it follows that, $m = \frac{y - y_1}{x - x_1}$

Cross multiplication gives,

$$y - y_1 = m(x - x_1)$$

This is the equation of a line with gradient m and passes through the point $A(x_1, y_1)$

Example 6.17

A straight line has a gradient of $\frac{3}{4}$ and passes through the point $(-1, 2)$. Find:

- Its equation in the form $y = mx + c$
- The x and y -intercepts.

Solution

- The equation of this line is $y - y_1 = m(x - x_1)$.

Given $m = \frac{3}{4}$ and the point $(x_1, y_1) = (-1, 2)$.

Substitute the values in the equation to get,

$$y - 2 = \frac{3}{4}(x - (-1))$$

$$4(y - 2) = 3(x + 1)$$

$$4y = 3x + 11$$

$$y = \frac{3}{4}x + \frac{11}{4}$$

Therefore, the equation of the straight line is $y = \frac{3}{4}x + \frac{11}{4}$.

- (b) The y -intercept is obtained when $x = 0$. Substituting $x = 0$ into the obtained equation of the line gives,

$$y = \frac{3}{4}(0) + \frac{11}{4}$$

$$= \frac{11}{4}$$

Thus, the point $\left(0, \frac{11}{4}\right)$ is the y -intercept.

The x -intercept is obtained by setting $y = 0$. Substitute $y = 0$ into the equation of the line to obtain:

$$0 = \frac{3}{4}x + \frac{11}{4}$$

$$3x = -11$$

$$x = -\frac{11}{3}$$

Thus, the point $\left(-\frac{11}{3}, 0\right)$ is the x -intercept.

Therefore, the x and y -intercepts are $\left(-\frac{11}{3}, 0\right)$ and $\left(0, \frac{11}{4}\right)$, respectively.

Example 6.18

An equation of a line has a gradient of $-\frac{1}{5}$. If one of the points satisfying the equation is $(2, 1)$, find its equation in the form $ax + by + c = 0$.

Solution

The equation of the line is given by

$$y - y_1 = m(x - x_1)$$

Given $m = -\frac{1}{5}$ and $(x_1, y_1) = (2, 1)$. It implies that,

$$y - 1 = -\frac{1}{5}(x - 2)$$

$$y - 1 = -\frac{1}{5}x + \frac{2}{5}$$

Thus,

$$y = -\frac{1}{5}x + \frac{7}{5}$$

$$x + 5y - 7 = 0$$

Therefore, the equation of the straight line is $x + 5y - 7 = 0$.

Example 6.19

Find the equation of a line in the form $ax + by + c = 0$, passing through the point $(3, 1)$ with gradient $-\frac{3}{5}$.

Solution

Given $(x_1, y_1) = (3, 1)$, $m = -\frac{3}{5}$.

From $y - y_1 = m(x - x_1)$, it follows that,

$$y - 1 = -\frac{3}{5}(x - 3)$$

$$5y - 5 = -3x + 9$$

Therefore, the equation of the line is $3x + 5y - 14 = 0$.

Example 6.20

Find the equation of a straight line in the form $ax + by + c = 0$ passing through the point $(3, -4)$ having the same gradient as the straight line $5x - 2y = 3$.

Solution

Given $(x_1, y_1) = (3, -4)$ and $5x - 2y = 3$. Rearranging the given equation in the form $y = mx + c$ gives, $y = \frac{5}{2}x - \frac{3}{2}$.

Comparing with $y = mx + c$, it implies that $m = \frac{5}{2}$

From $y - y_1 = m(x - x_1)$, it follows that,

$$y - (-4) = \frac{5}{2}(x - 3)$$

$$y + 4 = \frac{5}{2}x - \frac{15}{2}$$

Collecting like terms gives,

$$y = \frac{5}{2}x - \frac{23}{2}$$

$$2y = 5x - 23$$

$$5x - 2y - 23 = 0$$

Therefore, the equation of the straight line is $5x - 2y - 23 = 0$.

Finding the equation of a straight line given the gradient and x or y-intercepts

Given the gradient and x or y-intercept, the equation of a line can also be obtained directly by substituting these values in the equation, $y = mx + c$.

Example 6.21

Find the equation of a straight line whose gradient is $\frac{3}{2}$ and its y-intercept is (0, 3).

Solution

Given the gradient $m = \frac{3}{2}$ and the y-intercept (0, 3), then substitute (0, 3).

From $y = mx + c$, substitute $m = \frac{3}{2}$ and $c = 3$ in this equation to obtain

$$y = \frac{3}{2}x + 3.$$

Therefore, the equation of the line is $y = \frac{3}{2}x + 3$.

Example 6.22

A straight line which cuts the x-axis at the point (-2, 0) has a gradient of -2. Find its equation.

Solution

Given that $m = -2$ and x-intercept (-2, 0). Substitute these values into the equation $y = mx + c$. That is,

$$0 = -2(-2) + c$$

$$c = -4$$

From $y = mx + c$, it implies that,

$$y = -2x - 4$$

Therefore, the equation of the straight line is $y = -2x - 4$.

Example 6.23

Find the gradient and y -intercept in each of the following lines:

(a) $12x + 4y - 15 = 0$ (b) $24 - 8x - 3y = 0$

Solution

(a) Given $12x + 4y - 15 = 0$.

Make y the subject of the formula. Thus,

$$4y = -12x + 15$$

Dividing by 4 both sides of the given equation gives,

$$y = -3x + \frac{15}{4}$$

Compare this equation with $y = mx + c$ gives $m = -3$ and $c = \frac{15}{4}$. Thus,

$\left(0, \frac{15}{4}\right)$ is the y -intercept.

Therefore, the gradient is -3 and the y -intercept is $\left(0, \frac{15}{4}\right)$.

(b) Given $24 - 8x - 3y = 0$.

Make y the subject of the formula:

$$-3y = 8x - 24.$$

Dividing by -3 on both sides gives,

$$y = -\frac{8}{3}x + 8$$

Compare this equation with $y = mx + c$ to get $m = -\frac{8}{3}$ and $c = 8$.

Therefore, the gradient is $-\frac{8}{3}$ and the y -intercept is $(0, 8)$.

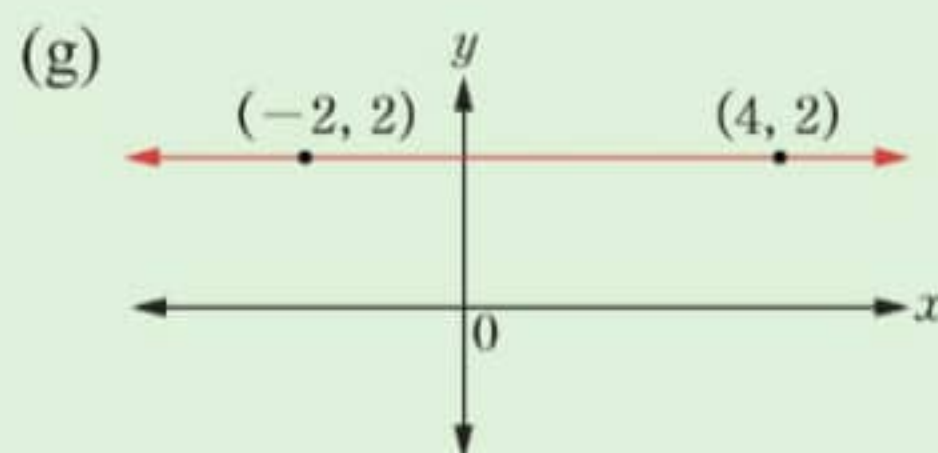
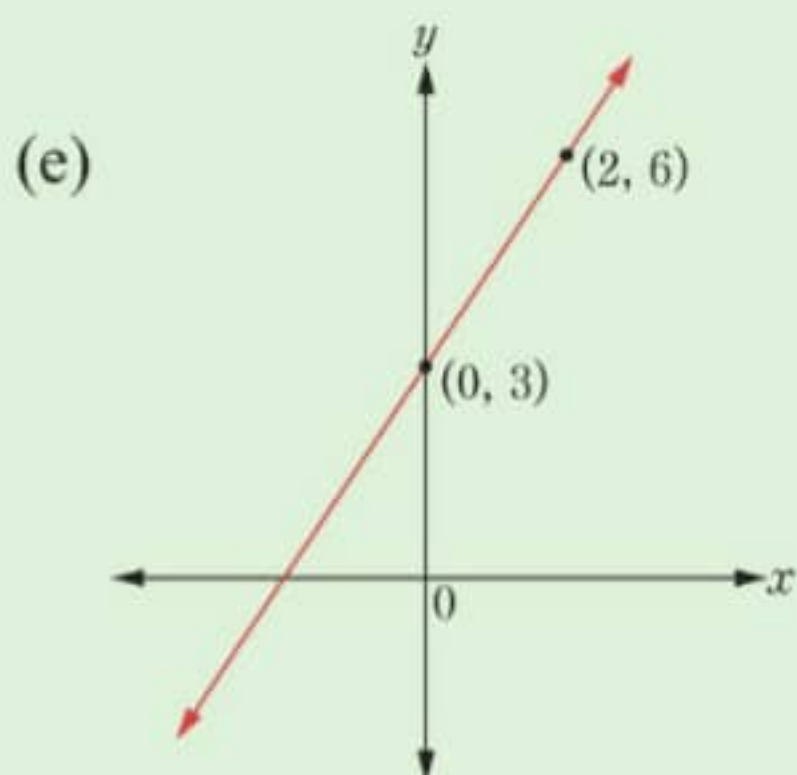
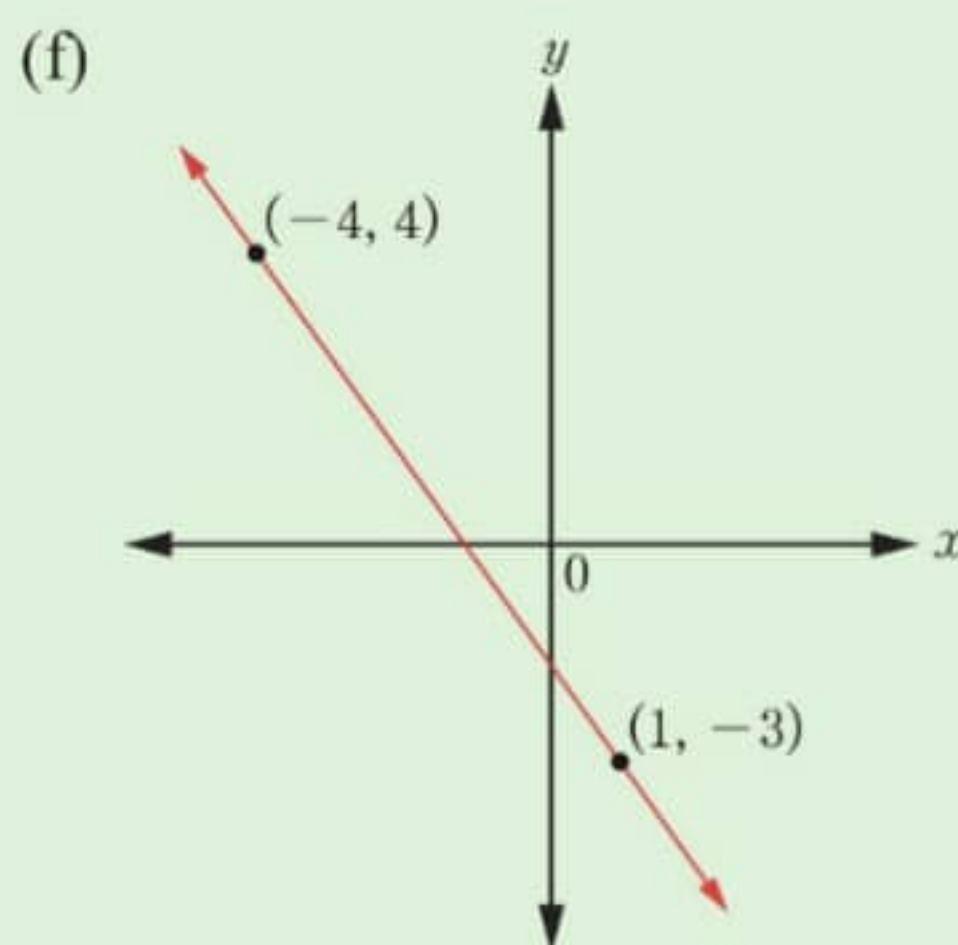
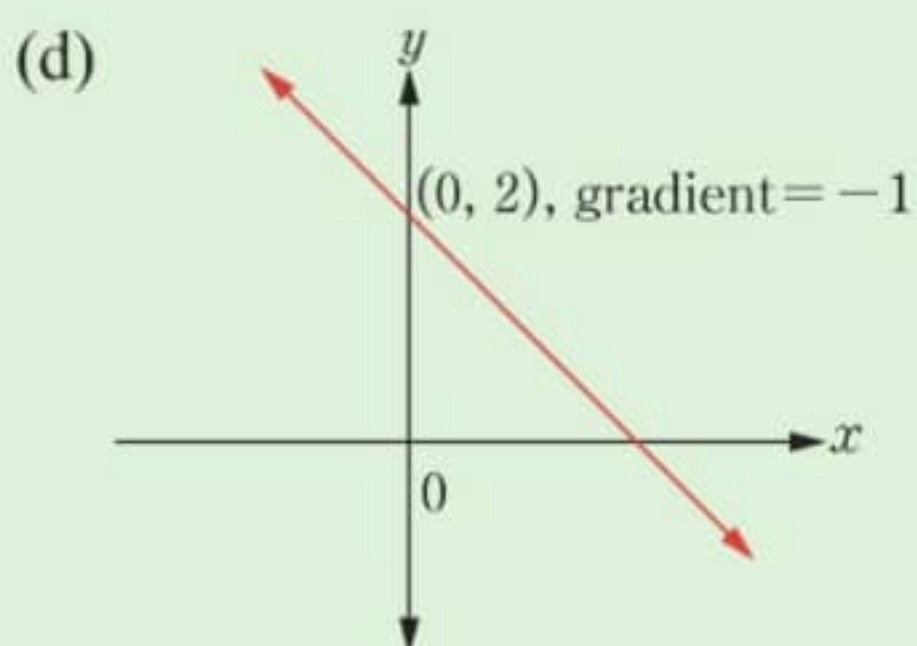
Exercise 6.5

1. Find the equation of the line using the given point and gradient:

- (a) $(2, 1)$, gradient 2 (d) $(-2, -4)$, gradient $\frac{3}{2}$
 (b) $(0, 5)$, gradient -2 (e) $(0, 0)$, gradient -3
 (c) $(1, -3)$, gradient 3

2. Find the equation of the line using the given information:

- (a) y -intercept $= (0, 2)$, gradient $= 1$
 (b) y -intercept $= (0, -16)$, gradient $= 4$
 (c) y -intercept $= (0, 7)$, gradient $= \frac{3}{4}$



3. In each of the following conditions, find the equation of the straight line in the indicated form:

- (a) Passing through the point $(3, 5)$ with gradient 2 in the form $y = mx + c$.
 (b) passing through the point $(4, 7)$ having a gradient of -3 , in the form $ax + by + c = 0$

- (c) $m = -\frac{5}{2}$, $c = 4$ in the form of $ax + by + c = 0$
- (d) Gradient $m = -\frac{6}{5}$ and passes through the point $(2, 2)$ in the form $y = mx + c$.
- (e) Passing through the point $(6, 2)$ and having the same gradient as the line $x + 3y - 13 = 0$ in the form of $ax + by + c = 0$.
4. Find the equation of a line passing through the point $(3, 2)$ with the same gradient as a line passing through the points $(-1, -3)$ and $(4, 7)$. Express your answer in the slope-intercept form.
5. Find the equation of a line with slope $\frac{2}{3}$ having the same y -intercept as the line $2x - 5y + 20 = 0$. Give your answer in general form.
6. Find the equation of a straight line:
- Through the point $(0, 5)$ with gradient 0 .
 - Through the point $(2, 0)$ with undefined gradient.
 - Joining the points $(0, 4)$ and $(3, 0)$. Express your answer in general form.

Graphing linear equations

The graph of a linear equation can be drawn after obtaining at least two points whose coordinates satisfy the given equation. Locate these points on an xy -plane and join them to obtain a straight line. The following are the common methods used to obtain the required points:

- Table of values
- Using x and y -intercepts

Graphing linear equations using table of values

The general linear equation has the form $ax + by + c = 0$. The table of values of x and y coordinates is easily constructed after re-writing the equation in the form $y = mx + c$. Choose at least two values of x and determine the corresponding values of y to obtain the points lying on the line. These points are presented in a table which is generally referred to as a table of values.

Example 6.24

Given the equation $x + y = 1$, complete the following table of values and draw the corresponding graph.

Table of values for $x + y = 1$

x	-2		0	1	2
y		2		0	

Solution

Given the equation $x + y = 1$. Re-writing it in the form $y = mx + c$ gives $y = -x + 1$.

Substitute the given values from the given table of values.

When $x = -2$, then $y = -(-2) + 1 = 3$.

When $y = 2$, $x = -1$.

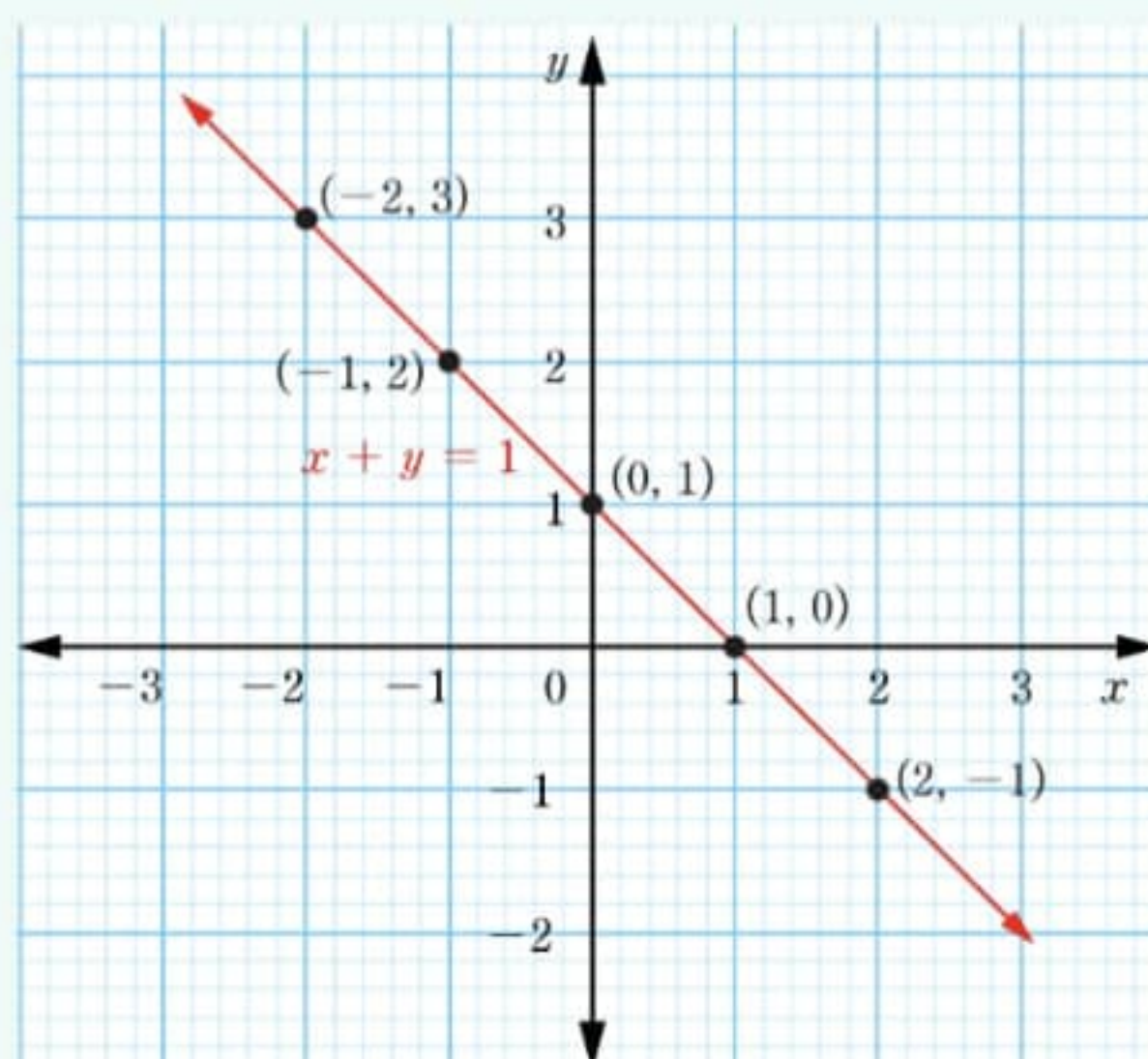
When $x = 0$, $y = 1$.

When $x = 2$, $y = -1$.

The following is the table of values for $x + y = 1$.

x	-2	-1	0	1	2
y	3	2	1	0	-1

Locate the set of ordered pairs from the table of values on the xy -plane. Join the points to form a straight line as shown in the following figure.



Example 6.25

Draw the graph of the equation $y - 2x = 2$.

Solution

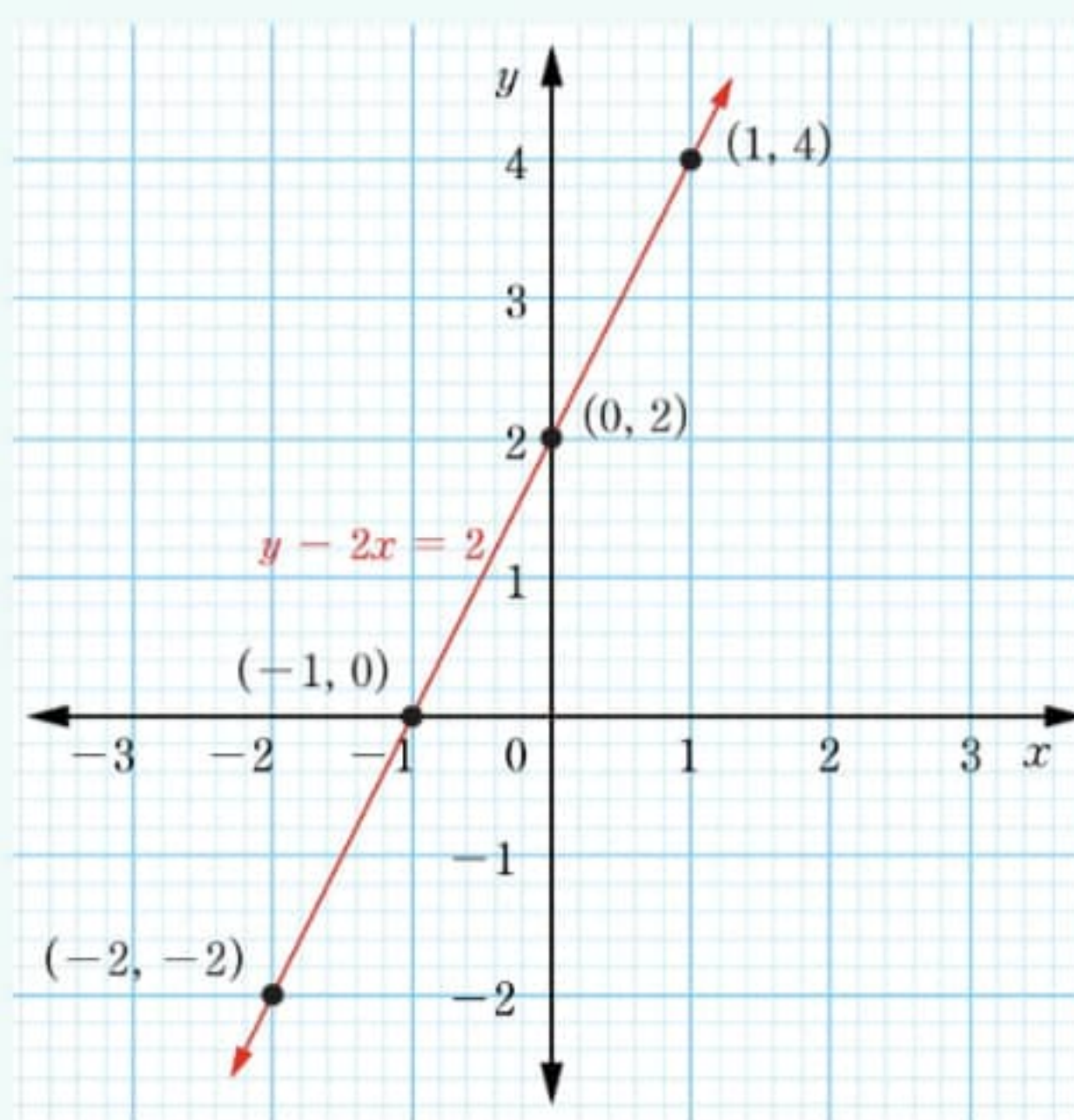
Given the equation $y - 2x = 2$ Re-write the equation in the form $y = mx + c$.

That is, $y = 2x + 2$.

The table of values for $y = 2x + 2$ is as follows:

x	-2	-1	0	1
y	-2	0	2	4

Therefore, the graph of the given equation is illustrated in the following figure.

**Example 6.26**

Draw the graph of the equation $5x - 4y + 7 = 0$.

Solution

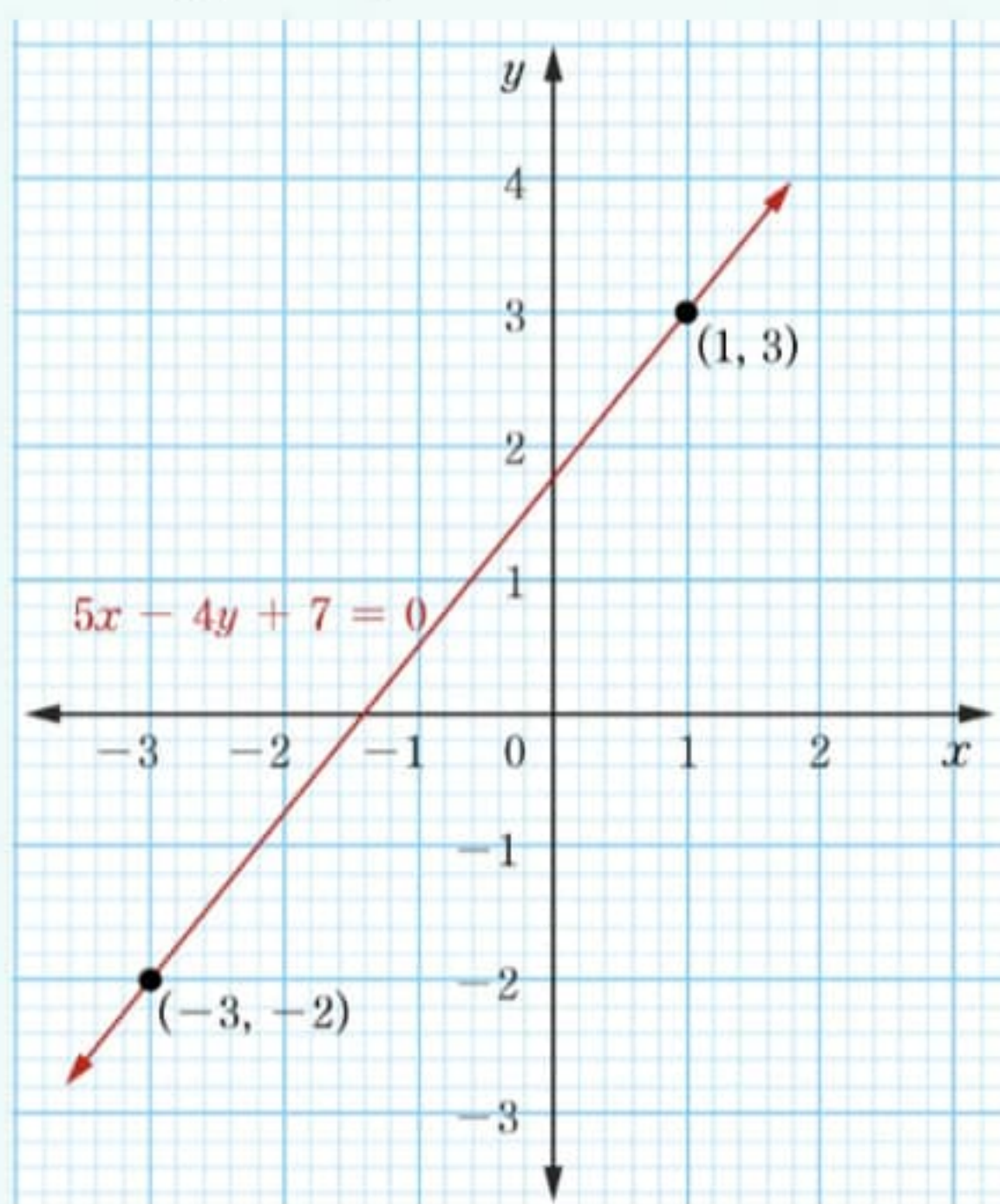
Given $5x - 4y + 7 = 0$. Re-write the equation in the form $y = mx + c$.

That is $y = \frac{5}{4}x + \frac{7}{4}$.

Table of values for $y = \frac{5}{4}x + \frac{7}{4}$

x	–	–2	–1	0	1
y	–2	$-\frac{3}{4}$	$\frac{1}{2}$	$\frac{7}{4}$	3

Therefore, the graph of the given equation is shown in the following graph.



Graphing linear equations using x and y -intercepts

Recall that x and y -intercepts are points on the x and y -axes, respectively. Thus, if x and y -intercepts are known, it is easy to draw the graph of the equation of a straight line by joining the two points.

Example 6.27

Draw the graph of the line $y = 2x + 3$.

Solution

Given the line $y = 2x + 3$. The x and y -intercepts are determined as follows:
For x -intercept, set $y = 0$. It implies that,

$$0 = 2x + 3$$

$$x = -\frac{3}{2}$$

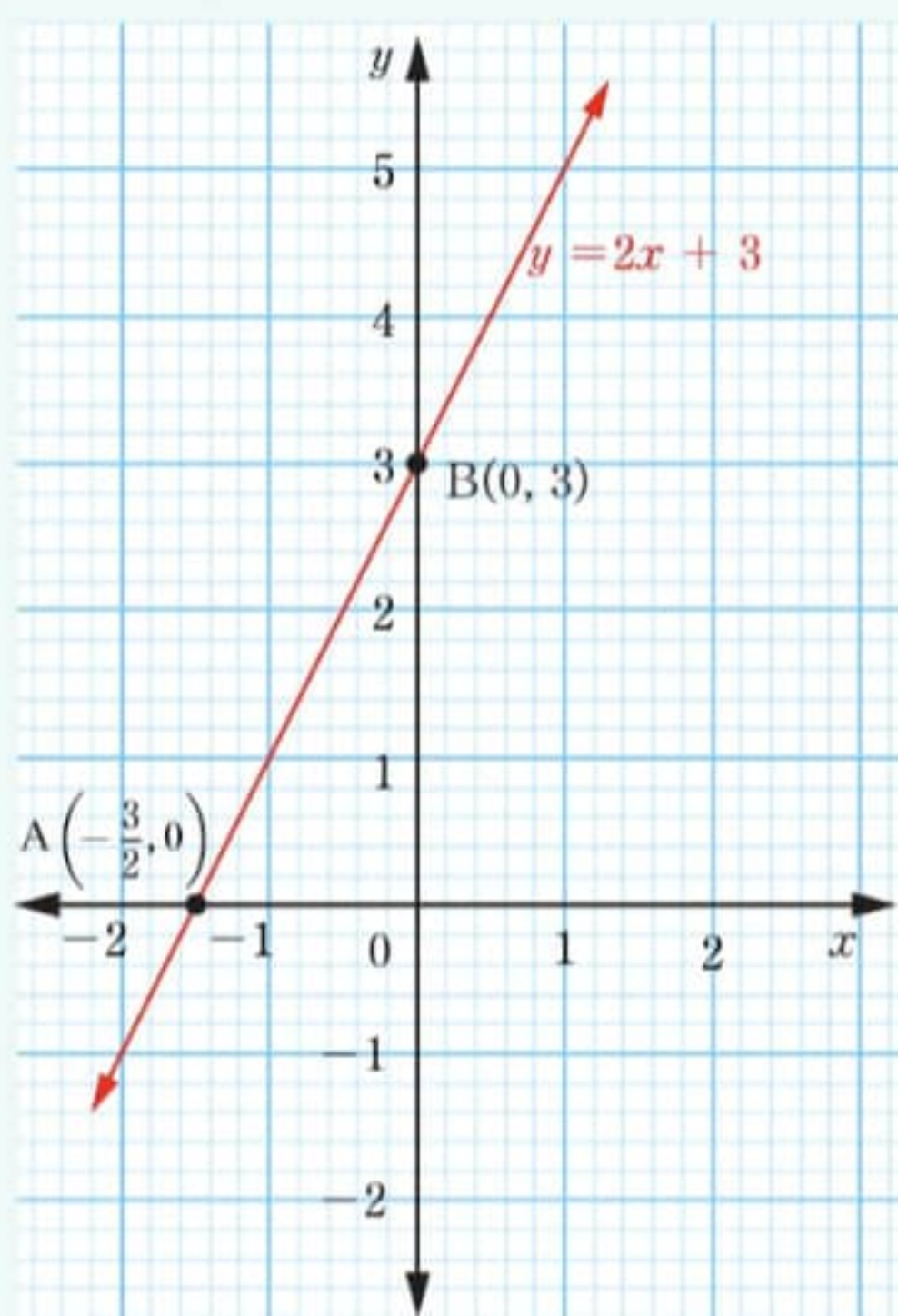
Thus, the x -intercept is $\left(-\frac{3}{2}, 0\right)$.

For y -intercept, set $x = 0$. It implies that,

$$\begin{aligned} y &= 2(0) + 3 \\ &= 3 \end{aligned}$$

Thus, the y -intercept is $(0, 3)$.

Locate the two points $\left(-\frac{3}{2}, 0\right)$ and $(0, 3)$ on the xy -plane and draw a straight line passing through these points as shown in the following figure.



Example 6.28

Use the x and y -intercepts to sketch the graph of $2x + 3y + 6 = 0$.

Solution

Given the equation $2x + 3y + 6 = 0$.

For y -intercept, set $x = 0$. It implies that

$$\begin{aligned} 3y + 6 &= 0 \\ y &= -2. \end{aligned}$$

Thus, the y -intercept is $(0, -2)$.

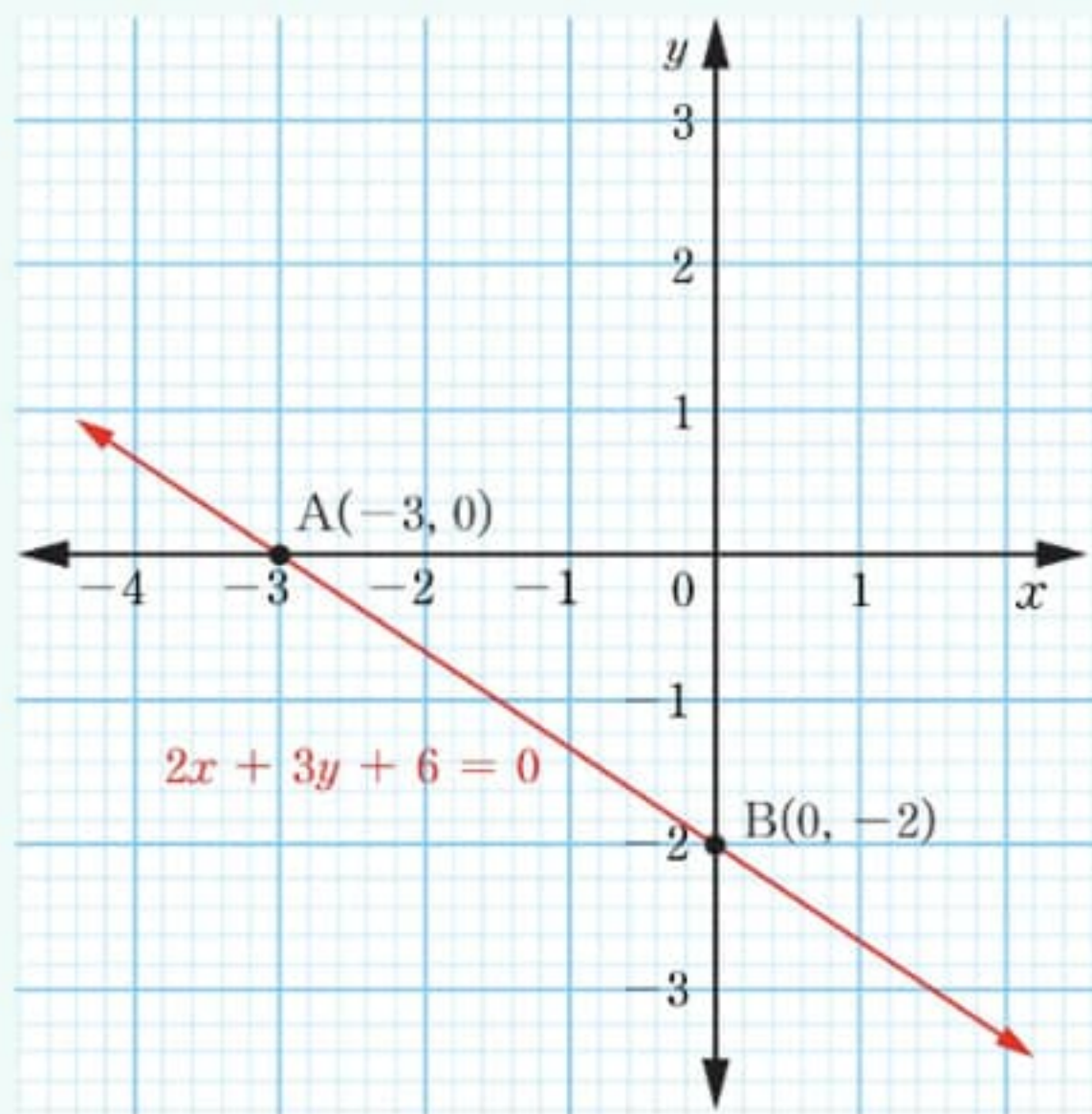
For x -intercept, $y = 0$. It implies that

$$2x + 6 = 0$$

$$x = -3$$

Thus, the x -intercept is $(-3, 0)$.

Draw a straight line passing through the x and y -intercepts to obtain the graph of the given equation as shown in the following figure.



Example 6.29

Use x and y -intercepts to sketch the graph of $-2x + 5y = 10$.

Solution

Given the equation $-2x + 5y = 10$.

For y -intercept, set $x = 0$. It implies that,

$$5y = 10$$

$$y = 2$$

Thus, the y -intercept is $(0, 2)$.

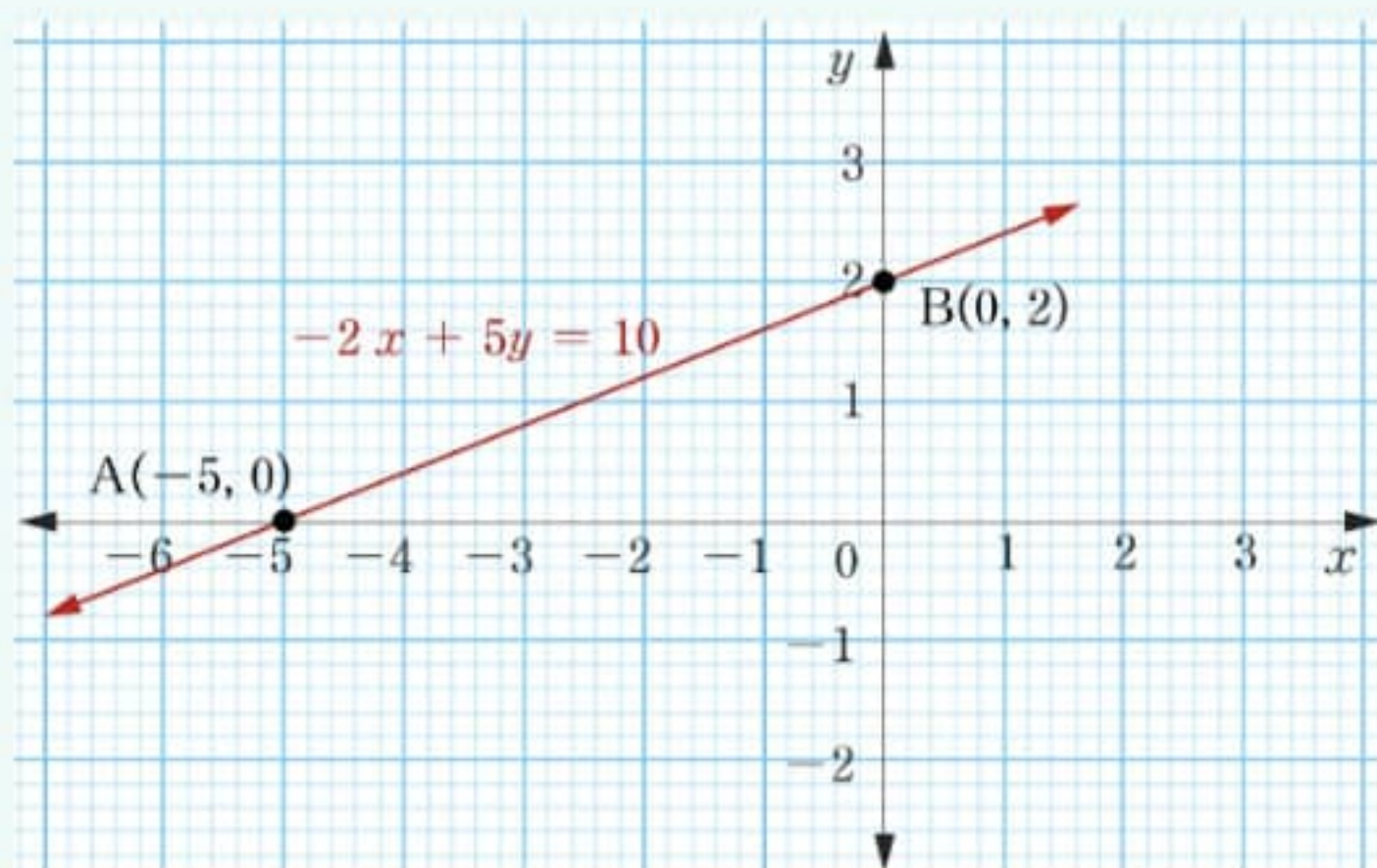
For x -intercept, $y = 0$. It implies that,

$$-2x = 10$$

$$x = -5$$

Thus, the x -intercept is $(-5, 0)$.

Locate the two points $(0, 2)$ and $(-5, 0)$ on the xy -plane and join them to obtain the straight line.



Engage in Activity 6.3 and learn how to use graphing software in graphing linear equations.

Activity 6.3: Graphing linear equations using graphing tools

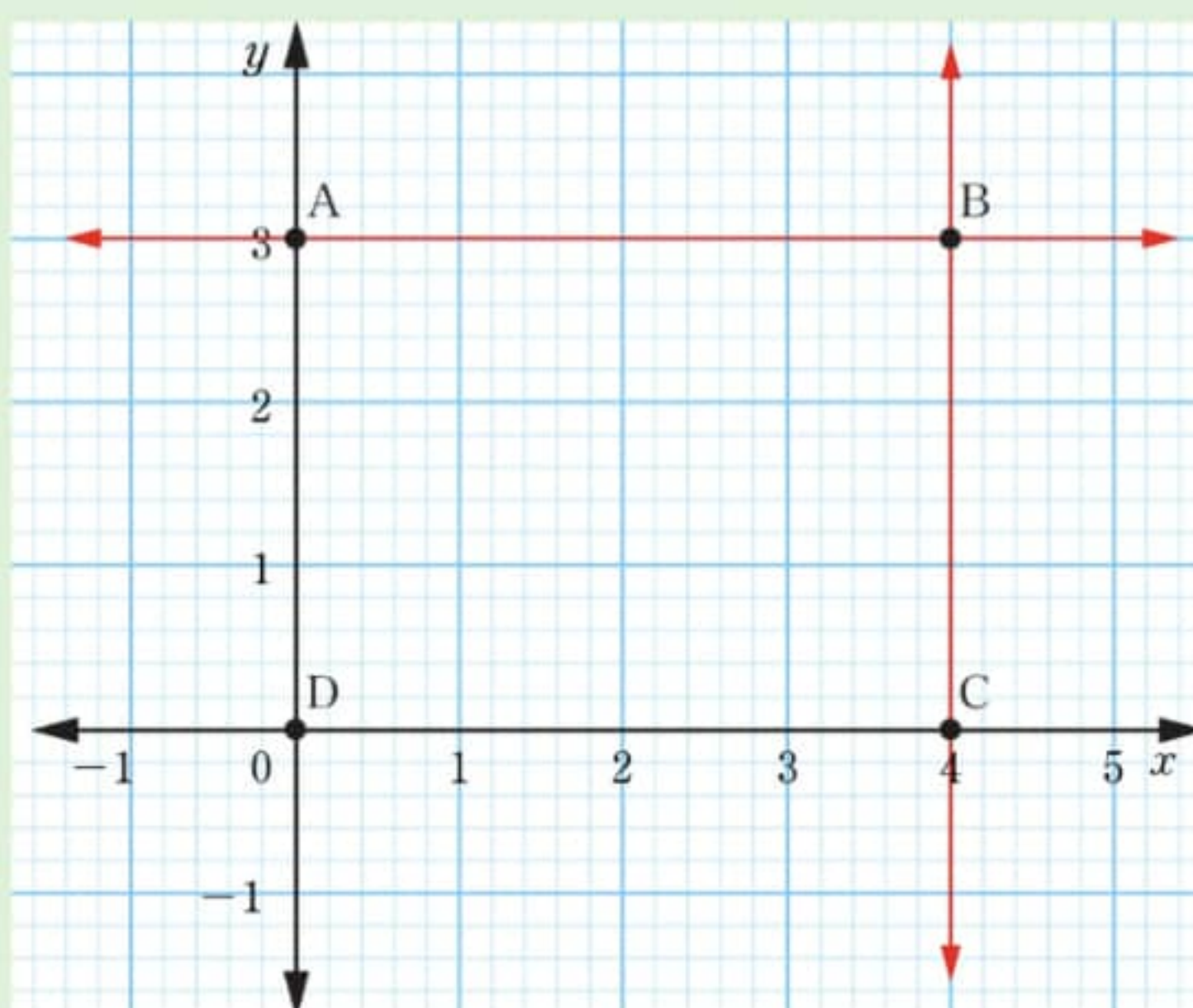
Write 5 linear equations of your choice and use a graphing software of your choice to draw their graphs.

Exercise 6.6

- Use a table of values to draw the graph of $2x + y - 1 = 0$ and hence determine the x and y -intercepts.
- Draw a graph for each of the following equations.

(a) $x + 3y = 2$	(e) $2x - 9 = 3y$
(b) $3x - 5y = 1$	(f) $y + 3x = 4$
(c) $7x - 2y = 28$	(g) $2x = y$
(d) $3x - y = 2$	

3. Use the following graph to answer the questions that follow.



- Find the gradient of each of the lines through the points A and B, and through the points B and C.
 - Find the equation of each of the lines through the points A and B, and through the points B and C.
 - Find the gradient and equation of the line passing through the points A and C. Hence, determine x and y -intercepts.
4. Draw the graph in each of the following linear equations by using x and y -intercepts.
- | | |
|-------------------|------------------|
| (a) $3x - 2y = 6$ | (e) $2x = y + 6$ |
| (b) $x - 2y = 4$ | (f) $2y + 4 = x$ |
| (c) $2x + y = 9$ | (g) $x - y = 1$ |
| (d) $x + y = 2$ | (h) $x + 2y = 2$ |
5. Sketch the graph of the equation $-\frac{y}{2} - \frac{x}{5} = 1$
6. Use x and y -intercepts to sketch the graph of the equation $\frac{2}{5}x + 0.1y = \frac{3}{5}$
7. Use table of values to draw the graph of $0.3x - 0.2y = 0.6$.
8. Draw the graph of $3x - 4y = 12$ by using table of values.

9. Two bicycles, one travelling at 3 m/s faster than the other bicycle start at the same time from the same point. Formulate a linear equation that represent this scenario. Use table of values to draw the graph of the equation formulated.
10. A bread is made using 500 g of wheat flour and 100 g of sugar. If the total cost of the two items is Tshs 800;
 - (a) write a linear equation that represent the cost of the items.
 - (b) draw the graph of the line in (a).

Solving linear simultaneous equations graphically

Solving linear simultaneous equations graphically helps to realize and understand the relationships between variables and their solutions. Engage in Activity 6.4 to learn this relationship.

Activity 6.4: Recognising the point of intersection of two linear equations

1. Use the given equations to complete the given tables of values.

(a) $y = 2x + 1$

x	-3		-1	0		2		4
y		-3		1	3		7	9

(b) $y = -x + 1$

x		-2		0		2		4
y	4	3	2		0		-2	-3

2. From task 1, use a graph paper to draw the graph of the two equations on the same xy -plane, then read and record the point where the two equations intersect.
3. Use graphing mathematical software of your choice to draw the two graphs on the same xy -plane and read the point of intersection.
4. Compare the results obtained in task 2 and that in task 3 and provide your conclusion.
5. Check if the point obtained in task 2 satisfies both equations and give your conclusion.

When solving linear simultaneous equations graphically, the graphs of the equations are drawn on the same xy -plane. The coordinates of the point of intersection of the lines is the solution of the system of equations. If the lines of the graphs drawn do not meet or intersect, then the system of simultaneous equations have no solution. In this case, the lines are said to be parallel.

Example 6.30

Determine graphically the solution of the following systems of simultaneous equations.

$$y = 2x - 1 \text{ and } y = -x + 2.$$

Solution

Given the two equations $y = 2x - 1$ and $y = -x + 2$. Construct the tables of values of the two equations as follows.

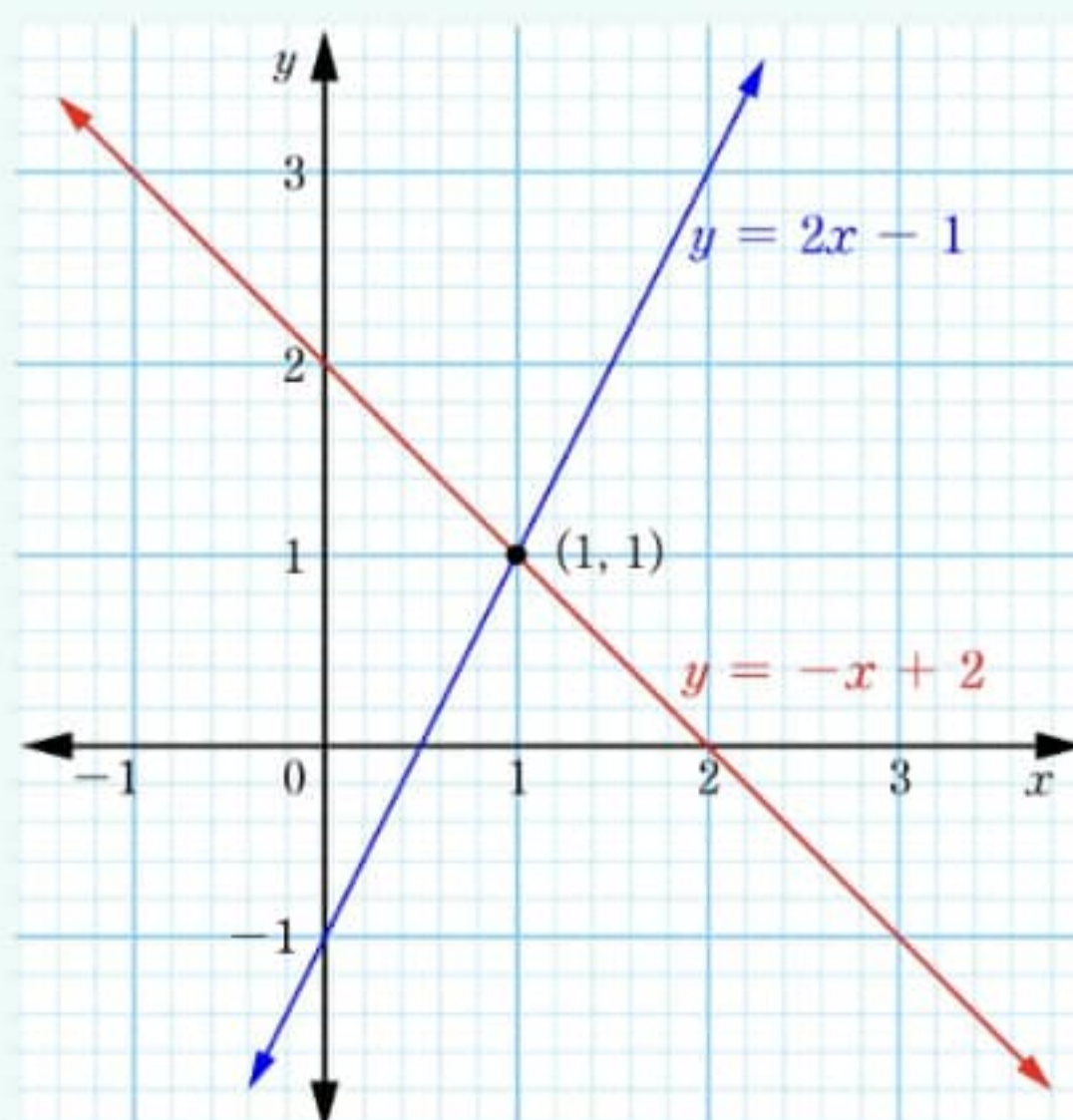
Table of values for $y = 2x - 1$

x	-1	0	1	2
y	-3	-1	1	3

Table of values for $y = -x + 2$

x	-1	0	1	2
y	3	2	1	0

Use the set of ordered pairs in each table to draw the graph of each of the equations on the same xy -plane and identify a point of intersection. Label each graph by its corresponding equation.



From the graph, the two lines intersect at the point $(1, 1)$.

Therefore, $x = 1$ and $y = 1$ is the solution of the system of simultaneous equations.

Example 6.31

Find graphically the solution of the following pair of linear simultaneous equations.

$$3x - y = 2 \text{ and } 2x - y = 3.$$

Solution

Given the two equations $3x - y = 2$ and $2x - y = 3$. Re-write the equation in the form $y = mx + c$ and construct a table of values of each equation as follows:

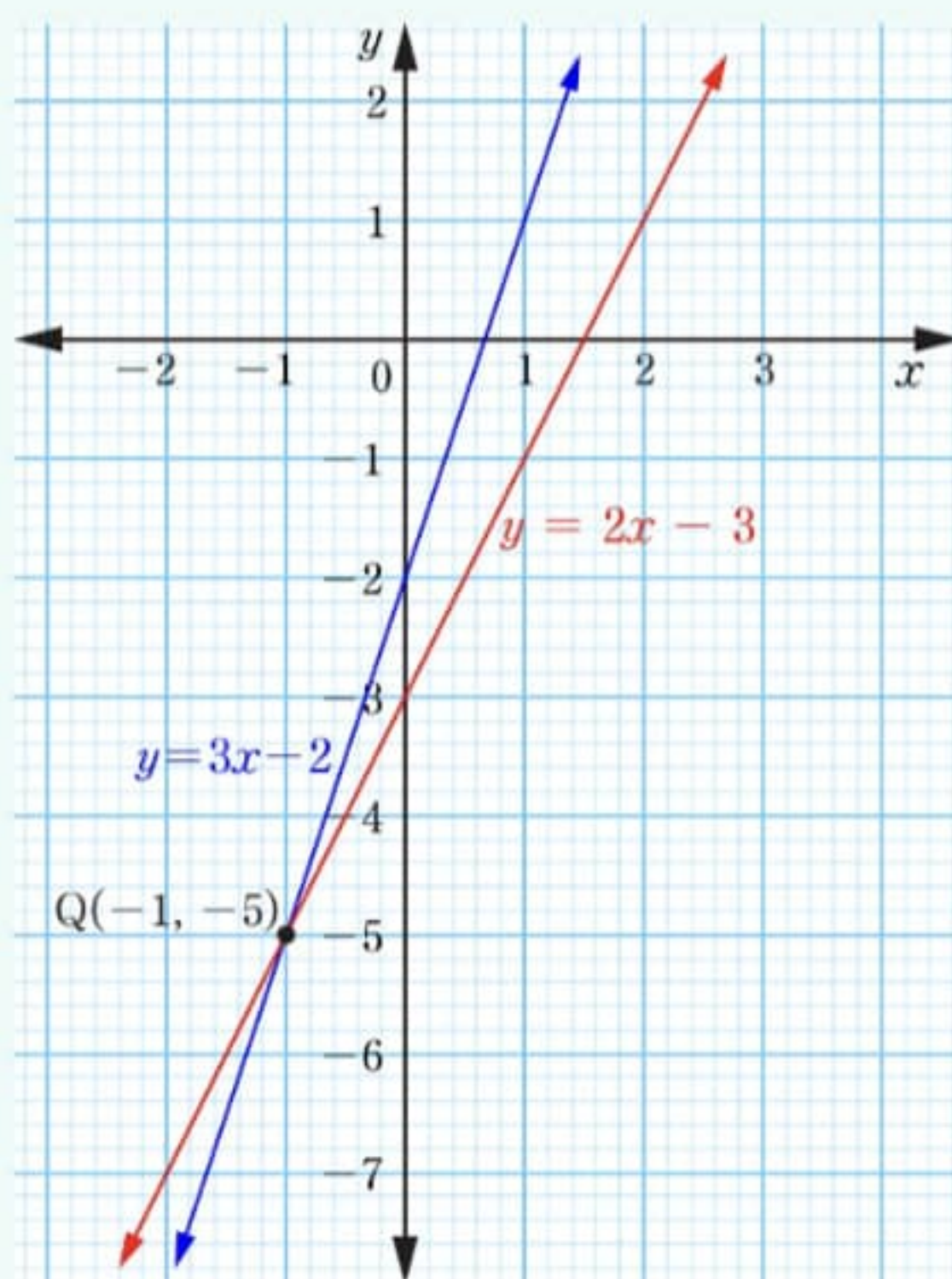
Table of values for $y = 3x - 2$

x	-2	-1	0	1	2
y	-8	-5	-2	1	4

Table of values for $y = 2x - 3$

x	-2	-1	0	1	2
y	-7	-5	-3	-1	1

Locate the ordered pairs of values for $y = 3x - 2$ and $y = 2x - 3$ on the same xy -plane and draw the graph of each linear equation and identify the point of intersection of the two lines. The following is the graph representing the two equations.



From the graph, the two lines intersect at a point $(-1, -5)$.

Therefore, $x = -1$ and $y = -5$ is the solution of the system of linear simultaneous equations.

Example 6.32

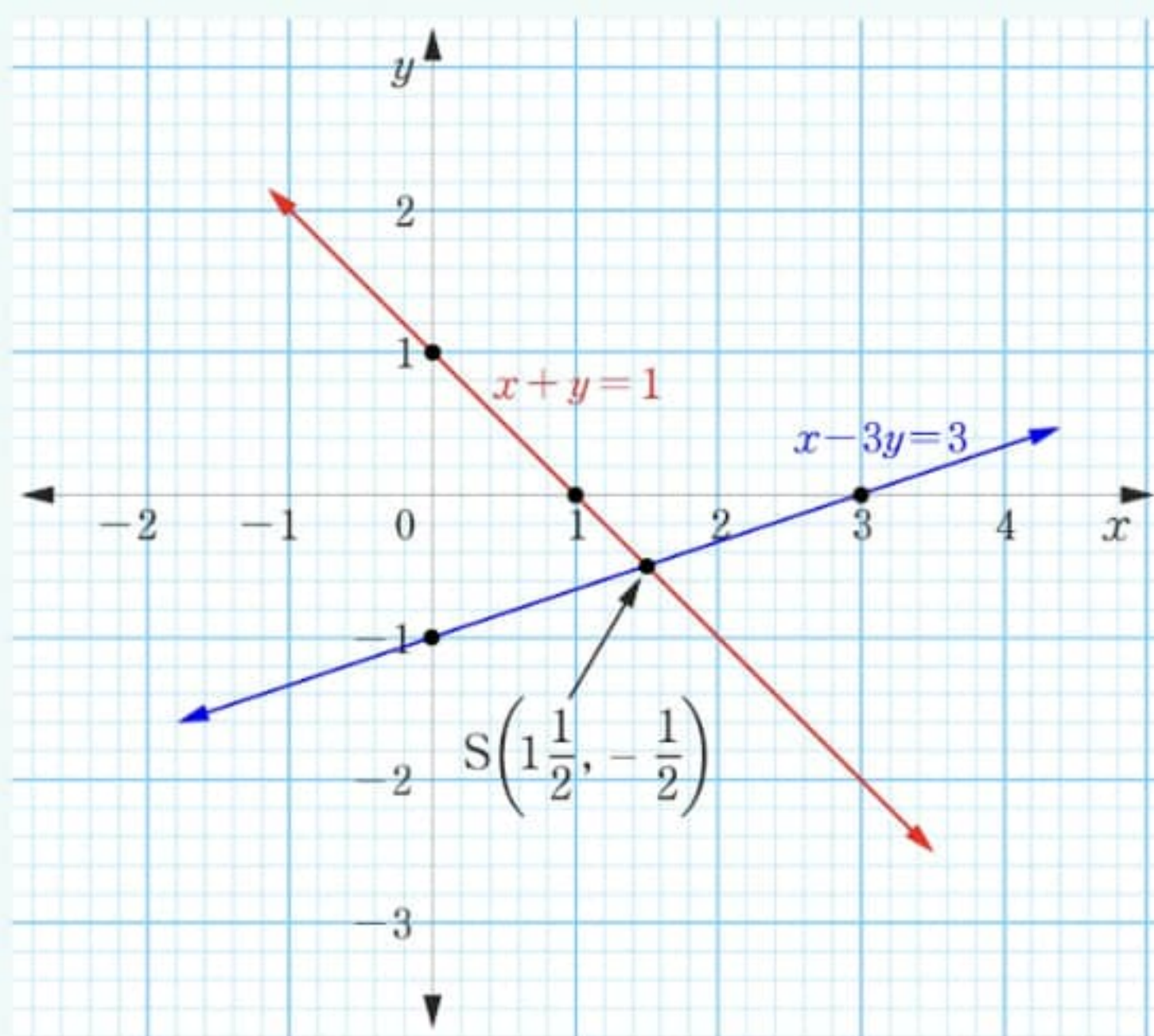
Solve graphically the following system of simultaneous equations.

$$x + y = 1 \text{ and } x - 3y = 3$$

Solution

The x and y -intercepts for $x + y = 1$ are $(1, 0)$ and $(0, 1)$, respectively. Also, x and y -intercepts for $x - 3y = 3$ are $(3, 0)$ and $(0, -1)$, respectively.

Use the intercepts to draw the graphs as shown in the following figure.



The two lines intersect at the point $S\left(1\frac{1}{2}, -\frac{1}{2}\right)$. Therefore, $x = 1\frac{1}{2}$ and $y = -\frac{1}{2}$ is the solution of the given system of linear simultaneous equations.

Example 6.33

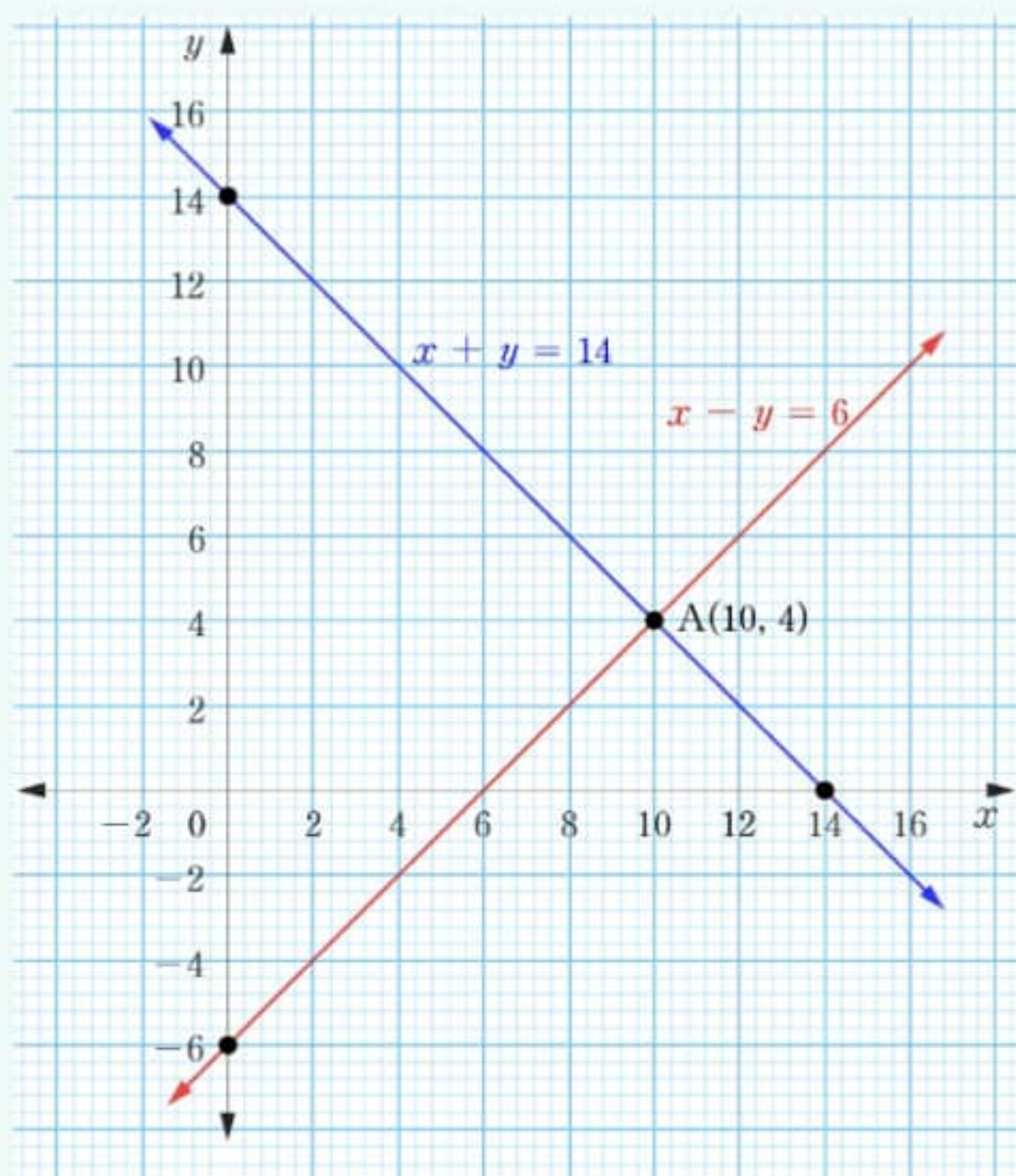
The average of two numbers x and y is 7 and three times y subtracted from x is 18. Find the numbers graphically.

Solution

Let x and y be the numbers. It follows that, the average of two numbers is

$$\frac{x + y}{2} = 7 \text{ and three times when } y \text{ is subtracted from } x \text{ is } 3(x - y) = 18.$$

Thus, the two equations are $x + y = 14$ and $x - y = 6$. The x and y -intercepts are $(14, 0)$ and $(0, 14)$ for $x + y = 14$ and $(6, 0)$ and $(0, -6)$ for $x - y = 6$. Use the intercepts to draw the graph as shown in the following figure.



From the figure, the two lines intersect at a point $(10, 4)$. That is, $x = 10$ and $y = 4$. Therefore, the numbers are 4 and 10.

Engage in Activity 6.5 to learn how to use mathematical software involving linear simultaneous equations graphically.

Activity 6.5: Solving linear simultaneous equations graphically by mathematical software

1. Identify a mathematical software of your choice.
2. Formulate 5 different linear equations and draw them on the xy -plane using the mathematical software identified in task 1.
3. Use the same software to draw systems of linear simultaneous equations of your choice on the same xy -plane and state the solution.

Exercise 6.7

Find graphically the solution in each of the following pair of linear simultaneous equations. Verify your answers by a calculator or any other graphing software of your choice.

$$1. \quad \begin{cases} 2x + y = 4 \\ x + y = 3 \end{cases}$$

$$7. \quad \begin{cases} y = -2x + 5 \\ y = x - 4 \end{cases}$$

$$2. \quad \begin{cases} x + y - 3 = 0 \\ x - y - 1 = 0 \end{cases}$$

$$8. \quad \begin{cases} 2y + 12 = 3x \\ 4y + 8 = 2x \end{cases}$$

$$3. \quad \begin{cases} x + y = 1 \\ 2x + 3y = 5 \end{cases}$$

$$9. \quad \begin{cases} y + 1 = 3x \\ y - 3 = x \end{cases}$$

$$4. \quad \begin{cases} x - 2y = -5 \\ x + 3y = 15 \end{cases}$$

$$10. \quad \begin{cases} 3y = 2x + 3 \\ x - \frac{2}{3}y = 1 \end{cases}$$

$$5. \quad \begin{cases} x - 2y = 1 \\ x + 2y = 5 \end{cases}$$

$$11. \quad \begin{cases} 3p + 4q = 7 \\ 5p + 6q = 11 \end{cases}$$

$$6. \quad \begin{cases} 2x + y = 6 \\ 6x - y = 6 \end{cases}$$

$$12. \quad \begin{cases} 2c + 2d = c - d \\ 2d + 2 = c + 1 \end{cases}$$

13. Ali paid Tshs 15,000 for 10 oranges and 35 mangoes. Peter went to the same fruit market and paid Tshs 8,800 for 16 oranges and 18 mangoes. Find graphically the price of an orange and a mango.

14. The sum of the number of boys and girls in a class is 36. If twice the number of girls exceeds the number of boys by 12, use graphical method to find the number of girls and boys in the class.

15. The sum of the ages of a mother and her daughter is 52 years. Eight years ago, the mother was eight times as old as her daughter. Use the graphical method to find their present age.

Chapter summary

1. The xy -coordinate plane or xy -plane is made up of two number lines intersecting at right angles.
2. The horizontal axis is called the x -axis.
3. The vertical axis is called the y -axis.
4. The point at which the two axes intersect is called the origin.
5. The two axes divide the xy -plane into four equal parts called quadrants.
6. Each coordinate point in the coordinate plane has two numbers called coordinates written as an ordered pair (x, y) .
7. The formula for calculating the gradient or slope is

$$m = \frac{\text{change in } y \text{ coordinates}}{\text{change in } x \text{ coordinates}}.$$
8. A vertical line has an undefined gradient and is parallel to the y -axis.
9. A horizontal line has a gradient whose value is 0 and is parallel to the x -axis.
10. The equation of a line can be written in the form $y = mx + c$, where m is the gradient and c is a constant.
11. The general equation of a line has the form $ax + by + c = 0$.
12. Graphs of linear simultaneous equations intersect at a common point whose coordinates give the solution of the equation.
13. Linear simultaneous equations in two unknowns have solution of an ordered pair (x, y) .

Revision exercise 6

1. Find the equation and y -intercept of each line passing through the following pair of points.

(a) $(4, 4), (-3, 2)$	(d) $(-2, -2), (4, 1)$
(b) $(-7, 2), (2, 7)$	(e) $\left(\frac{4}{7}, 6\right), \left(\frac{3}{2}, 6\right)$
(c) $\left(\frac{5}{6}, 2\right), \left(6, \frac{1}{6}\right)$	(f) $(-155, -541), (541, 155)$

2. In each of the following pairs of points, state whether the line represented by the points is vertical, horizontal, or slanted.

(a) $(1, 0), (0, 0)$	(e) $(2, 8), (-2, 8)$
(b) $(1, 2), (2, 1)$	(f) $\left(\frac{3}{4}, 1\right), \left(\frac{7}{3}, 4\right)$
(c) $(3, -2), (3, 5)$	(g) $(0.5, -1), (0.5, -2)$
(d) $(-4, 2), (-4, -1)$	(h) $(4, 2), (0, -3)$
3. Draw a straight line represented by each of the following pair of points and state its gradient.

(a) $(0, 0), (1, 3)$	(d) $(7, 5), (2, 3)$
(b) $(-1, -1), (8, 8)$	(e) $(2.5, -3), (6.5, -4)$
(c) $(0, 1), (1, 0)$	(f) $(3, 6), (6, 3)$
4. Draw a graph of each of the following equations.

(a) $x + 6y = 5$	(d) $x - y = 1$
(b) $5x = 3y + 8$	(e) $-4x + 3y = 10$
(c) $4x + 2y = 28$	(f) $5x - y = 4$
5. Re-write each of the following equations in the form $y = mx + c$ and determine their gradients and y-intercepts.

(a) $4x + y = 8$	(d) $3x + 4y = 23$
(b) $6y = x + 2$	(e) $3x + 4y + 4 = 0$
(c) $2x - y = 2$	(f) $3y + 4x + 9 = 0$
6. Draw the graph of each of the following equations.

(a) $x - 2y = 4$	(c) $5x = 3y + 8$
(b) $2x + 3y = 9$	(d) $6y = x + 2$
7. Draw straight lines from each of the following information.

(a) Passing through the point $(2, -1)$ and with gradient $\frac{1}{2}$
(b) Passing through $(3, 1)$ with the y-intercept $(0, 3)$
(c) Passing through the point $(1, 4)$ with the gradient -4
(d) Passing through the point $(-2, 0)$ and the y-intercept is $\left(0, \frac{1}{3}\right)$

8. Solve graphically each of the following linear simultaneous equations.

(a) $\begin{cases} x - y = -3 \\ x + 2y = 3 \end{cases}$

(c) $\begin{cases} 2x + y = 7 \\ 2x - 5y = 4 \end{cases}$

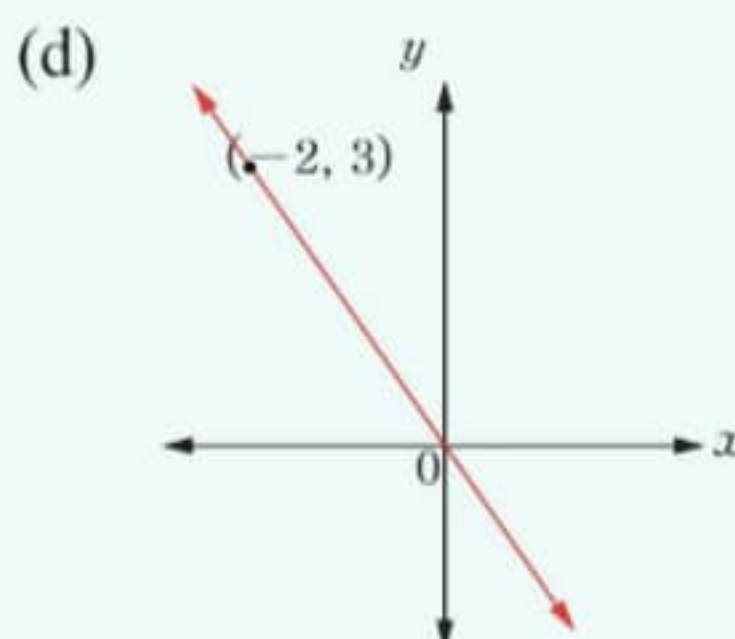
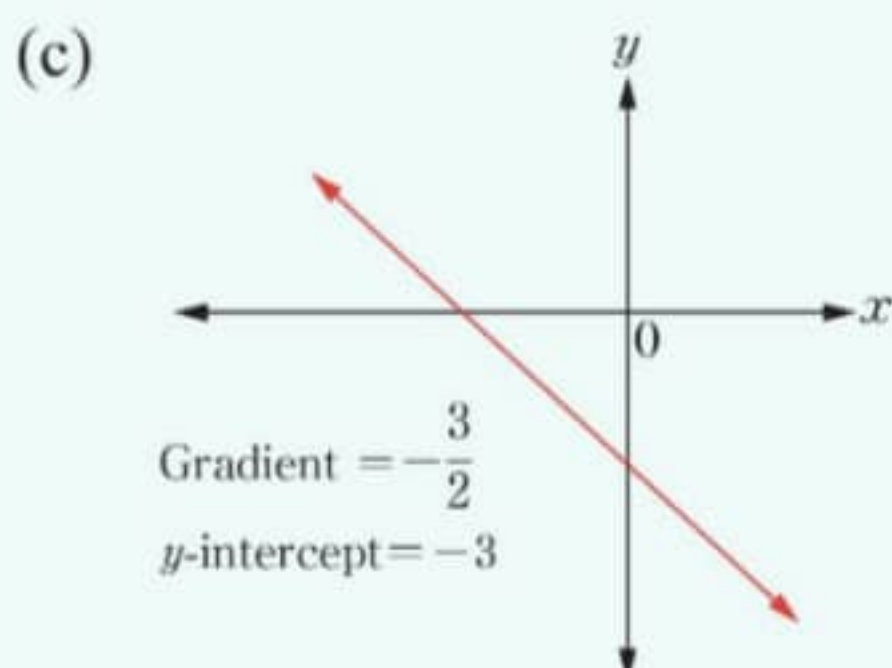
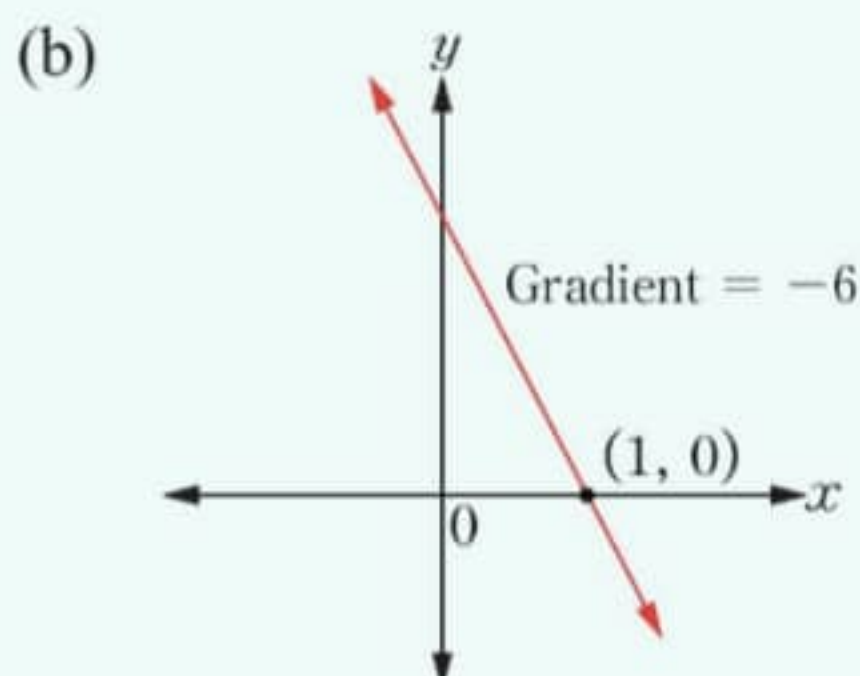
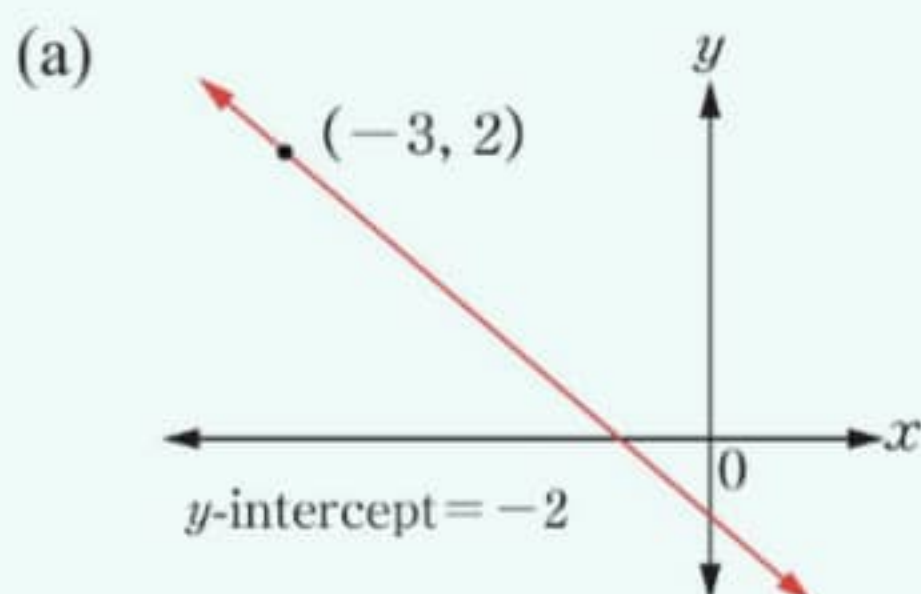
(e) $\begin{cases} y = \frac{1}{2}x + 4 \\ y = 2x - 2 \end{cases}$

(b) $\begin{cases} 3 + 2y - 3x = 0 \\ 5x = 5 + y \end{cases}$

(d) $\begin{cases} y = \frac{1}{2}x + \frac{1}{2} \\ y = x - 1 \end{cases}$

(f) $\begin{cases} y = 3x - 2 \\ y = x + 2 \end{cases}$

9. Find the equation of the line in each of the following graphs.



10. Find graphically the solution of the following system of linear simultaneous equations.

(a) $5y = 6x + 30$ and $y = \frac{1}{5}x + 2$

(b) $\begin{cases} \frac{x}{2} + \frac{y}{2} = 1 \\ x = 1 + y \end{cases}$

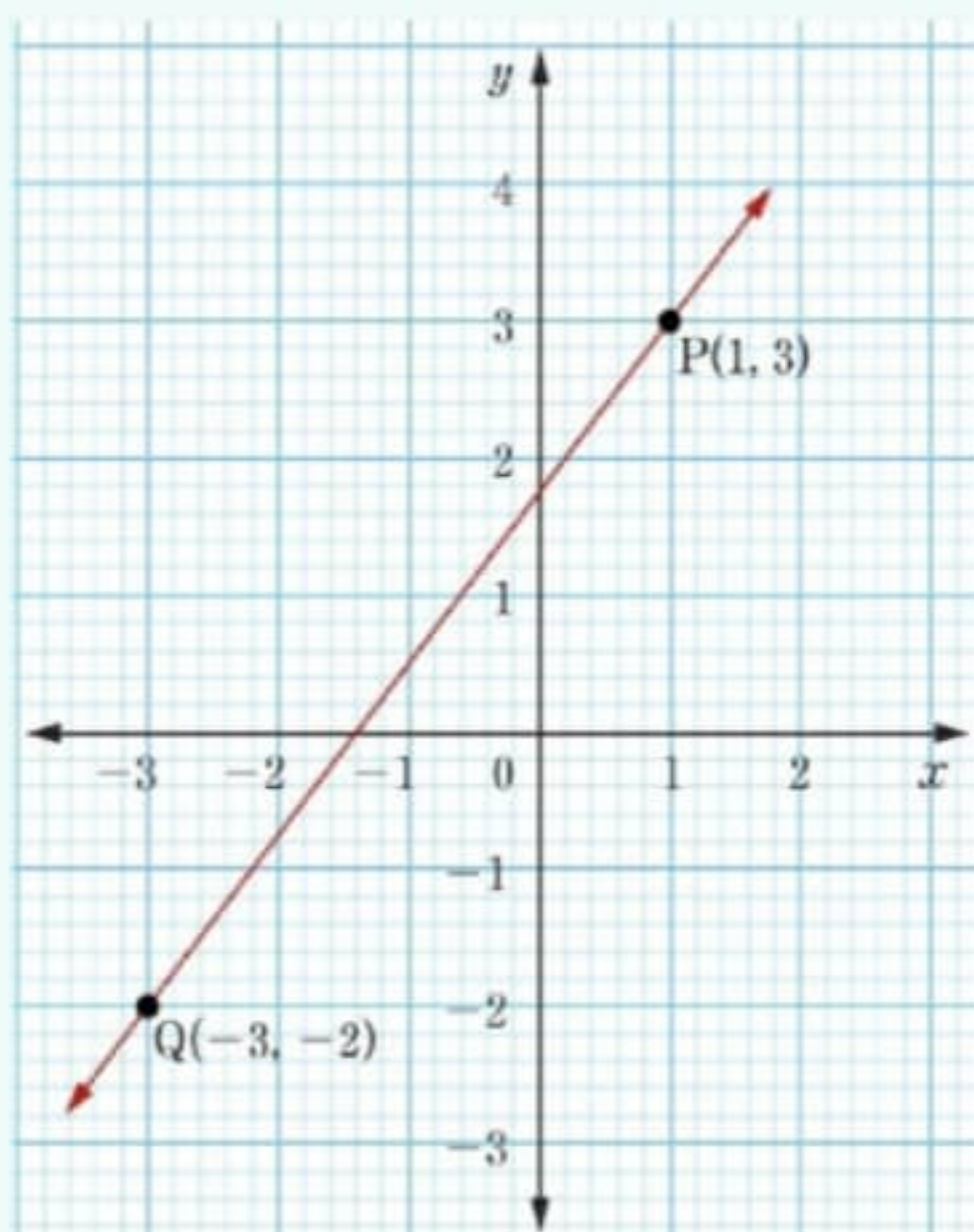
(c) $2x - y = 5$ and $x = 1 - y$

(d) $3x + 4y = 3$ and $2x - 5y + 5 = 0$

11. Determine the equation of a straight line whose gradient is $\frac{3}{5}$ and passes through the point of intersection of two lines $3x - 2y = 2$ and $2x - 3y = 3$.
12. (a) Find the equation of a straight line whose gradient is 2 and passes through the point (3,5).
(b) Determine the values of m and c such that the straight line $y = mx + c$ will pass through the points $(-1, 4)$ and $(3, 5)$.
(c) Find the value of y if the points A (3, y), B (0, 4), and C $(-2, 8)$ are in the same line.
(d) Find the value of x so that the straight line joining the points A($x, -5$) and B(2, 1) has the same slope as the line joining the points C(1, 5) and D $(-2, -1)$.
13. Find the equation for each of the following lines.
(a) Passing through the points (3, 3) and (1, 4) in the form $ax + by = c$.
(b) Passing through the point (3, -4) with gradient which is a negative of the reciprocal of the gradient of the line $5x - 2y = 3$.
14. Kiwengu is travelling by his car from Tanga. Before he starts the journey, he makes sure that the fuel tank is full. After travelling a distance of 50km, the odometer reaches 35 litres and when he travels further 150 km, the odometer read 25 litres. If x is the distance travelled and y is the odometer reading;
(a) Draw a line on the xy -plane to represent the given information
(b) Determine the x and y -intercepts and give the meaning of these information.
(c) Write the equation of the line drawn in (a)
(d) What is the slope of the line and what does it signify?
15. A family want to buy 8 loaves of bread. White bread costs Tshs 3,000 per loaf and brown bread costs Tshs 3,500 per loaf. The family can spend Tshs 25,000 for that shopping. Find the number of each type of bread the family can buy.
16. Juma is thinking of two numbers. Two times the first number is 3. Three times the first number is equal to the second number plus six. Find the two numbers graphically by using any mathematical software of your choice.
17. A cafe sales tea and cakes. The first customer bought a cup of tea and two cakes for Tshs 800. The second customer bought a cup of tea and a cupcake for Tshs 500. Use any mathematical software of your choice to determine graphically or algebraically the price of a cup of tea and a cake.

18. (a) Draw the graph of a straight line $y = 3x + 5$. By using construction method, draw a line perpendicular to the line drawn in (a) at point $(0, 5)$.
(b) Calculate the slope of the line drawn in (b).
(c) How are the slopes in (a) and (b) related?
(d) Measure and record the angle between the lines.
(e) What conclusion can you make about the product of the gradients of the two perpendicular lines?
19. Without drawing a graph, find the point of intersection between a line passing through the points $(0, 1)$ and $(3, 7)$ and a line $y = 4x + 7$.
20. Find the value of x so that $A(1, 3)$, $B(-2, -3)$, and $C(x, 7)$ are the points on the same straight line.
21. Which of the following points lie on the line $3x + 4y - 12 = 0$?
(a) $A(0, 2)$ (b) $B(4, 0)$ (c) $C(2, 4)$
(d) $D\left(\frac{4}{3}, 2\right)$ (e) $E(-3, 2)$
22. (a) Points A , B , and C are the vertices of a right-angled triangle at C . If the coordinates of A and C are $A(0, -3)$ and $C(5, -7)$, what are the coordinates of point B ?
(b) A parallelogram $ABCD$ has vertices at $A(1, -3)$, $B(6, -2)$, $C(3, 4)$, and $D(x, y)$. What are the coordinates of vertex D ?
23. Find the equation of a line whose y -intercept is the same as the that of a line $4x + y + 6 = 0$ and the product of their gradients is -1 .
24. A line through the point $(3, 1)$ has the same gradient as the line $16x - 4y = 23$. Find the equation of the line.
25. Two roads whose equations are $2x + 3y = 7$ and $x - 3y + 4 = 0$ intersect at point P . A new road is to be constructed from $A(5, 5)$ to point P . Find the equation of the new road in the form $ax + by = c$.
26. The line passing through $A(1, k)$ and $B(k + 9, -2)$ has the same gradient as the line passing through $C(3, 4)$ and $D(-2, 3)$. Find the value of k .
27. A doctor advice the patient to take 500 mg of calcium and 1,200 micrograms of vitamin A by drinking milk and orange juice. A unit of milk contains 38 mg of calcium and 56 micrograms of vitamin A while a unit of orange juice contains 5 mg of calcium and 60 micrograms of vitamin A. How many units of each should the patient take?

28. Calculate the gradient and the equation of the line represented by the following figure.



29. Sketch the graph of the straight line represented by $3x - 4y - 24 = 0$.
30. Draw the graph of a straight line represented by each of the following equations.
- (a) $14x + 7y = 25$ (b) $3y = 4x - 12$ (c) $15 = 3x - 15y$
31. Plot the graph of a straight line whose equation is $4x + y + 6 = 0$.
32. Find x and y -intercepts of the straight line $3x - 2y + 10 = 0$.
33. Determine the gradient and the y -intercept in each of the following lines.
- (a) $3x - y = 12$
- (b) $7x - 2y + 3 = 0$
- (c) $ax + by + c = 0$
- (d) $\frac{x}{a} + \frac{y}{b} = 1$
34. Plot the points $A(1, -1)$, $B(4, -1)$, and $C(1, 3)$ on the xy -plane and join the points by line segments.
35. Find the equation of a line through the point $A(4, -3)$ having the same slope as line $x + 4y - 1 = 0$.

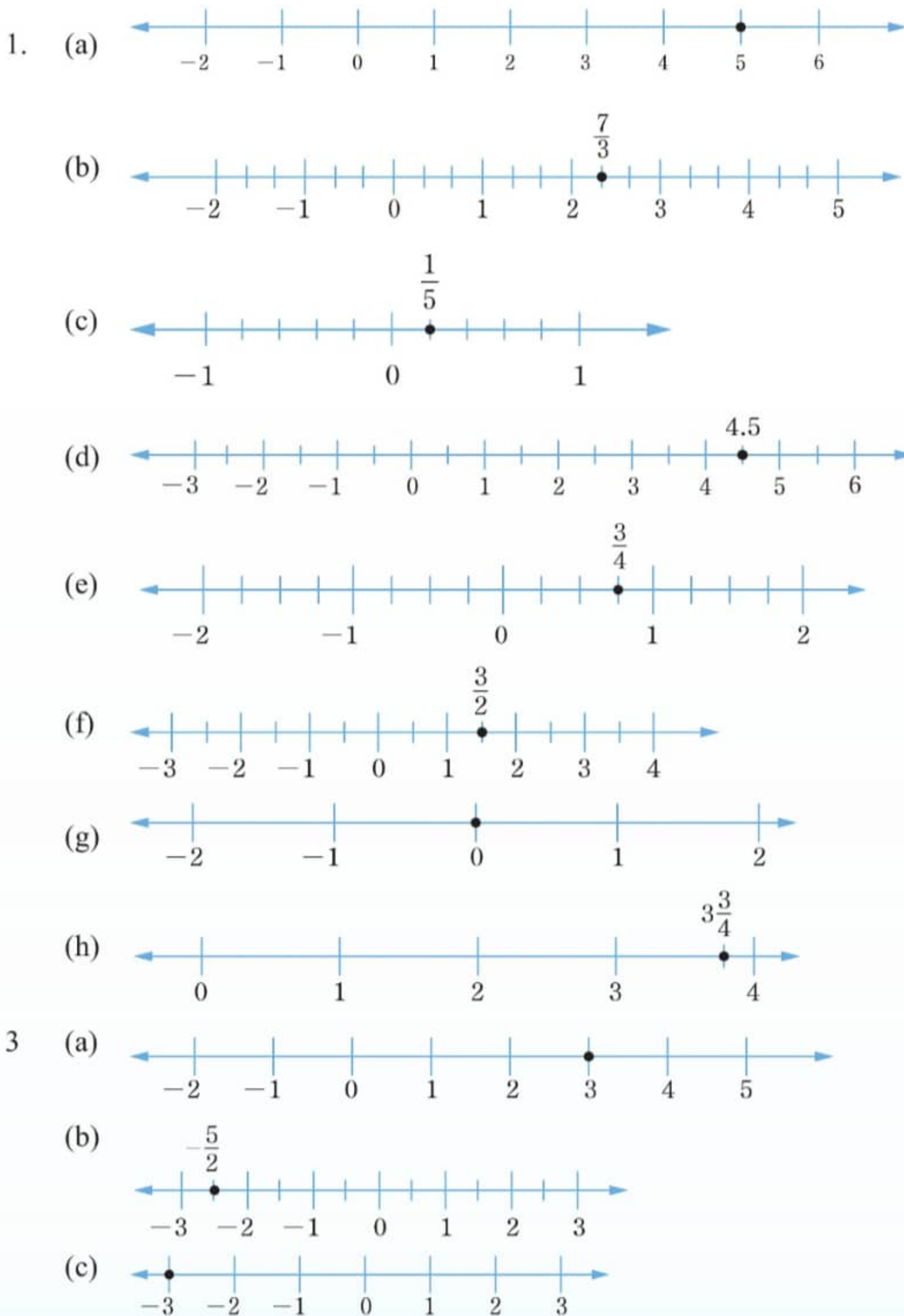
Project: Locating places

Create a map on an xy -plane which shows location of different important places of your choice such as school compound by showing important buildings, home premises, and many other places of your interest.

Answers to Odd-Numbered Questions

Chapter Two

Exercise 2.1



5. (a) $\frac{\quad}{250}$ (b) $\frac{4}{1}$ (c) $\frac{27}{1}$ (d) $\frac{28}{5}$

(e) $-\frac{3}{100}$ (f) $\frac{17}{4}$ (g) $-\frac{49}{8}$ (h) $-\frac{8}{3}$

7. Some of the fractions which are equivalent to $\frac{7}{10}$ are $\frac{14}{20}$, $\frac{21}{30}$, $\frac{28}{40}$, $\frac{35}{50}$, $\frac{42}{60}$

9. $-\frac{6}{35}$

Exercise 2.2

1 (a) $\frac{1}{9}$ (b) $\frac{8}{9}$ (c) $\frac{80\ 137}{999}$ (d) $\frac{1\ 027}{330}$

(e) $\frac{859}{66}$ (f) $\frac{31}{90}$ (g) $\frac{241}{333}$ (h) $\frac{22}{9}$

3. $\frac{16}{17}$

5. (a) $\frac{73}{99}$ (b) $\frac{341}{90}$ (c) $\frac{23}{90}$

7. $\frac{932}{495}$

9. 14

Exercise 2.3

1. (a) $0.\dot{8}5714\dot{2}$ (c) $0.\dot{7}\dot{2}$ (e) $1.\dot{9}2307\dot{6}$
(b) $1.8333\dots$ (d) $1.1\dot{3}$ (f) $1.\dot{6}1538\dot{4}$

3. (a) $\frac{1}{3}$ is greater than $0.\dot{1}\dot{2}$

7. $0.\dot{3}\text{m}$

9. $130.\dot{9}\dot{0}\text{ km}$

Exercise 2.4

1. $\sqrt{11}$, $3+\sqrt{7}$, $\sqrt{12}$, $3\sqrt{3}-\sqrt{5}$, $1.101001000100001\dots$, $-\sqrt{7}$ and $\frac{\pi}{2}$ are irrational because when expressed as decimals, the decimal digits continue infinitely without any repeating pattern.

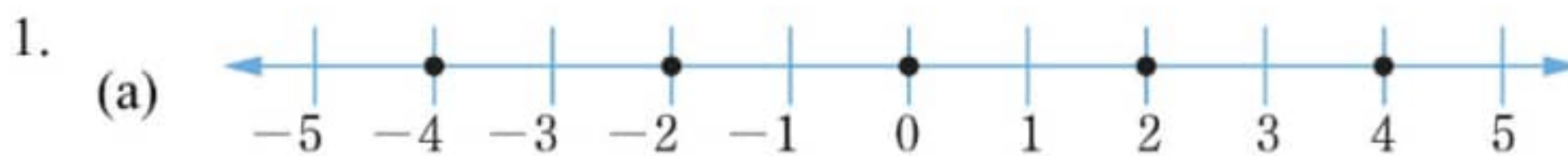
0.65 and $\sqrt{9}$ are rational numbers because they can be expressed in the form $\frac{a}{b}$.

3. (a) Whole numbers $0, 9$
 (b) Integers $0, -5, 9$
 (c) Rational numbers $-5, -3.5, 0, \frac{3}{4}, 9, -4.3\dot{2}\dot{5}$
 (d) Irrational numbers $\sqrt{3}, \sqrt{5}, \pi, \frac{\sqrt{2}}{2}$.
5. $\frac{10}{\sqrt{\pi}}m$

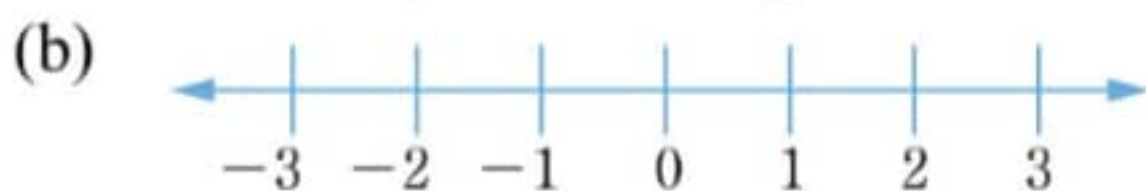
Exercise 2.5

1. (a) True (b) True (c) False (d) False (e) True
 (f) True (g) False (h) False (i) True (j) False
3. (a) 1, 2, 3, 4, 5
 (b) 0, 1, 2, 3, 4
 (c) 5, 7, 9, 11, 13, ...
 (d) -2, -1, 0, 1, 2, 3, 4
5. 0, 1, 2, 3, 4, 5.
7. Ester's height < Nancy's height and Nancy's height > Hyasinta's height.
9. -7 and -8

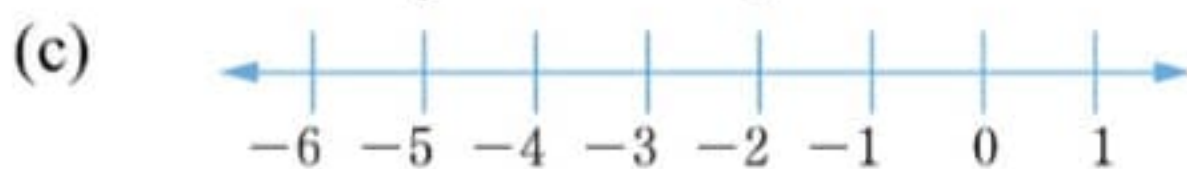
Exercise 2.6



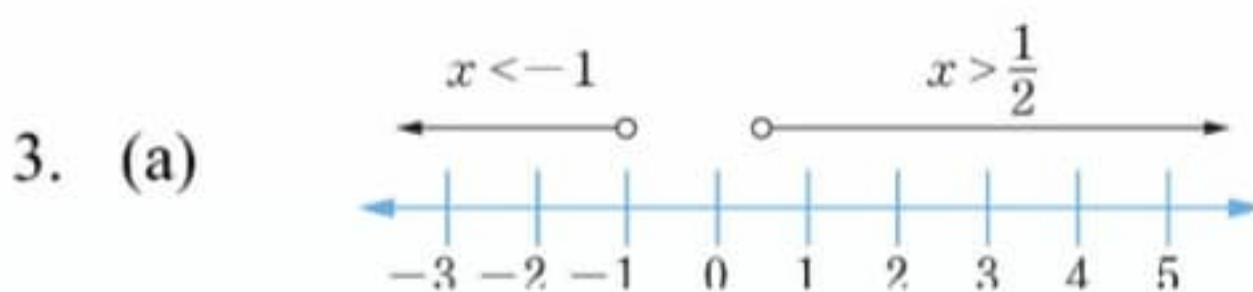
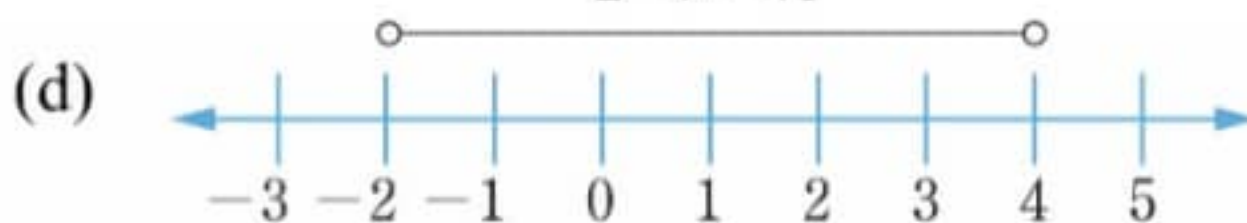
$$-2 \leq k \leq 1$$



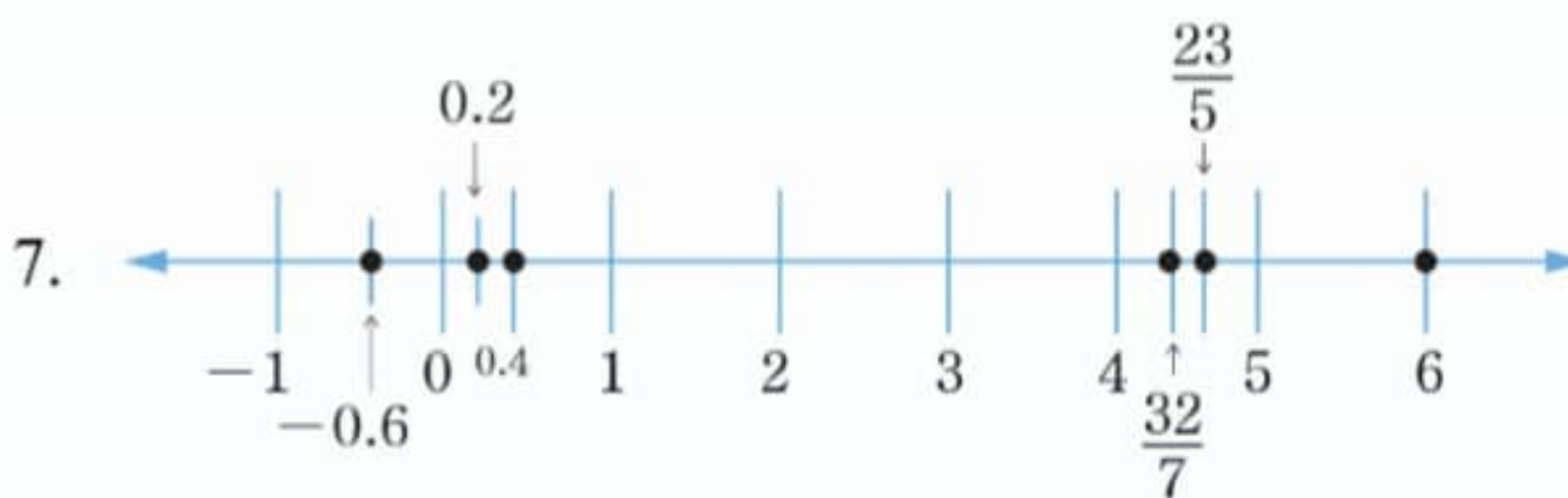
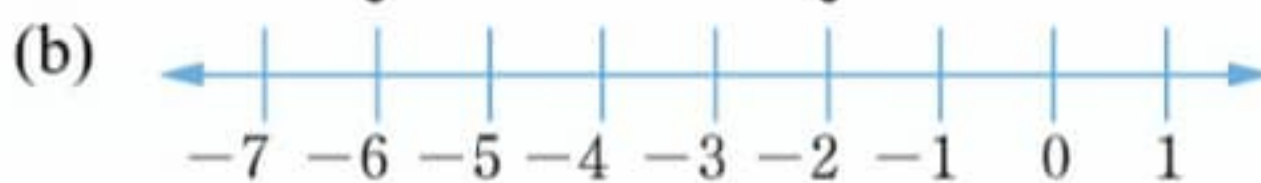
$$-5 < y \leq -2$$

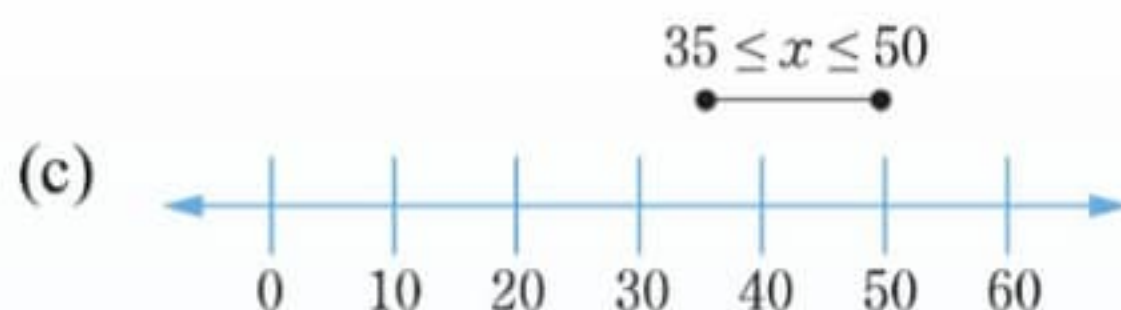
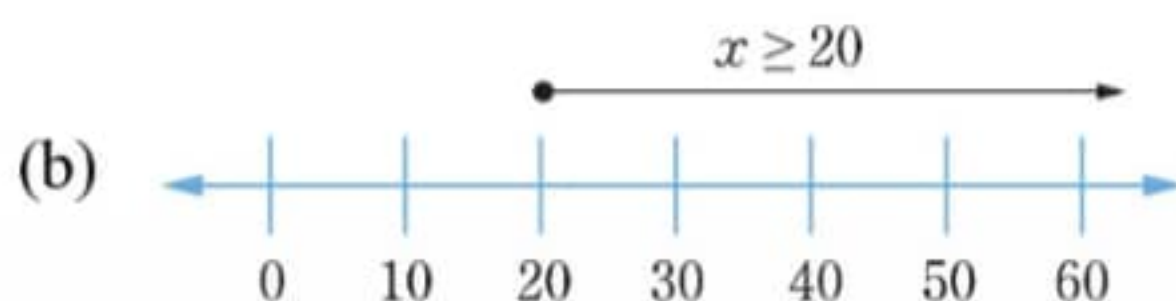


$$-2 < x < 4$$



$$-6 \leq x \leq -2$$





Exercise 2.7

1. (a) 62 (b) $\sqrt{2}$ (c) 8 (d) $\frac{4}{7}$ (e) 12.5 (f) pq

3. -12, -4, -2, 5, 9, 10, 12, 37

5. (a) -6 (b) -2 (c) -11 (d) 6 (e) -60

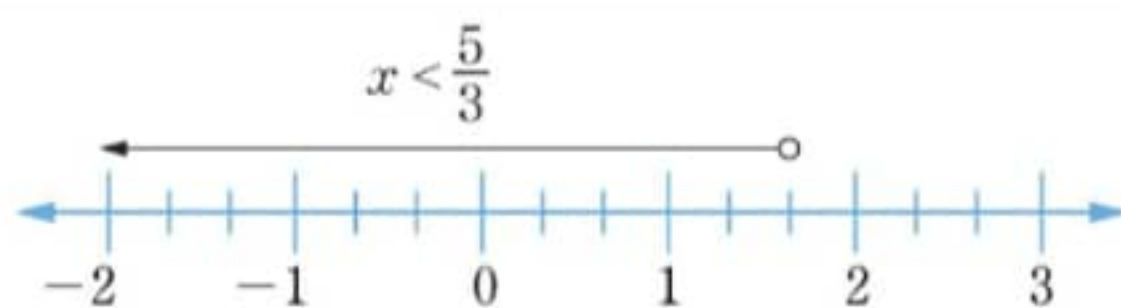


9. Lightest cone is 155 g, the heaviest cone is 165 g

Revision Exercise 2

- 1 (a) $\frac{13}{2}$ (b) $-\frac{102}{11}$ (c) $\frac{9}{11}$ (d) $\frac{387}{2}$ (e) $\frac{23}{1}$ (f) $\frac{2207}{9}$
 (g) $-\frac{43}{10}$ (h) $\frac{5174}{45}$ (i) $\frac{3}{250}$ (j) $\frac{41}{333}$

3.



5. $-\frac{11}{4} < x \leq \frac{3}{4}$

7 (a) $\frac{19}{8}$ (b) $\frac{7}{12}$ (c) $\frac{20}{33}$ (d) $\frac{17}{2}$ (e) $\frac{718}{175}$

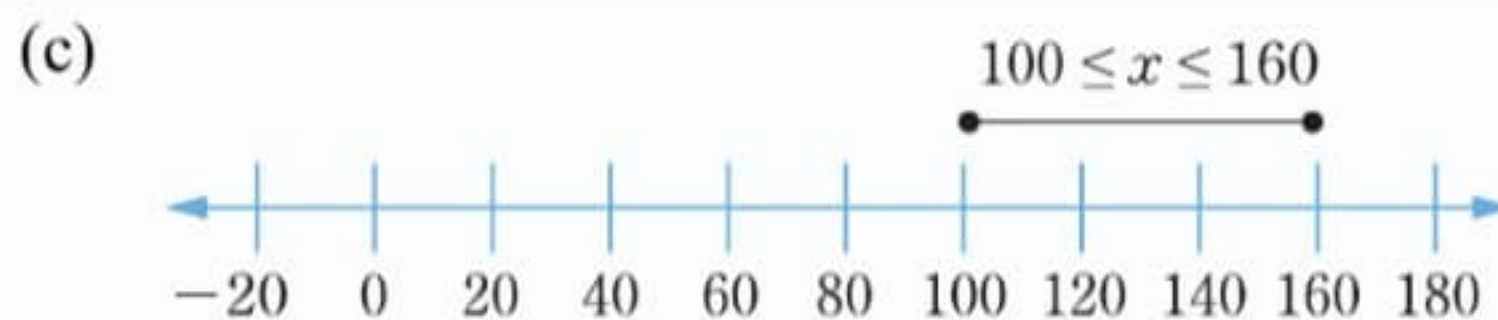
(f) $\frac{269}{22}$ (g) $\frac{50}{163}$ (h) $\frac{235}{67}$ (i) $\frac{4}{15}$

9. (a) $0.\dot{5}$ (b) $1.\overline{27}$ (c) $4.8\dot{3}$ (d) $0.\dot{0}7\dot{4}$ (e) $0.41\overline{6}$
(f) $0.4\dot{5}$ (g) $1.\overline{21}$ (h) $0.\dot{1}4\dot{8}$ (i) $0.4\dot{6}$ (j) $0.0\overline{76923}$

11. Rational numbers are 0.75 , $\frac{90}{12007}$, $\sqrt{7}$ and $\frac{1}{\sqrt{2}}$.
12. Irrational numbers are13. $5 \leq x \leq 6$ where x is a rational number.

15. (a) $\frac{211}{90}$ (b) $\frac{21284}{999}$ (c) $\frac{17}{99}$

17. 31 tomatoes 19. Tshs 3 787 600 21. 2692.3 min or 44.9 hrs



25. (a) $0.2\dot{2}$ (b) $0.0\dot{1}$ (c) $0.6\dot{5}$ (d) $0.\dot{2}$

Chapter Three

Exercise 3.1

1. (a) 8,000 (b) 12,000 (c) 14,000 (d) 101,000 (e) 18,000
(f) 2,350,000 (g) 60,000 (h) 10,000 (i) 1,235,000
3. (a) 30 (b) 20 (c) 30 (d) 10
(e) 50 (f) 30 (g) 20 (h) 310
5. 237,000 pupils
7. (a) 157.9 (b) 10.0 (c) 0.3 (d) 294.5
9. (a) 688.9341 (b) (i) 688.93 (ii) 688.9 (iii) 700
11. (a) 12,000 m (b) 38.3 m

Exercise 3.2

1. (a) 4 000.0 (b) 19.9 (c) 0.008
(d) 25.3545 (e) 0.0100 (f) 10.010
3. (a) 0.08 (b) 5.07 (c) 1.70 (d) 0.72
(e) 12.05 (f) 3.35 (g) 3.61 (h) 0.01 (i) 72.79
5. (a) 3.1 (b) 0.7 (c) 250.7
(d) 0.1 (e) 0.7 (f) 10.4
7. 18.03
9. (a) 0.0028 (b) 20.04 (c) 0.172
(d) 6.010 (e) 2.1
11. 9.860 m²

Exercise 3.3

1. (a) 5 (b) 10 (c) 3 (d) 8
(e) 8 (f) 3 (g) 7 (h) 5
3. (a) 1.923 (b) 0.714
5. (a) 0.612 (b) 100 (c) 35.51 (d) 7.44 (e) 16.6
(f) 493.0
7. (a) 12.78 (b) 320 (c) 0.00714 (d) 37.20 (e) 40
(f) 399.6
9. (a) 20,000 (b) 25,000
(b) 25,000 will be reported because it is the nearest value to the actual value 24,582

Revision exercise 3

1. (a) 8.65 (d) 31.78
(b) 1.05 (e) 19.67
(c) 0.34 (f) 0.45
3. (a) 5,00,000
(b) 5,268,000
(c) 5,267,900
5. (a) 86.5 (b) 86.46 (c) 90 (d) 86
7. (a) (i) 17.8 (ii) 17.1 (iii) 2.0 (iv) 0.0
(b) (i) 23.75 (ii) 23.08 (iii) 0.05 (iv) 0.08
9. (a) 0.69 (b) 0.95 (c) 0.09
11. (a) 35.00 (b) 35.0
15. (a) 8,700 m
(b) Tshs 86,000
(c) (i) 10,000 (ii) Tshs 40,000,000
17. 1.85 gm
19. (a) 240 m
21. (a) Tshs 7,000
(b) Tshs 270,000
(c) Tshs 360,000

Chapter Four**Exercise 4.1**

- 1 (a). 2:3 (h) 14:5 (o) 1 000:1
(b) 4:5 (i) 28:1 (p) 8:7
(c) 2:1 (j) 3:5 (q) 34:1
(d) 11:5 (k) 7:4 (r) 5:9
(e) 4:3 (l) 1:4 (s) 3:2
(f) 5:3 (m) 17:100 (t) 8:7
(g) 16:9 (n) 50:1
3. 3:10
5. 15:12:10
7. 36 oranges

Exercise 4.2

1. (a) 70, 30 (c) 6, 6, 4
(b) 45, 30 (d) $\frac{2}{5}, \frac{1}{10}$
3. Juma - Tshs 30,000, Ali - Tshs 18,000, Aisha - Tshs 12,000.
5. 2.1 kg
7. 40 kg of A, 32 kg of B
9. 50 kg
11. 9 100 litres
13. There are 312 coins for Tshs 500 coins, 1 170 coins for Tshs 200 coin and 3,900 coins for Tshs 100 coin.
15. $\frac{9n^2}{2}$

Revision exercise 4

1. (a) 5% (b) 0.5% (c) 9.8% (d) 75% (e) $66\frac{2}{3}\%$
3. Tshs 1,824,000 9. 60 15. 5.89 cm
5. 27 kg, 108 kg, 135 kg 11. 57.6 kg
7. 8.89 grams of chlorine 13. 20 cm 17. Tshs 4,200,000
19. (a) 7:9 (b) 56.25%
21. $\frac{2}{5}$ and $\frac{7}{5}$ 23. $\frac{2}{3}$ 25. Tshs 88,000

Chapter Five**Exercise 5.1**

1. (a) $5n + 3k + 2x$, 3 terms, coefficients are 5, 3 and 2 for n , k , and x , respectively.
(b) $6x - 3y - 5z$, 3 terms, coefficients are 6, -3 and -5 for x , y , and z , respectively.
(c) $\frac{41}{4}x$, 1 term, the coefficient is $\frac{41}{4}$.

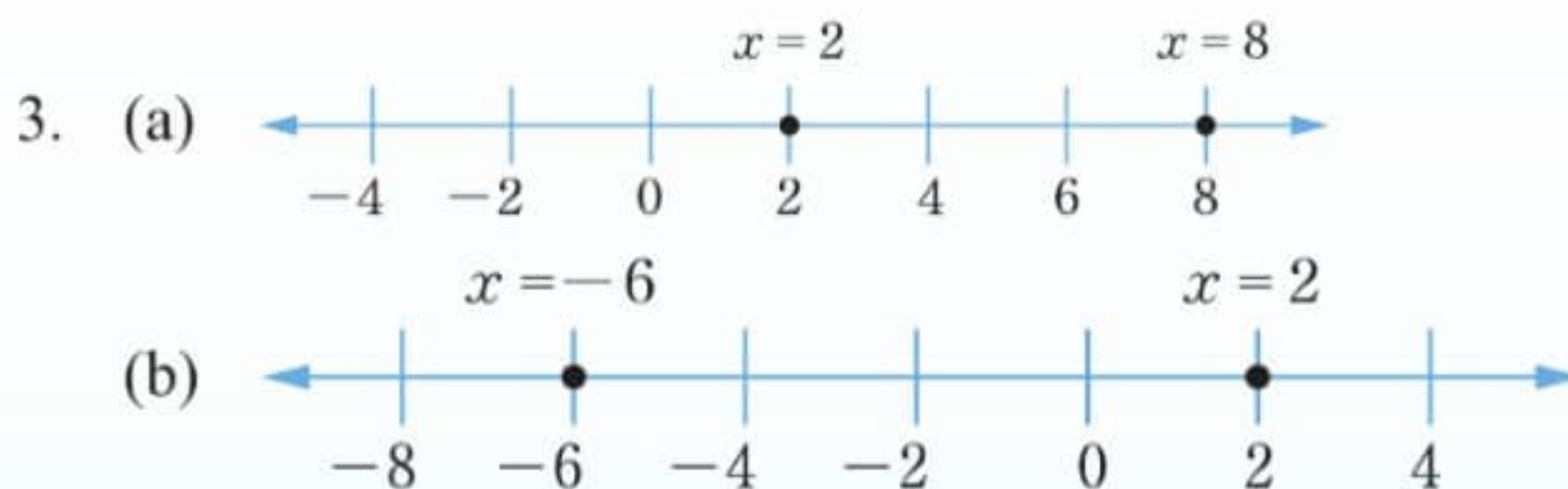
3. (a) $5z$ (b) $2x - 3y$ (c) $\frac{3}{2} - \frac{4}{3}b$ (g) $\frac{28 + 34y}{15}$
 (d) $\frac{3}{4} + \frac{3}{4}d$ (e) $\frac{a + 3b}{8}$ (f) $\frac{m - 4}{12n}$
5. (a) $4x^2 + 7x - 15$ (b) $5p^2q - 3q$
 (c) $4xy^2 + 2xy$ (d) $10x - 16y$
7. (a) 22 (b) 2.4 (c) 16 (d) -
9. (a) $6a$ (b) $-2x$ (c) $18y^2 - 75$ (d) $5n$

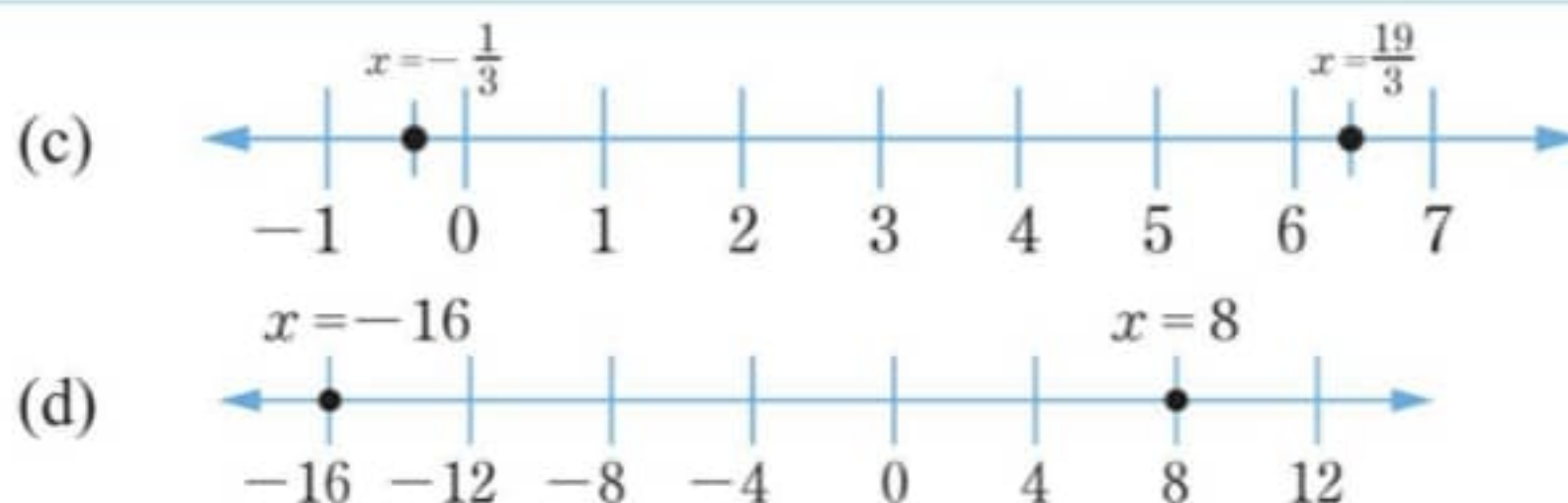
Exercise 5.2

1. $5x = 20$ 7. $2s + 8 = 3s - 7$ 13. $4x = 2 + \frac{1}{4}y$
 3. $x + 21 = 125$ 9. $5x + 8 = 4(x + 8)$
 5. $99 - m = 63$ 11. $5x + 6y = 15$

Exercise 5.3

1. (a) $x = 13$ (h) $x = 40$ (o) $x = 10$
 (b) $x = 16$ (i) $x = 10$
 (c) $x = 5$ (j) $x = 20$
 (d) $x = 25$ (k) $x = \frac{25}{24}$
 (e) $x = \frac{13}{2}$ (l) $x = 6$
 (f) $x = 8$ (m) $x = \frac{27}{7}$
 (g) $x = 10$ (n) $x = 6$





5. (a) $p = 15$ (b) $r = 6.75$ (c) $x = -\frac{1}{4}$ (d) $y = 0$
7. (a) $x = 6$ (b) $x = -\frac{1}{4}$ (c) $x = 2$
8. (a) $x = 66^\circ$ (b) $x = 47.5^\circ$

Exercise 5.4

1. 12 years 7. m 20 or 28 13. 9
 3. 18 and 19 9. 16 15. The number is 4
 5. $y = 10$ 11. Daughter 4 years and mother 32 years

Exercise 5.5

1. $x = 2, y = 1$ 7. $x = -1, y = 2$ 13. $u = 2, v = -\frac{1}{2}$
 3. $x = 4, y = 2$ 9. $x = 8, y = 4$ 15. $m = \frac{1}{3}, n = 7$
 5. $x = 3, y = 1$ 11. $x = -1, y = 3$

Exercise 5.6

1. $x = 3, y = 2$ 9. $a = -1, b = 23$
 3. $x = 8, y = 2$ 11. $x = 12, y = 3$
 5. $x = 2, y = 3$ 13. $x = 6, y = 10$
 7. $x = 2, y = 2$ 15. $x = 8, y = 5$

Exercise 5.7

1. 69 and 40
3. 16 girls and 20 boys
5. The cost of one pencil is Tshs 500 and one pen is Tshs 800
7. 70 and 80
9. $x = \frac{4}{5}, y = \frac{38}{5}$

Exercise 5.8

1. (a) 1, 2, 3, 4, 5 (b) 0, 1, 2, 3, 4 (c) 5, 7, 9, 11, ...
(d) -2, -1, 0, 1, 2, 3, 4
3. (a) $y > 2$ or $y < -2$ (b) $-3 \leq y \leq -1$ (c) $-3 \leq y \leq 3$
(d) $-\frac{5}{2} < y < \frac{5}{2}$ (e) $y > 0$ or $y < -4$
5. (a) $x > 4$ (b) $x \leq 5.7$ (c) $x \geq 3$ (d) $y < -7$
7. (a) $\frac{3}{8} \leq x \leq \frac{5}{8}$ (c) $y < -\frac{13}{3}$ or $y > -\frac{5}{3}$
(b) $x \geq \frac{10}{9}$ or $x \leq \frac{8}{9}$ (d) $b \geq -4$
9. (a) $1 \leq x \leq \frac{11}{3}$ (c) $-35 < c < -\frac{5}{9}$
(b) $y < -1$ or $y > 5$ (d) $x \leq -2$ or $x \geq \frac{6}{5}$

Exercise 5.9

1. Possible width > 5 cm
3. $x < \frac{2}{3}$
5. $x \geq 3$
7. $x < 2$
9. The smallest integer is 0 and the largest integer is 4

Revision exercise 5

1. (a) $16a$ (c) $-0.7x + 4.1y$
(b) $7m$ (d) $5x + \frac{3}{4}$
3. (a) $10m - n$ (c) $9x + y$
(b) $-6k - 5n - 5$ (d) $-5x + 5y$
5. (a) $120abmn$ (b) $-42mn$ (c) $-ax + 6bx$
7. (a) 80 (b) -4 (c) $\frac{15}{2}$
9. (a) $x = \frac{15}{13}$ (b) $x = -6$ (c) $x = 30$ (d) $x = 10$
(e) $x = \frac{7}{8}$ (f) $x = \frac{5}{2}$ (g) $x = \frac{25}{4}$ (h) $x = \frac{29}{19}$
11. Age of father is 55 years and James is 11 years.
13. (a) $x = 3, y = 6$ (b) $x = \frac{11}{5}, y = \frac{22}{5}$
(c) $x = 2, y = 6$ (d) $x = 10, y = 7$
15. (a) $x = 10, y = 11$ (b) $x = 8, y = 5$
(c) $x = \frac{3}{2}, y = \frac{5}{4}$ (d) $x = 2, y = 6$
17. $x = 15, y = 32$
19. (a) $x > \frac{5}{2}$ (b) $x \leq 1$ (c) $x < -1$
(d) $x \geq \frac{1}{4}$ (e) $x \geq \frac{9}{8}$ (f) $x < -2$
21. $x = 3, y = \frac{5}{2}$
23. $\frac{3}{8}$ 25. $x + y = 102$
27. (a) $x = 4$ or $x = \frac{16}{5}$ (c) $x = \frac{7}{3}$ or $x = -\frac{17}{7}$
(b) $x = \frac{3}{7}$ or $x = -\frac{1}{19}$ (d) $x = -\frac{19}{7}$ or $x = -\frac{17}{11}$

29. (a) $x = 1, y = 3$

(c) $x = 3, y = 4$

(b) $x = 3, y = 2$

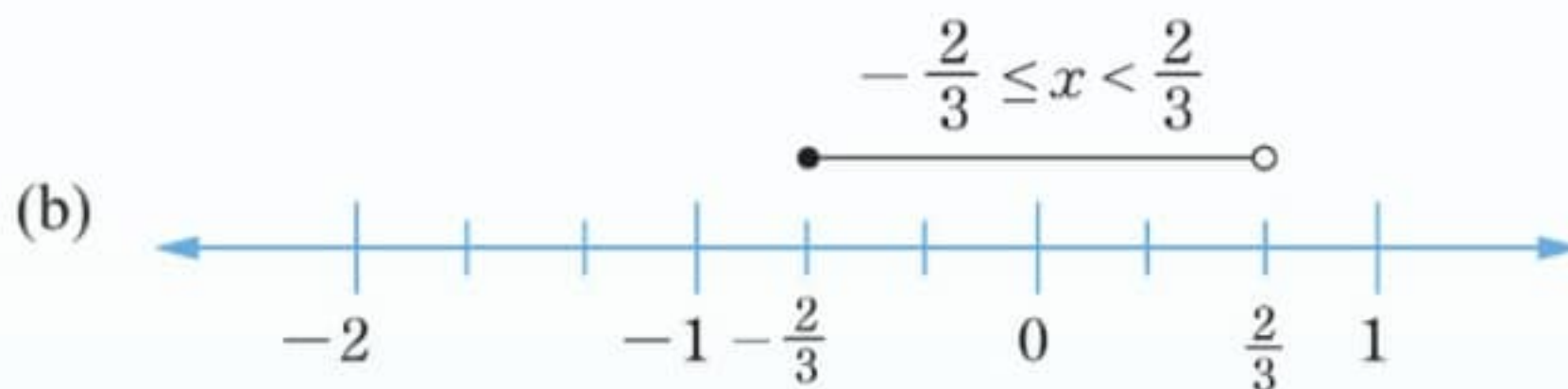
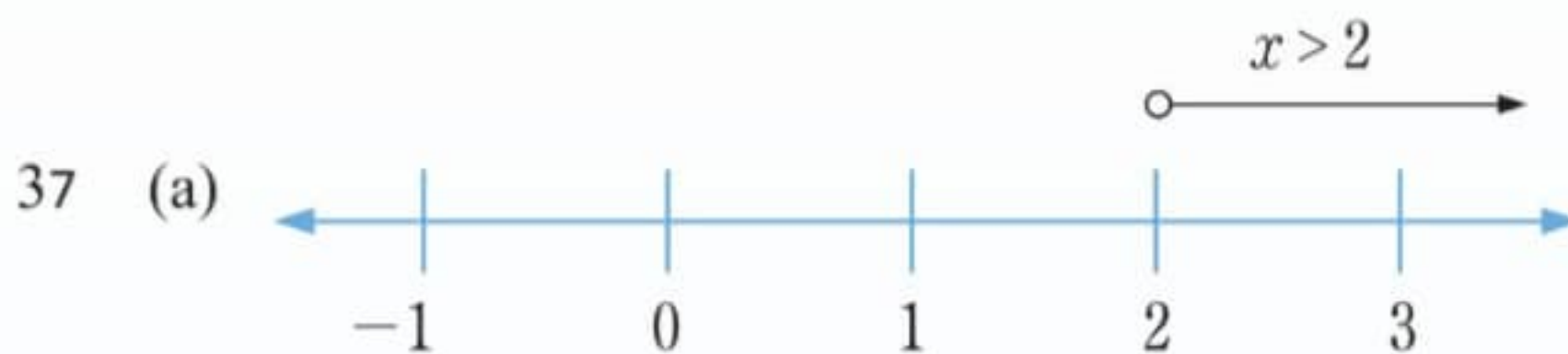
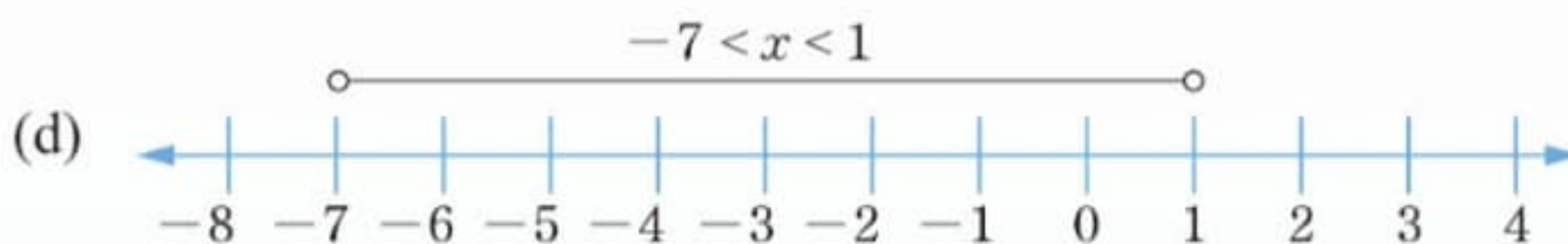
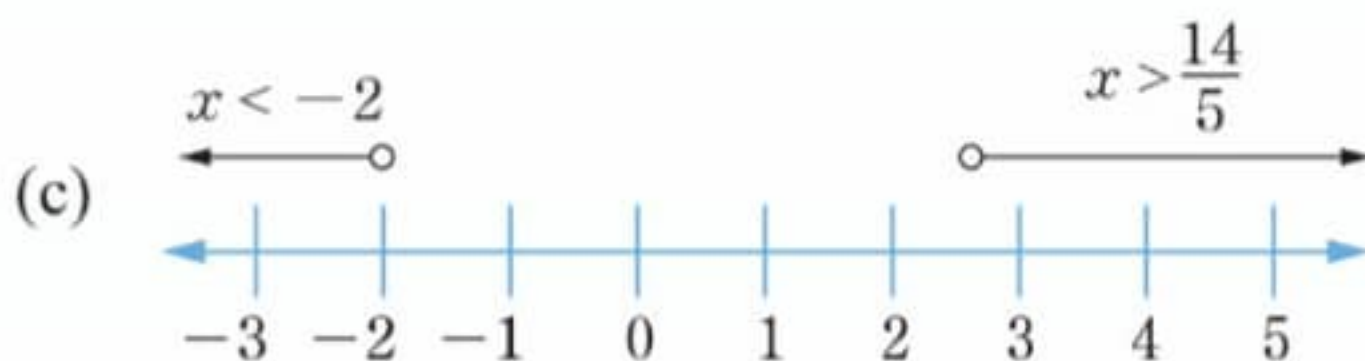
(d) $x = 0, y = 4$

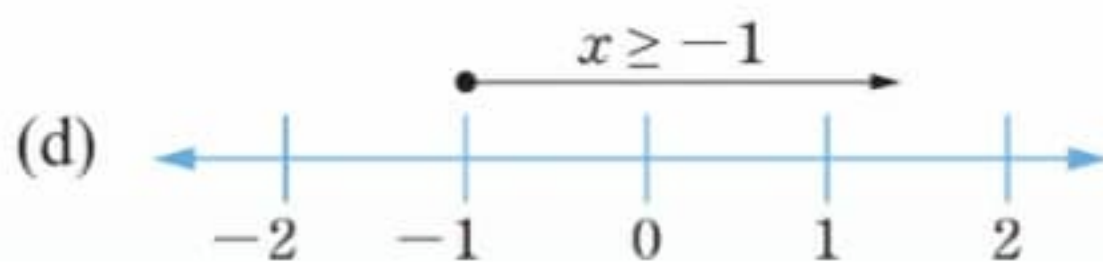
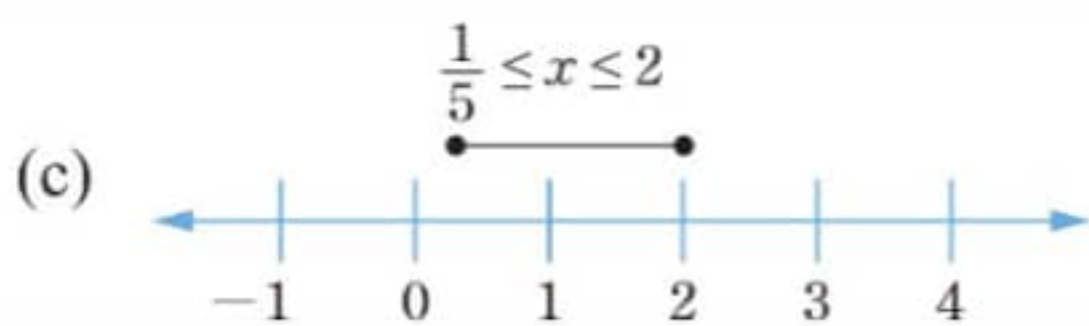
31. (a) $-\frac{22}{7} < x < 4$

(c) $\frac{10}{3} \leq p \leq \frac{14}{3}$

(b) $x \leq -\frac{7}{5}$ or $x \geq -\frac{3}{5}$

33. 20 cm



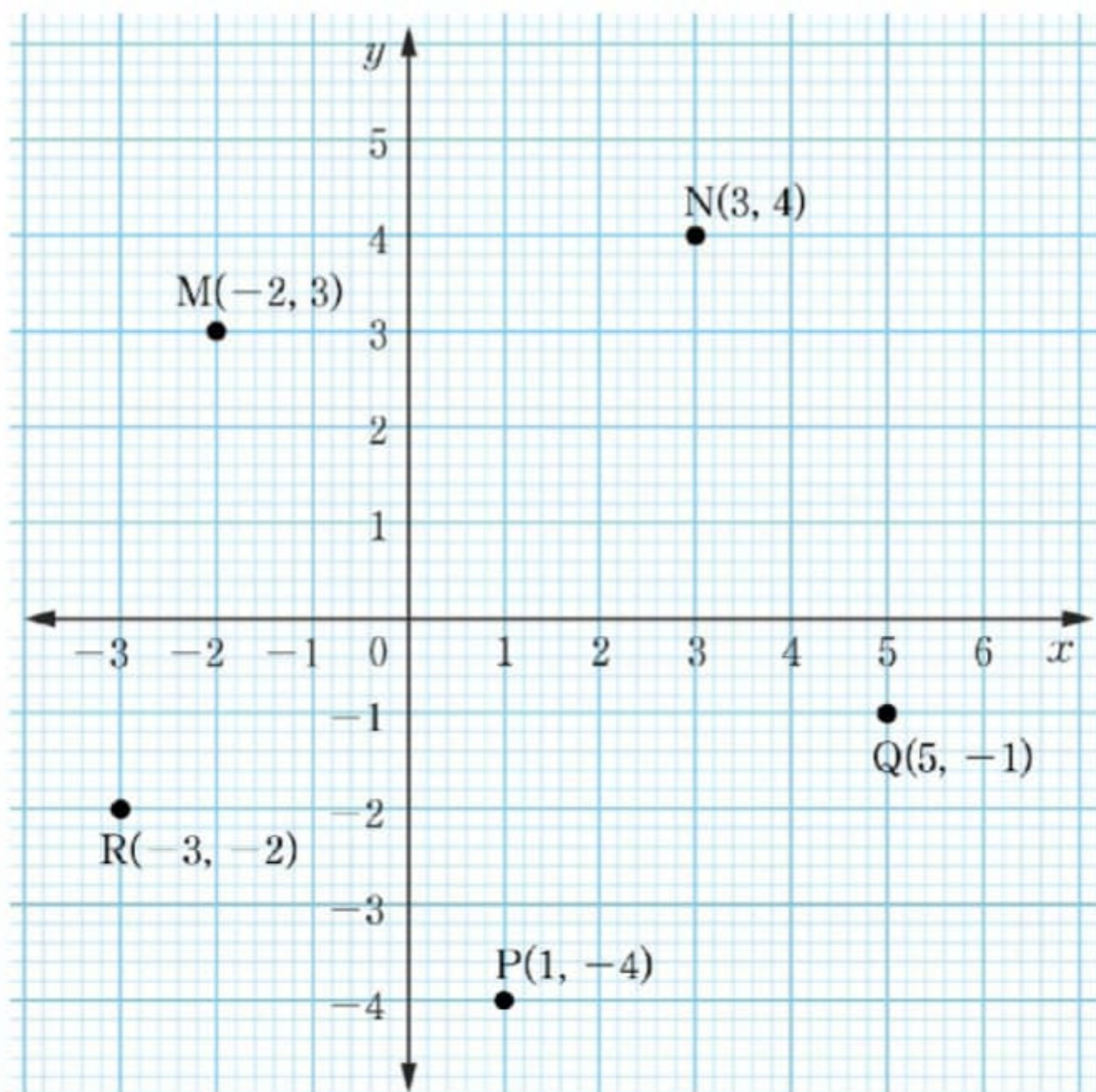


39. (a) $x = -1, y = 3$ (c) $x = 1, y = 0$
 (b) $x = 2, y = -1$ (d) $x = 0, y = 2$

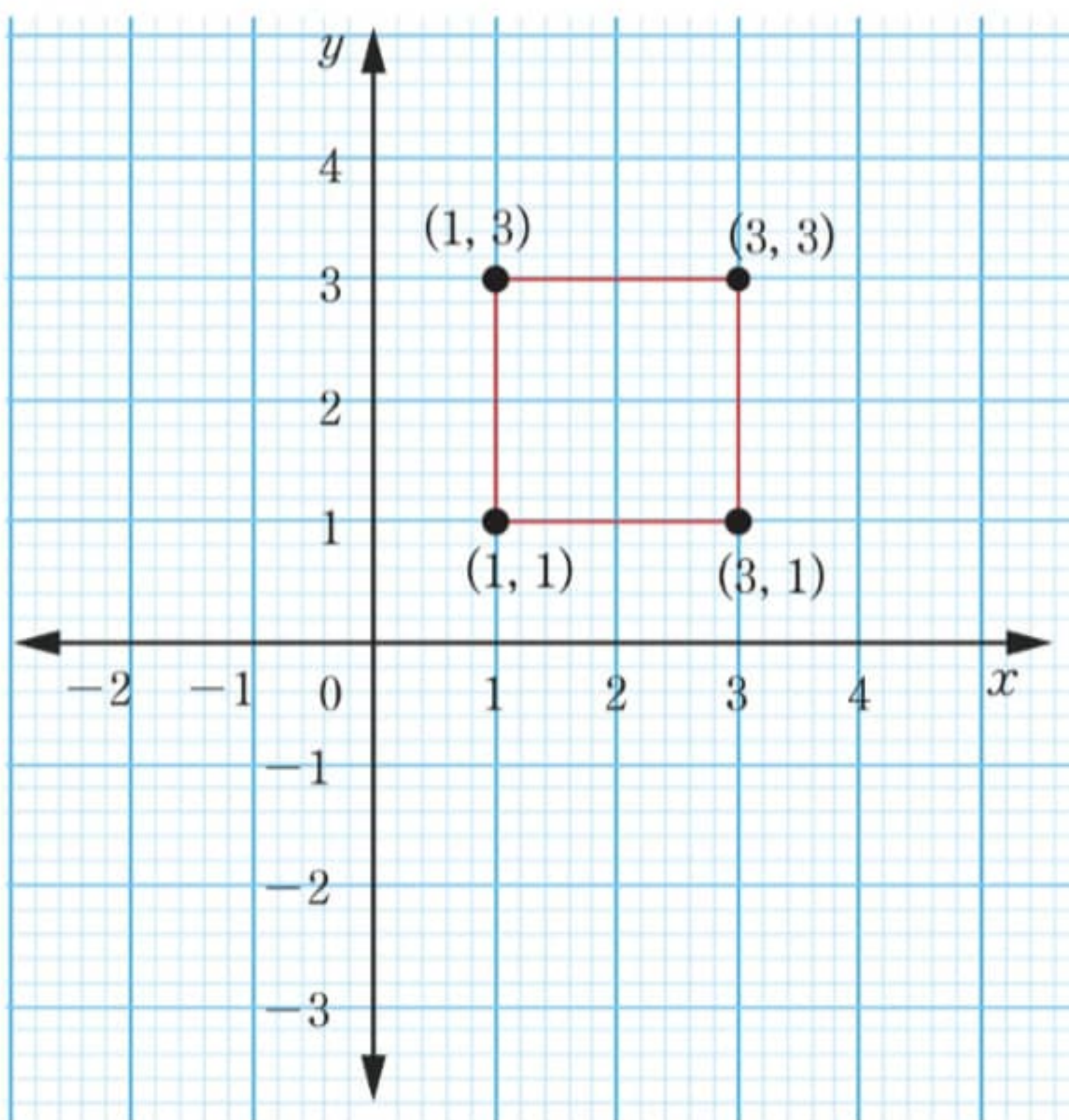
Chapter Six

Exercise 6.1

1.

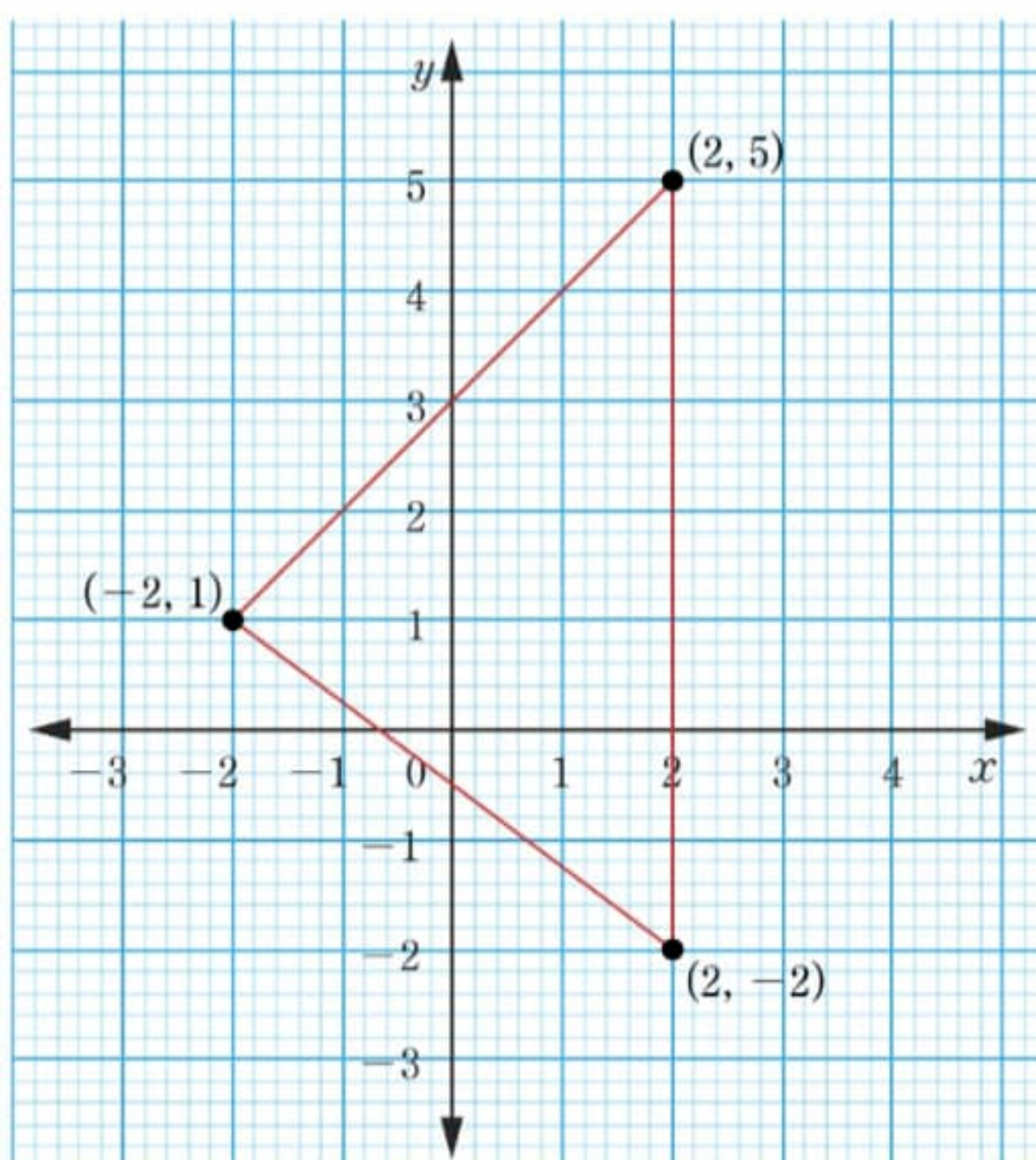


3. (a)



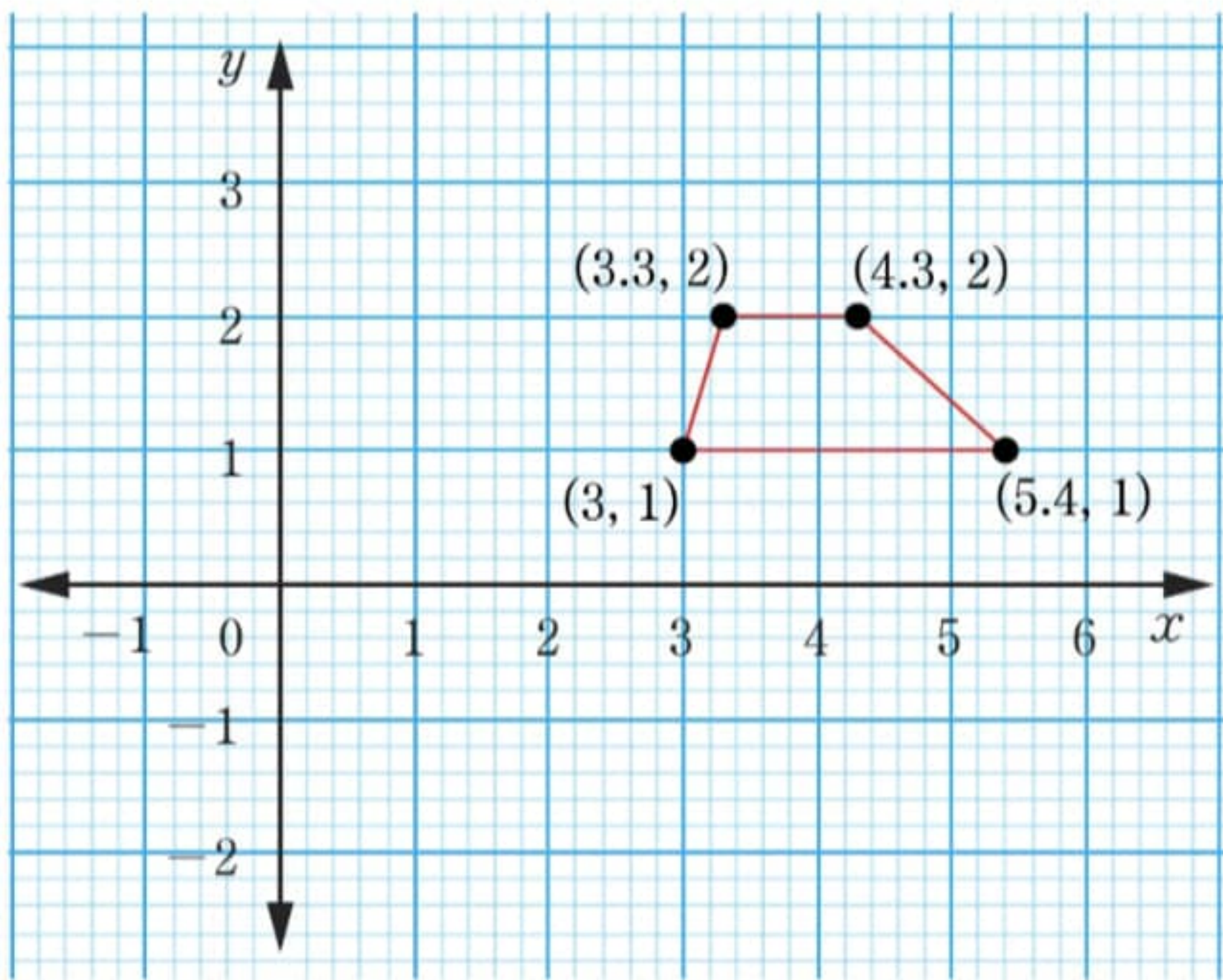
Square

(b)



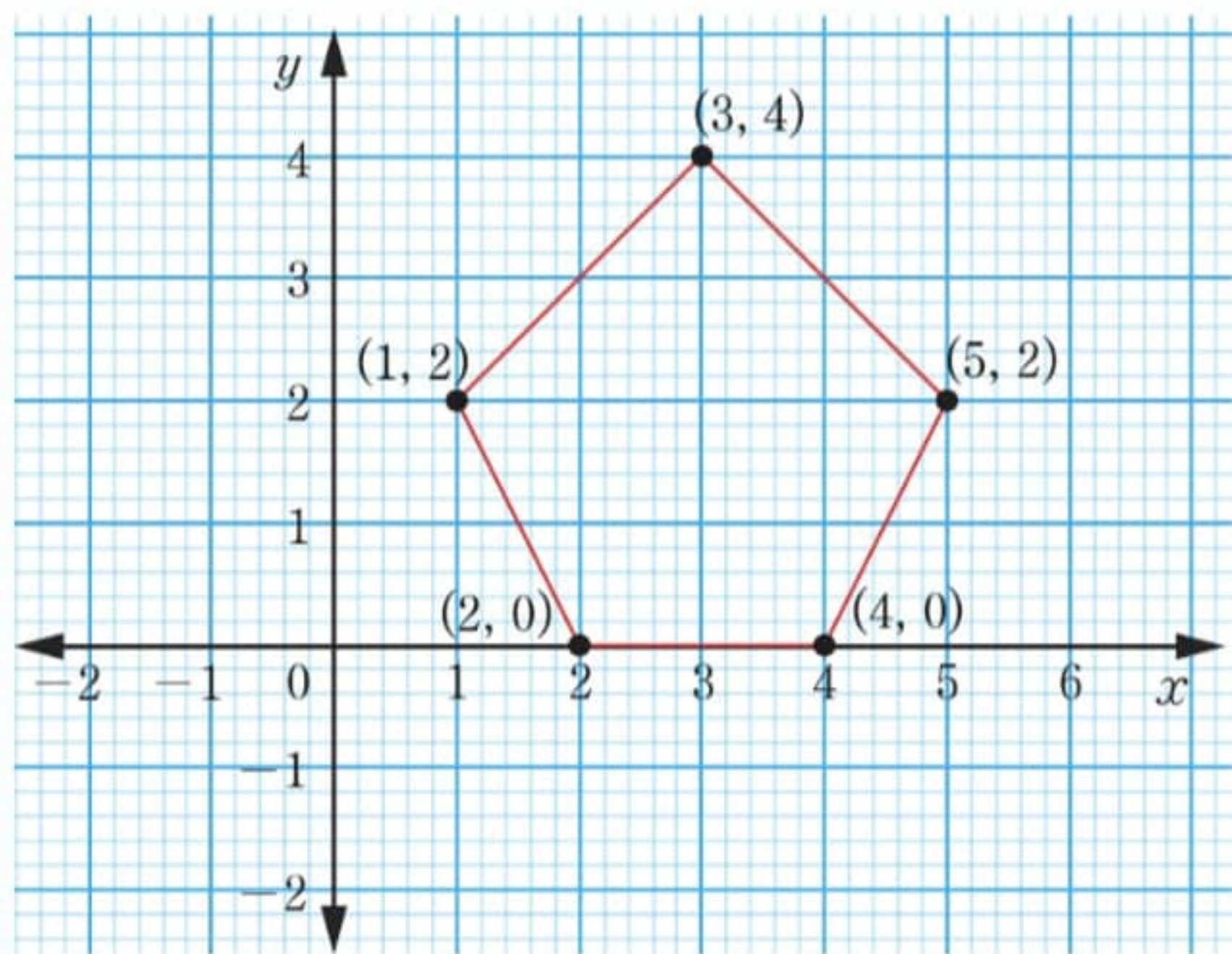
Triangle

(c)



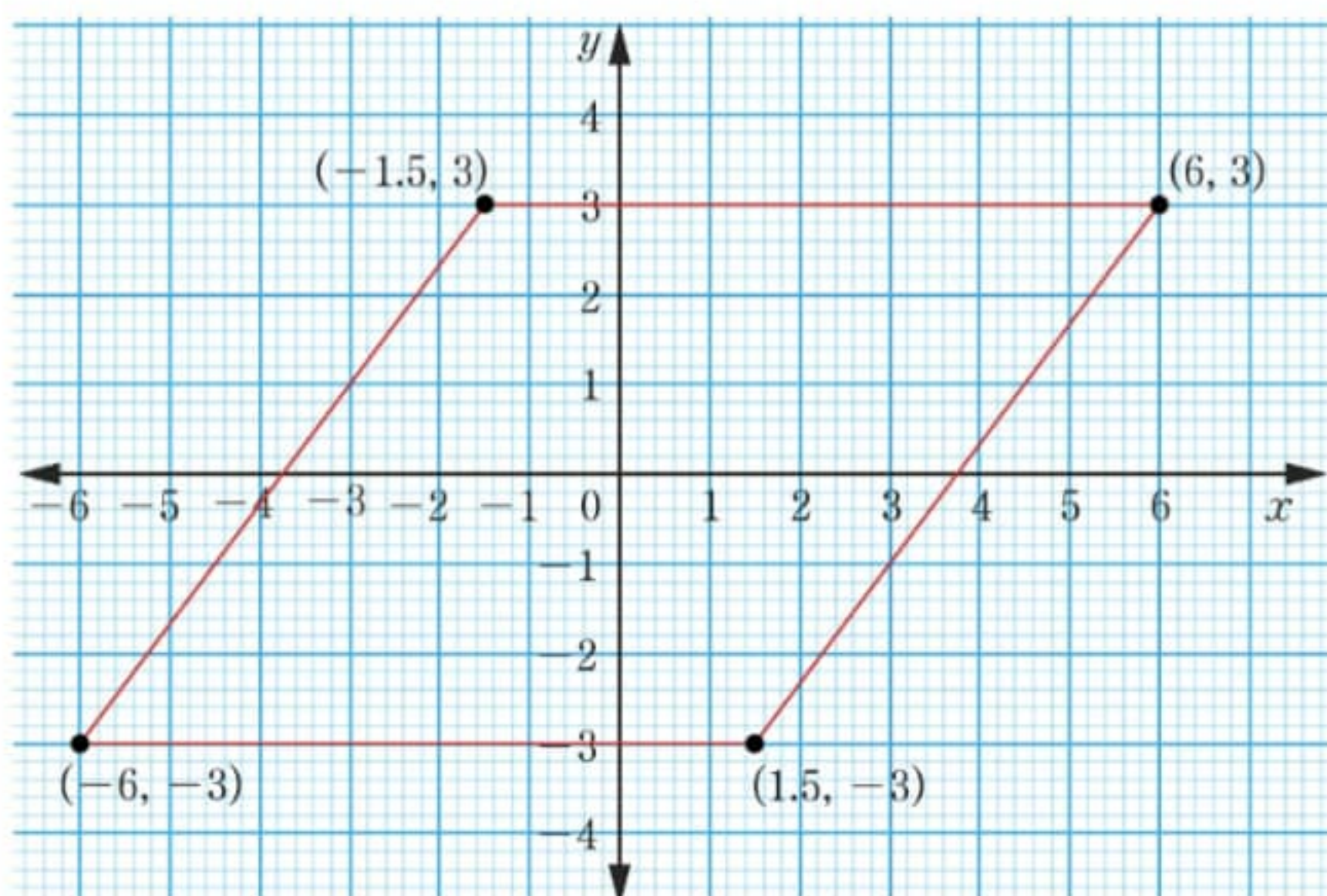
Trapezium

(d)



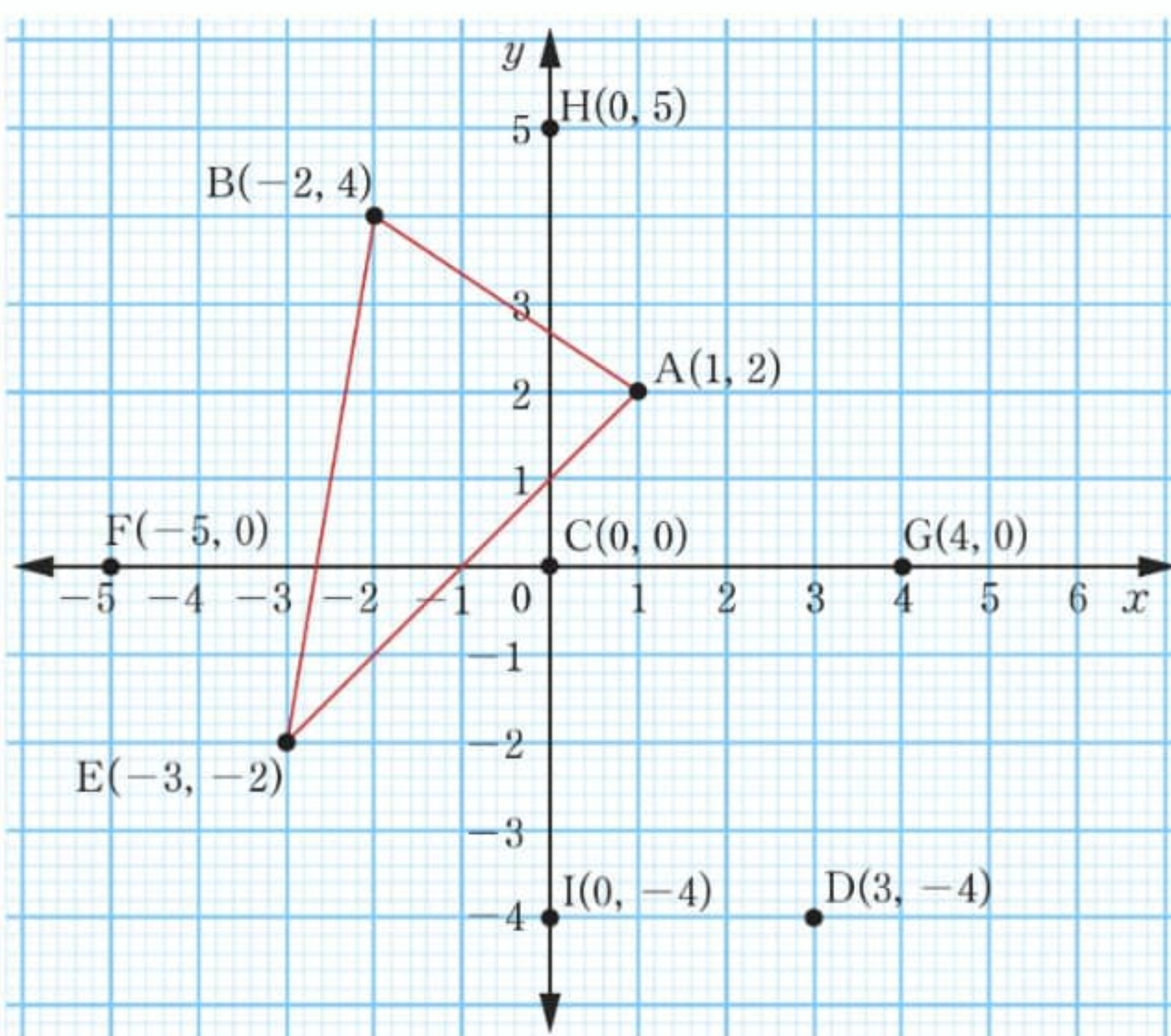
Pentagon

(e)

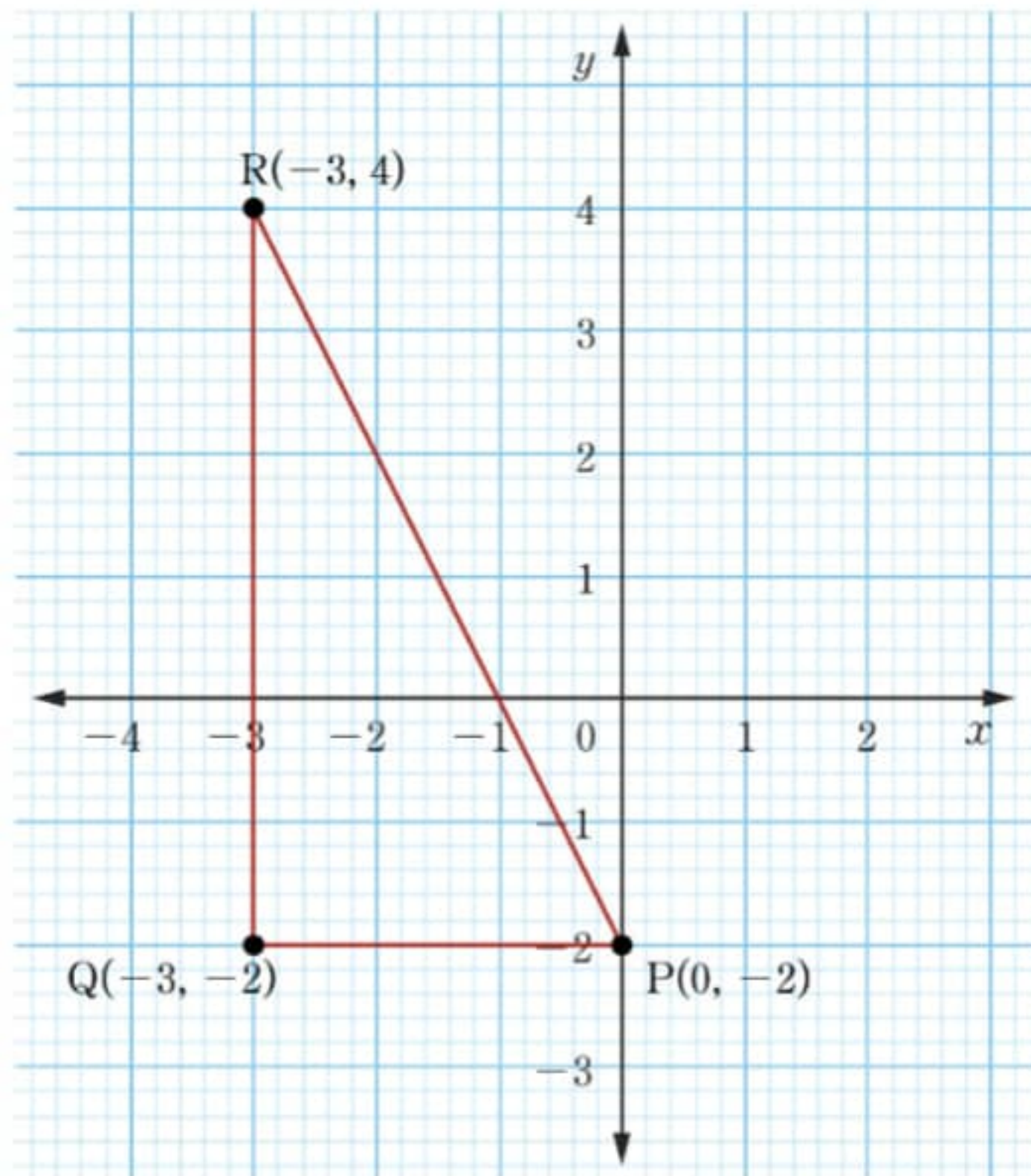


Parallelogram

5. (a)

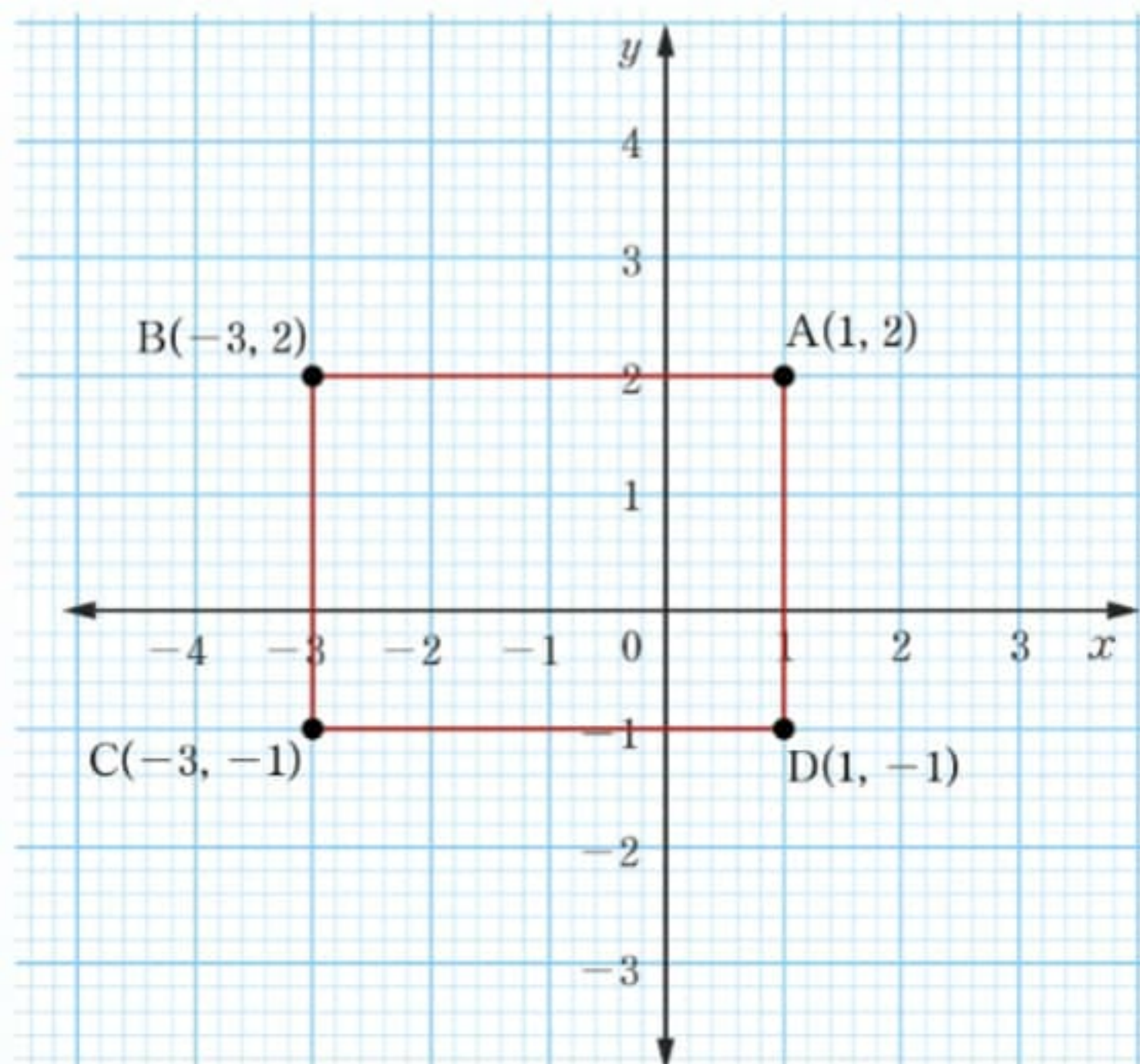


7. (a)



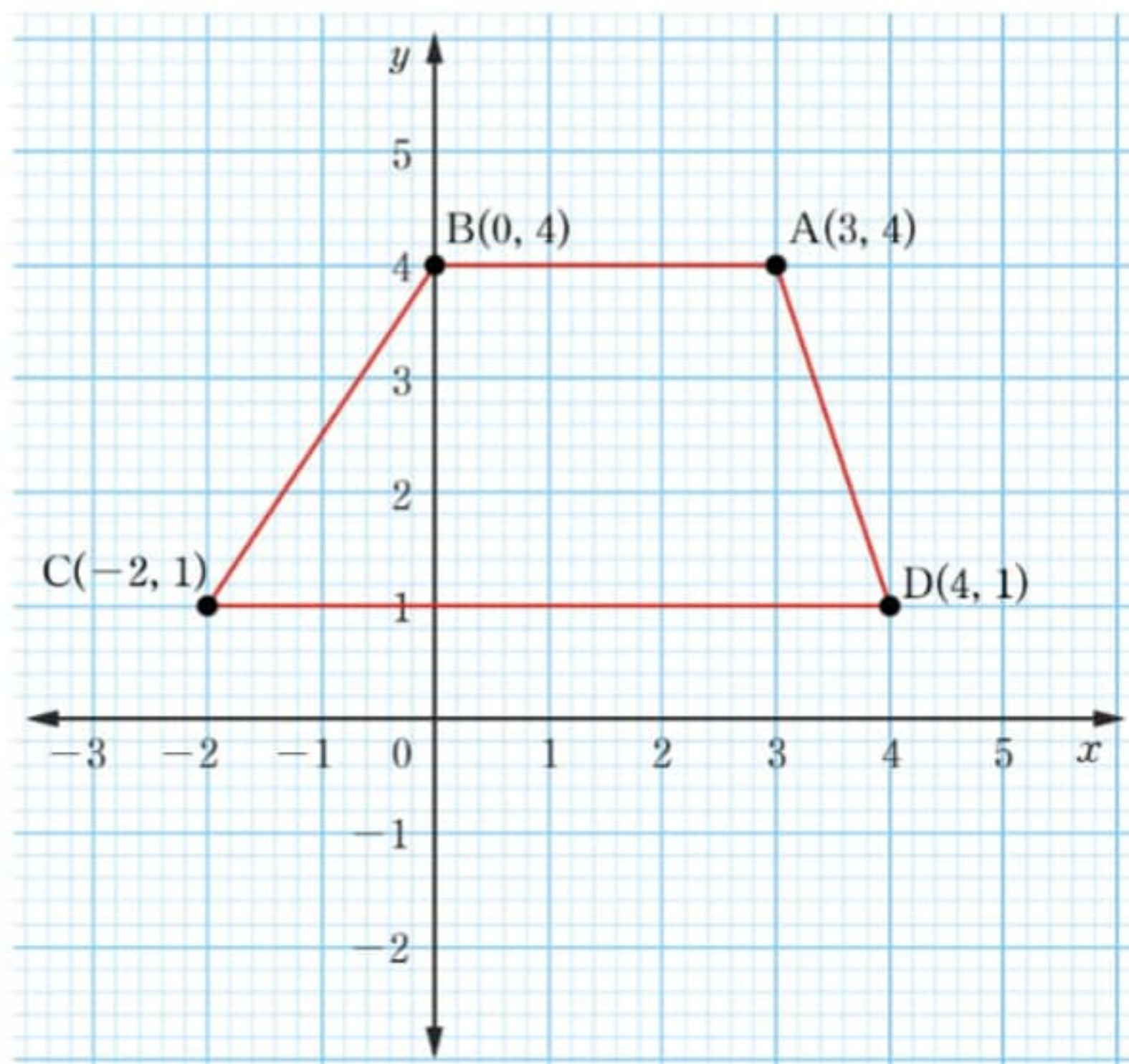
Triangle

(b)



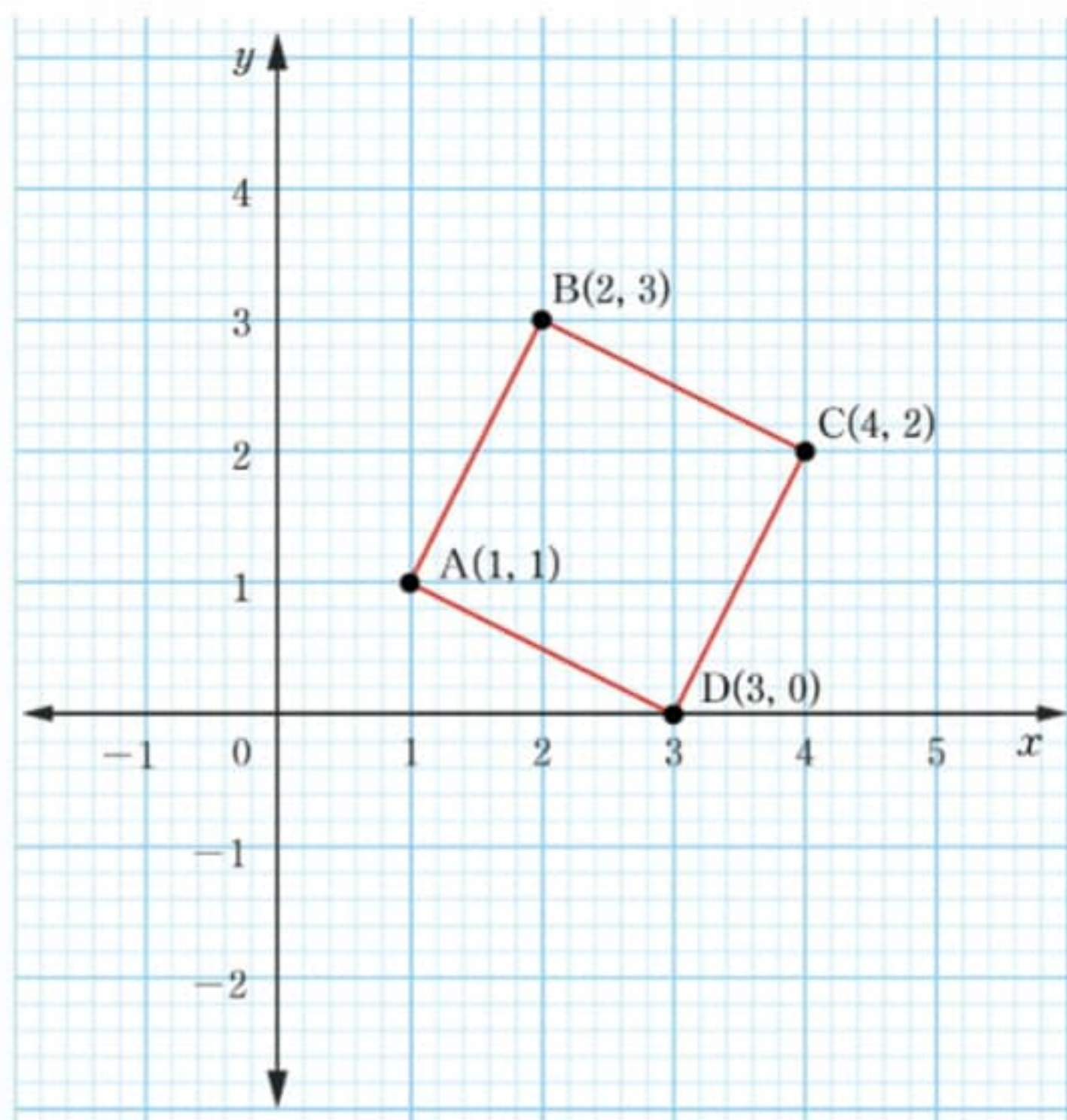
Rectangle

(c)



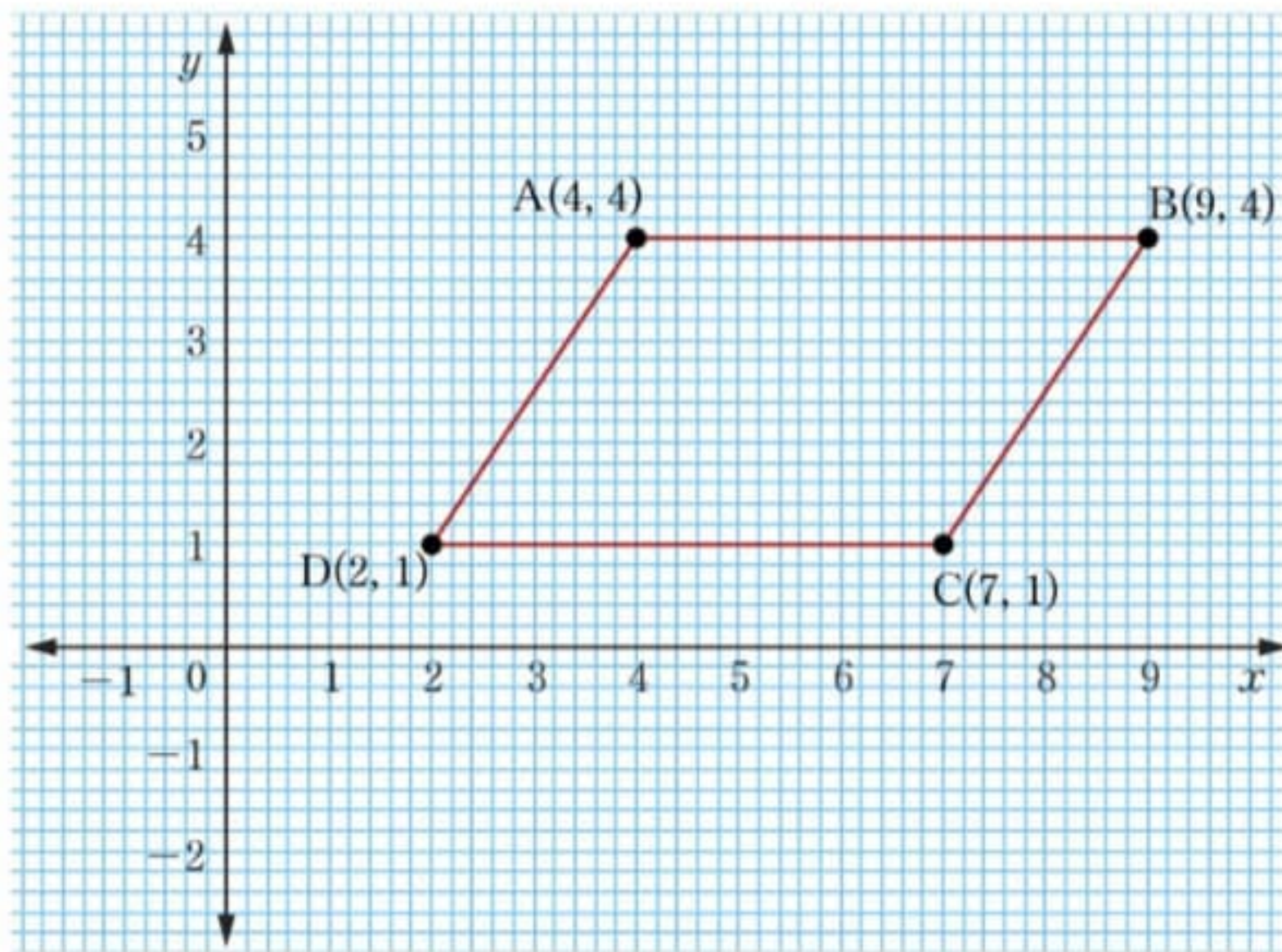
Trapezium.

9. (a)



D(3, 0)

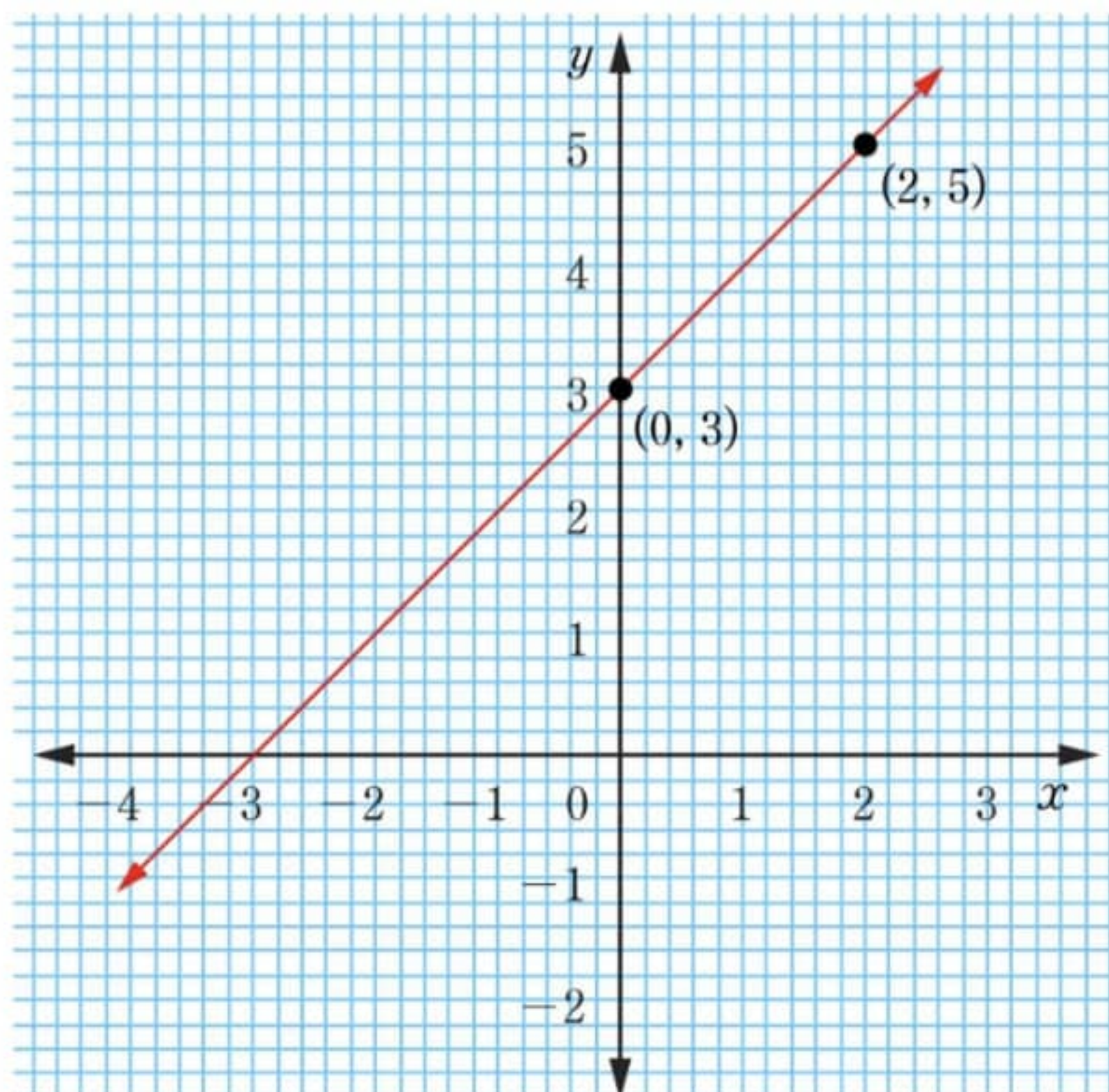
(b)



C (7, 1)

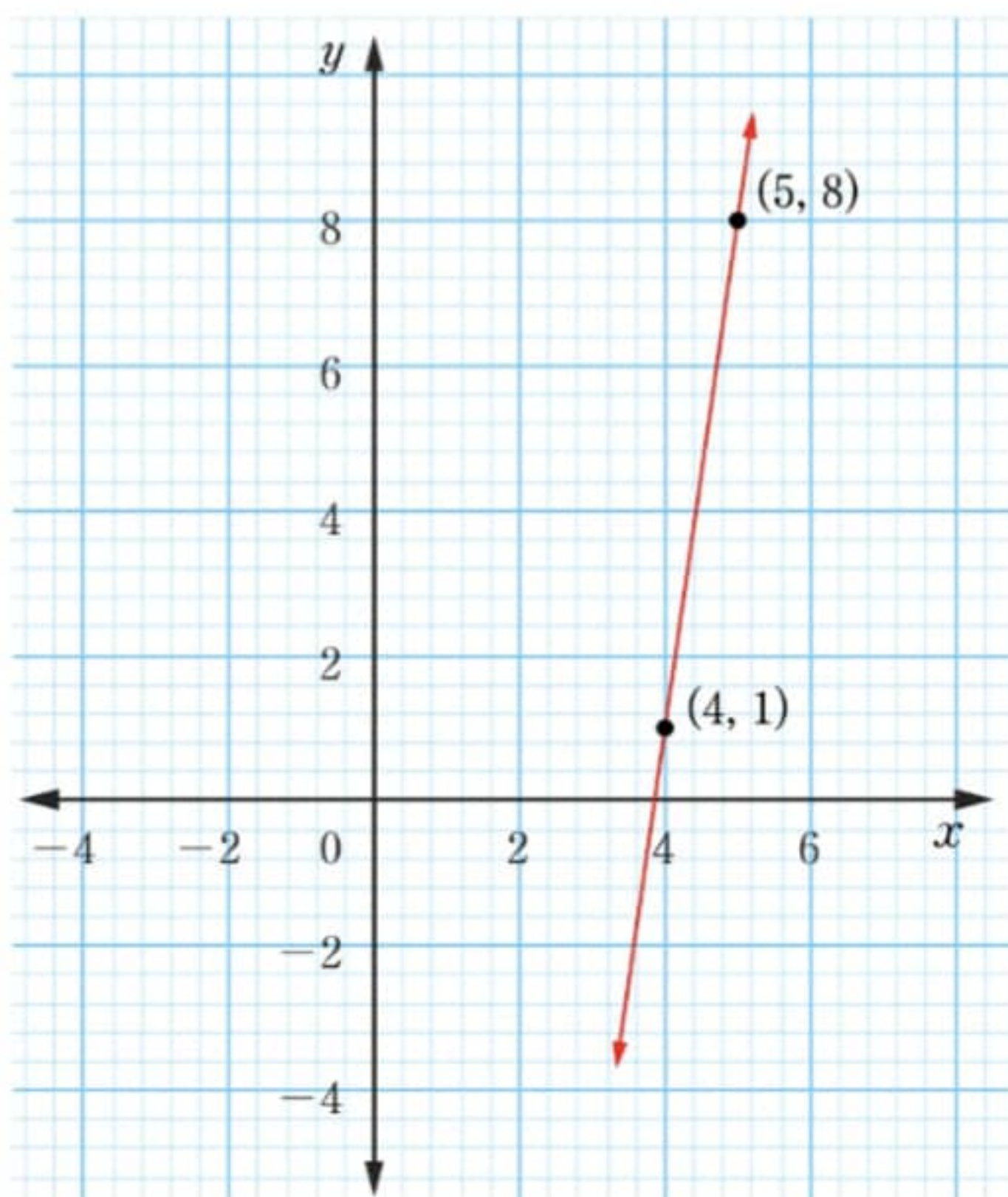
Exercise 6.2

1. (a)



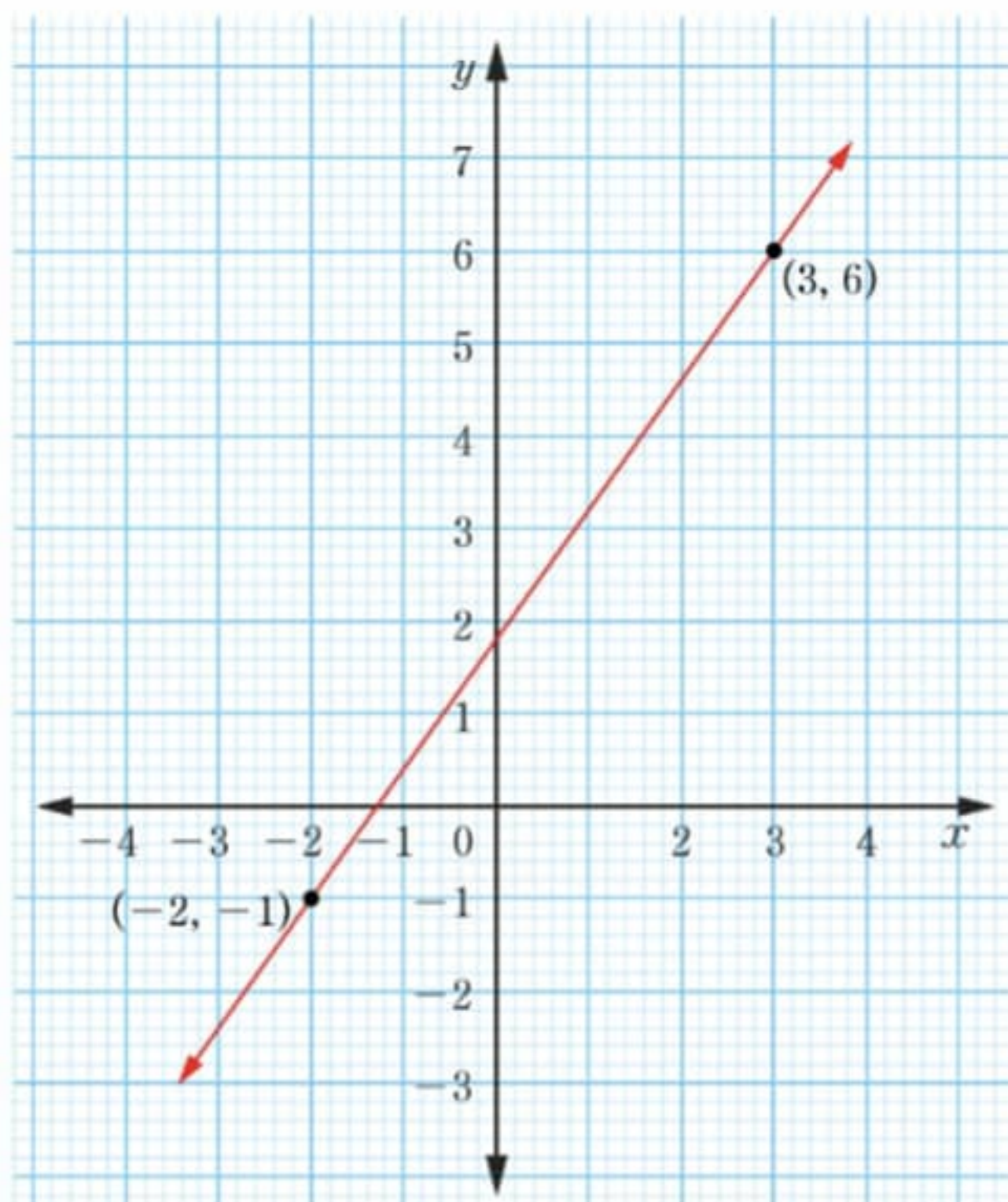
Gradient is positive

(b)



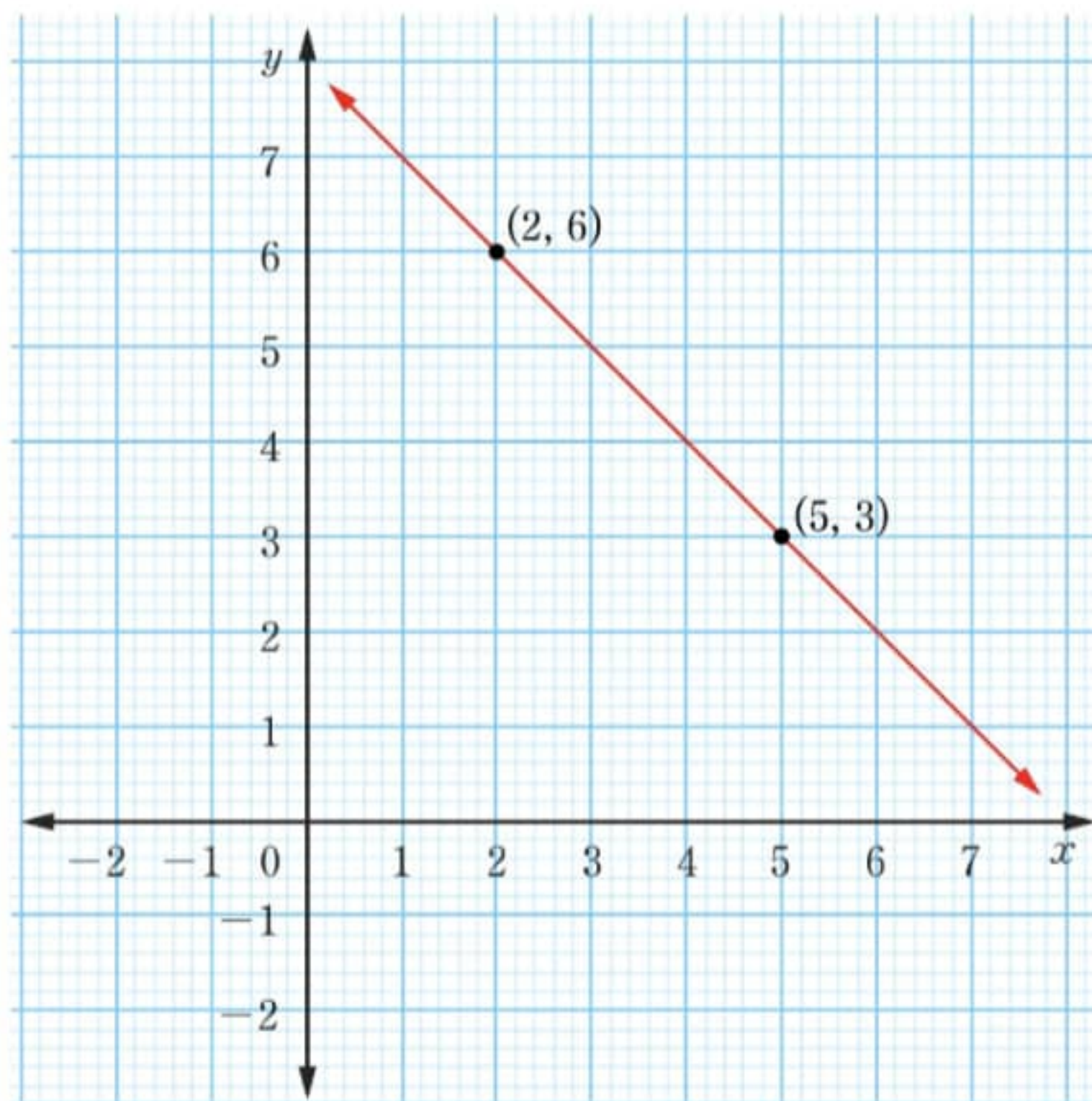
Gradient is positive

(c)



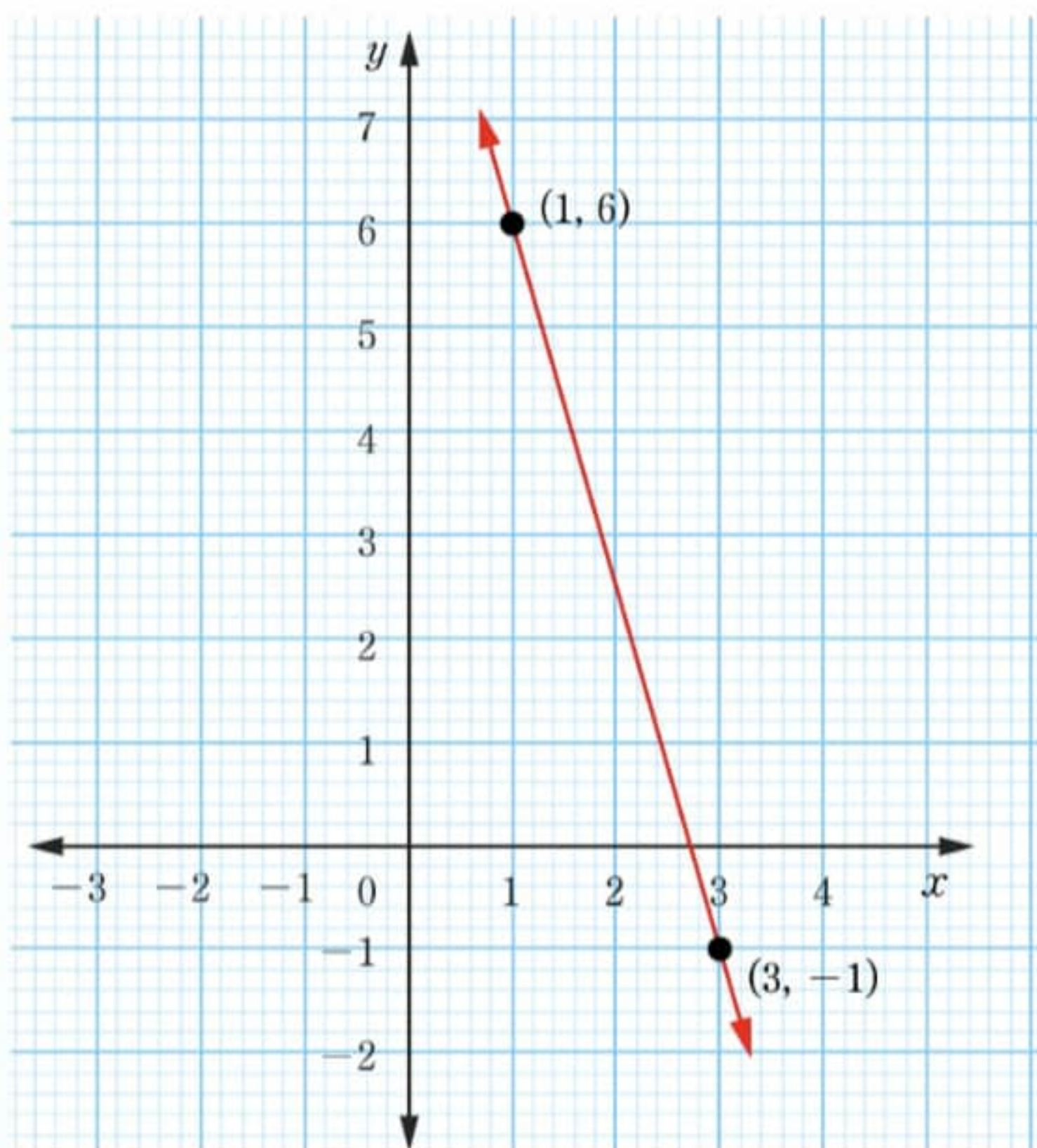
Gradient is positive

(d)



Gradient is negative

(e)



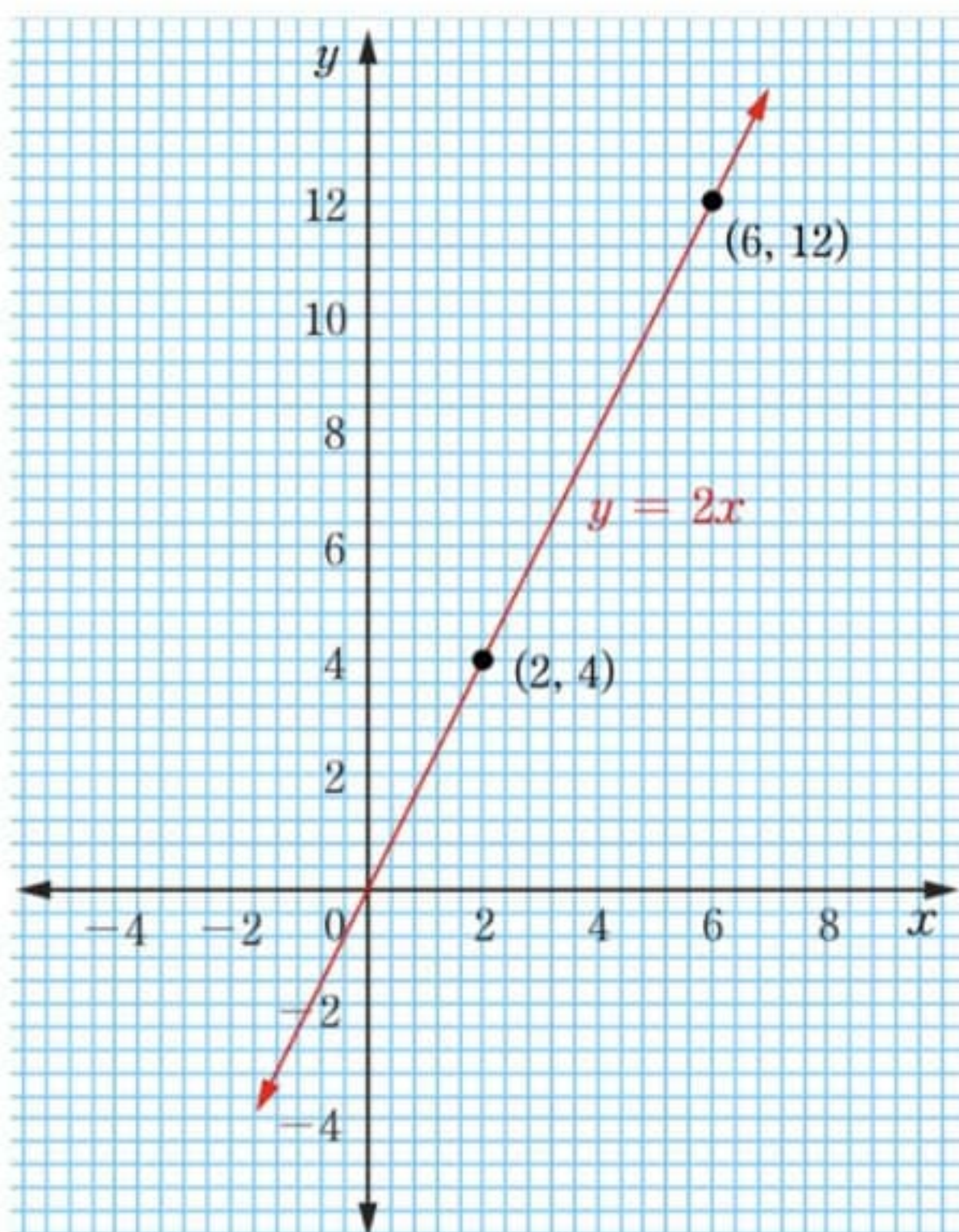
Gradient is negative

3. (a) $P(3, -2)$, $Q(0, 4)$, $R(-4.5, -2)$
 (b) Gradient of line PQ is -2 , line PR is 0 , and line QR is $\frac{4}{3}$.
 (c) Isosceles triangle.
5. $p = 1$ 7. (a) $k = 2$ (b) $k = 3$
9. $y = -2$ 11. $x = \frac{23}{3}$

Exercise 6.3

1. (a) $-4x + 5y = 17$ (d) $x + y = 7$ (g) $3x + 4y = 0$
 (b) $-2x + 5y = 5$ (e) $y = 0$ (h) $-2x + 3y = 20$
 (c) $-2x + 3y = 11$ (f) $2x + 5y = -14$ (i) $3x + 7y = -12$
 (j) $x + y = 0$
3. (a) $y = -10x + 48$, $\left(\frac{24}{5}, 0\right)$ and $(0, 48)$
 (b) $y = \frac{5}{4}x + \frac{7}{4}$, $\left(-\frac{7}{5}, 0\right)$ and $\left(0, \frac{7}{4}\right)$
 (c) $y = \frac{3}{4}x + \frac{21}{4}$, $(-7, 0)$ and $\left(0, \frac{21}{4}\right)$
 (d) $y = -x + 5$, $(5, 0)$ and $(0, 5)$
 (e) $y = -\frac{2}{3}x + \frac{7}{3}$, $\left(\frac{7}{2}, 0\right)$ and $\left(0, \frac{7}{3}\right)$
 (f) $y = -\frac{21}{2}x - \frac{77}{4}$, $\left(\frac{11}{6}, 0\right)$ and $\left(0, -\frac{77}{4}\right)$
5. $k = \frac{10}{3}$

7. (a)



- (b) (i) Gradient is 2 which signifies that 2 acres farm uses 1 bag of fertilizer
 (ii) $y = 2x$ and x and y -intercepts are $(0, 0)$
 (c) (i) Nyanda will need 12 bags of fertilizer
 (ii) The size of Myoma farm is 150 acres

Exercise 6.4

1. (a) $3x + y - 10 = 0$ (c) $y - 4 = 0$ (e) $3x - 5y + 14 = 0$
 (b) $x + y - 5 = 0$ (d) $x - 5y - 19 = 0$
3. (a) $5x - y - 14 = 0$ (c) $11x - 3y - 39 = 0$
 (b) $3x + 5y - 14 = 0$ (d) $x + 5y - 2 = 0$
5. (a) $y = -\frac{7}{4}x + \frac{11}{4}$, $m = -\frac{7}{4}$ and $\left(0, \frac{11}{4}\right)$
 (b) $y = -\frac{14}{3}x + 4$, $m = -\frac{14}{3}$ and $(0, 4)$
 (c) $y = 2x - 5$, $m = 2$ and $(0, -5)$

(d) $y = -\frac{4}{5}x + 8$, $m = -\frac{4}{5}$ and $(0, 8)$

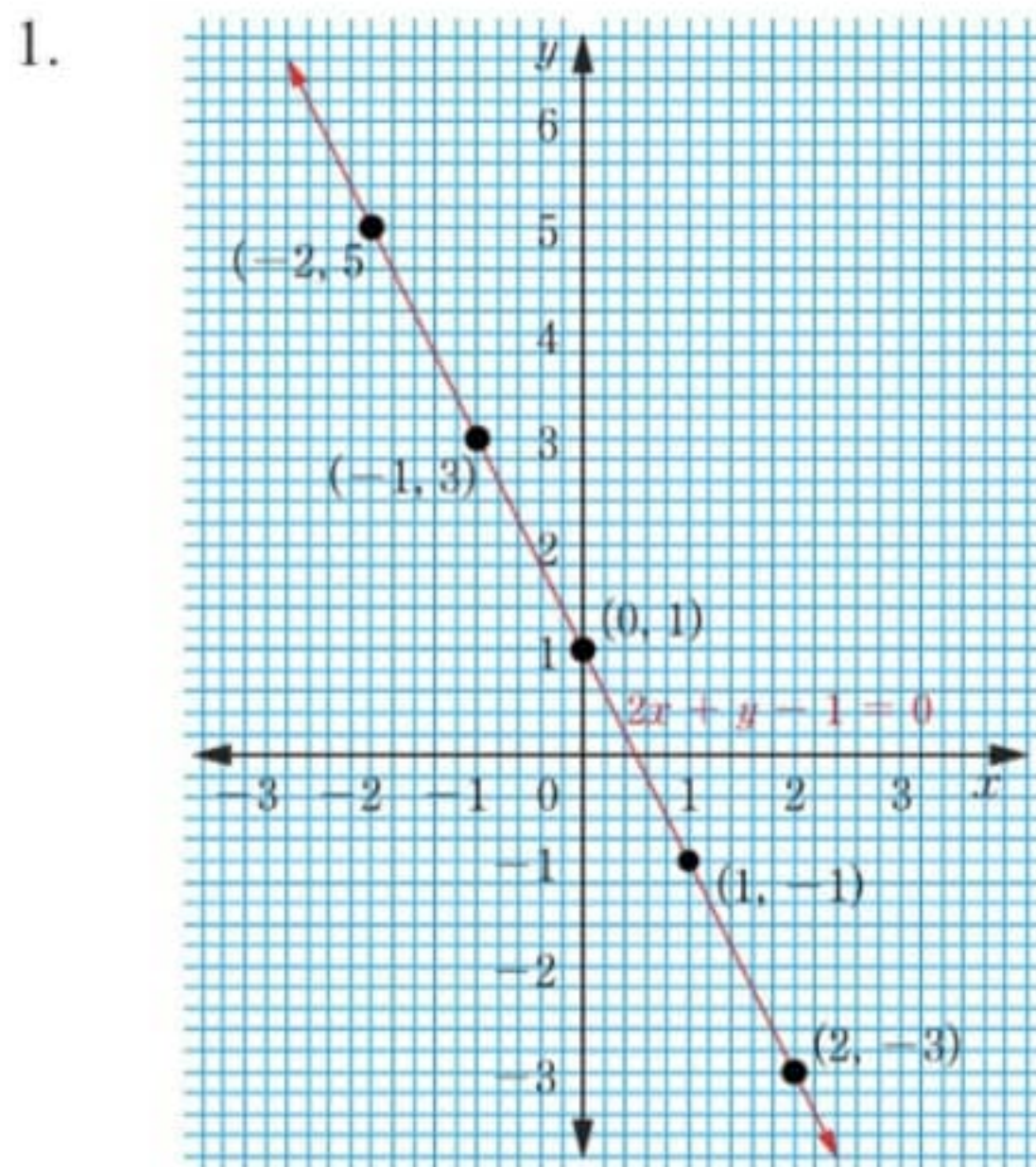
(e) $y = 8x$, $m = 8$ and $(0, 0)$

(f) $y = 3x + \frac{70}{13}$, $m = 3$ and $\left(0, \frac{70}{13}\right)$

Exercise 6.5

1. (a) $y = 2x - 3$ (c) $y = 3x - 6$ (e) $y = -3x$
- (b) $y = -2x + 5$ (d) $y = \frac{3}{2}x - 1$
3. (a) $y = 2x - 1$ (c) $5x + 2y - 8 = 0$ (e) $x + 3y - 12 = 0$
- (b) $3x + y - 19 = 0$ (d) $y = -\frac{6}{5}x + \frac{22}{5}$
5. $2x - 3y + 12 = 0$
6. (a) $y = 5$ (b) $x = 2$ (c) $4x + 3y - 12 = 0$

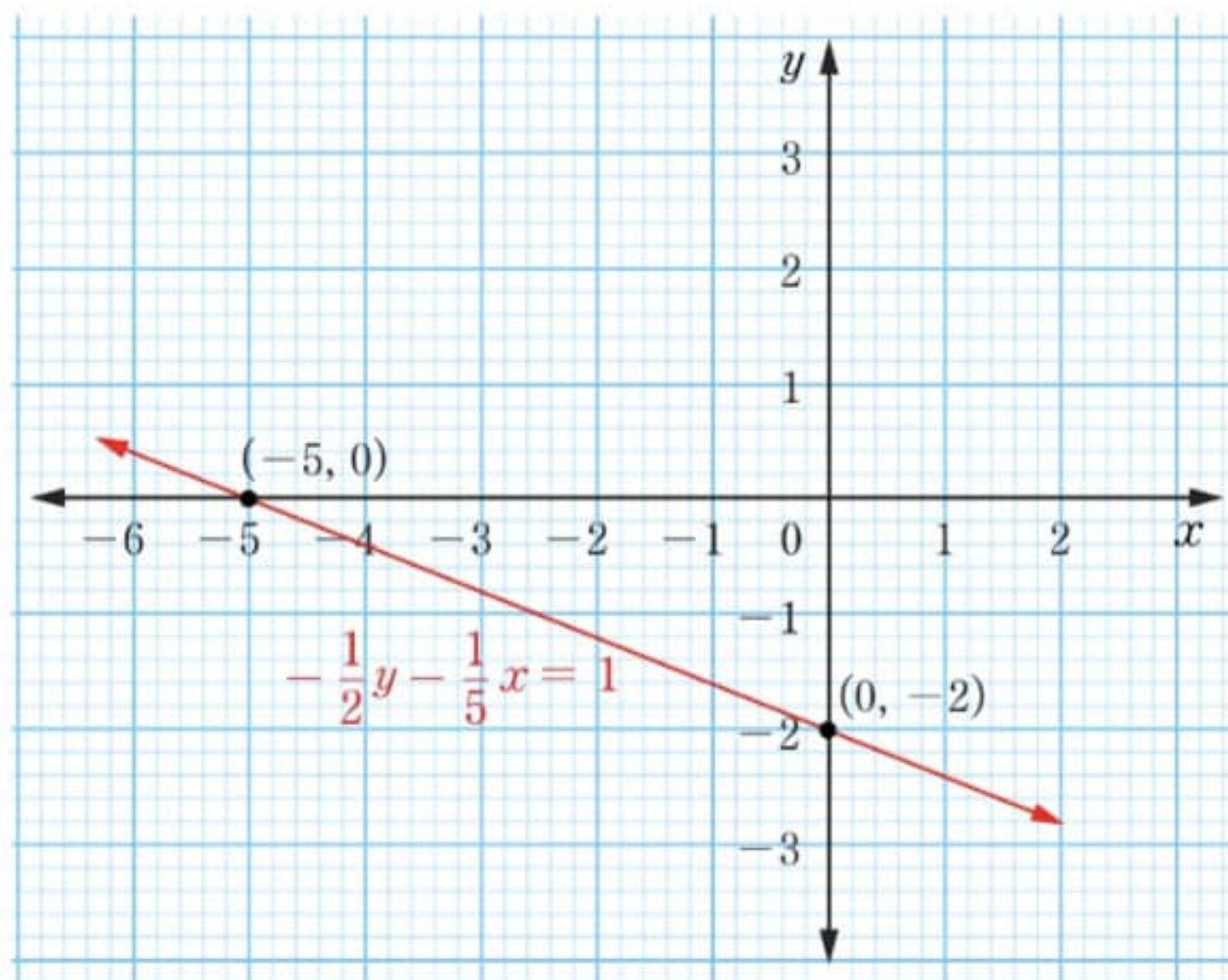
Exercise 6.6



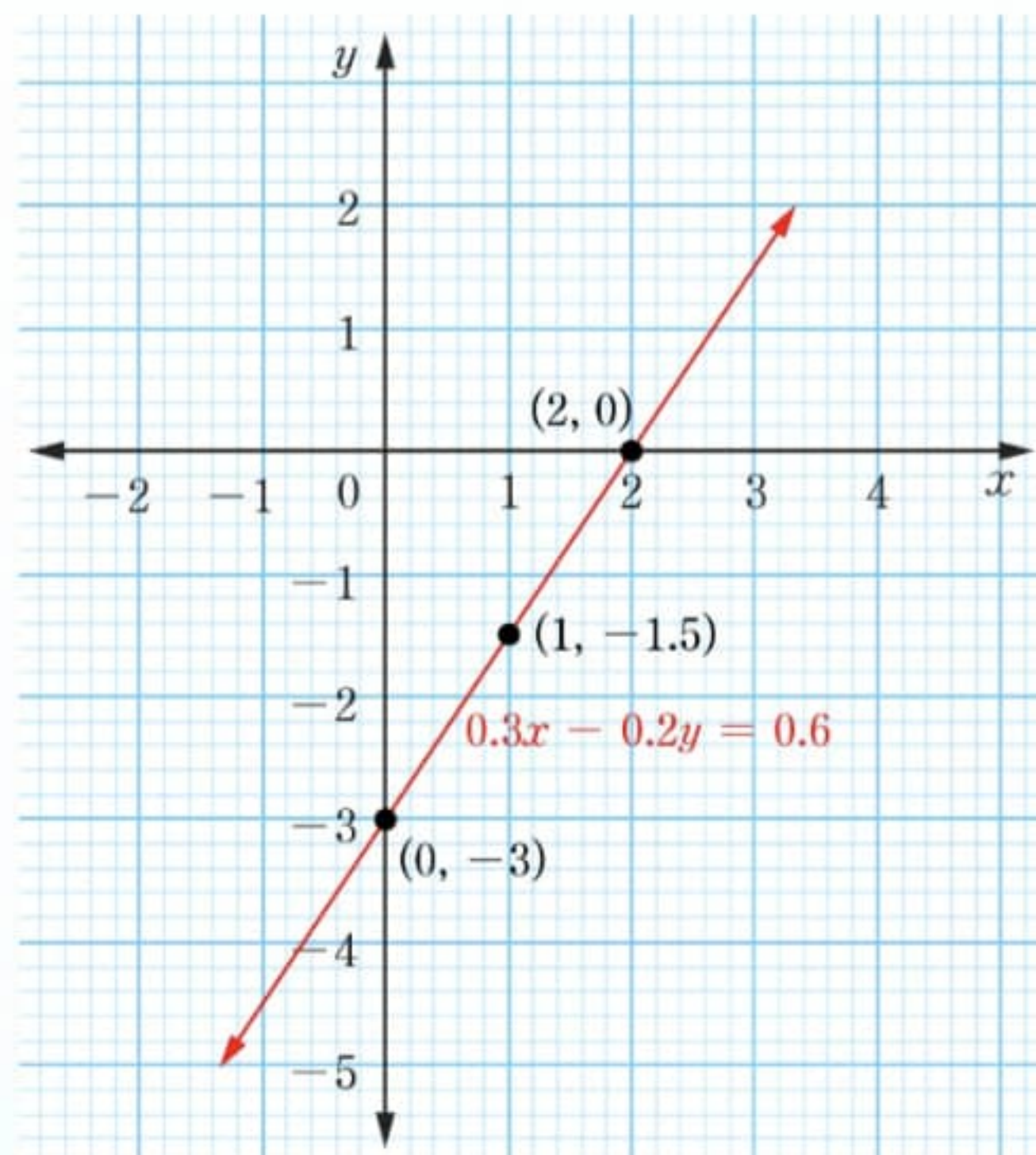
y-intercept is $(0, 1)$, x-intercept is $(0.5, 0)$

3. (a) The gradient of the line through the points A and B is $m = 0$ and the gradient of the line through B and C is undefined.
- (b) The equation through the points A and B is $y = 3$ and through points B and C is $x = 4$
- (c) $y = -\frac{3}{4}x + 3$, Slope $= -\frac{3}{4}$, y-intercept is $(0, 3)$, x-intercept is $(4, 0)$.

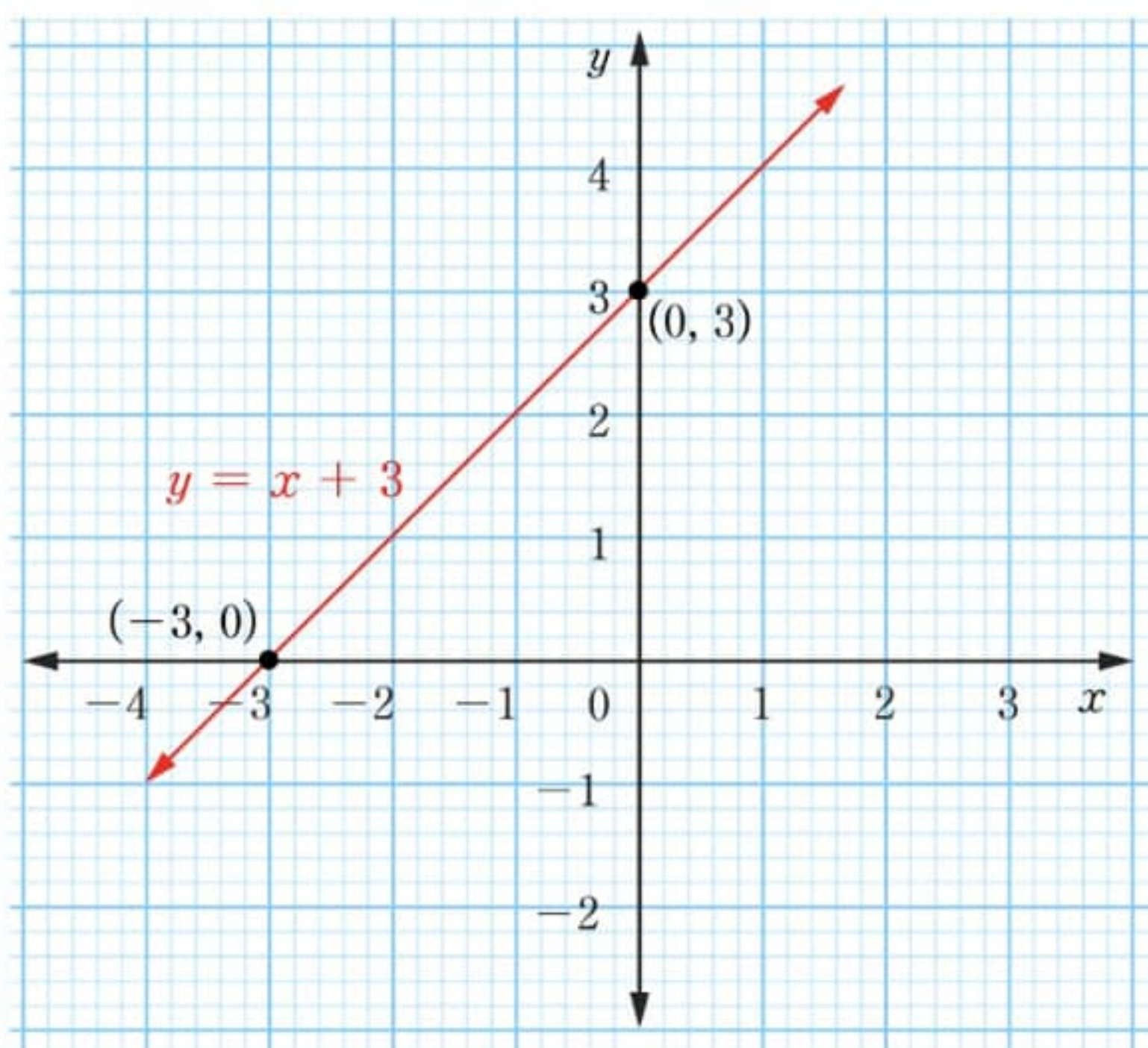
5.



7.



9.

**Exercise 6.7**

1. $x = 1, y = 2$
3. $x = -2, y = 3$
5. $x = 3, y = 1$
7. $x = 3, y = -1$
9. $x = 2, y = 5$
11. $p = 1, q = 1$
13. An orange costs Tshs 100 and mango costs Tshs 400
15. Mother is 40 years and daughter 12 years.

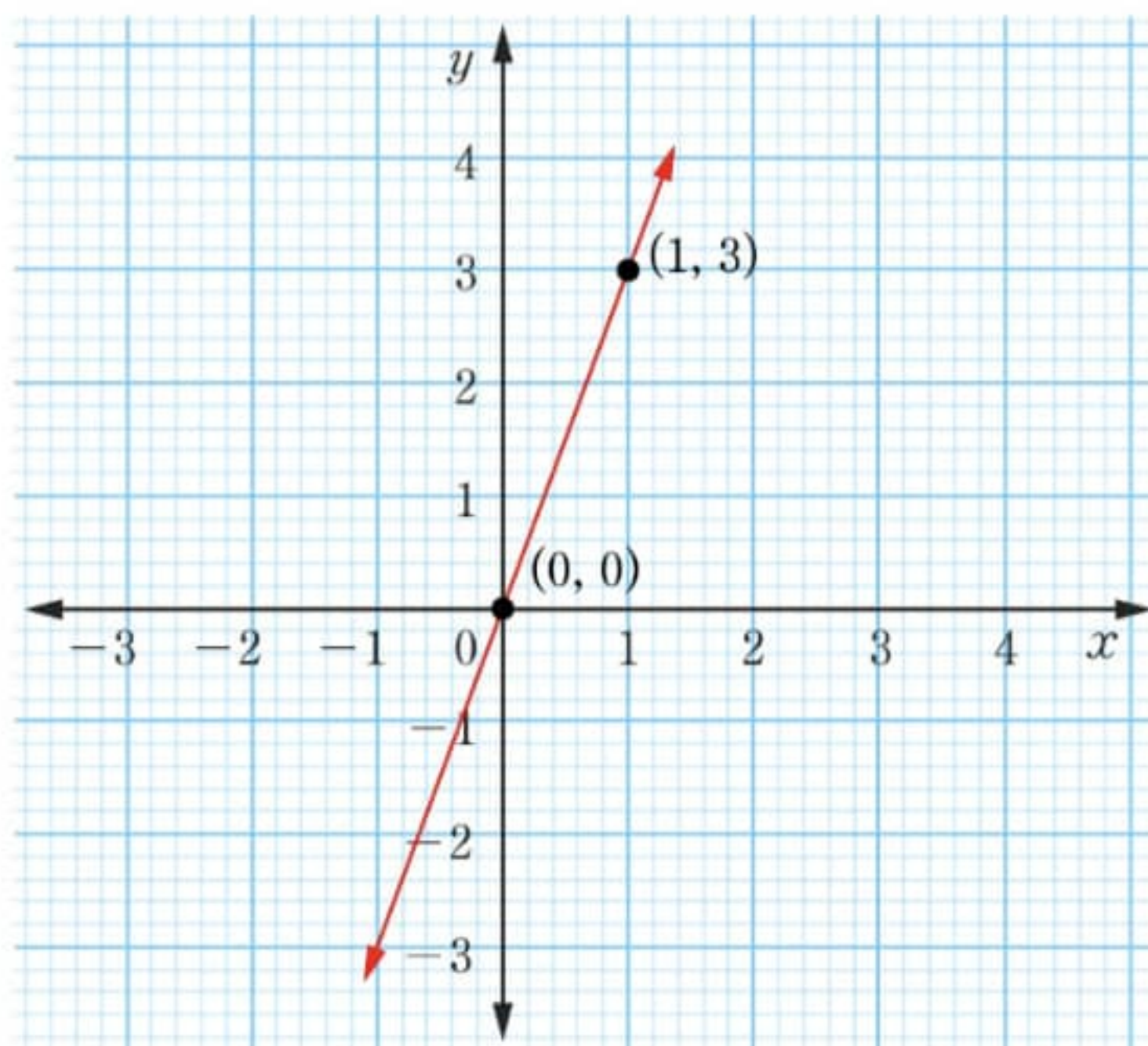
Revision exercise 6

1. (a) $2x - 7y = -20$, $y\text{-intercept} = \left(0, \frac{20}{7}\right)$
- (b) $5x - 9y = -53$, $y\text{-intercept} = \left(0, \frac{53}{9}\right)$
- (c) $66x + 186y = 427$, $y\text{-intercept} = \left(0, \frac{427}{186}\right)$
- (d) $y = \frac{x}{2} - 1$, $y\text{-intercept} = (0, -1)$

(e) $y = 6$, y -intercept = $(0, 6)$

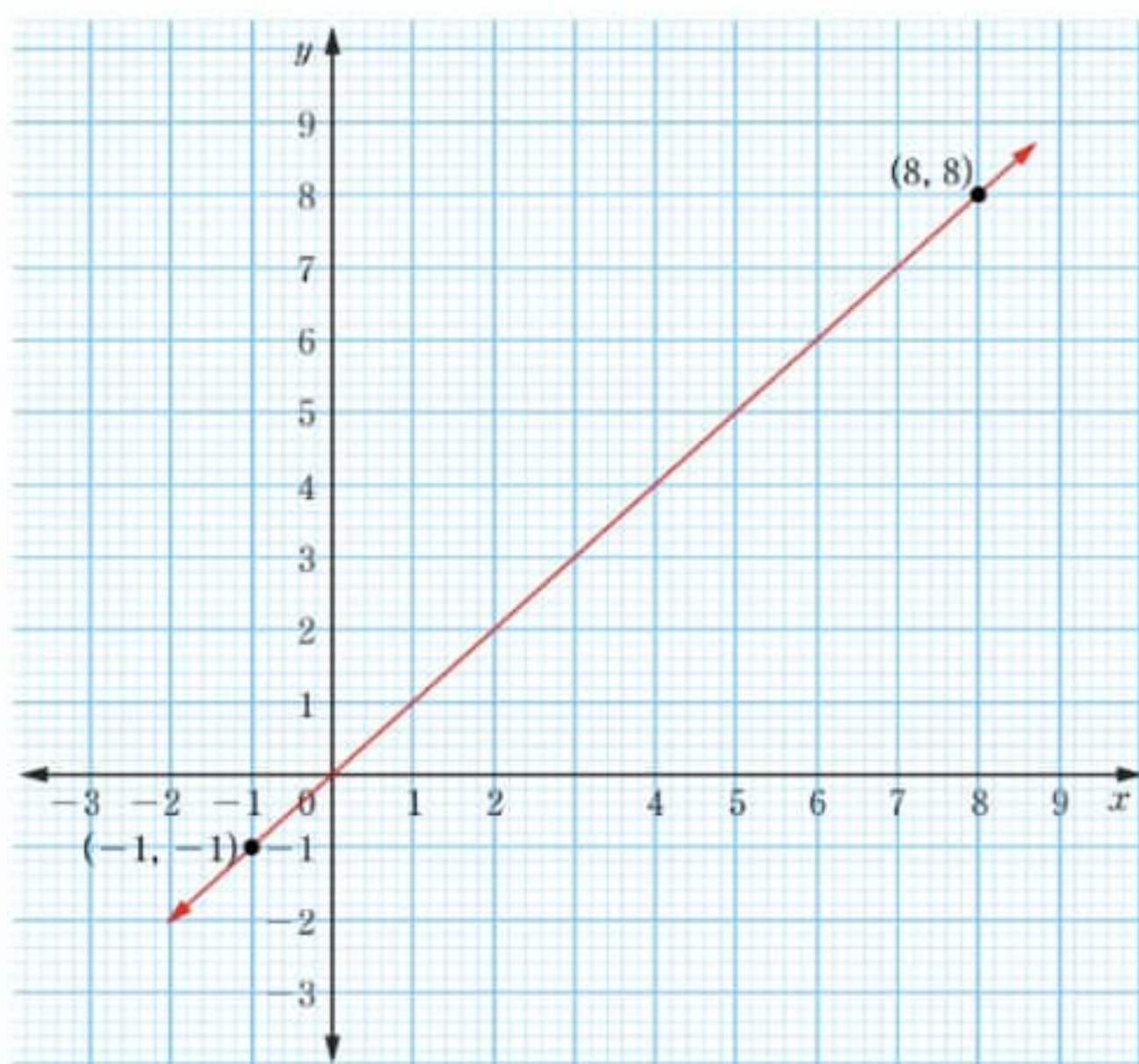
(f) $x - y = 386$, y -intercept = $(0, -386)$

3. (a)



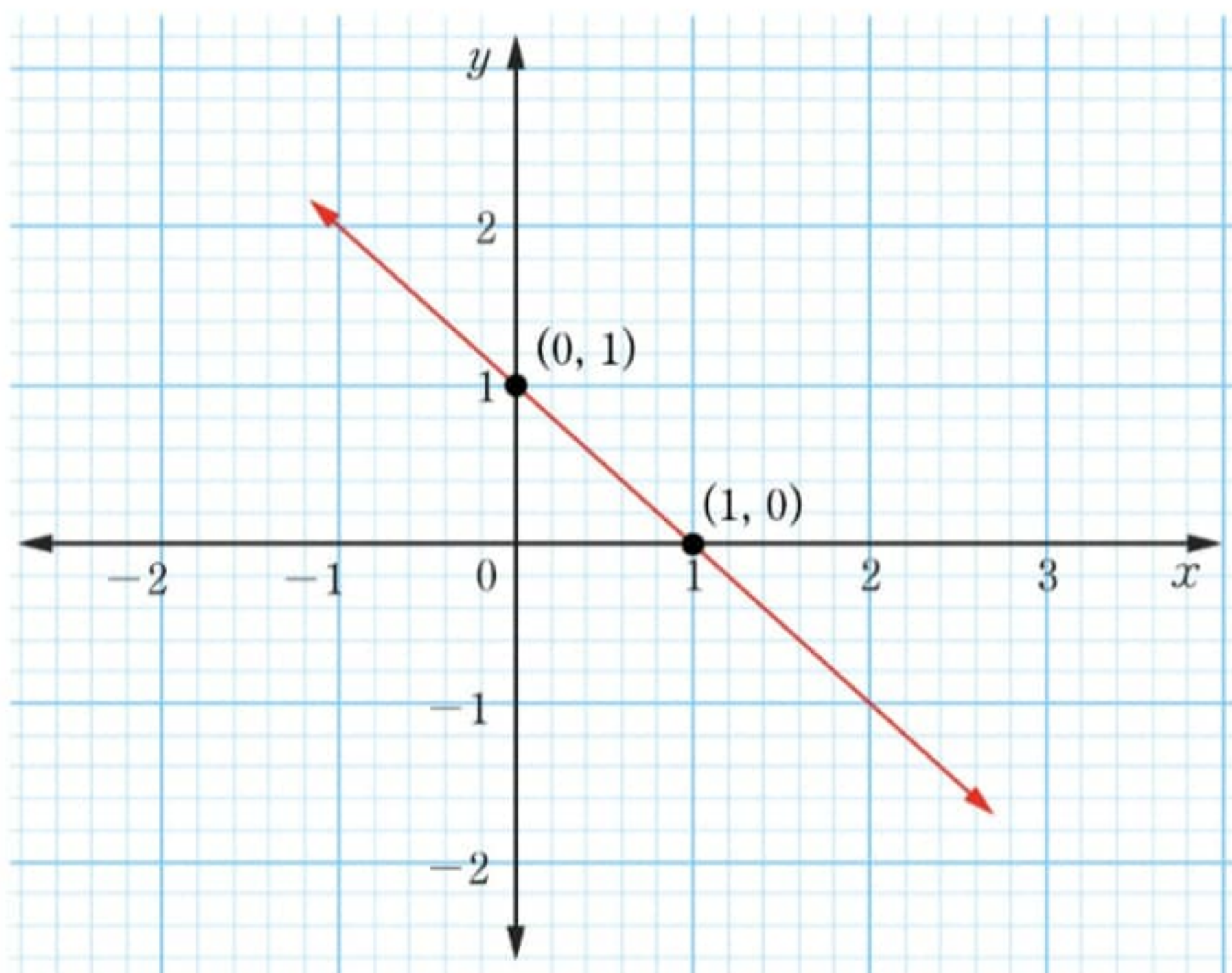
$m = 3$

(b)



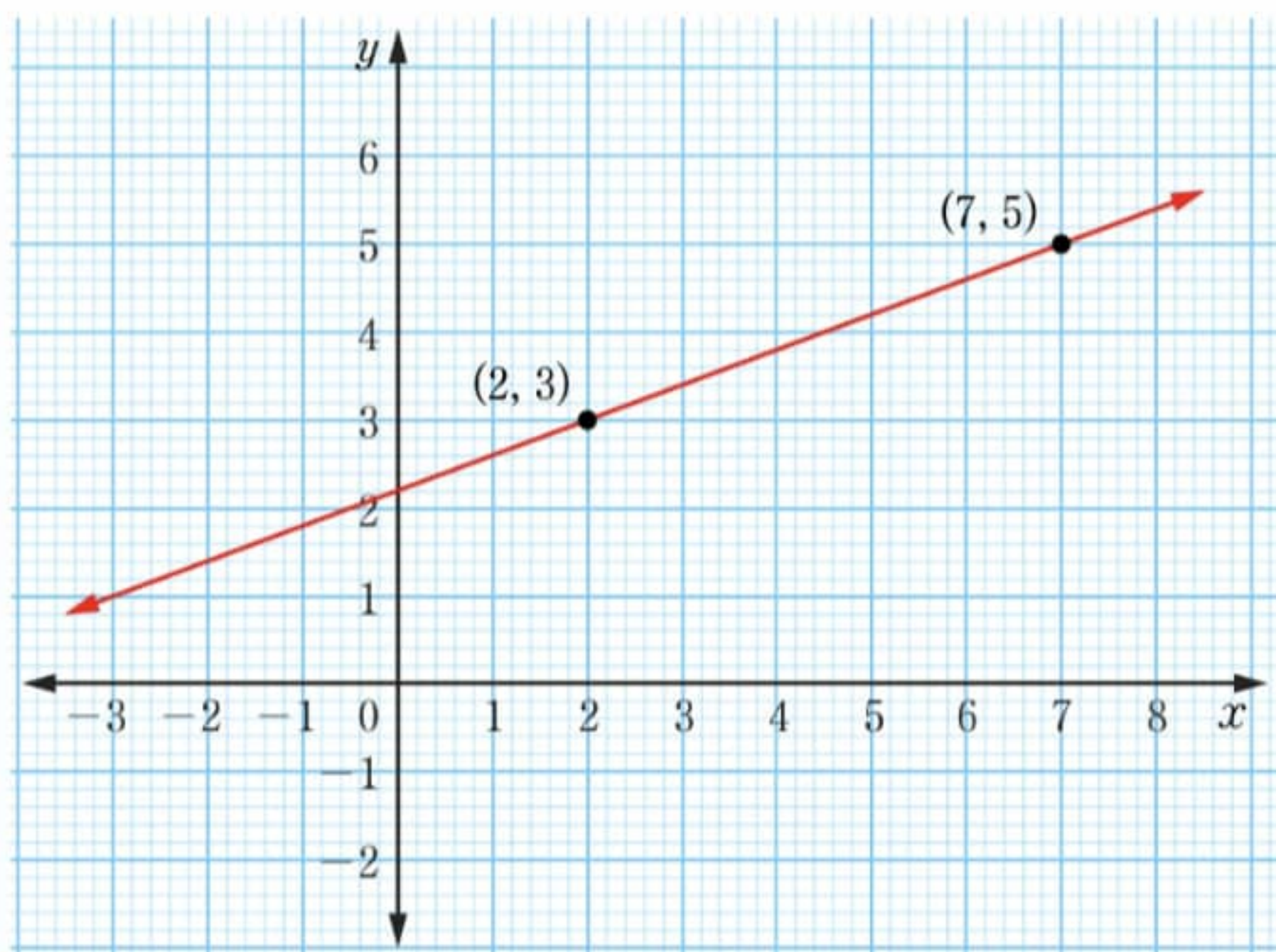
$m = 1$

(c)



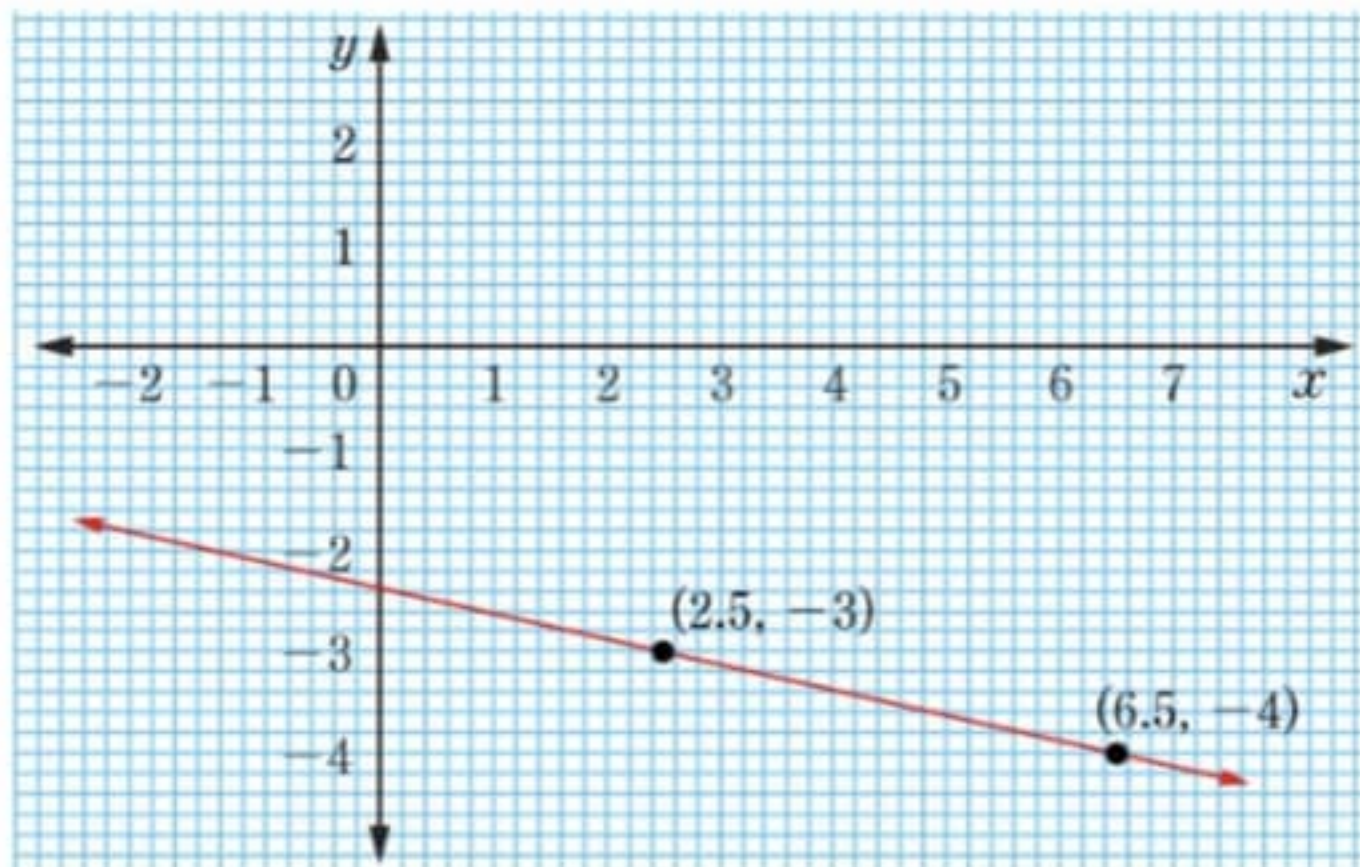
$$m = -1$$

(d)



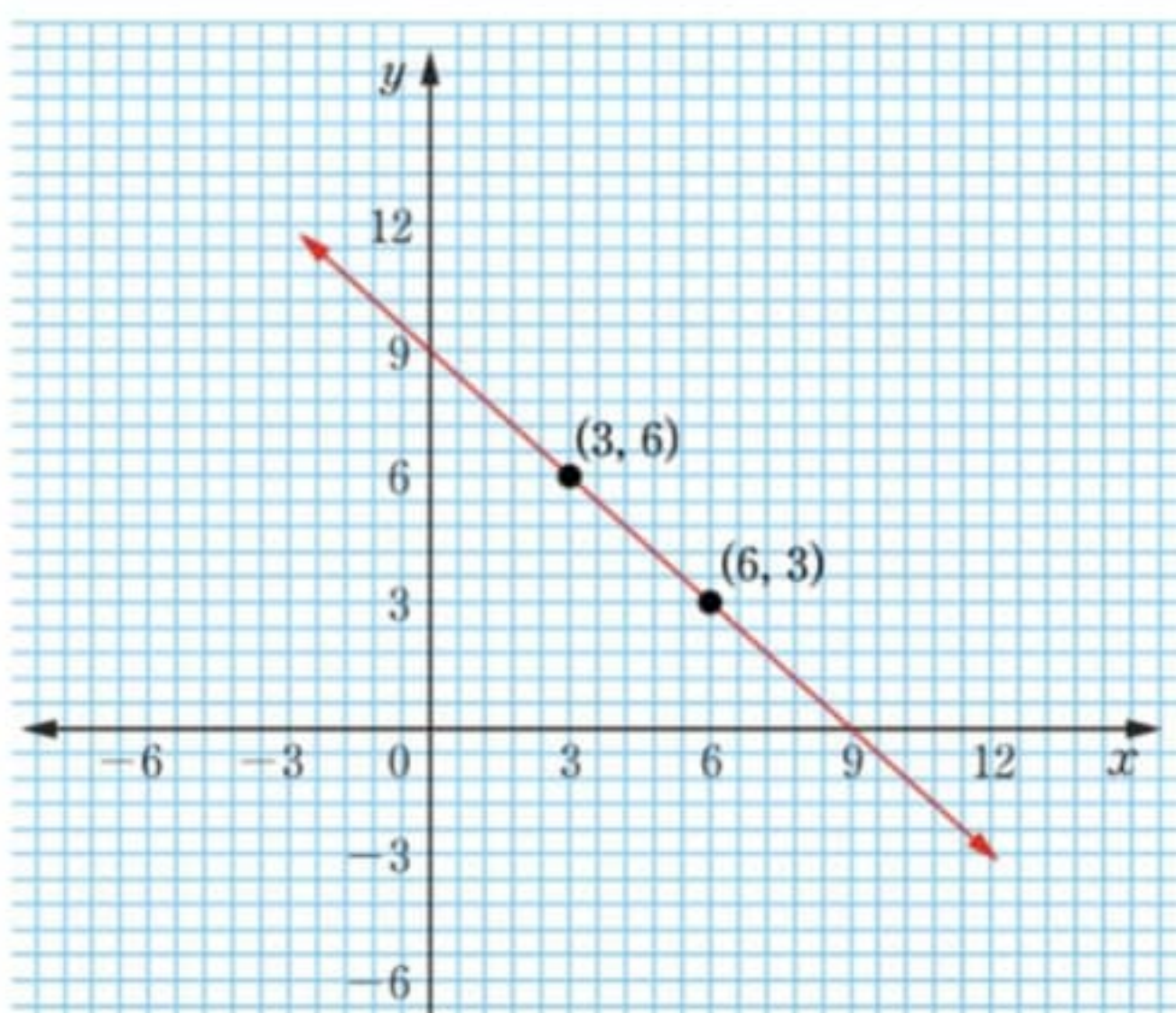
$$m = -$$

(e)



$$m = -\frac{1}{4}$$

(f)



$$m = -1$$

5. (a) $y = -4x + 8$, $m = -4$, $(0, 8)$

(b) $y = \frac{1}{6}x + \frac{1}{3}$ $m = \frac{1}{6}$, $\left(0, \frac{1}{3}\right)$

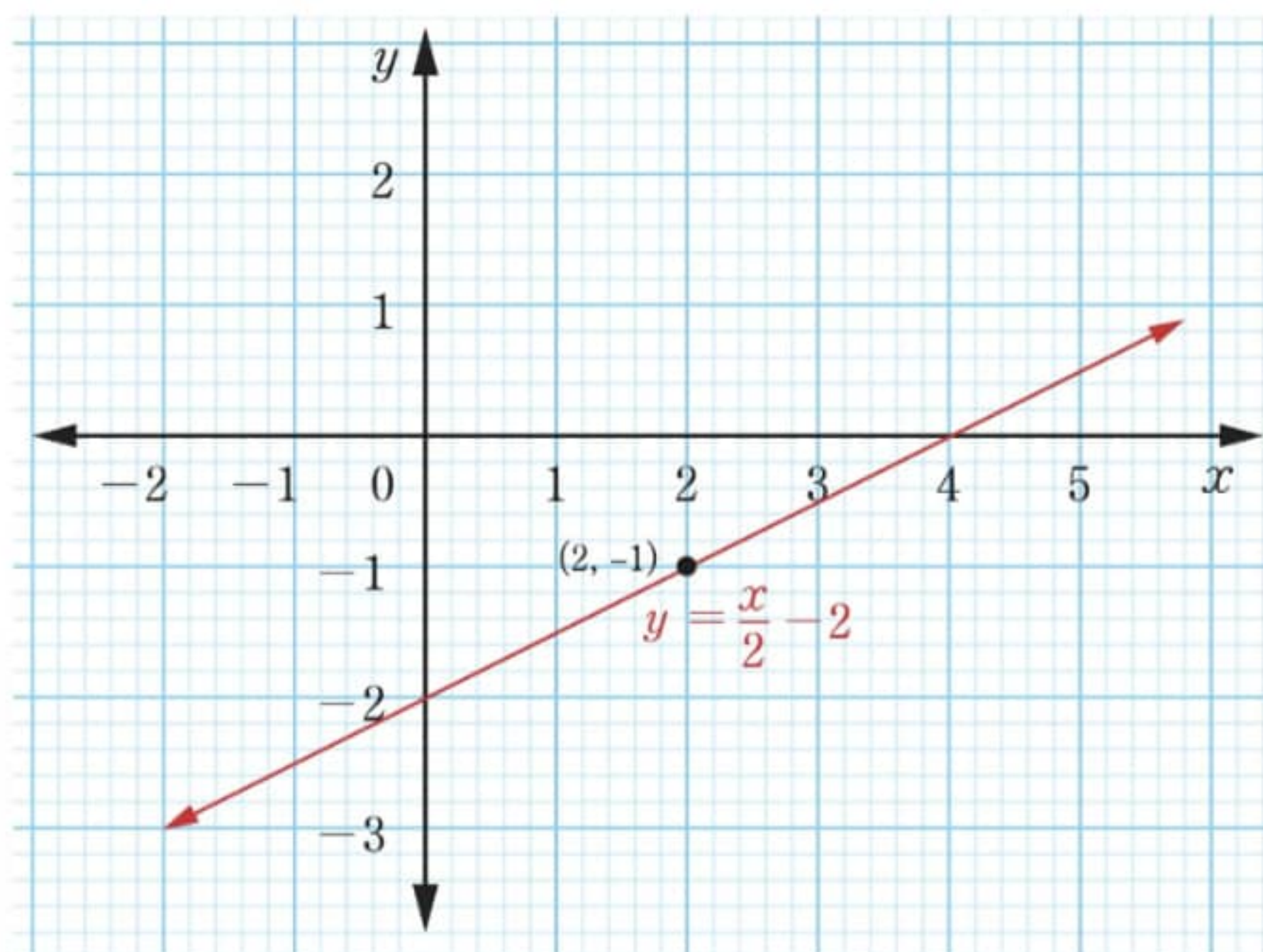
(c) $y = 2x - 2$, $m = 2$, $(0, -2)$

(d) $y = -\frac{3}{4}x + \frac{23}{4}$ $m = -\frac{3}{4}$, $\left(0, \frac{23}{4}\right)$

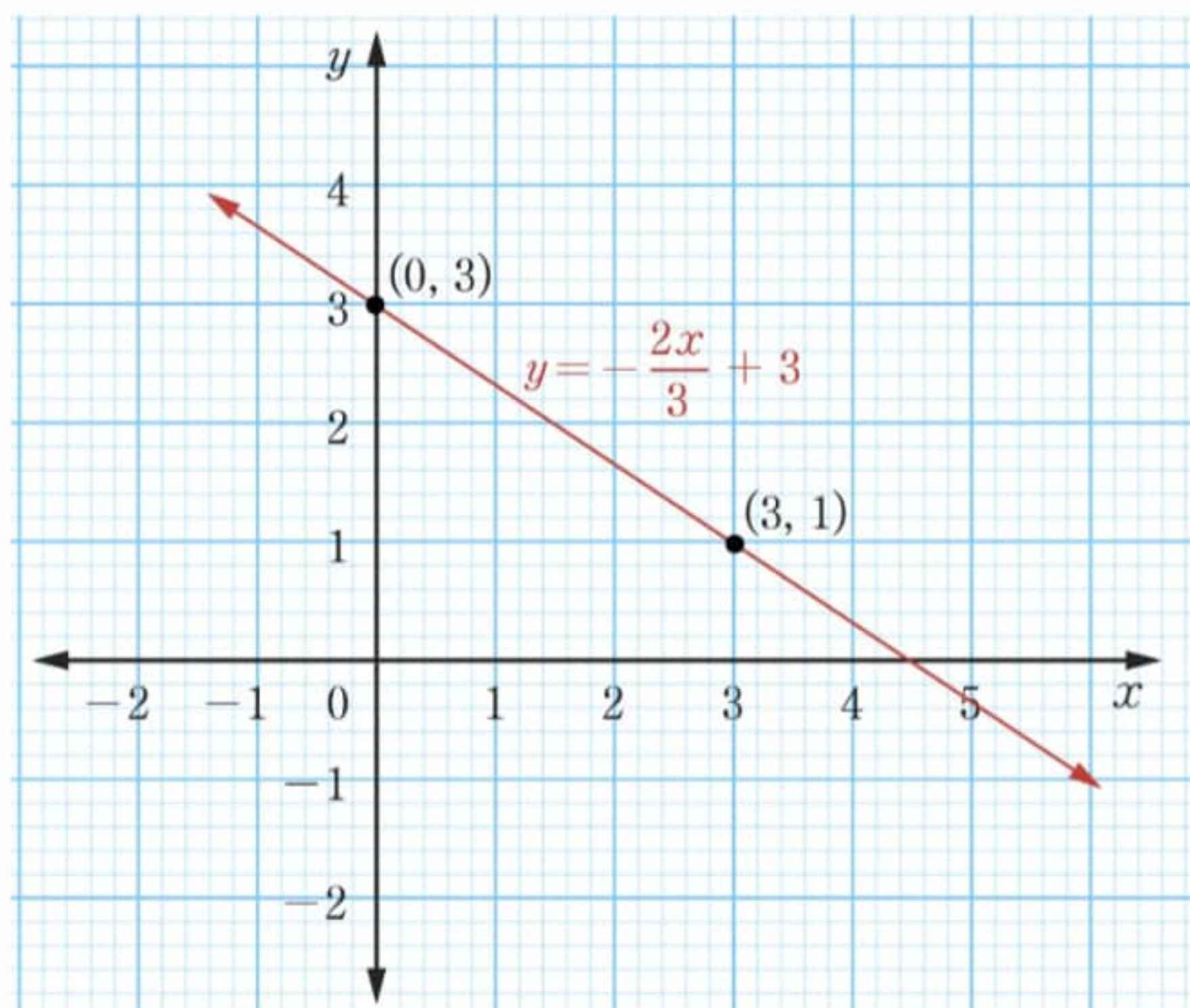
(e) $y = -\frac{3}{4}x - 1$ $m = -\frac{3}{4}$, $(0, -1)$

(f) $y = -\frac{3}{4}x - 3$ $m = -\frac{3}{4}$, $(0, -3)$

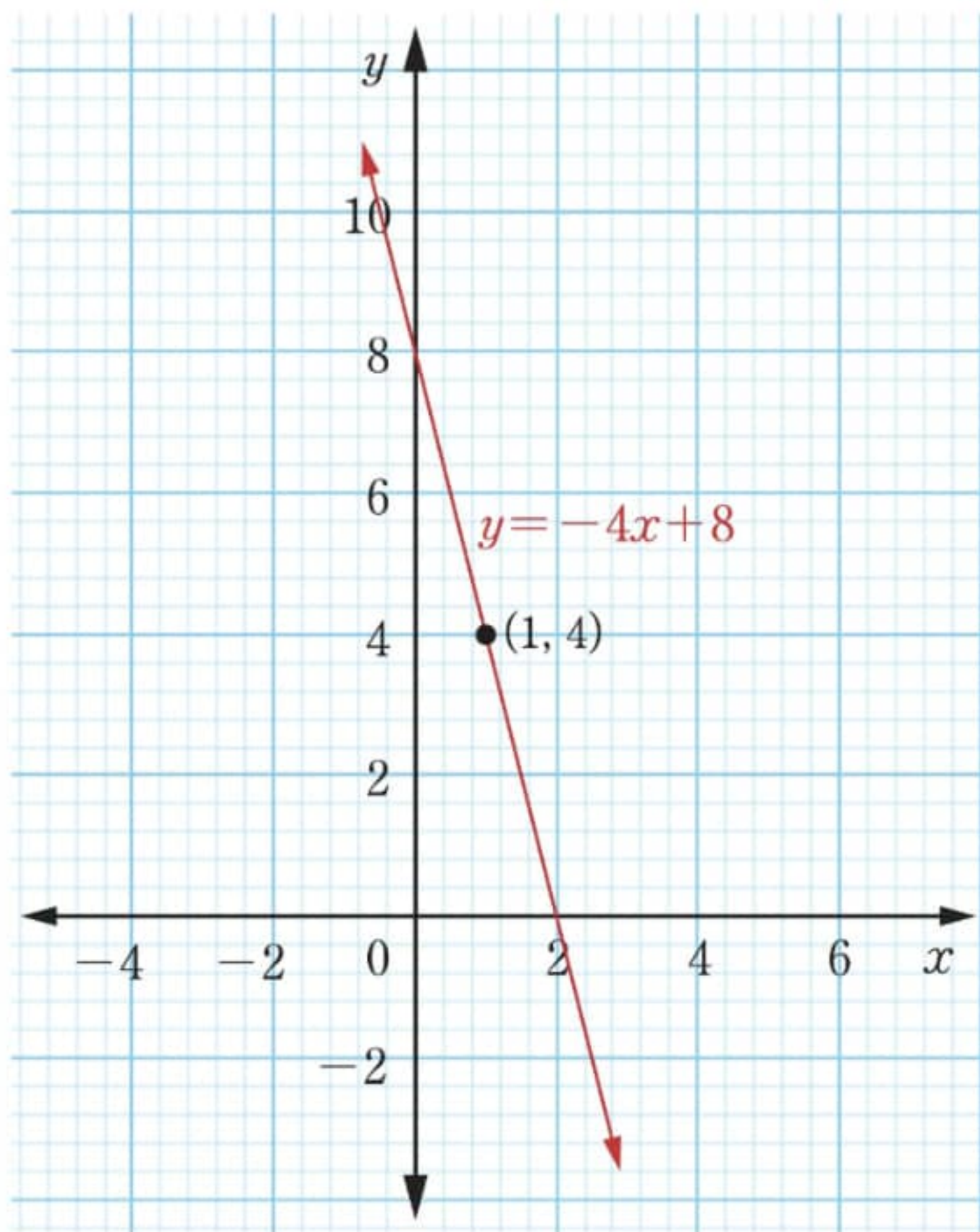
7. (a)



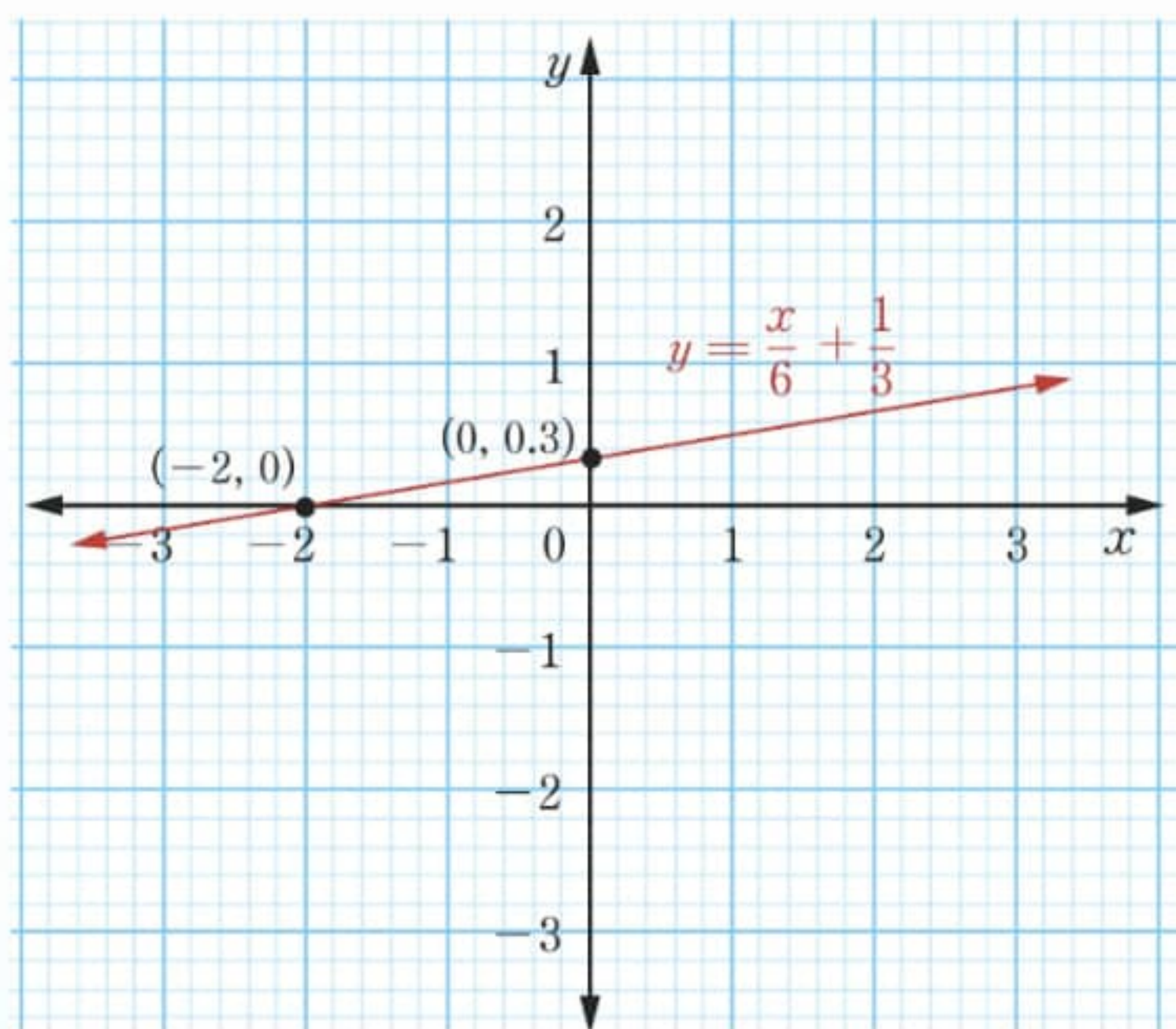
(b)



(c)



(d)



9. (a) $y = -\frac{4}{3}x - 2$ (c) $y = -\frac{3}{2}x - 3$
(b) $y = -6x + 6$ (d) $y = -\frac{3}{2}x$
11. $3x - 5y - 5 = 0$
13. (a) $x + 2y = 9$ (b) $y = -\frac{2}{5}x - \frac{14}{5}$

15. 6 white breads and 2 brown breads

17. The cost of one cup of tea is Tshs 200 and the cost for a cake is Tshs 300

19. $(-3, -5)$

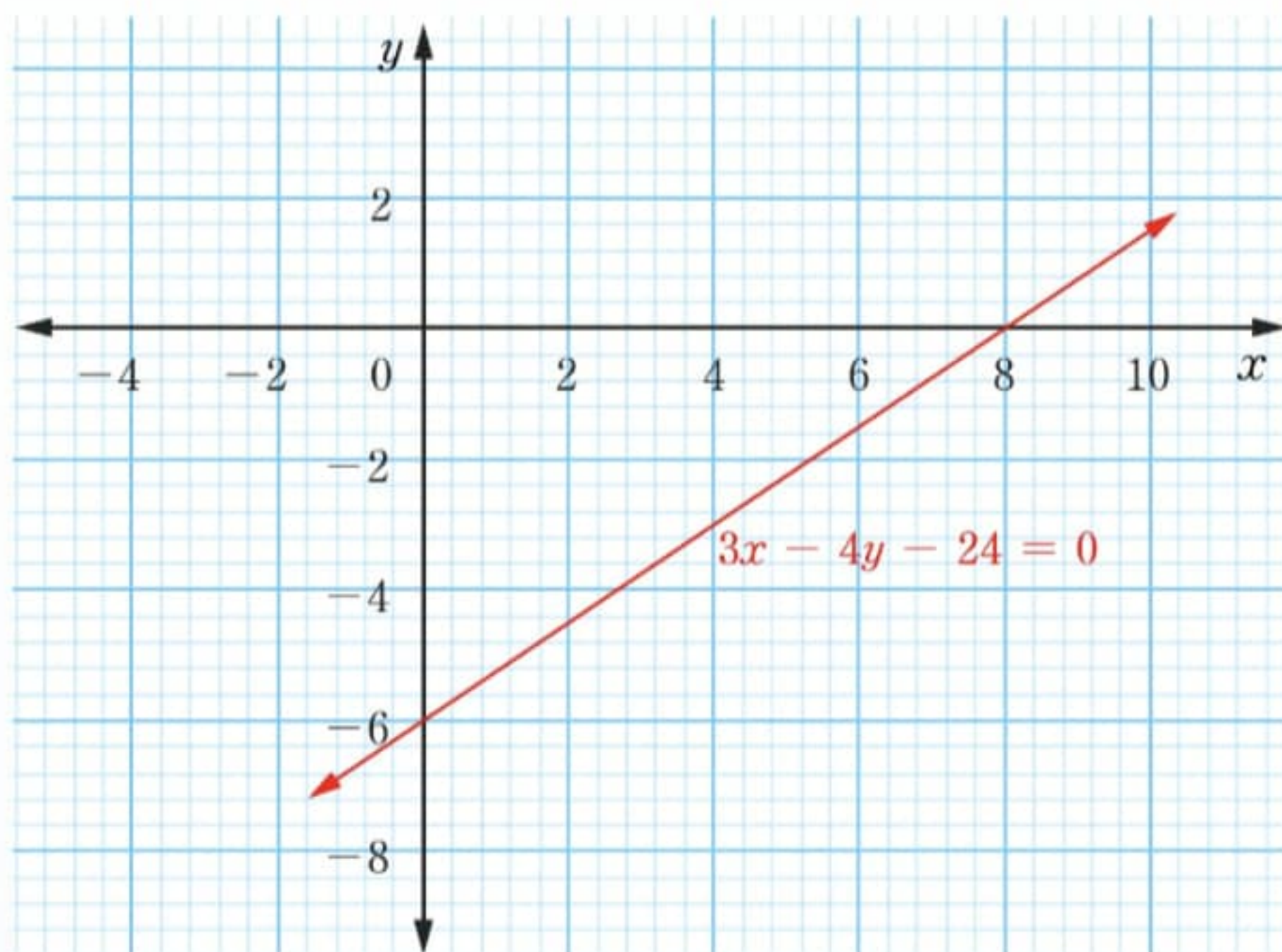
21. (b) $(4, 0)$ and (d) $\left(\frac{4}{3}, 2\right)$

23. $x - 4y - 24 = 0$

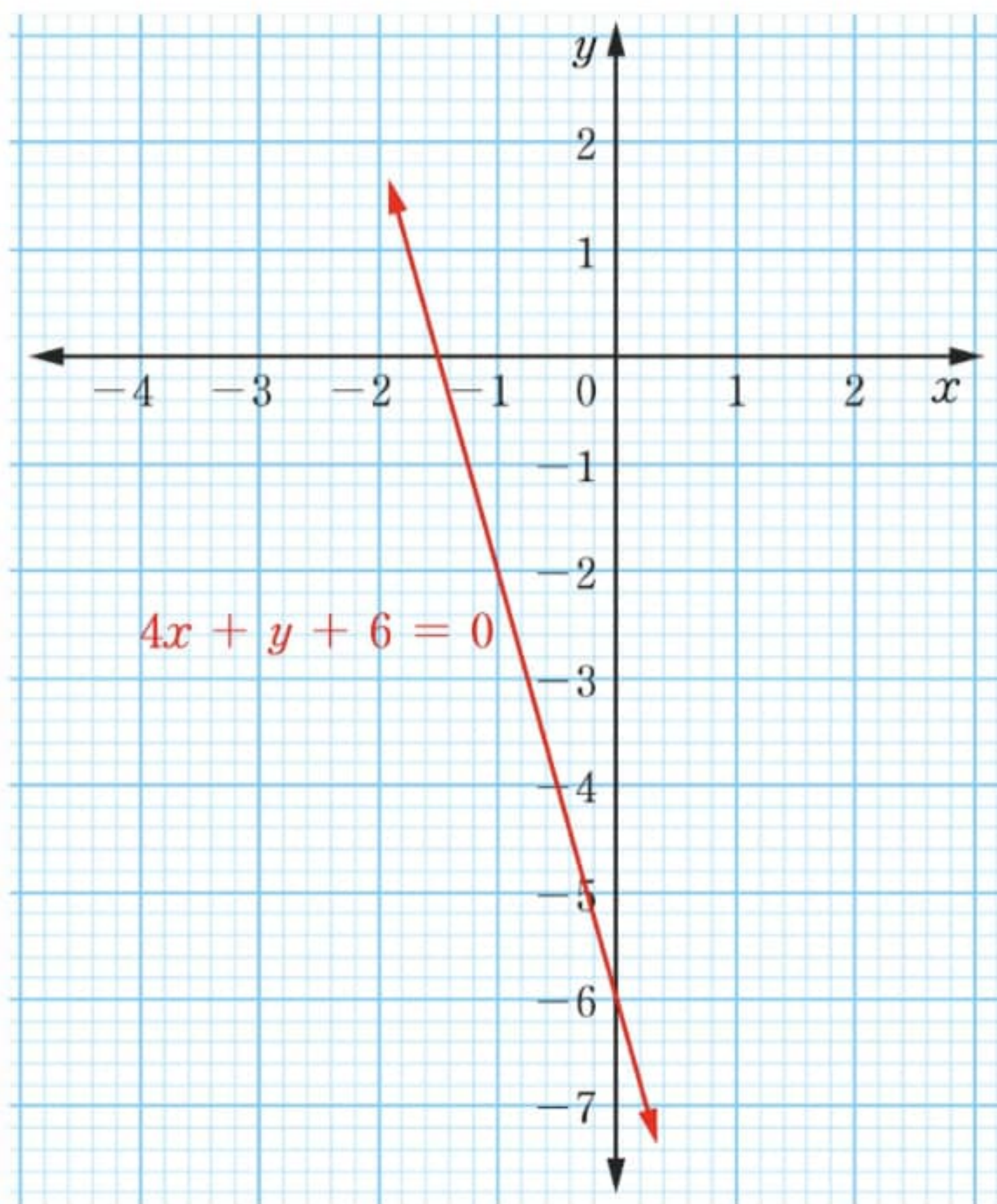
25. $5x - 6y = -5$

27. 12 units of milk and 8.8 units of juice

29.



31.



33. (a) Gradient = 3, y -intercept is $(0, -12)$

(b) Gradient = $\frac{7}{2}$, y -intercept is $\left(0, \frac{3}{2}\right)$

(c) Gradient = $-\frac{a}{b}$, y -intercept is $\left(0, -\frac{c}{b}\right)$

(d) Gradient = $-\frac{b}{a}$, y -intercept is $(0, b)$

35. $x + 4y + 8 = 0$

Glossary

Algebra:	A branch of Mathematics that substitutes letters for numbers. It deals with representing numbers through variables.
Coefficient:	A number that is written along with a variable or it is multiplied by the variable.
Constant:	A value or number that never changes in an expression.
Decimal:	A fraction of which its denominator is a multiple of 10, 100, 1 000, and so on.
Integer:	A number with no fractional part (no decimals).
Natural numbers:	Counting numbers starting with 1, 2, 3, 4, 5, ...
Number line:	A straight line marked with equal intervals, and labelled with numbers.
Variable:	A letter used to represent a numerical value that changes.
Whole numbers:	All non-negative numbers including zero which are not fractions and decimals.

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